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Fiscal-Monetary Interactions: The Role of Debt Maturity

Prepared by Luca Bettarelli, Francesco Frangiamore, Davide Furceri, Anh Dinh Minh Nguyen, Francesco Scianna

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Authorized for distribution by Era Dabla-Norris
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Authors' email addresses:	luca.bettarelli@unipa.it; DFurceri@imf.org; francesco.frangiamore@unipa.it; ANguyen3@imf.org; francesco.scianna@unipa.it

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Luca Bettarelli
University of Palermo

Davide Furceri
International Monetary Fund

Francesco Frangiamore
University of Palermo

Anh Dinh Minh Nguyen
International Monetary Fund

Francesco Scianna
University of Palermo *

September 29, 2026

Abstract

This paper examines how the maturity structure of sovereign debt shapes fiscal–monetary interactions in the United States. Using exogenous monetary policy shocks and nonlinear local projections, we show that debt maturity influences both the fiscal consequences of monetary tightening and its macroeconomic transmission. When the government relies more heavily on short-term debt, contractionary monetary policy shocks lead to large and persistent increases in public debt, more pronounced deteriorations in fiscal and financial conditions, and significantly larger declines in output, inflation, and private investment. These effects are substantially attenuated when debt maturity is longer. We rationalize the findings with a model in which a constrained intermediary holds both private credit and long-duration government debt and carries its bond portfolio at amortized cost. The intermediary's opportunity cost of lending therefore depends in part on the yield locked in when the bonds were acquired, rather than solely on their one-period holding return, allowing sovereign debt maturity to directly influence the price of private credit.

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*Emails: Bettarelli (luca.bettarelli@unipa.it); Furceri (DFurceri@imf.org); Frangiamore (francesco.frangiamore@unipa.it); Nguyen (ANguyen3@imf.org); Scianna (francesco.scianna@unipa.it).
Corresponding author: Davide Furceri

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1 Introduction

> "Many governments have responded to the changing interest rate environment by significantly shortening the average maturity of the debt they issue. This means that the fiscal consequences of monetary tightening are now being felt faster and more acutely: the insulation that longer average maturities once provided is being progressively eroded."
>
> *Isabel Schnabel*, Member of the Executive Board of the ECB
> Speech at the Fifth Annual Charles Goodhart Lecture, May 7, 2026

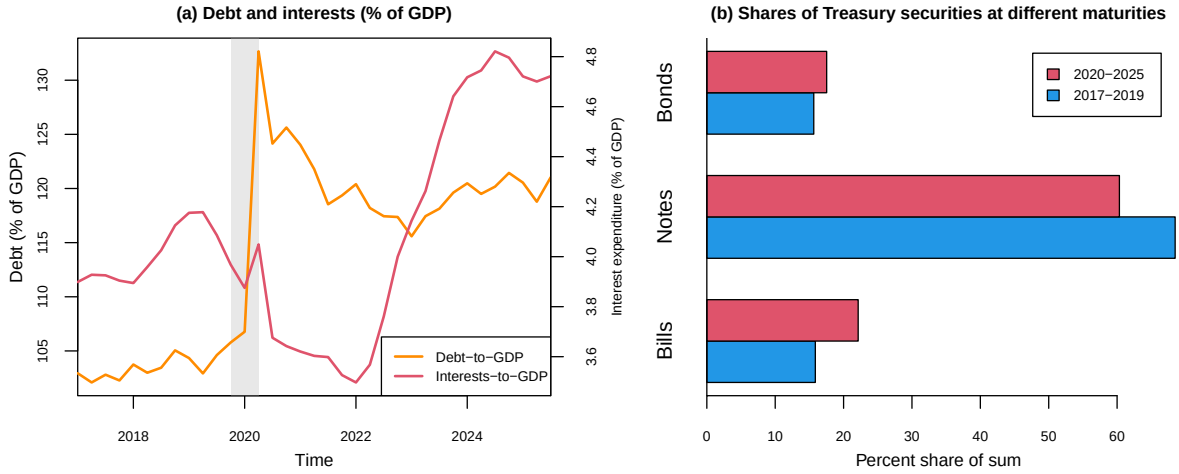
Over the past decade, the interaction between monetary policy and public debt dynamics has re-emerged as a central macroeconomic issue (Hofmann, Lombardi, Mojon, and Orphanides, 2021; Campos, Cardoso-Costa, Gomes, and Jacquinot, 2025; Cantore and Leonardi, 2025). Following the Global Financial Crisis and the COVID-19 pandemic, sovereign debt ratios in advanced economies increased to levels not observed since the post-war period (Arslanalp, Eichengreen, and Simpson-Bell, 2025; IMF, 2026), while central bank balance sheets expanded substantially through large-scale asset purchases (Blanchard, 2019; Cochrane, 2022). The subsequent inflation surge in 2021-2022 forced central banks to implement one of the sharpest tightening cycles in recent decades, rapidly increasing sovereign financing costs. In this environment, concerns about fiscal dominance and the limits of monetary policy normalization have re-emerged prominently in both policy discussions and recent academic work (Cochrane, 2022; Campos et al., 2025). As public debt and refinancing needs rise, higher policy rates increasingly affect sovereign interest expenditures and debt dynamics, potentially creating tensions between price stability objectives and fiscal sustainability.

An important but relatively underexplored dimension of these interactions concerns the maturity structure of sovereign debt. The fiscal sensitivity of monetary policy depends not only on the level of public debt (Sterk and Tenreyro, 2018), but also on the speed at which changes in policy rates are transmitted into sovereign refinancing costs. When the government relies heavily on short-term debt issuance, monetary tightening feeds more rapidly into interest expenditures and rollover needs, increasing public finances' exposure to higher interest rates and reinforcing the transmission of policy shocks through fiscal and financial channels. Shorter sovereign debt maturity therefore reduces the insulation provided by long-duration debt, potentially amplifying the effects of policy shocks on debt dynamics, financial conditions, and real activity (Bhandari, Evans, Golosov, and Sargent, 2021). More broadly, these mechanisms suggest that sovereign debt management can itself shape the strength of monetary transmission by linking the maturity structure of public debt to macroeconomic stabilization outcomes. In this sense, sovereign rollover exposure affects not only the fiscal consequences of monetary tightening, but also the propagation of monetary policy across the broader economy.

Recent issuance patterns suggest that these mechanisms have become quantitatively more important in the post-pandemic period. US public debt rose from about 105 percent of GDP in 2019 to nearly 130 percent after the COVID-19 recession, while interest expendi-

tures increased sharply following the subsequent monetary tightening cycle, reaching almost 5 percent of GDP by 2024 (Figure 1, panel a). At the same time, the composition of Treasury issuance has shifted toward shorter maturities: the share of Treasury bills increased substantially in 2020–2025 relative to 2017–2019, from about 17 percent to 22 percent, while the share of longer-term notes declined markedly and the share of bonds remained broadly unchanged, with only a slight increase (Figure 1, panel b). Moreover, recent estimates suggest that, absent an increase in longer-dated Treasury issuance, bills would account for 28 percent of total issuance by 2028 (The Economist, 2026).

FIGURE 1: RECENT DEVELOPMENTS IN U.S. PUBLIC DEBT



Note: Panel (a) shows the recent evolution of public debt and government interest outlays, both expressed as shares of GDP. The shaded area denotes the COVID-19 recession. Panel (b) reports the shares of Treasury bills, notes, and bonds in their combined total, computed separately for two periods: 2017–2019 and 2020–2025.

This change in debt composition is important because yields on short-term government securities adjust more rapidly to changes in monetary policy rates. As a result, a larger share of short-term debt increases the exposure of sovereign financing costs to monetary tightening and raises rollover risk, making monetary policy rates a more important component of fiscal dynamics. In this paper, we quantify these mechanisms along two complementary dimensions of fiscal-monetary interactions. We first investigate how monetary policy shocks affect fiscal outcomes under different degrees of sovereign rollover exposure. We then examine how the maturity structure of public debt shapes the transmission of monetary policy to macroeconomic and financial variables. To this end, we empirically focus on the U.S. economy using quarterly data from 1969Q1 to 2019Q4. Our baseline measure of exogenous monetary policy is the Romer and Romer (2004) narrative shock, purged of its predictable component following Miranda-Agrippino and Ricco (2021). This measure is well suited to our long historical sample, which is dominated by periods when monetary policy operated mainly through changes in the short-term policy rate. By contrast, high-frequency surprises—used in our robustness checks—are better suited to periods when communication, forward guidance, and unconventional policies play a larger role. Because our analysis requires a consistent measure of policy shocks over a sample extending to the 1960s, the Romer

and Romer shock is better aligned with our empirical objective. Sovereign debt maturity is measured using the share of Treasury bills in total marketable Treasury debt, computed from U.S. Treasury data. To estimate the dynamic effects of monetary policy shocks, we employ a local projections (LP) framework à la Jordà (2005), complemented by a smooth-transition regime-switching framework in which the transmission of monetary policy varies with the degree of sovereign rollover exposure. Specifically, the economy transitions smoothly between low and high short-term debt regimes according to the share of Treasury bills outstanding. Our framework therefore allows us to study how the maturity structure of sovereign debt alters the fiscal and macro-financial propagation of monetary policy shocks.

Results show that the unconditional effects of an exogenous contractionary monetary policy shock, raising the federal funds rate by 100 basis points on impact, on public debt dynamics are relatively limited and emerge gradually over time. In the linear (unconditional) specification, real public debt does not respond significantly on impact but increases over the medium run, rising by about 4 percent after three years. Similarly, the response of the debt-to-GDP ratio remains statistically insignificant throughout the horizon, although the estimated impulse responses are positive in the medium term. These effects, however, depend critically on the maturity structure of government debt. In periods characterized by a high share of Treasury bills, contractionary monetary policy shocks lead to persistent increases in both real public debt and the debt-to-GDP ratio: real debt rises by about 10 percent, while the debt-to-GDP ratio increases by roughly 3 percentage points in the medium run. This is particularly relevant today given that the share of T-bills is currently high, at around 22%, and is expected to reach 28% by 2028 (The Economist, 2026). By contrast, when the share of T-bills is low, the responses of both debt variables are small and statistically indistinguishable from zero.

To better understand the fiscal transmission mechanism, we examine the responses of government revenues, primary expenditures, and the effective interest rate. The results show that, when the share of short-term debt is high, revenues decline more sharply, primary expenditures increase temporarily, and the effective interest rate rises more strongly, thereby contributing to a deterioration in fiscal conditions. By contrast, when the T-bill share is low, revenues improve temporarily, expenditures decline, and the contemporaneous response of effective rates is considerably more muted.

Shorter debt maturity also amplifies the broader macroeconomic effects of monetary tightening. When the T-bill share is high, output falls by about 2 percent, price (GDP deflator) declines by roughly 2 percent over the medium run, and private investment contracts by about 6 percent; these responses are weak or insignificant when the T-bill share is low. Financial conditions tighten more sharply as well: the excess bond premium rises by up to 0.2 percentage point, while Aaa and Baa corporate yields increase by about 0.7 and 0.9 percentage point, respectively. Overall, the results point to a state-dependent fiscal-financial mechanism through which greater sovereign rollover exposure magnifies monetary transmission.

Our findings are robust across alternative specifications and identification strategies. Two checks directly address concerns about the exogeneity and independent role of debt maturity. First, using debt maturity measures lagged by two periods yields quantitatively similar results, and maturity does not systematically respond to monetary policy shocks. Second, the baseline estimates remain stable when controlling for potential determinants of

maturity—including initial debt, interest rates, inflation, quantitative easing, and differential linear and quadratic trends—as well as recessions, the Volcker disinflation, the 1970s, and the Global Financial Crisis. Thus, the nonlinear effects do not merely reflect high-debt periods, distinct macroeconomic environments, central bank balance sheet operations, secular trends, or major historical episodes. Beyond these key exercises, the results are also robust to a broad set of alternative specifications and identification strategies, such as the use of alternative local projection specifications and using the monetary shocks as instruments. In addition, the results remain quantitatively similar when monetary policy shocks are identified using high-frequency surprises within a proxy SVAR framework following Bauer and Swanson (2023), covering the period 1963Q1-2025Q3. The findings are also robust to using monthly rather than quarterly data and to alternative nonlinear specifications, including linear interaction models and regime splits based on the historical distribution of the Treasury bill share. Similar results emerge when maturity is measured by the share of privately held short-term debt, available from 1975Q1 to 2019Q4. Overall, the evidence consistently identifies sovereign debt maturity as an independent determinant of the fiscal and macroeconomic effects of monetary tightening.

Finally, we develop a theoretical model to rationalize these findings, and in doing so identify a mechanism the existing literature has assumed away. The model combines the Gertler and Karadi (2011, 2013) intermediary with the Bernanke, Gertler, and Gilchrist (1999) financial accelerator, nested following Rannenberg (2016): households fund constrained intermediaries, intermediaries allocate between one-period private credit and long-duration government bonds subject to an enforcement constraint, and entrepreneurs borrow under costly state verification. Government debt is a geometric-coupon perpetuity à la Woodford (2001), so average maturity is governed by a single parameter.

Our departure is a single assumption about the intermediary's balance sheet: how its government bond portfolio is carried. In Gertler and Karadi (2013) and Sims and Wu (2021) the portfolio is carried at market, so the return recognized in net worth is the one-period holding return, which is pinned close to the policy rate whatever the bond's duration. Maturity therefore has almost no independent role. We instead carry the portfolio at amortized cost, so what the intermediary recognizes is the yield locked in at acquisition, and that yield does depend on duration. What makes the carrying basis consequential is an asymmetry in the enforcement constraint. The assets a banker could divert are valued at market, since he would sell them; but his continuation value is built on book net worth, because that is what accumulates at the recognized return. The carrying basis determines the size of the gap between the two, and therefore which return (the locked-in yield or the one-period holding return) the intermediary uses to price new bonds and, through the binding constraint, private credit.

Because private loans and government bonds compete for the same scarce pledgeable capacity, the required return on private credit is tied to the government yield with a pass-through coefficient exceeding one due to larger risk of diverting private assets, following Gertler and Karadi (2011, 2013). Debt maturity therefore enters the price of external finance directly. The shorter the maturity, the larger the pass-through of policy rate into the yield, into the lender's opportunity cost, and into the external finance premium. This channel operates alongside the more familiar rollover channel that raises the effective cost of servicing outstanding debt. This rollover exposure combined with a larger contraction in GDP under

short maturity leads to a larger increase debt and debt-to-GDP ratio.

Three theoretical results emerge. First, the model reproduces the key empirical findings: relative to long-maturity debt, short-maturity debt generates a larger contraction in output and investment, a greater rise in private borrowing costs, and a larger increase in the debt-to-GDP ratio. Second, the maturity effect operates entirely through financial intermediaries. When the enforcement constraint is removed, the short- and long-maturity economies are identical in all real variables; when the financial accelerator is removed instead, roughly 70 percent of the maturity gap remains. Maturity matters because a constrained intermediary holds a long-duration asset, not because firms are leveraged. Third, consistent with this mechanism, mark-to-market accounting attenuates the maturity effect: valuing the portfolio at market prices eliminates most of the maturity gap, rendering duration nearly irrelevant. We further show that the maturity gap declines monotonically with the share of the securities portfolio recorded at market value—or, equivalently, increases with the share carried at amortized cost. Thus, the greater the reliance on mark-to-market accounting, the smaller the role of debt maturity.

This paper contributes to the literature on monetary-fiscal interactions and fiscal dominance by showing that the transmission of monetary policy depends crucially on sovereign rollover exposure and the maturity composition of public debt. While a large body of work studies the macroeconomic implications of public debt and fiscal-monetary interactions (Sargent and Wallace, 1981; Leeper, 1991; Cochrane, 2022; Bianchi, Faccini, and Melosi, 2023), recent empirical contributions have also examined how monetary tightening affects fiscal dynamics and debt accumulation (Campos et al., 2025; Bouscasse and Hong, 2026). Existing contributions on debt maturity, however, have mainly focused either on the interaction between government debt supply and asset prices (Greenwood and Vayanos, 2014; Greenwood, Hanson, and Stein, 2015; Krishnamurthy and Vissing-Jorgensen, 2012), or on theoretical aspects of optimal debt management and interest-rate risk (Bhandari et al., 2021). Empirical evidence on how sovereign debt maturity shapes the macroeconomic effects of monetary policy remains comparatively limited. The paper closest to ours is Andreolli (2021), who shows that longer debt maturity dampens the transmission of monetary policy to output through a financing channel. Relative to this contribution, our paper focuses on sovereign rollover exposure associated with short-term debt issuance and emphasizes its implications not only for output dynamics, but also for sovereign debt accumulation, fiscal adjustments, sovereign borrowing costs, and broader macro-financial transmission mechanisms. More generally, the paper shows that the maturity structure of public debt is a key determinant of the state-dependent transmission of monetary policy shocks.

On the theoretical side, the paper connects two literature branches that have developed separately. The intermediary literature following Gertler and Karadi (2011, 2013) and Sims and Wu (2021) studies constrained intermediaries holding long government bonds, but carries those bonds at market, which suppresses any role for duration. The banking literature has established that capital requirements bind on book values and that the speed of loss recognition is itself a policy instrument (Begenau, Bigio, Majerovitz, and Vieyra, 2026); there the wedge between book and market value arises from the delayed recognition of credit losses on loans, a delay that does not depend on maturity. We identify a second and distinct source: a security carried at amortized cost is never re-marked, so a valuation loss does not enter book capital at all, and how large that loss is depends on duration. It is this wedge

that makes the maturity structure of public debt a determinant of monetary transmission.

Our model architecture is similar to Rannenberg (2016), who combines the intermediary constraint of Gertler and Karadi (2011) with the costly-state-verification friction of Bernanke et al. (1999). However, his asset side has one-period private claims, similar to Gertler and Karadi (2011), so its balance sheet has no duration and the maturity of public debt cannot matter as in our model. Relative to Andreolli (2021), whose financing channel operates in the primary market for corporate bond issuance, our channel instead operates through the lender's balance sheet: government debt maturity reaches the price of private credit because a constrained intermediary holds both assets against a common constraint. Finally, the model carries a fuller fiscal block than is usual in this literature, including different tax and spending measures. This allows the fiscal consequences of a tightening, including the deterioration in fiscal balance, the debt path and the effective financing cost, to be traced alongside the credit channel, and therefore better captures the response of debt dynamics.

The remainder of this paper is organized as follows. Section 2 describes the data and empirical methodology used. Section 3 presents the empirical findings, while Sections 4 and 5 illustrate the theoretical model. Section 6 concludes.

2 Data and Methodology

2.1 Data

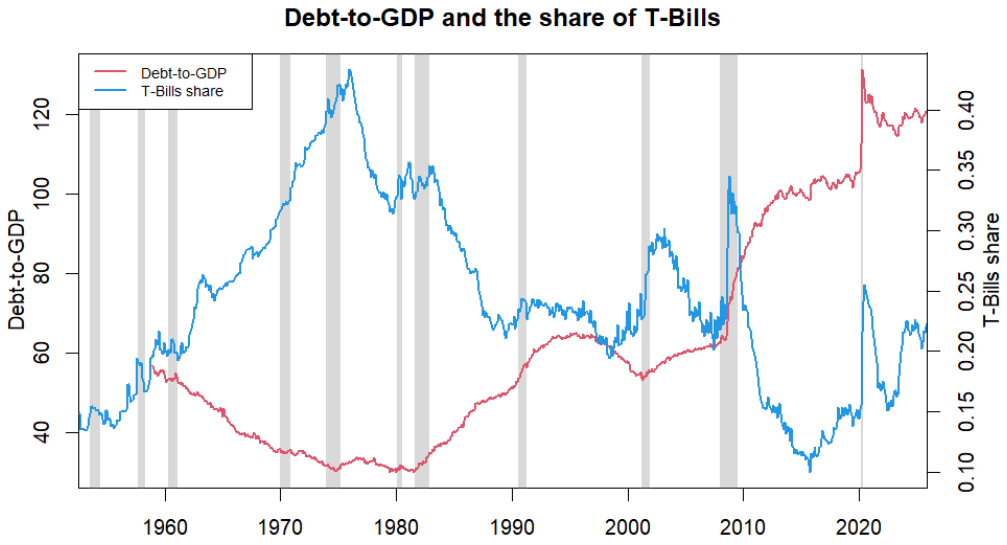
We gather data for the United States from several sources. We construct our main dataset at quarterly frequency since our primary interest lies in the response of fiscal variables, in particular public debt and government budgetary aggregates. We also collect monthly data, which are used to identify monetary policy shocks and to conduct robustness exercises. The dataset covers the period 1963Q1–2025Q3.

The key variable in our analysis is the share of short-term debt in total debt, which we use to study the nonlinear transmission of monetary policy shocks depending on the maturity structure of public debt. We compute this share as the ratio of outstanding Treasury bills to total marketable Treasury debt, defined as the sum of Treasury bills, notes, bonds, Treasury Inflation-Protected Securities (TIPS), and Floating Rate Notes (FRNs). These data are obtained from the U.S. Treasury.[1] Figure 2 plots the share of short-term debt over a long historical period, together with public debt as a percentage of GDP. The figure highlights two key features. First, debt maturity was relatively short during several historical episodes, including the oil crises of the 1970s and 1980s and the first decade of the 2000s. In those periods, however, public debt relative to the size of the economy remained comparatively low, and substantially below the levels observed more recently. Second, the share of short-term debt has begun to rise again, but this time in an environment characterized by a much higher stock of public debt. This combination is central to the challenge currently facing the U.S. economy, as discussed in the Introduction.

As for the monetary policy shocks, in the baseline analysis we use the narrative series

[1] As a robustness check, we also use a measure of the privately held short-term debt share, obtained from HAVER. Figure A1 in the Appendix compares this alternative measure with the T-Bills share used in the baseline analysis.

Figure 2: Historical Evolution of U.S. Debt-to-GDP and the Share of Short-Term Debt



Note: Shaded areas indicate NBER recessions.

of Romer and Romer (2004).[2] The RR shocks are particularly well suited to our analysis because they provide a historically consistent measure of exogenous monetary policy actions over a long sample period in which policy measures were predominantly conventional. This is especially important in our setting, as we need to cover a long historical period during which the federal funds rate was the main operational instrument of the Federal Reserve and in which there is sufficient variation in the maturity structure of public debt. This choice is particularly relevant because our empirical strategy requires exploiting long-run variation in the share of short-term public debt. A long sample allows us to capture changes in debt maturity structure that unfold gradually over time and to study whether monetary policy transmission differs across periods with different degrees of short-term debt exposure. However, these shocks are predictable from factors extracted from a large cross section of U.S. time series as shown by Miranda-Agrippino and Ricco (2021). We therefore purge the Romer and Romer shocks of this predictable component using the ten dynamic monthly factors, following Miranda-Agrippino and Ricco (2021). The factors are extracted from the large panel of time series in the Federal Reserve Bank of St. Louis FRED-MD database (McCracken and Ng, 2016), using the approach developed by Bai and Ng (2021) to handle missing observations in factor extraction. Specifically, we regress the Romer and Romer shock on the first lag of the ten factors, as in Miranda-Agrippino and Ricco (2021), and use the residual as our cleaned narrative shock.

However, to complement the narrative measure, we also employ high-frequency monetary policy surprises from Bauer and Swanson (2023) as a robustness check, thereby incorporating these more recent measures of monetary policy shocks. These surprises are measured in narrow windows around FOMC announcements and therefore exploit market-based revisions

[2]We use the series updated by Miguel Acosta up to 2019, available at this link.

in expectations at the time policy news is released. Using both approaches allows us to assess whether our results are robust to alternative identifying assumptions and sources of variation. Accordingly, the empirical analysis reports estimates based on both narrative monetary policy shocks and high-frequency surprise measures.

Finally, we additionally gather data on our main response variables of interest: the debt-to-GDP ratio, obtained from FRED, and the level of public debt, obtained from the U.S Treasury. Furthermore, we include key macroeconomic variables—real GDP, private fixed investment, and the GDP deflator—from FRED, as well as fiscal variables such as government revenues, expenditures, and interest payments obtained from the Bureau of Economic Analysis (BEA).[3] We also source from FRED the federal debt held by the Federal Reserve as a percentage of GDP, a variable that captures central bank balance sheet policies such as quantitative easing (QE), which we use as a robustness check in Section 3.2.8. In addition, we consider the following financial variables: the Excess Bond Premium (EBP) of Gilchrist and Zakrajšek (2012), downloaded from the Federal Reserve (Favara, Gilchrist, Lewis, and Zakrajšek, 2016), which is standard in the monetary policy literature and relevant for the transmission of monetary policy shocks; the Aaa and Baa bond yields, which capture financing conditions in high- and medium-grade corporate bond markets and variation in credit risk across the investment-grade spectrum, and whose source is Moody's via FRED; and the effective interest rate, obtained from HAVER, which captures sovereign financial conditions and is particularly relevant for our analysis of the transmission of monetary policy shocks to public debt.

As mentioned above, our main dataset is constructed at quarterly frequency because the fiscal variables and the debt-to-GDP ratio, which are central to our analysis, are originally available at quarterly frequency. However, we also construct a monthly dataset for robustness purposes. In this dataset, quarterly variables are replaced by monthly estimates of GDP and investment taken, respectively, from S&P and Barrett (2025), while government expenditure, revenues, and debt are collected from Haver and are originally available from the U.S. Treasury. This allows us to assess whether our results are robust to the frequency of the data and, at the same time, to exploit the higher-frequency variation in financial variables, which are typically available at monthly frequency.

Tables A1 and A2 in the Appendix report descriptive statistics and sample availability for all variables used in the quarterly- and monthly-frequency analyses, respectively. Our baseline sample is 1969Q1–2019Q4, as determined by the availability of the RR narrative shocks; the longer sample is used in the robustness exercises based on high-frequency monetary policy surprises, which extend from 1963Q1 to 2025Q3.

2.2 Methodology

To assess the effects of monetary policy on macroeconomic variables, we estimate impulse response functions through local projections (Jordà, 2005). Compared to the Vector Autoregression (VAR) models proposed by Sims (1980), local projections (LPs) present some

[3]In the analysis, we convert all nominal variables into real terms using the GDP deflator. This is done for government debt, revenues, primary expenditure (i.e., expenditures minus interest payments), budget balance (revenues minus expenditures), and private fixed investments.

interesting properties that are well-suited for our particular goal. First, they do not require the imposition of dynamic restrictions which are implicit in a VAR. Secondly—and of particular relevance here—LPs are particularly well-suited for capturing non-linearities in the data, as regime-switching LP formulations do not require explicit assumptions about the transition process across regimes. More generally, Montiel Olea, Plagborg-Møller, Qian, and Wolf (2025) show that impulse responses estimated through LPs can equivalently be interpreted as those obtained from a VAR model that includes a large number p of lags, exhibiting lower bias at the expense of higher variance.

In particular, for our baseline we estimate the following h equations:

$$y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)} \tag{1}$$

where y_{t+h} is the variable of interest (log of real debt, debt-to-GDP, log of real GDP and deflator, excess bond premium, etc.) h quarters (months) after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ (twelve in the monthly specification) of dependent variable and shock. Standard errors are adjusted following Newey and West (1987) and are thus heteroskedasticity and autocorrelation (HAC) robust. The coefficient $\beta^{(h)}$ denotes the average (percent) response of the variables of interest h quarters (months) after a 100 bps exogenous monetary policy tightening shock.

Then, to assess how the transmission of monetary policy shocks is affected by the maturity of public debt, we follow Auerbach and Gorodnichenko (2012) and Ramey and Zubairy (2018) and estimate the following regime-switching LP model:

$$\begin{aligned} y_{t+h} = \alpha^{(h)} &+ F(z_{t-1}) \Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell} \Big) \\ &+ \big(1 - F(z_{t-1})\big) \Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell} \Big) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)} \end{aligned} \tag{2}$$

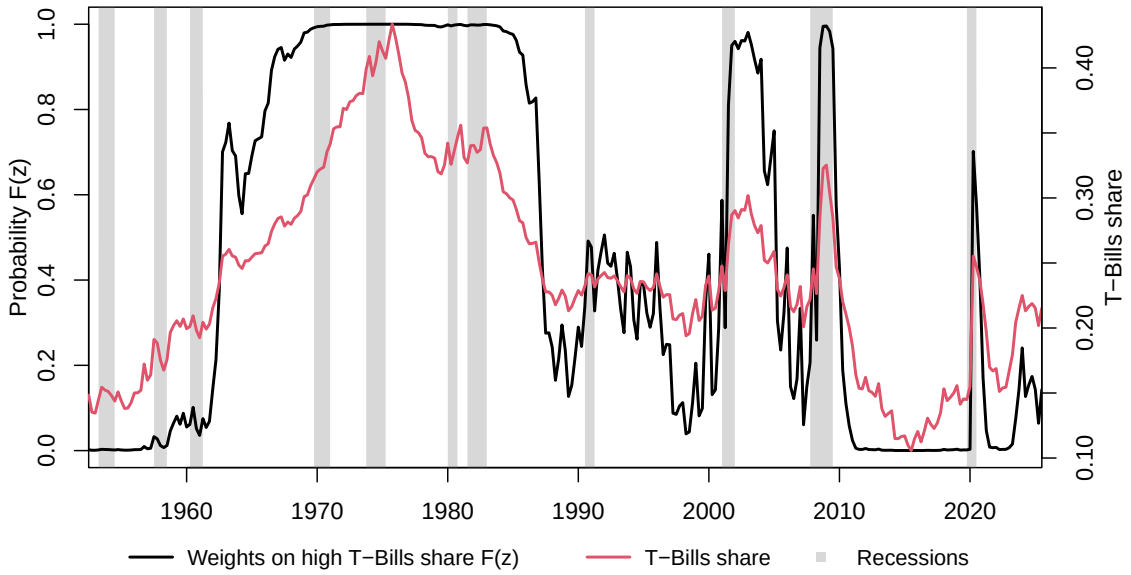
where $F(z_{t-1})$ is an indicator of the regime, dated at $t-1$ following Auerbach and Gorodnichenko (2012), in order to avoid contemporaneous feedback from monetary policy shocks to the share of short-term debt. Specifically, it is a smooth transition function as used in Tenreyro and Thwaites (2016), specified as

$$F(z_t) = \frac{exp(\theta \frac{z_t - \mu_z}{\sigma_z})}{1 + exp(\theta \frac{z_t - \mu_z}{\sigma_z})}, \tag{3}$$

where θ is a parameter determining how abruptly the economy transitions from one regime to another (here set to an intermediate value of 5), z_t is the share of T-Bills over total public debt, μ_z is its mean and σ_z is its standard deviation. Thus, the smooth transition function $F(z_t)$ ranges between zero and one, and can be interpreted as the probability of being in a high short-term debt share regime. Figure 3 reports this probability, together with the T-Bills share. The probability of being in a high short-term debt share regime was very high

and stable throughout the 1970s and the first half of the 1980s, a period characterized by the oil crises, although it does not appear to be closely related to the recessions occurring in those years, as the probability remained constantly close to one throughout the period. Over the same period, the T-Bills share reached its historical peak in the sample, close to 40%, at the end of the 1970s, and remained relatively high, at around 30%, before gradually declining after the recession of the early 1980s. Following this episode, the U.S. economy entered a regime characterized by a very low probability of a high short-term debt share. Although the T-Bills share increased temporarily during the 1990s, the probability of being in a high short-term debt share regime remained low. It increased again during the early-2000s recession and the Global Financial Crisis (GFC), periods that also coincided with relative peaks in the share of Treasury bills. Both measures then declined markedly in the post-GFC period, with the probability of being in a high short-term debt-share regime falling close to zero. However, they rose sharply again during the COVID-19 crisis in 2020.

FIGURE 3: PROBABILITY OF BEING IN A HIGH SHORT-TERM DEBT SHARE REGIME



Note: The black line shows the probability of being in a high short-term debt share regime, given by $F(z_t)$, where $F(z_t) = \exp\left(\theta \frac{z_t - \mu_z}{\sigma_z}\right) / \left[1 + \exp\left(\theta \frac{z_t - \mu_z}{\sigma_z}\right)\right]$, with $\theta = 5$, and z_t denotes the T-Bills share. The T-Bills share is shown by the red line and reported on the secondary vertical axis. The shaded areas indicate NBER recessions.

3 Empirical results

3.1 Baseline

Figure 4 reports our main results on the dynamic response of real public debt and the debt-to-GDP ratio to a monetary policy shock that raises the federal funds rate by 100 basis points on impact. The unconditional response of public debt is relatively limited: real

debt increases significantly only after approximately three years, by almost 4 percent. The response of the debt-to-GDP ratio is even weaker and is statistically indistinguishable from zero at most horizons. Taken together, these results suggest that, on average, monetary policy shocks have only a muted effect on public debt dynamics when the maturity structure of public debt is not taken into account.

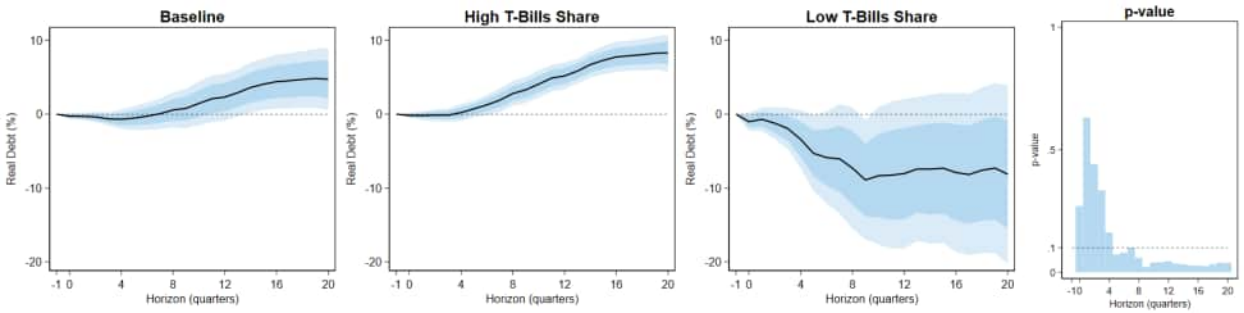
A markedly different picture emerges once we condition on the share of short-term debt. In periods characterized by a high share of short-term debt, measured by the share of T-bills, both real debt and the debt-to-GDP ratio increase persistently following a monetary contraction. The effects are quantitatively large and statistically significant across most horizons: real debt rises by about 10 percent, and the debt-to-GDP ratio increases by approximately 3 percentage points in the medium run—about 1.5 times the standard deviation of debt-to-GDP changes. By contrast, when the T-bill share is lower, the responses of both real debt and the debt-to-GDP ratio are largely statistically indistinguishable from zero.[4] Importantly, the difference between the two regime-specific responses is itself statistically significant, as shown by the p-value from the test of equality between the two impulse responses reported in the last column of each panel.

The results show that monetary tightening worsens public debt dynamics when rollover exposure is high, consistent with a stronger pass-through of policy rates to financing costs under shorter debt maturities. When the T-bill share is low, real debt and the debt-to-GDP ratio instead decline after the shock, though the effects are generally not statistically significant, suggesting that longer maturities partly insulate public finances from short-term rate fluctuations.

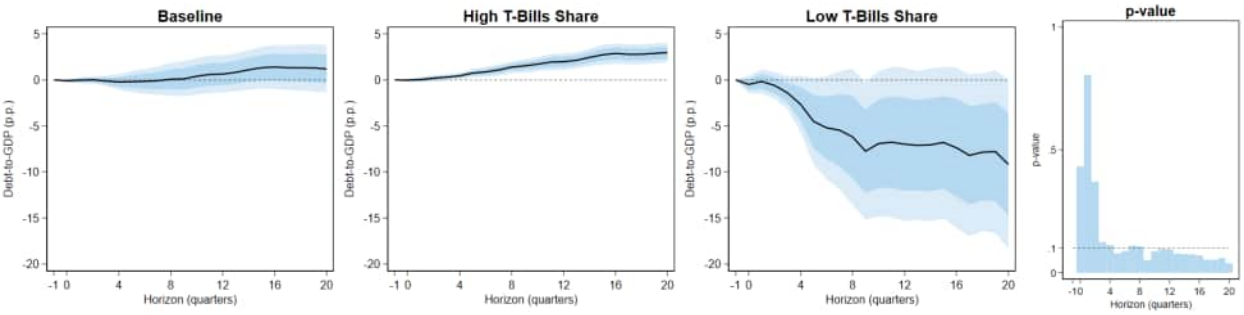
[4]Figure A2 in the Appendix shows that the PDFs and CDFs of monetary policy shocks are broadly similar across regimes, suggesting that our results are not driven by differences in the distribution or sign asymmetry of shocks between high- and low-T-bill-share periods.

FIGURE 4: THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



Panel B. Debt-to-GDP



Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

Figure 5 reports the results for the drivers of public debt. In the linear model, revenues decline following the contraction in economic activity, falling by about 2 percent after ten quarters, while real primary expenditure exhibits a limited and imprecisely estimated response. Effective interest rates increase significantly on impact, with the response peaking at around 0.25 percentage point after approximately six quarters. They subsequently decline persistently, although this decline is not statistically significant.

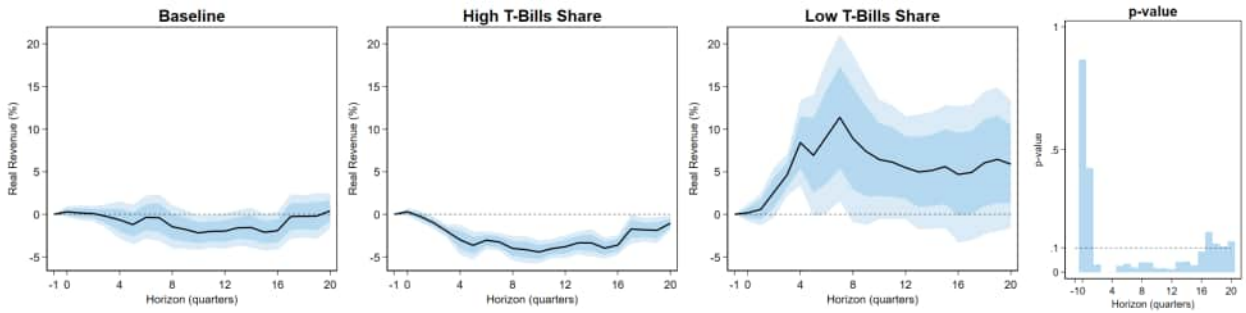
The state-dependent analysis based on the share of short-term debt reveals substantial asymmetries. When the T-bill share is high, revenues fall more sharply and persistently, reaching a decline of about 4 percent, consistent with the stronger contraction in economic activity documented below. Primary expenditure, by contrast, rises temporarily, peaking at about 1 percent—smaller than the revenue decline—thereby contributing to a deterioration in the primary balance. This deterioration reflects both the endogenous response

of fiscal variables to weaker activity through automatic stabilizers, such as unemployment insurance and transfers, and possibly a fiscal response to the contractionary effects of monetary tightening.[5] At the same time, the effective interest rate increases more strongly on impact, consistent with tighter financing conditions. By contrast, when the T-bill share is low, revenues temporarily improve, primary expenditure declines, and the effective interest rate response is more muted. The last columns of the chart show that these regime-specific differences are highly statistically significant. Overall, the results suggest that a larger share of short-term debt amplifies the fiscal consequences of monetary tightening through weaker revenues, higher primary expenditure, and tighter financing conditions.

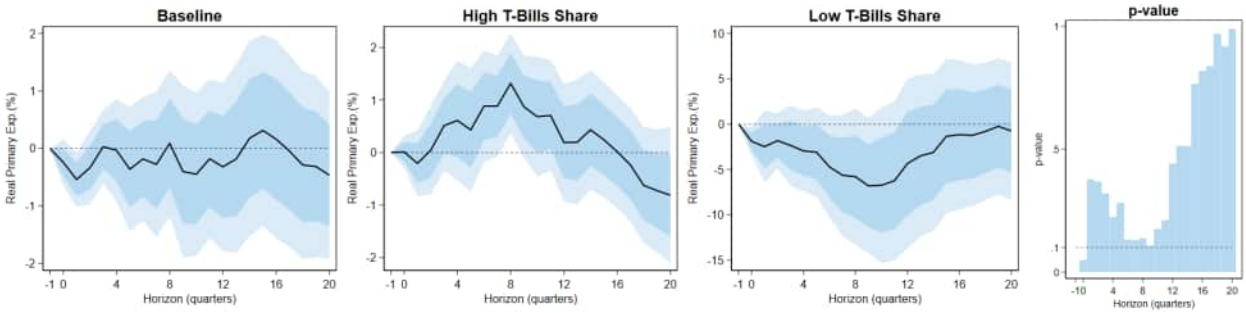
[5]Given these two mechanisms, if fiscal authorities systematically react more aggressively when the T-Bill share is high, then part of the stronger output response that we attribute to debt maturity could instead reflect a different fiscal stance. However, in section 3.2.8 we show that the results are robust to controlling for state-dependent fiscal conditions through interactions between the budget balance and the T-Bill share. This suggests that the stronger effects observed in the high short-term debt regime are not driven by differences in fiscal policy stance across states, but rather by the maturity structure of public debt itself.

FIGURE 5: THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

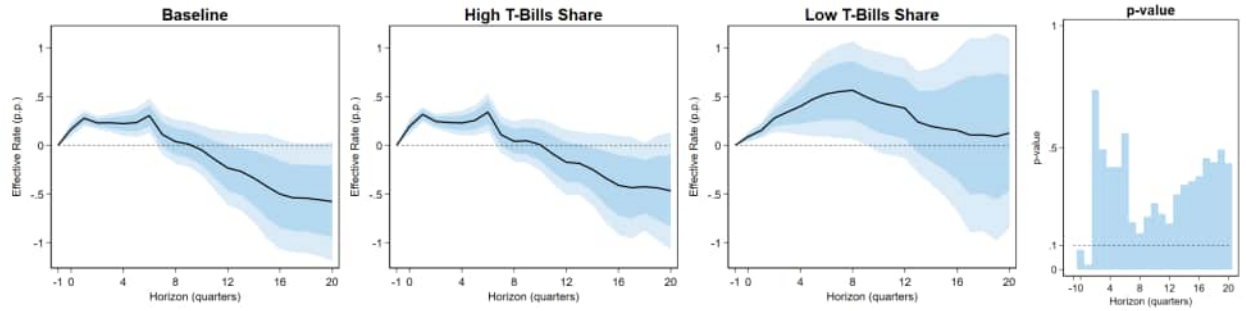
Panel A. Real Revenue



Panel B. Real Primary Expenditure



Panel C. Effective interest rate



Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

Next, we look at the effects of the monetary policy shock on key macroeconomic outcomes.

Figure 6 reports the dynamic responses of real GDP, real private fixed investment and the GDP deflator to the same exogenous monetary policy tightening. The unconditional responses from the linear local projections indicate that output declines moderately, with a peak response of about -1 percent after roughly one and a half years. Real private investment follows the drop in economic activity and declines persistently after a monetary tightening, although the response is statistically significant only at very short horizons. The GDP deflator, instead, falls more persistently and substantially, reaching a decline of around 2 percent after five years. These patterns are consistent with standard monetary transmission mechanisms and empirical evidence.

However, the results from the model that conditions the responses on the maturity structure of public debt reveal important heterogeneity in the transmission of monetary policy shocks to output and prices. In particular, monetary policy shocks have more contractionary effects when the maturity of public debt is shorter. Real GDP falls sharply and significantly when the share of short-term debt is high, reaching a peak decline of about 2 percent after five quarters. By contrast, GDP does not respond significantly when the share of short-term debt is low. The difference between the two regime-specific responses is highly statistically significant in the short to medium run, as shown by the p-values reported in the last column of panel A of Figure 6.

Consistent with the transmission to economic activity, private investment declines substantially when the share of T-bills is high, with the response reaching a trough of about 5 percent after approximately one and a half years. By contrast, the response is not statistically significant when the share of T-bills is low. The difference between the responses across the two regimes is statistically significant over the short to medium run, consistent with the evidence for real GDP, as indicated by the p-values reported in the last column of Panel B in Figure 6.

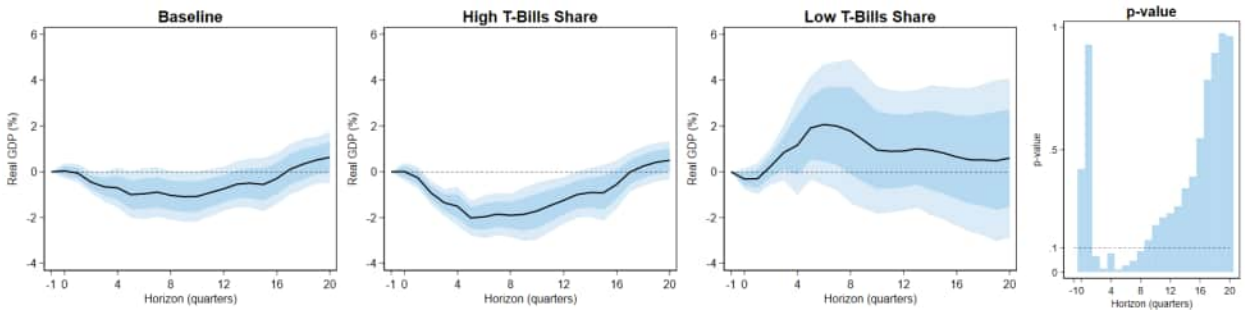
The response of prices displays a similar pattern. The GDP deflator declines significantly only when the share of T-bills is high, possibly reflecting the stronger contraction in aggregate demand associated with the larger decline in economic activity in this regime. However, the difference relative to the low-T-bill regime is not statistically significant due to the large confidence intervals in the low regime, as indicated by the p-values reported in the last column of Panel C. [6]

Overall, these findings are consistent with Andreolli (2021), who shows that industrial production falls more strongly after a monetary policy tightening when the duration of public debt is shorter, while the response of prices is not clearly related to the maturity structure of public debt.

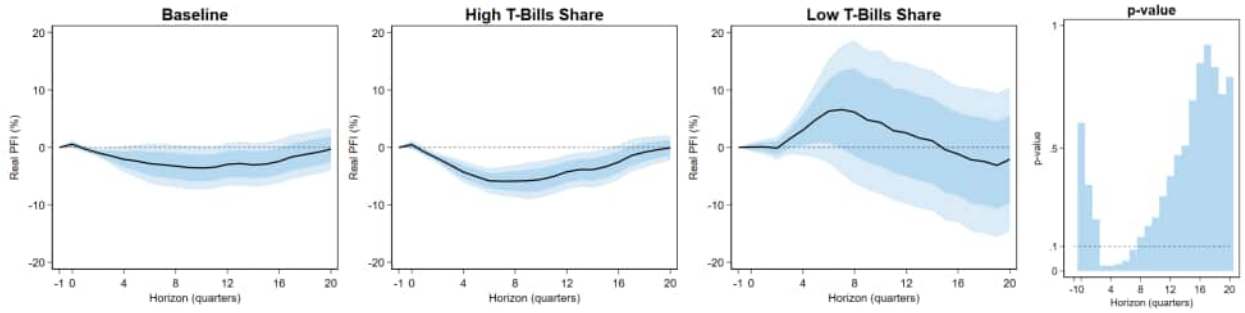
[6] As discussed above, the differences across regimes are statistically significant for real GDP, but not for the GDP deflator. This result does not carry over to the structural shocks identified through high-frequency surprises that we use as a robustness check, where the differences are statistically significant for both variables; see Figure A26 in the Appendix.

FIGURE 6: THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

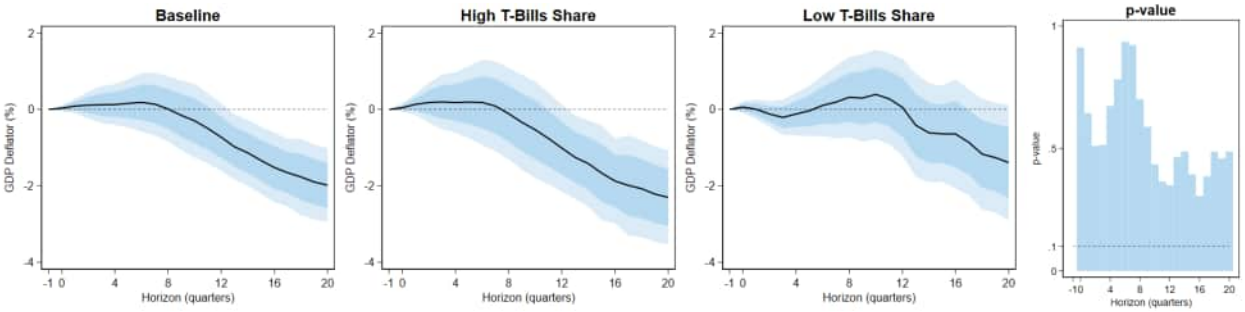
Panel A. Real GDP



Panel B. Real Private Fixed Investments



Panel C. GDP Deflator



Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and the GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

Finally, we report the responses of key financial variables in Figure 7. In the linear specification, the EBP increases on impact and then quickly returns to baseline, consistent with an

immediate but short-lived tightening of financial conditions. Consequently, borrowing costs for the private sector increase for both high-grade and lower-grade borrowers, as reflected in the rise in Aaa- and Baa-rated corporate bond yields. However, state-dependent local projections reveal substantial heterogeneity across debt maturity regimes. When the T-bill share is high, the EBP rises more persistently, peaking at about 0.2 percentage point after roughly five quarters. This prolonged tightening of financial conditions leads to a significant contraction in private investment, consistent with Gilchrist and Zakrajšek (2012), as we have seen in Figure 6.[7] In contrast, the effects are statistically weaker when the T-bill share is low. Consistent with this, private sector borrowing conditions tighten substantially when the share of T-bills is high (i.e., when debt maturity is shorter). By contrast, when the T-bills share is low, there is a modest and very short-lived easing that quickly dissipates.

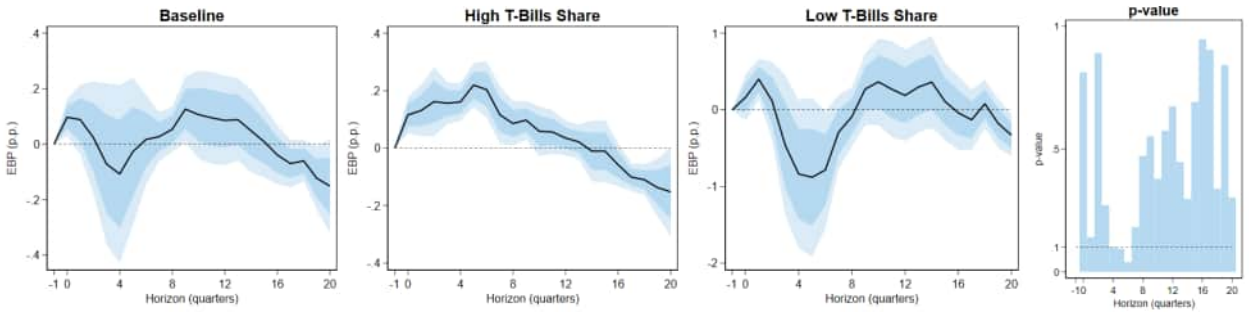
Overall, the empirical results are consistent with two complementary channels through which shorter debt maturity amplifies the effects of monetary policy shocks, as rationalized in the theoretical model below. First, the *debt-refinancing channel* (or debt rollover channel) implies that, when the share of short-term government debt is higher, a larger fraction of outstanding debt must be rolled over at higher interest rates following a monetary tightening. This raises government interest expenditures more quickly and worsens debt dynamics. Second, the *fiscal-financial channel* operates through the greater sensitivity of sovereign borrowing costs to policy-rate changes: higher reliance on short-term debt raises financing costs more broadly across the economy, tightening financial conditions and weakening firms' incentives and ability to invest. This, in turn, slows capital accumulation and reduces aggregate demand.[8] These channels are formalized in the next section.

[7] In the monthly frequency analysis, the results for the EBP are even stronger. In particular, besides the initial rapid response that is temporarily reabsorbed, a statistically significant medium-term increase emerges after about two years. Moreover, the non-linear responses are larger than in the quarterly analysis, with the peak response of the EBP in the high T-bill share regime reaching about 0.4 percentage point, nearly 0.2 percentage point larger than in the quarterly frequency analysis (see Figure A31).

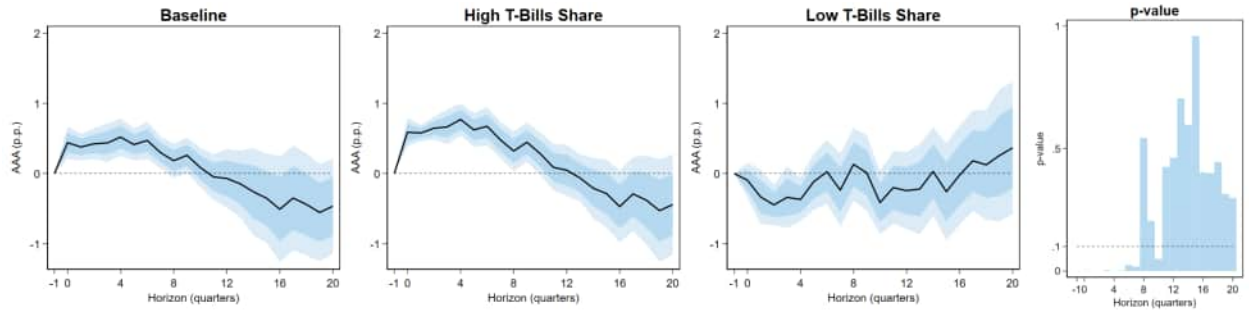
[8] Figure A3 in the Appendix reports the response of nominal debt. Following a 100bps monetary policy tightening, nominal debt increases persistently in the high-T-bill-share regime by about 7%, whereas the response is largely not statistically significant in the low-T-bill-share regime. The differences between the two regimes are statistically significant at several horizons. Thus, the stronger responses of real debt and the debt-to-GDP ratio under a high T-bill share cannot be attributed exclusively to denominator effects arising from lower prices and output. Instead, the results indicate that differential nominal debt accumulation plays an important role, consistent with the channels highlighted above.

FIGURE 7: THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

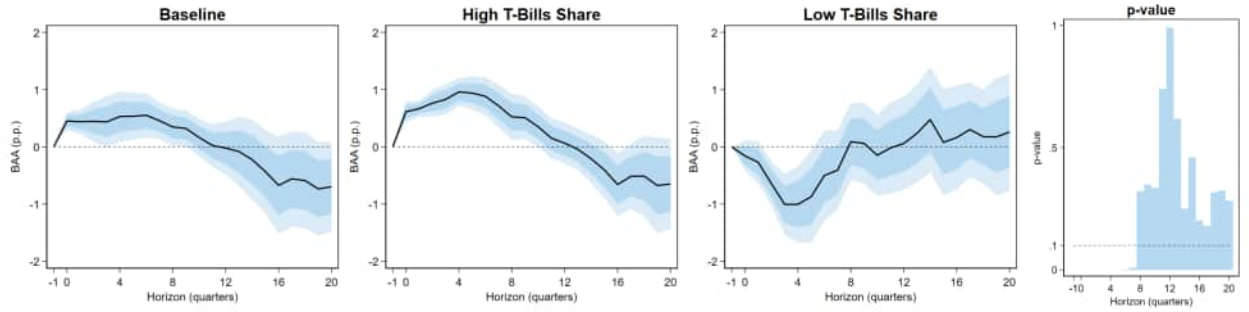
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



Panel C. Baa Corporate Bond Yield



Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

3.2 Robustness checks

We conduct several robustness checks to assess the stability of our baseline empirical results. Overall, the evidence shows that our main findings are robust across a range of alternative specifications and data choices.

3.2.1 Alternative LP specification

First, we examine whether our findings depend on the specific local-projection specification employed in the baseline. In particular, since our baseline uses an LP in levels, we also estimate an LP in long differences, so that the model takes the following form:

$$y_{t+h} - y_{t-1} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}, \tag{4}$$

where the dependent variable is now the cumulative growth at horizon $t+h$ of the variable of interest. Accordingly, the estimated smooth transition-LP model becomes:

$$\begin{aligned} y_{t+h} - y_{t-1} = \alpha^{(h)} &+ F(z_{t-1}) \Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell} \Big) \\ &+ \big(1 - F(z_{t-1})\big) \Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell} \Big) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}. \end{aligned} \tag{5}$$

We also assess whether our results depend on the specific lag structure of the baseline specification by re-estimating the original local projections with fewer lags and with more lags, specifically with two or six lags of shock and dependent variable. Figures A4 to A15 in the Appendix indicate that the responses are virtually identical to the baseline and remain as significant as before.

3.2.2 Instrumental Variable

Second, we replicate our analysis with a two-stage least squares instrumental variable approach. In fact, the monetary policy shock series constructed by Romer and Romer (2004) is a generated regressor, and Wooldridge (2010) shows that generated instruments avoid the inference problems that Pagan (1984) highlighted for generated regressors. The following LP-IV is estimated:

$$\begin{aligned} y_{t+h} &= \alpha^{(h)} + \beta^{(h)} \widehat{\Delta FFR}_t + \sum_{\ell=1}^{4} \theta_{\ell}^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}, \\ \text{where } \widehat{\Delta FFR}_t &= \gamma + \delta MPS_t + \sum_{\ell=1}^{4} \iota_{\ell}^{\prime} W_{t-\ell} + \nu_t. \end{aligned} \tag{6}$$

The same approach applies to the smooth-transition local projection described in Eq. (2), where the federal funds rate is again instrumented with the monetary policy shock series.

We estimate these specifications both for the LP in levels and for the LP in long differences as a further robustness exercise. As reported by Tables A3 and A4 in the Appendix, the instrument is strong, with the Kleibergen-Paap rk Wald F statistic consistently above the Stock-Yogo 10% critical value for all variables[9].

Figures A16 to A23 in the Appendix show that our results remain broadly robust to this approach, both when using the LP in levels and the LP in long differences. Across all variables, the estimated impulse responses remain qualitatively similar to those obtained in the baseline analysis, with the point estimates preserving both their sign and overall dynamic pattern. In particular, the responses continue to indicate stronger effects of monetary policy on debt and fiscal dynamics in periods characterized by a high share of Treasury bills outstanding.

Although some differences between regimes are no longer statistically significant at conventional levels when using the LP in levels (they are significant when using the LP in long differences), this largely reflects the wider confidence intervals associated with instrumental variable estimation. Nevertheless, the point estimates continue to move in the expected direction and remain economically meaningful. Importantly, the responses in the high Treasury-bill-share regime are estimated considerably more precisely than those in the low-share regime. Confidence bands are generally narrower and the estimated effects are more clearly distinguishable from zero, thereby reinforcing our conclusions.

3.2.3 Alternative monetary policy shock series

Third, we use a high-frequency instrument to recover an alternative structural monetary policy shock series that we use as a robustness check. The "external instrument" (or "proxy SVAR") procedure is the one described by Ramey (2016), following Mertens and Ravn (2014). Consider a model with three endogenous variables: industrial production (Y_1), the consumer price index (Y_2) and the monetary policy rate (Y_3). To account for the extreme macroeconomic volatility introduced by the COVID-19 pandemic without discarding recent data, we augment the system with a vector of strictly exogenous variables, X_t, following the approach proposed by Ng (2021). The vector X_t includes a dummy variable for March 2020, a step-shift dummy taking the value of one from March 2020 onwards, and the current value plus twelve lags of the log ratio of COVID-19 deaths. The following structural model captures the dynamics of the vector of endogenous variables $Y_t = [Y_{1t}, Y_{2t}, Y_{3t}]$:

$$Y_t = B(L)Y_t + \Gamma X_t + \Omega\epsilon_t, \tag{7}$$

where L is the lag operator, $B(L) = B_0 + \sum_{k=1}^{p} B_k L^k$, Γ is the matrix of coefficients for the exogenous COVID-19 variables, and ϵ_t are the structural shocks that can affect all the variables in the system. We can rewrite the model in terms of the innovations of the reduced form VAR:

$$A(L)Y_t = \Phi X_t + \eta_t, \tag{8}$$

where $A(L) = I - \sum_{k=1}^{p} A_k L^k$ and $\eta_t = [\eta_{1t}, \eta_{2t}, \eta_{3t}]$ are the reduced form VAR innovations, which can be linked to the unobserved structural shocks ϵ_t through the equation

$$\eta_t = B_0\eta_t + \Omega\epsilon_t. \tag{9}$$

[9]Except for the deflator, for which it only exceeds the 15% critical value.

We are interested in the structural monetary policy shock ϵ_{3t}, but we only observe the reduced form innovations η_t. Instead of imposing unrealistic theoretical restrictions to identify the structural shock, we can use an external instrument Z_t for which $E[Z_t\epsilon_{3t}] \neq 0$ and $E[Z_t\epsilon_{it}] = 0$ for $i = 1, 2$, meaning that the instrument is relevant for the monetary policy structural shock of interest, but is not correlated with the other structural shocks in the system. The orthogonalized high-frequency measure of monetary policy shocks constructed by Bauer and Swanson (2023) is well suited to our scope, and we use the updated version that goes from 1988m2 to 2023m12. The three-step procedure to recover the structural monetary policy shock series of interest is the following:

1. We estimate the reduced form VAR to obtain estimates of the reduced form residuals, η_t.

2. We regress η_{1t} and η_{2t} on η_{3t} using the external instrument Z_t as the instrument. These regressions yield unbiased estimates of the effects of monetary policy shocks on innovations to industrial production and CPI. We then define the residuals of these regressions to be ν_{1t} and ν_{2t}.

3. We regress η_{3t} on η_{1t} and η_{2t} using the estimated ν_{1t} and ν_{2t} as instruments. This yields unbiased estimates of the effects of innovations to industrial production and CPI on innovations to the policy rate, effectively leaving as a residual the true structural monetary policy shock (MPS_t).

This methodology has been used by Cloyne, Ferreira, Froemel, and Surico (2023), among others. To mitigate concerns related to the zero lower bound and forward guidance periods, we use the one-year government bond rate as the relevant monetary policy rate instead of the federal funds rate. This choice also allows us to construct a structural monetary policy shock series starting in 1963, broadly in line with the series from Romer and Romer (2004), which begins in 1969. By contrast, alternative measures—such as the two-year government bond rate—are only available from the 1970s onward, preventing us from capturing the increase in T-Bills share that occurred during the 1960s. Importantly, the first-stage Kleibergen-Paap rk Wald F statistic for the instrument in the second step is equal to 10.9, exceeding the critical value of 10 proposed as a rule of thumb by Stock and Watson (2012), below which the instrument would be considered weak.

Figures A24 to A27 in the Appendix show that the main results discussed above are unchanged when we use this alternative monetary policy shock series. This approach allows us to extend the baseline sample from 1963Q1 to 2025Q3. However, to ensure consistency and comparability with the baseline specification, the smooth transition function is still constructed from 1969Q1 onward. The resulting shock series does not display standard empirical puzzles and, in line with well-established findings in the literature, yields a faster and more precisely estimated response of prices relative to narrative-based measures.

3.2.4 Alternative frequency

Fourth, we examine the sensitivity of our results to the frequency of the dataset. As discussed above, in the baseline analysis the monetary policy shocks are identified at the monthly

frequency and then aggregated to the quarterly level by summation. This aggregation is necessary because the fiscal variables, which are central to our analysis, are available at the quarterly frequency. As a robustness check, we repeat the analysis using monthly estimates for the variables originally observed at quarterly frequency. Monthly estimates for GDP and investment are taken, respectively, from S&P and Barrett (2025), while monthly information on fiscal variables is sourced from the U.S. Treasury. Figures A28 to A35 in the Appendix show that our findings remain robust when the analysis is conducted at the monthly frequency.

3.2.5 Alternative interaction specification

Fifth, we assess whether our state-dependent results are driven by the specific smooth-transition local projection specification. To this end, in addition to varying the value of the parameter θ to alternative values commonly used in the literature (1.5 and 3), we estimate two alternative local projection models that allow the response to monetary policy shocks to vary with the share of T-bills. The first replaces the smooth transition function with a more standard regime-splitting approach based on a dummy variable. Specifically, we distinguish between periods in which the T-bill share is above its historical median and periods in which it is below its historical median:

$$\begin{aligned} y_{t+h} = \alpha^{(h)} &+ D_{t-1}\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) \\ &+ \big(1 - D_{t-1}\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} D_{t-1} + \varepsilon^{(h)}_{t+h}. \end{aligned} \tag{10}$$

Here, D_{t-1} denotes the regime indicator. In this specification, it is a dummy variable equal to one when the share of T-bills in total public debt is above its sample median, and zero otherwise. Figures A36 to A47 in the Appendix show that the results are essentially unchanged. Public debt increases significantly following a monetary policy tightening when the T-bill share is above its historical median, while it is not significantly affected when the T-bill share is below the median. The other transmission channels discussed in Section 3 are also preserved.

The second alternative specification is a standard linear interaction model, given by

$$y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \gamma^{(h)} MPS_t T_{t-1} + \delta^{(h)} T_{t-1} + \sum_{\ell=1}^{4} \theta'^{(h)}_{\ell} W_{t-\ell} + \epsilon^{(h)}_{t+h}, \tag{11}$$

where T_{t-1} is the share of T-bills in total public debt. In this specification, $\beta^{(h)}$ captures the effect of a monetary policy tightening shock in the hypothetical case in which the T-bill share is equal to zero, while $\gamma^{(h)}$ measures how this effect varies with the T-bill share. Hence, $\gamma^{(h)}$ captures the differential impact of monetary policy shocks as the maturity composition of public debt changes. Figures A48 to A51 in the Appendix show that the results are consistent with our baseline evidence. The smooth transition function is preferable to both alternatives. Unlike the dummy approach, it allows the effects of monetary policy shocks

to vary continuously with the state variable, rather than imposing a discrete split at an arbitrary threshold. Unlike a linear interaction, it also allows responses to vary nonlinearly with the level of debt maturity.

3.2.6 Alternative measure of debt maturity

Sixth, we use an alternative measure of short-term debt exposure, namely the privately held short-term debt share, which is defined as the amount of privately held Treasury debt of all types maturing within one year relative to total privately held Treasury debt outstanding. Because this measure is based on remaining maturity rather than on the original classification of the security, it captures the full range of debt that must be refinanced over the subsequent year. While this measure has the advantage of providing a broader measure of near-term refinancing and rollover exposure, it has the disadvantage of being available only from 1975 onward. Nevertheless, the shorter sample provides an additional robustness check, as it allows us to assess whether the results hold over a different sample period. Figures A52 to A55 show that the results remain unchanged. This confirms that our findings are not specific to the use of the T-bill share as the baseline measure of debt maturity, but instead capture a broader role of short-term debt exposure in shaping the transmission of monetary policy shocks to public debt dynamics.

3.2.7 Debt maturity response to monetary policy

Seventh, we further strengthen the assumption that the maturity structure of government debt is predetermined with respect to the monetary policy shock. In the baseline analysis, we use the T-bill share lagged by one quarter. To further mitigate concerns about potential contemporaneous feedback between monetary policy and debt maturity, we re-estimate the baseline specification by interacting the monetary policy shock with the smooth transition function lagged by two quarters. Figures A56–A59 show that the results remain qualitatively unchanged, thereby providing further support for our baseline findings.

We also examine whether the maturity structure of debt responds endogenously to monetary policy shocks. This is of particular importance, as Gonçalves, Herrera, Kilian, and Pesavento (2024) show that the validity and interpretation of the state-dependent LP estimator depends on whether the state of the economy (e.g., the T-bills share in our case) evolves exogenously with respect to the economy or responds endogenously to the macroeconomic shock of interest. Figure A60 shows that the responses of both the T-bill share and the privately held short-term debt share to a 100bps exogenous monetary policy shock are statistically indistinguishable from zero.[10]

Taken together, these findings alleviate concerns about the endogeneity of the state variable and address the critique raised by Gonçalves et al. (2024).

[10]These results concern the response of debt maturity to unexpected monetary policy innovations and therefore do not directly address whether the Treasury adjusts its issuance strategy in anticipation of future interest-rate movements. Testing the latter hypothesis would require a measure of expected monetary policy or anticipated changes in interest rates. Nevertheless, the absence of a significant response indicates that the maturity structure does not adjust mechanically following unexpected monetary policy changes.

3.2.8 Correlates of debt maturity

Finally, a potential concern with our baseline specification is that the share of T-bills in total government debt may be correlated with other initial fiscal and macroeconomic conditions, including the debt-to-GDP ratio, the interest rate, inflation, the outstanding stock of T-bills relative to GDP, as well as the occurrence of recessions or other historical periods such as the Volcker disinflation (see Figure 2). If periods characterized by a higher T-bill share are systematically associated with different initial levels of these variables or with particular historical periods, the interaction between monetary policy shocks and the T-bill share may partly capture heterogeneity arising from these initial conditions rather than from the maturity composition of government debt per se.

To address this concern, we augment our baseline specification by including an additional interaction term between the monetary policy shock and, in turn, each of the following initial conditions: the debt-to-GDP ratio, the federal funds rate, inflation, and the outstanding stock of T-bills relative to GDP. The resulting specification is as follows:

$$\begin{aligned} y_{t+h} = \alpha^{(h)} &+ F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) \\ &+ \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) \\ &+ \psi^{(h)} MPS_t\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}, \end{aligned} \tag{12}$$

where $F(r_{t-1})$ is a smooth transition function based, in turn, on the initial debt-to-GDP ratio, federal funds rate, inflation rate, or stock of T-bills relative to GDP, and $\overline{F(r_{t-1})}$ is its mean. Centering the interaction term at its sample mean ensures that the coefficients on MPS_t in the high- and low-regime equations retain a clear interpretation as the marginal effect of a monetary policy shock at the average level of the additional fiscal or macroeconomic condition level. Since the interaction term is equal to zero when $F(r_{t-1}) = \overline{F(r_{t-1})}$, the coefficients $\beta^{(h)}_{\text{high}}$ and $\beta^{(h)}_{\text{low}}$ are not mechanically shifted by the average level of debt, short-term debt, interest rate, or inflation. Figures A61 to A76 in the Appendix show that results remain qualitatively and quantitatively similar to those obtained in the baseline specification, suggesting that our baseline results are not driven by the correlation between the T-bills share and the initial levels of these macroeconomic variables.

In this specification, the coefficient $\psi^{(h)}$ is of independent interest, as it captures how the response of y_{t+h} to a tightening monetary policy shock varies with the initial level of debt, short-term debt, interest rate or inflation, thereby providing additional evidence on fiscal-monetary interactions that we show in Figures A77 to A92 in the Appendix. Results show that contractionary monetary policy shocks lead to a larger increase in real debt and the debt-to-GDP ratio when public debt is already high, a result that is consistent with a debt accumulation channel, whereby higher initial debt amplifies the effect of rising interest rates on debt servicing costs. This mechanism is accompanied by a stronger deterioration in macroeconomic and fiscal conditions, as evidenced by larger declines in real GDP, the GDP deflator, real private fixed investment, and real revenues, together with a larger increase in the excess bond premium and real primary expenditure. The coefficient is negative for

effective interest rate, which is consistent with the slow adjustment of average financing costs on outstanding debt, reflecting the fact that a large share of debt continues to be serviced at previously contracted rates even as market interest rates rise. Overall, the results indicate that high initial debt amplifies the adverse effects of monetary policy shocks on both fiscal sustainability and economic activity. These effects are confirmed and become even more pronounced when considering the initial level of short-term debt. A larger stock of short-term debt increases the amount of debt that must be refinanced at higher interest rates following a monetary tightening, accelerating the pass-through of policy rates to government borrowing costs. The evidence therefore provides further support for the rollover-cost channel and indicates that the fiscal consequences of monetary tightening depend not only on the overall level of government debt but also on its short-term refinancing exposure.[11]

We further assess whether our results are driven by particular macroeconomic or historical episodes that may be correlated with debt maturity. In addition to recessions, identified using the same NBER dummy employed to highlight recession periods in Figure 2, we control separately for the 1970s, the Volcker disinflation, the 1981–1982 recession, and the Global Financial Crisis. Since these variables are binary indicators, the specification is slightly adjusted, as a smooth-transition function is unnecessary:

$$\begin{aligned} y_{t+h} = \alpha^{(h)} &+ F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) \\ &+ \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) \\ &+ \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon^{(h)}_{t+h}, \end{aligned} \tag{13}$$

where D_{t-1} denotes, in turn, the dummy for recessions or for each of the historical episodes considered, and $\bar{D}$ is its mean. Figures A93 to A112 in the Appendix show that our baseline results are robust to controlling for these episodes, suggesting that the estimated nonlinear effects are not simply capturing particular macroeconomic environments or historical periods associated with high or low debt maturity.

A further concern is that the estimated heterogeneous effects attributed to the maturity composition of government debt may partly reflect central bank balance sheet operations. In particular, quantitative easing (QE) and quantitative tightening (QT) may affect the amount and maturity composition of Treasury securities held by the private sector, as well as term premia and broader financial conditions. To address this concern, we augment the baseline specification by controlling for the interaction between the monetary policy shock and two measures of quantitative easing: a smooth transition function for the level of federal debt

[11] With regards to initial interest rate levels, results show that contractionary monetary policy shocks have stronger adverse effects on economic activity, prices, and the fiscal base when monetary policy starts from an accommodative stance (low initial interest rate). Turning to initial inflation, the results suggest that higher inflation is associated with smaller increases in real debt following a contractionary monetary policy shock, as inflation erodes the real value of outstanding liabilities and supports nominal revenues. The smaller increase in the debt-to-GDP ratio is consistent with the more positive responses of real GDP and real private investment: when initial inflation is high, the same 100bps monetary policy tightening produces a smaller increase in the real interest rate, thereby attenuating its contractionary effects on investment and economic activity.

held by Federal Reserve Banks as a percentage of GDP following Eq. (12), and a dummy equal to one during the periods 2008Q4-2014Q4 and 2020Q1-2022Q1 following Eq. (13). Results for these exercises are reported both for the updated narrative shock and for the structural shock series identified through high-frequency surprises in a proxy SVAR, as the latter enables us to extend the analysis to the post-pandemic QE period. Figures A113 to A128 in the Appendix show that the estimated responses remain qualitatively similar to those in the baseline specification. This suggests that the heterogeneity associated with the T-bill share is not driven by the effects of central bank balance sheet policies on term premia and financial conditions.

When discussing the results on fiscal dynamics, we noted that the deterioration in the primary balance after a monetary tightening when the T-Bills share is high may be due to automatic stabilizers and discretionary fiscal policy reaction. To assess whether the stronger responses observed in the high short-term debt regime are driven by differences in fiscal policy rather than by debt maturity itself, we augment the baseline specification with interactions between the budget balance and the T-bills-share regime. If fiscal authorities systematically react more aggressively when the share of short-term debt is high, part of the estimated asymmetry could reflect differences in fiscal stance across regimes rather than the maturity structure of public debt. The estimated model becomes:

$$\begin{aligned} y_{t+h} = \alpha^{(h)} &+ F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \gamma^{(h)}_{\text{high}} \Delta BB_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) \\ &+ \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \gamma^{(h)}_{\text{low}} \Delta BB_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \varepsilon^{(h)}_{t+h} \end{aligned} \tag{14}$$

where ΔBB_t is the first difference of budget balance-to-GDP. Figures A129 to A132 in the Appendix show that our results remain qualitatively unchanged after controlling for state-dependent fiscal conditions, suggesting that the stronger effects documented in the high short-term debt regime primarily reflect the role of debt maturity.

A final concern is that the estimated heterogeneous effects may capture differences in underlying trends across periods characterized by high and low T-bill shares. To address this concern, we augment the baseline specification by including centered linear and quadratic time trends, both separately and interacted with the smooth transition function, as follows:

$$\begin{aligned} y_{t+h} = \alpha^{(h)} &+ F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) \\ &+ \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) \\ &+ \gamma^{(h)}(t - \bar{t})_t + \iota^{(h)}(t^2 - \bar{t^2})_t + F(z_{t-1})\Big(\gamma^{(h)}_{\text{high}}(t - \bar{t})_t + \iota^{(h)}_{\text{high}}(t^2 - \bar{t^2})_t\Big) + \varepsilon^{(h)}_{t+h} \end{aligned} \tag{15}$$

This specification allows the outcome variables to follow distinct nonlinear trends depending on the maturity composition of government debt. Figures A133 to A136 in the Appendix show that the estimated responses remain qualitatively similar to those in the baseline specification, suggesting that the heterogeneity associated with the T-bill share is not driven by differential time trends.

Taken together, these robustness checks indicate that our findings are not driven by the specific monetary policy shock series, the frequency of the data, the functional form used to model state dependence, the potential endogenous response of debt maturity to monetary policy shocks, the initial level of key macroeconomic variables (debt, short-term debt, interest rates, and inflation), central bank balance sheet policies such as QE, differential time trends, changes in the budget balance, or the occurrence of recessions and major historical episodes, including the 1970s, the Volcker disinflation, the 1981–1982 recession, and the Global Financial Crisis. The central result remains stable: monetary policy tightening has substantially larger effects on public debt and fiscal dynamics when the government relies more heavily on short-term debt.

4 Theoretical model

This section develops a theoretical model with the purpose of providing a coherent economic rationale for the empirical findings, qualitatively replicating the differential fiscal-macro impacts of monetary policy across debt maturity regimes. In particular, the model speaks to three sets of findings: when government debt has shorter average maturity, (i) debt is more sensitive to monetary policy shocks; (ii) the borrowing rate faced by the private sector increases more; and (iii) a monetary tightening produces larger and more persistent contractions in output and investment.

The economy is a closed New Keynesian economy with two distinct financial frictions. Households lend to financial intermediaries; intermediaries lend to entrepreneurs and hold government debt; and entrepreneurs buy capital from the loans.

$$\text{households} \longrightarrow \underbrace{\text{financial intermediaries}}_{\text{enforcement constraint}} \longrightarrow \underbrace{\text{entrepreneurs}}_{\text{costly state verification}} \longrightarrow \text{capital.} \tag{16}$$

Financial intermediaries (FIs) face the Gertler-Karadi moral hazard/enforcement constraint which limits how much they can borrow and hence how they allocate funds between private credit and long-term government bonds (Gertler and Karadi, 2011). Meanwhile, entrepreneurs face the Bernanke-Gertler-Gilchrist costly-state-verification (CSV) problem, so the external-finance premium depends on entrepreneurial leverage (Bernanke et al., 1999). We therefore follow Rannenberg (2016) to combine different leverage constraints into a single framework.

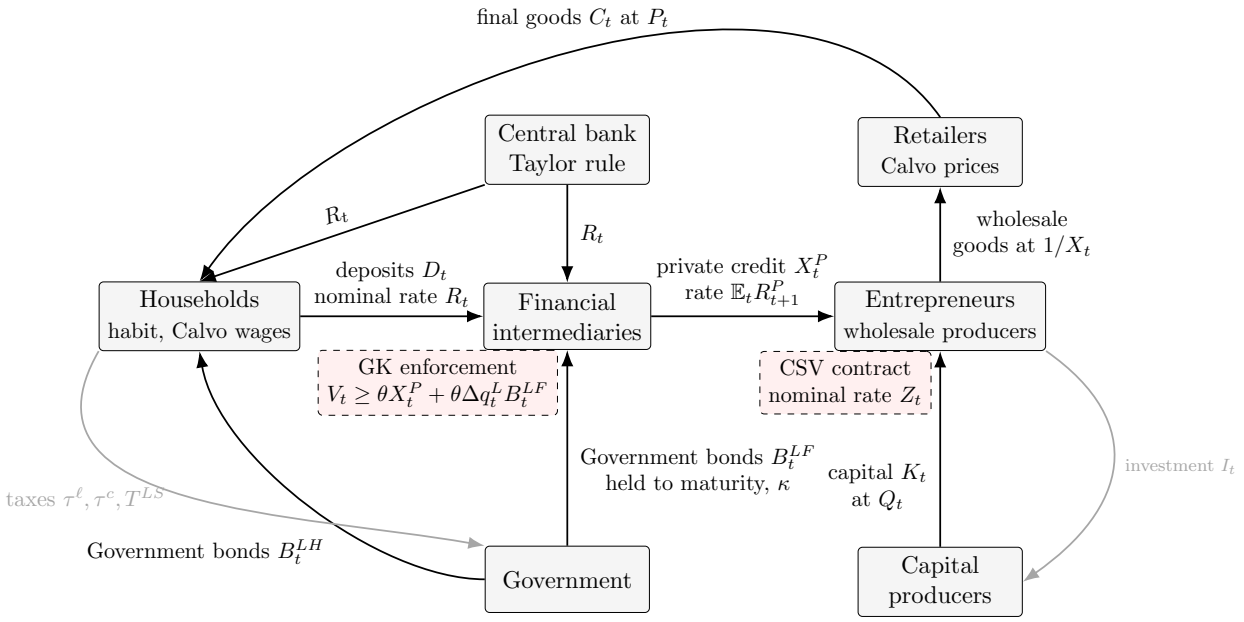
A conceptually central extension of our paper is the accounting treatment of the intermediary's bond portfolio, and it is what makes debt maturity matter. The framework we inherit from Gertler and Karadi (2013) already has constrained intermediaries allocating between private credit and long-term government bonds, with bonds absorbing a fraction Δ of the pledgeability a private loan absorbs, which in turn ties the required returns between these two assets. However, in Gertler and Karadi (2013) and Sims and Wu (2021), the intermediary values its portfolio at market and can sell it, so the relevant object is the one-period expected holding return. Consequently, the return is pinned down close to the policy rate regardless of the bond's duration, and maturity therefore has limited impacts on the cost of credit.

We instead assume that the intermediary recognizes the yield locked in at acquisition, which is consistent with held-to-maturity classification, under which securities are carried at amortized cost and net worth is in book equity. Because the entrepreneur's lender is this constrained intermediary, private loans and government bonds compete for the same scarce capacity, therefore requiring their returns to be tied together. The shorter the maturity of public debt, the larger and faster the pass-through of a monetary tightening to the government bond yield, hence to the lender's opportunity cost, and hence to the external finance premium. As a result, maturity enters the price of external finance for entrepreneurs, which in turn affects the transmission and propagation of monetary shocks to the real economy.

Government debt is modeled as perpetuities that pay 1 unit initially, with payments gradually declining over time, following Woodford (2001). Each bond issued in period t delivers a geometrically decaying coupon stream $\{1, \kappa, \kappa^2, ...\}$ in subsequent periods, where the decay parameter $\kappa \in [0, 1]$ controls average maturity. This setting allows us to consider monetary transmission under short- and long-maturity government debt by varying a single calibration of coupon decaying rate.

The economy is populated by a continuum of households, a continuum of entrepreneurs who produce the wholesale goods, financial intermediaries, a competitive capital producer, monopolistically competitive retailers, a government, and a central bank. Households save in one-period nominal claims (intermediary deposits and a short-term monetary policy bond in zero net supply) and in long-term government bonds, subject to a debt-elastic pricing wedge. Households are monopolistic suppliers of differentiated labor and reset nominal wages subject to Calvo frictions. In the baseline, financial intermediaries fund one-period private credit and a held-to-maturity portfolio of long-term government bonds with deposits and accumulated net worth, subject to the enforcement constraint. Entrepreneurs combine internal net worth with external finance to purchase capital, facing a financial friction that gives rise to an endogenous external finance premium. A competitive capital producer transforms consumption goods into capital subject to adjustment costs. Retailers purchase wholesale goods from entrepreneurs, differentiate them, and set prices subject to Calvo nominal rigidities, generating the real effects of monetary policy. The government finances expenditure by issuing long-term bonds and adjusts lump-sum taxes and spending through automatic stabilizers. The central bank sets the short-term nominal interest rate according to a Taylor rule. Figure 8 summarizes the architecture and should be read alongside the description above.

Figure 8: Model Architecture: Agents, Assets, and Frictions



Note: Solid arrows denote flows of funds and goods in the direction shown; gray arrows denote the corresponding real counterflows; dashed boxes denote the two financial frictions. Retailers differentiate the wholesale good and sell final goods to households at the price index P_t. Households fund intermediaries with one-period nominal deposits at the policy rate and are the holders of long-term government bonds, subject to a debt-elastic pricing wedge. Financial intermediaries allocate deposits and net worth between one-period private credit and a held-to-maturity portfolio of geometric-coupon government bonds (κ governs duration), subject to the Gertler–Karadi enforcement constraint in which government bonds absorb a fraction Δ of the pledgeability capacity absorbed by private loans. Entrepreneurs finance capital purchases from capital producers with net worth and intermediary credit under costly state verification with a nominal, non-state-contingent promised rate, so aggregate surprises generate loan losses that feed back into intermediary net worth. Retailers differentiate the wholesale good under Calvo pricing; the central bank follows a Taylor rule; the government issues the long bond and operates automatic stabilizers and debt-stabilizing tax rules.

4.1 Road map

Before presenting the full model, we briefly outline how the key empirical findings map to the theoretical mechanisms developed below.[12] Table 1 provides this mapping. The real effects of a monetary policy shock are modeled through the standard New Keynesian framework with price and wage stickiness. Government debt is a perpetuity with a decaying coupon stream, so average maturity is governed by a single parameter. The pass-through of monetary policy to the effective borrowing rate operates via the bond maturity structure, and its consequences for entrepreneurial net worth and investment are captured by the financial accelerator. A key departure is that the lender is an intermediary facing a binding enforcement constraint and carrying its government bonds at amortized cost, so its required return on private credit

[12]The full set of equilibrium conditions is reported in Appendix A.

is tied to the government bond yield locked in at acquisition rather than the policy rate. This is the mechanism through which maturity shapes the macroeconomic response. Finally, the fiscal consequences, including revenues, expenditures, and the evolution of debt and debt-to-GDP, are governed by the automatic stabilizers and the debt dynamics equation.

Table 1: Mapping Empirical Results to Model Mechanisms

Empirical Findings	**Model Mechanism**
Monetary policy shock	New Keynesian model with price/wage stickiness
Government bond maturity regime	Perpetuity bond with a decaying coupon stream
Government effective rate	Pass-through via bond maturity structure
Corporate borrowing cost	GK enforcement constraint & BGG accelerator
Investment	GK enforcement constraint & BGG financial accelerator
Macro effects under different maturity	Lender's opportunity cost + amortized cost
Debt and Debt-to-GDP	Debt dynamics equation
Revenue and Expenditure	Automatic stabilizers and fiscal rules

4.2 Households

There is a continuum of households indexed by $i \in [0,1]$. At each period t, the household i chooses consumption C_{it}, one-period nominal claims A_{it}, government bond holdings $\mathcal{B}^H_{it}$, and supplies a differentiated labor input H_{it} to maximize the following lifetime utility function

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\log(C_{it} - bC_{it-1}) - \chi_H \frac{H_{it}^{1+1/\eta}}{1+1/\eta} \right], \tag{17}$$

where $C_{i,t}$ denotes consumption, $H_{i,t}$ the labor input of household i, $\beta \in (0,1)$ the subjective discount factor, $b \in (0,1)$ the internal habit consumption, and $\eta > 0$ the Frisch elasticity of labor supply.

At each period, the household decision is subject to the following period budget constraint:

$$(1+\tau^C_t)P_tC_{it} + A_{it} + \zeta^L_t\, q^L_t \mathcal{B}^H_{it} \le R_{t-1}A_{it-1} + \left(1+\kappa q^L_t\right)\mathcal{B}^H_{it-1} + (1-\tau^l_t)W_{it}H_{it} + \Pi^{div}_{i,t} - T_{i,t}, \tag{18}$$

where P_t is the price level and W_{it} is the nominal wage rate from providing labor input H_{it}, Π^{div}_t collects dividends net of endowment given to entering intermediaries. The household pays consumption and labor-income taxes, with respective tax rates of τ^c_t and τ^ℓ_t, as well as lump-sum taxes $T_{i,t}$. Household holds government bond at the price of q^L_t, purchased subject to a small proportional transaction/convenience factor ζ^L_t. The wedge ζ^L_t distorts the bond Euler equation and is a parsimonious stand-in for preferred-habitat/limited-arbitrage forces in the Treasury market (Greenwood and Vayanos, 2014; Krishnamurthy and Vissing-Jorgensen, 2012; Mian, Straub, and Sufi, 2025), and it is what allows the supply of government debt to affect its yield.[13]

[13] It is rebated lump sum and therefore does not absorb resources.

Households hold short-term (one-period) claims from intermediary deposits and monetary policy bonds, $A_t = B_t^{mp} + D_t$, paying the gross nominal return R_t from period t to t+1. The gross nominal interest rate of R_t is set by the central bank. Also, the monetary bond instrument is assumed in zero net supply and used only for conducting monetary policy. Therefore, the total of one-period claims held by households is equal to deposits to intermediaries, or $A_t = D_t$. The household does not directly hold private loans, but does trade in long-term government bonds alongside the financial intermediaries.

Let Λ_t denote the marginal utility of consumption. The household's first order condition regarding consumption/savings are:

$$\Lambda_t\big(1+\tau_t^c\big) = \frac{1}{C_t - bC_{t-1}} - \beta b \mathbb{E}_t \frac{1}{C_{t+1} - bC_t}. \tag{19}$$

Denote by $M_{t,t+1} = \beta \frac{\Lambda_{t+1}}{\Lambda_t}$ the real stochastic discount factor, and $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$ the gross inflation. The Euler equations for the one-period claim and for the long-term bond are respectively:

$$1 = \mathbb{E}_t\left[M_{t,t+1}\frac{R_t}{\Pi_{t+1}}\right]. \tag{20}$$

and

$$\zeta_t^L\, q_t^L = \mathbb{E}_t\left[M_{t,t+1}\frac{1+\kappa q_{t+1}^L}{\Pi_{t+1}}\right], \tag{21}$$

where the wedge ζ_t^L is:

$$\zeta_t^L = \bar{\zeta}^L\Big(\frac{q_t^L B_t^L/Y_t}{\bar{q}^L \bar{B}^L/\bar{Y}}\Big)^{\eta_d}. \tag{22}$$

with $\bar{\zeta}^L = \bar{R}^L/\bar{R}$ and $B_t^L = \mathcal{B}_t^L/P_t$. By defining the one-period market holding return as $R_{t+1}^{LM} = \frac{1+\kappa q_{t+1}^L}{q_t^L}$, and combining (20) and (21), to the first order,

$$\mathbb{E}_t \widehat{R}_{t+1}^{LM} - \widehat{R}_t = \widehat{\zeta}_t^L = \eta_d\big(\widehat{q}_t^L + \widehat{B}_t^L - \widehat{Y}_t\big), \tag{23}$$

where $\hat{X}_t$ reflects the log deviation of variable X_t from its steady state. The term $\bar{\zeta}^L$ aims to generate a constant steady-state term premium, while the elasticity η_d governs its time variation. The expected excess holding return on government debt over the short rate is increasing in the debt outstanding as percent of GDP. This debt-elastic wedge is the model's term-premium/debt-supply channel: when a tightening raises the debt burden, the required yield on government bonds rises *over and above* the expected path of policy rates, and, through the intermediary block below, this feeds into private borrowing costs.

Each household is a monopolistic supplier of differentiated labor, H_{it}, at the corresponding wage rate W_{it} as in Erceg, Henderson, and Levin (2000). Labor services are aggregated with elasticity ϵ_w by a labor contractor who then sells the aggregated labor service to intermediate good producers for nominal wage W_t. Only a fraction $1-\theta_w$ of households can reset their nominal wage each period. The optimal reset wage W_t^* satisfies

$$\sum_{k=0}^{\infty}(\beta\theta_w)^k \mathbb{E}_t\left\{H_{i,t+k|t}\left[\Lambda_{t+k}\big(1-\tau_{t+k}^{\ell}\big)\frac{W_t^*}{P_{t+k}} - \frac{\epsilon_w}{\epsilon_w - 1}\chi_H H_{i,t+k|t}^{1/\eta}\right]\right\} = 0, \tag{24}$$

where $H_{i,t+k|t} = \left(\frac{W_t^*}{W_{t+k}}\right)^{-\epsilon_w} H_{t+k}$ is the labor demand faced at $t+k$ by a household that last reset its wage at t. The aggregate wage index evolves according to

$$W_t^{1-\epsilon_w} = (1-\theta_w)(W_t^*)^{1-\epsilon_w} + \theta_w \left(W_{t-1}\right)^{1-\epsilon_w}. \tag{25}$$

4.3 Financial intermediaries

We follow Gertler and Karadi (2011, 2013) to model financial intermediaries and depart from that literature in one respect: how the intermediary's government bond portfolio is valued on its balance sheet. There is a continuum of financial intermediaries, indexed by j, owned by households. Each funds a portfolio of private and treasury bonds with household deposits and its own net worth. Its ability to raise deposits is limited by a costly-enforcement constraint of the Gertler–Karadi type: the intermediary can divert a fraction of the assets it carries, and depositors lend only up to the point at which continuing to operate is at least as valuable as absconding.

An intermediary survives into the next period with probability $\sigma_F \in (0,1)$, independent of history, thus not depending on how long the person has been a banker. Otherwise, the intermediary exits with probability $1-\sigma_F$ and becomes a worker. Exiting bankers are assumed to remit their book net worth to households. A similar number of workers randomly become bankers, keeping the relative proportion of each type fixed. Entrants receive a start-up endowment X^F from households, so that sector's net worth is not degenerate.[14]

4.3.1 The government bond: market return and book return

The government bond bought by financial intermediaries is a nominal perpetuity with price of q_t^L, modeled à la Woodford (2001). A bond issued in period t pays a geometrically decaying coupon stream $\{1, \kappa, \kappa^2, \ldots\}$ with $0 \leq \kappa \leq 1$ in subsequent periods. Its gross yield to maturity (YTM) satisfies

$$q_t^L = \sum_{j=1}^{\infty} \frac{\kappa^{j-1}}{(R_t^L)^j} = \frac{1}{R_t^L - \kappa}, \qquad R_t^L = \kappa + \frac{1}{q_t^L}, \tag{26}$$

and its one-period market holding return is

$$R_{t+1}^{LM} = \frac{1 + \kappa q_{t+1}^L}{q_t^L}. \tag{27}$$

The decay parameter for the coupon payments of the bonds decides the average duration of bond: the smaller the κ, the shorter the bond average maturity.[15] Intermediate values of κ therefore parameterize the full spectrum from short- to long-maturity government debt, enabling a model-based analysis of monetary policy transmission under different debt structures. A persistent policy tightening has almost full pass-through to short-maturity yield,

[14] Both flows are contained in Π_t^{div} in households' budget constraint.

[15] Two polar cases illustrate the role of κ: when $\kappa = 0$, the bond collapses to a one-period instrument (perpetual roll-over at par), and $\kappa \to 1$ approaches a consol with infinite maturity.

but moves the longer yield only by the duration-weighted average of expected future short rates together with the debt-supply wedge.

Given FI holdings of duration bonds, fluctuations in the bond prices would affect the assets' valuation of FIs, and therefore their net worth, which through the enforcement constraint, governs the size of the portfolio they can carry. Whether they do so on the balance sheet, however, depends on debt securities classification and associated accounting convention: at market value or at book value (amortized cost).

We conceptualize this by assuming that a share of the intermediary's government portfolio ($\chi \in [0,1]$) is held to maturity and carried at amortized cost and the remaining $1-\chi$ is available for sale and carried at market.[16] The average return on the portfolio is

$$R_{t+1}^{LC} \equiv (1-\chi)\, R_{t+1}^{LM} + \chi R_t^{LB}. \tag{28}$$

- $\chi = 0$: $R_{t+1}^{LC} = R_{t+1}^{LM}$ valuation changes are fully recognized and the block reproduces the mark-to-market formulation of Gertler and Karadi (2013) and Sims and Wu (2021).
- $\chi = 1$: $R_{t+1}^{LC} = R_t^{LB}$ the position is carried at amortized cost, and the return recognized in net worth R_t^{LB} is the (predetermined) book yield locked in at acquisition. This will be our baseline model.
- $0 < \chi < 1$: Intermediate values capture a mixed securities portfolio. In section 5.3.3, we will extend our model to consider this case.

Two useful properties are worth noting. First, if the maturity of government bond is only one period or $\kappa = 0$, we obtain:

$$R_{t+1}^{LC} = R_t^{LB} = R_{t+1}^{LM} = \frac{1}{q_t^L} \tag{29}$$

regardless the value of χ, so the distinction does not matter in this case.

Second, in the deterministic steady state, the gross YTM and the one-period return are equal $\bar{R}^{LB} = \bar{R}^{LM}$, so

$$\bar{R}^{LC} = \bar{R}^{LB} = \bar{R}^{LM} \qquad \text{for every } \chi. \tag{30}$$

The accounting regime has no effect whatever on the steady state. Any difference in impulse responses across χ is attributable to the convention itself rather than to a shifted steady state, making the comparison across χ interpretable.

4.3.2 Amortized cost and the unrealized gain

Marking the securities book to market is the standard assumption in the intermediary-DSGE literature. This applies to trading securities and available for sales (in some cases), but not for HTM portfolios which are carried at amortized cost and never marked at all. Additionally, for the great majority of institutions, bank capital requirements are written on book value (Begenau et al., 2026).

[16] The share χ can be endogenous, since FIs could reclassify securities in response to the regulatory filter and to the rate environment. However, we leave this endogenous reaction for future extension to focus on our key channel.

Therefore, for the *baseline* model, we consider a book value convention for the FI government portfolio whose position is carried at amortized cost. Let $\mathcal{J}^L_{jt}$ denote new purchases of government bonds and $\mathcal{B}^L_{jt}$ the total holdings, both in nominal face (coupon) units, of intermediary j at time t. The latter includes the new purchase $\mathcal{J}^L_{jt}$ and the previous government holdings $\mathcal{B}^L_{jt-1}$, which decays by a factor κ:

$$\mathcal{B}^{LF}_{jt} = \mathcal{J}^L_{jt} + \kappa \mathcal{B}^{LF}_{jt-1} \tag{31}$$

Let $\mathcal{A}^{LB}_{jt}$ denote the nominal carrying value of intermediary j's government bond portfolio. New purchases in nominal value $\mathcal{J}^L_{jt}$ enter at the market price q^L_t; inherited positions are amortized at rate κ and are not marked to market. According to amortized-cost accounting, $\mathcal{A}^{LB}_{jt}$ evolves as:

$$\mathcal{A}^{LB}_{jt} \;=\; \kappa \mathcal{A}^{LB}_{jt-1} + q^L_t \Big(\mathcal{B}^{LF}_{jt} - \kappa \mathcal{B}^{LF}_{jt-1} \Big), \tag{32}$$

the second term being $q^L_t \mathcal{J}^L_{jt}$ by (31). In real terms with $B^{LF}_{jt} = \mathcal{B}^{LF}_{jt}/P_t$, $A^{LB}_{jt} = \mathcal{A}^{LB}_{jt}/P_t$, (32) becomes $A^{LB}_{jt} = \kappa \frac{A^{LB}_{jt-1}}{\Pi_t} + q^L_t \big(B^{LF}_{jt} - \kappa \frac{B^{LF}_{jt-1}}{\Pi_t} \big)$.

The return recognized on this portfolio R^{LB}_t between t and $t+1$ is coupon plus the change in carrying value, per unit of book at t:

$$R^{LB}_{jt} = \frac{\kappa \mathcal{A}^{LB}_{jt} + \mathcal{B}^{LF}_{jt}}{\mathcal{A}^{LB}_{jt}} = \kappa + \frac{\mathcal{B}^{LF}_{jt}}{\mathcal{A}^{LB}_{jt}} = \kappa + \frac{B^{LF}_{jt}}{A^{LB}_{jt}}. \tag{33}$$

It is known at t, which is the sense in which amortized cost insulates net worth: an innovation to q^L_{t+1} has no effect on the return recognized in $t+1$ on a position already on the books.

The unrealized gain/loss. Define the accumulated unrealized gain as the market value of the portfolio less its carrying value,

$$U_{jt} \;=\; q^L_t B^{LF}_{jt} - A^{LB}_{jt} = \frac{\kappa}{\Pi_t} \Big(q^L_t B^{LF}_{jt-1} - A^{LB}_{jt-1} \Big). \tag{34}$$

where the second equality follows from substituting the real form of (32). New purchases do not affect U_{jt} because they enter at the price paid, so they carry no gain/loss. Hence, U_{jt} is not a date-t choice: it is determined by the inherited position and the current price q^L_t, so it moves one-for-one with q^L_t but is independent of the portfolio decision taken at t.

Combining (33) and (34), together with R^L_t defined in (26):

$$R^{LB}_{jt} A^{LB}_{jt} = \underbrace{R^L_t A^{LB}_{jt}}_{\text{market yield on carrying value}} + \underbrace{\frac{U_{jt}}{q^L_t}}_{\text{embedded gain/loss}}. \tag{35}$$

A portfolio carried below its market value delivers above-market coupons *relative to its carrying amount*, and the excess is recognized as book income as the position amortizes toward par. In other words, $R^{LB}_{jt} > R^L_t$ once $U_{jt} > 0$. Therefore, HTM accounting does not eliminate the gain or loss; it recognizes it gradually over time rather than at once.

4.3.3 The intermediary's problem

Intermediary j holds one-period (nominal) private bonds $\mathcal{X}^P_{j,t}$ and a government-bond portfolio with book value as $\mathcal{A}^{LB}_{jt}$, funded by its book net worth $\mathcal{N}^F_{j,t}$ and deposits $\mathcal{D}_{j,t}$ (all in nominal terms), so that

$$\mathcal{X}^P_{j,t} + \mathcal{A}^{LB}_{jt} = \mathcal{N}^F_{j,t} + \mathcal{D}_{j,t}. \tag{36}$$

A financial intermediary accumulates net worth until stochastically exiting. For surviving FI, the real net worth evolves as:

$$\mathcal{N}^F_{j,t+1} = \Big[R^P_{t+1}\,\mathcal{X}^P_{j,t} + R^{LB}_t\,\mathcal{A}^{LB}_{j,t} - R_t\,\mathcal{D}_{j,t}\Big], \tag{37}$$

in which R^P_{t+1} is the gross nominal return realized on the diversified private loan portfolio made at t, which can differ from the return expected at origination as in Rannenberg (2016), R_t is the nominal gross interest rate on deposits, set by the central bank. R^{LB}_t is the gross nominal book yield of government bond portfolio.

The objective of an FI is to maximize its expected terminal (real) net worth discounted by the stochastic discount factor of the household. Taking into account the assumption that intermediaries survive each period with probability σ_F, its objective function at the end of period t is:

$$V_{jt} = \max_{\{X^P_{jt}, B^{LF}_{jt}, D_{jt}\}} (1-\sigma_F)\mathbb{E}_t \sum_{k=1}^{\infty} (\sigma_F)^{k-1} M_{t,t+k} N^F_{jt+k} \tag{38}$$

where $N^F_{jt} = \frac{\mathcal{N}^F_{jt}}{P_t}$ and $X^P_{jt} = \frac{\mathcal{X}^P_{jt}}{P_t}$, and $D_{jt} = \frac{\mathcal{D}_{jt}}{P_t}$ are real net worth, private credit, and deposit.[17]

We assume a costly enforcement problem facing financial intermediaries at the end of each period. An FI can choose to abscond with some assets at the end of a period instead of continuing as an FI, in which case depositors can recover only the remaining fraction of the FI's assets. Accordingly, depositors are willing to lend to FIs only if it is not optimal for the intermediary to divert funds. We assume that a FI can only abscond a fraction θ of its private bonds and a fraction $\theta\Delta$ of its government bonds, with $0 \leq \Delta \leq 1$, as in Gertler and Karadi (2013). The restriction of Δ implies that it is easier for the FI to divert private bonds than government bonds. A banker who absconds sells what he takes, so the diverted bonds are valued at the market price rather than at carrying value. The costly enforcement problem is therefore expressed as:

$$V_{jt} \geq \theta X^P_{jt} + \theta\Delta q^L_t B^{LF}_{jt} \tag{39}$$

which is equivalent to

$$V_{jt} \geq \theta X^P_{jt} + \theta\Delta A^{LB}_{jt} + \theta\Delta U_{jt}. \tag{40}$$

by using (34).

[17] N^F_{jt+k} is accounted at book value, so excludes the gain U_{jt+k} associated with outstanding government bond portfolio at that date, while the weights $(1-\sigma_F)\sigma_F^{k-1}M_{t,t+k}$ combine the exit hazard with the household's real stochastic discount factor: V_{jt} is thus a market valuation of book net worth. However, the valuation effect is not thereby lost. By (35), capital gain/loss U_{jt} is spread over time rather than recognized at once, which is therefore captured in V_{jt}.

All financial intermediaries are identical, behaving in the same way. Let λ_t be the multiplier on (40). The value function is linear in book net worth and in the unrecognized gain:

$$V_{jt} = \nu_t N_{jt}^F + \xi_t U_{jt}, \tag{41}$$

where ν_t is the marginal value of book net worth and ξ_t the marginal value of an unrecognized gain, expressed as below, respectively:

$$\nu_t = (1+\lambda_t)\,\mathbb{E}_t\Big[\widetilde{M}_{t,t+1}\frac{R_t}{\Pi_{t+1}}\Big], \tag{42}$$

$$\xi_t = (1+\lambda_t)\left(\big(R_t-\kappa\big)\mathbb{E}_t\Big[\frac{\widetilde{M}_{t,t+1}}{\Pi_{t+1}}\Big] + \sigma_F\kappa\,\mathbb{E}_t\Big[M_{t,t+1}\frac{\xi_{t+1}}{\Pi_{t+1}}\Big]\right). \tag{43}$$

The first-order conditions for private credit X_{jt}^P and for new government bond acquisitions are:

$$\mathbb{E}_t\left[\widetilde{M}_{t,t+1}\frac{(R_{t+1}^P - R_t)}{\Pi_{t+1}}\right] = \frac{\lambda_t}{1+\lambda_t}\,\theta, \tag{44}$$

$$\mathbb{E}_t\left[\widetilde{M}_{t,t+1}\frac{(R_t^L - R_t)}{\Pi_{t+1}}\right] \;+\; \Xi_t = \frac{\lambda_t}{1+\lambda_t}\,\theta\Delta, \tag{45}$$

where

$$\widetilde{M}_{t,t+1} = M_{t,t+1}\Omega_{t+1} \tag{46}$$

$$\Omega_t = 1 - \sigma_F + \sigma_F\theta\phi_t \tag{47}$$

$$\phi_t = \frac{\nu_t}{\theta} = \frac{1+\lambda_t}{\theta}\mathbb{E}_t\left[\widetilde{M}_{t,t+1}\frac{R_t}{\Pi_{t+1}}\right] \tag{48}$$

$$\Xi_t = \frac{\sigma_F\kappa}{q_t^L}\,\mathbb{E}_t\Big[M_{t,t+1}\xi_{t+1}\frac{q_{t+1}^L - q_t^L}{\Pi_{t+1}}\Big]. \tag{49}$$

where Ξ_t is the discounted value of the unrecognized gain that a purchase creates.

We focus on an equilibrium in which the intermediary enforcement constraint binds, $\lambda_t > 0$. When the constraint in (40) binds,

$$X_{jt}^P + \Delta A_{jt}^{LB} = \phi_t N_{jt}^F + \Big(\frac{\xi_t}{\theta} - \Delta\Big)U_{jt}, \tag{50}$$

with $\phi_t = \frac{\nu_t}{\theta}$.

Aggregating over intermediaries, sectoral net worth evolves as

$$N_t^F = \frac{\sigma_F}{\Pi_t}\Big[R_t^P X_{t-1}^P + R_{t-1}^{LB}A_{t-1}^{LB} - R_{t-1}D_{t-1}\Big] + X^F, \tag{51}$$

with X^F the (constant) transfer to entering bankers.

The link between the private lending cost and the government borrowing cost. Combining (44) and (45), we can derive the private-government return link as:

$$\mathbb{E}_t\big[\widetilde{M}_{t,t+1}\frac{\big(R_{t+1}^P - R_t\big)}{\Pi_{t+1}}\big] = \frac{1}{\Delta}\mathbb{E}_t\big[\widetilde{M}_{t,t+1}\frac{\big(R_t^L - R_t\big)}{\Pi_{t+1}}\big] + \frac{1}{\Delta}\Xi_t. \tag{52}$$

or

$$\mathbb{E}_t\big[\widetilde{M}_{t,t+1}\frac{R^P_{t+1}}{\Pi_{t+1}}\big] = \frac{1}{\Delta}\mathbb{E}_t\big[\widetilde{M}_{t,t+1}\frac{R^L_t}{\Pi_{t+1}}\big] + \frac{\Delta-1}{\Delta}\mathbb{E}_t\big[\widetilde{M}_{t,t+1}\frac{R_t}{\Pi_{t+1}}\big] + \frac{1}{\Delta}\Xi_t. \tag{53}$$

This is the central equilibrium relation of the paper: the required return on private lending is priced off the government bond yield, with pass-through coefficient $1/\Delta > 1$, plus the embedded-gain wedge Ξ_t. The term Ξ_t captures the value the intermediary attaches to future changes in the price of a bond it buys. To the first order, it is proportional to the expected change in the bond price, so the government return that the intermediary prices combines the yield locked in at acquisition with part of the expected one-period holding return. Maturity enters through the yield. With one-period debt, R^L_t tracks the policy rate almost one-for-one, amplified by the debt-supply wedge, and passes directly into the lender's opportunity cost. At ten-year duration the yield on new acquisitions moves only by the duration-weighted average of expected future short rates, so pass-through to the private borrowing cost is correspondingly slower. The maturity structure of public debt thus governs the strength of monetary transmission to private credit, which is the model's *fiscal–financial channel*.

4.4 Entrepreneurs: Capital Acquisition and the Financial Accelerator

Entrepreneurs operate the wholesale technology

$$Y_t = A_t \tilde{H}_t^{1-\alpha} K_{t-1}^{\alpha} \tag{54}$$

where $\tilde{H}_t = H_t^{\omega}(H_t^e)^{1-\omega}$ combines household labor H_t with the unit of labor each entrepreneur supplies inelastically, $H_t^e = 1$, whose income gives entering entrepreneurs a positive net worth. Selling at the relative price $1/X_t$, cost minimization gives $r_t^k = \alpha Y_t/(X_t K_{t-1})$, $w_t = (1-\alpha)\omega Y_t/(X_t H_t)$ and entrepreneurial labor income $W_t^e = (1-\alpha)(1-\omega)Y_t/X_t$.

At the end of time t, after the production returns have been realized, entrepreneur j has a net worth of N_{jt}. This net worth is used to partially finance the purchase of capital K_{jt} at the current real price Q_t, with the remainder L_{jt} borrowed from a financial intermediary

$$L_{jt} = Q_t K_{jt} - N_{jt}. \tag{55}$$

Entrepreneur j faces an idiosyncratic shock, resulting in the ex-post nominal gross return on capital at period $t+1$ as $\psi^j R^K_{t+1}$, where $\psi^j \sim F(\psi)$ is an i.i.d. idiosyncratic shock with support on $[0,\infty)$, $\mathbb{E}\{\psi^j\} = 1$, and $R^K_{t+1} = R^k_{t+1}\Pi_{t+1}$ is the aggregate (cross-firm average) nominal gross return to capital, realized within period $t+1$.

Entrepreneurs default when ψ^j falls below a cutoff $\bar{\psi}_{t+1}$ at which the promised repayment exhausts project resources:

$$\bar{\psi}_{t+1}\, R^K_{t+1}\, Q_t K_{jt} = Z_t\, L_{jt}, \tag{56}$$

Consequently, the net worth accumulation N_{jt} of a non-defaulting entrepreneur at time t+1, before exit and entry, is

$$N_{jt} = \frac{1}{\Pi_{t+1}}\left[\psi^j R^K_{t+1} Q_t K_{jt} - Z_t L_{jt}\right], \tag{57}$$

and zero for a defaulting one, where Z_t is the promised gross nominal non-default loan rate by entrepreneur j for the loan. Unlike the original BGG contract, in which the loan rate adjusts ex post to guarantee the lender a state-by-state return, we follow Rannenberg (2016) in assuming that Z_t is fixed at origination and is *not* contingent on the period-t+1 aggregate state. An unexpected aggregate shock therefore moves the realized default rate and, with it, the realized return on the lender's diversified loan portfolio.

Default threshold. The optimal contract is characterized by a cutoff $\bar{\psi}^j_{t+1}$ such that if $\psi^j \geq \bar{\psi}^j_{t+1}$ the entrepreneur repays the loan; if $\psi^j < \bar{\psi}^j_{t+1}$ the firm defaults. The cutoff satisfies

$$\overline{\psi}^j_{t+1} = \frac{Z_t L_{jt}}{R^K_{t+1} Q_t K_{jt}} \tag{58}$$

Entrepreneur's Optimal Financial Contract. The entrepreneur j's expected earnings are the difference between the earnings on capital investment given all realizations of idiosyncratic shocks exceeding the threshold value and the costs of loans during non-default stage:

$$\mathcal{E}^j_{t+1} = \mathbb{E}_t \left[\left(\int_{\bar{\psi}^j_{t+1}}^{\infty} \psi \, f(\psi) d\psi - \left(1 - F(\bar{\psi}^j_{t+1})\right) \bar{\psi}^j_{t+1} \right) R^k_{t+1} Q_t K_{jt} \right]. \tag{59}$$

where $f(\psi)$ is the density function of ψ and $F(\overline{\psi}^j_{t+1})$ is the probability of default at time $t+1$ defined by

$$F(\overline{\psi}^j_{t+1}) = \int_0^{\overline{\psi}^j_{t+1}} f(\psi) d\psi.$$

The entrepreneur maximizes net earnings in (59) by choosing $\bar{\psi}^j$ and K_{jt} subject to the intermediary's participation constraint, which requires the expected return on the loan to cover the intermediary's required return $\mathbb{E}_t R^P_{t+1}$. The maximization problem of the entrepreneur is summarized as below:

$$\max_{\{K_{jt}, \bar{\psi}^j_{t+1}\}} \; \mathbb{E}_t \left[(1 - \Gamma(\bar{\psi}^j_{t+1})) R^K_{t+1} Q_t K_{jt} \right], \tag{60}$$

subject to

$$\mathbb{E}_t \left[R^K_{t+1} Q_t K_{jt} (\Gamma(\bar{\psi}^j_{t+1}) - \mu G(\bar{\psi}^j)) \right] = \mathbb{E}_t (R^P_{t+1} L_{jt}). \tag{61}$$

where $\mathbb{E}_t R^P_{t+1}$ is the expected nominal diversified return required by the constrained intermediary. We also use the standard definition $\Gamma(\bar{\psi}) \equiv \bar{\psi} \int_{\bar{\psi}}^{\infty} f(\psi) \, d\psi + \int_0^{\bar{\psi}} \psi f(\psi) \, d\psi$, reflecting the share of capital returns going to the lender, and $G(\bar{\psi}) \equiv \int_0^{\bar{\psi}} \psi f(\psi) \, d\psi$, capturing default events.

This is the precise sense in which the lender's opportunity cost is tied to the government bond yield: the right-hand side of the participation constraint is not the risk-free rate, as in the original BGG model, but the intermediary's required return, which is priced off the government yield via (53). Because the participation constraint holds in expectation rather than state by state, the *realized* return on the loan portfolio satisfies, to first order,

$$\hat{R}^P_{t+1} \;=\; \underbrace{\mathbb{E}_t \hat{R}^P_{t+1}}_{\text{expected at origination}} \;+\; \chi_b \left(\hat{R}^K_{t+1} - \mathbb{E}_t \hat{R}^K_{t+1} \right), \tag{62}$$

where $\chi_b = 1 - \varepsilon_A \in (0,1)$ and ε_A is the elasticity of the lender's net repayment share with respect to the cutoff. Unexpectedly weak project returns or unexpectedly low inflation are absorbed partly by intermediary net worth. This is the loan-loss channel through which the contract feeds into the intermediary's balance sheet.

For the constraint condition in (61), $\mu G(\bar{\psi})$ is the associated monitoring costs, therefore the left-hand side term $\Gamma(\bar{\psi}^j) - \mu G(\bar{\psi}^j)$ reflects the share of gross returns accruing to the lender net of bankruptcy costs. The right-hand side is the intermediary's required return, which is tied to the government bond return rather than to the risk-free short rate. This is a deviation from the original BGG setting, in which the lender's opportunity cost is the risk-free rate on a one-period bond.

The first-order conditions of the contract deliver the familiar external-finance-premium schedule. In log-linear form,

$$\mathbb{E}_t \widehat{R}^K_{t+1} - \mathbb{E}_t \widehat{R}^P_{t+1} \;=\; \nu_{lev} \left(\widehat{Q}_t + \widehat{K}_t - \widehat{N}_t \right), \qquad \nu_{lev} > 0 : \tag{63}$$

the expected nominal project return exceeds the intermediary's required nominal return by a premium increasing in entrepreneurial leverage. Monetary policy affects investment both through the usual aggregate-demand and asset-price channels and through $\mathbb{E}_t R^P_{t+1}$, whose sensitivity to the policy rate is governed by the maturity of government debt via (53).

4.4.1 Evolution of Net Worth

So far we have established the optimizing decision of a representative entrepreneur. We now assume that each period, after returns are realized, a fraction $1-\gamma$ of entrepreneurs exits and consumes its residual equity. Each period new entrepreneurs enter in sufficient numbers so that the supply of entrepreneurial labor remains constant. The aggregate net worth accumulating at the end of time t is as follows

$$N_t = \gamma \left[\big(1 - \mu G(\bar{\psi}_t)\big) R^k_t Q_{t-1} K_{t-1} - \tfrac{R^P_t}{\Pi_t} (Q_{t-1} K_{t-1} - N_{t-1}) \right] + W^e_t. \tag{64}$$

4.5 Capital Producer

Entrepreneurs buy capital from a representative, competitive capital producer who uses a portion of final goods purchased from retailers as investment goods I_t to produce capital goods. The production, which is subject to quadratic investment adjustment costs $S(.)$, generates $(1 - S(\frac{I_t}{I_{t-1}}))I_t$ capital goods. These goods are sold at a real price Q_t per unit at the end of period t.

The adjustment cost function, similar to Smets and Wouters (2007) and Christiano, Eichenbaum, and Evans (2005), is specified as

$$S\left(\frac{I_t}{I_{t-1}}\right) = \frac{\phi_s}{2}\left(\frac{I_t}{I_{t-1}} - 1\right)^2, \tag{65}$$

where $\phi_s = S''(1)$ is the adjustment parameter. At the steady state, $S = S' = 0$.

The produced capital goods combine with the existing capital stock to generate new capital goods. In other words, the capital accumulation process is described by

$$K_t = (1-\delta)K_{t-1} + (1 - S(\frac{I_t}{I_{t-1}}))I_t, \tag{66}$$

and the producer's optimality condition delivers the standard Tobin's-q relation linking the capital price to current and expected investment growth, to first order:

$$\widehat{Q}_t = \phi_s\big[(\widehat{I}_t - \widehat{I}_{t-1}) - \beta\mathbb{E}_t(\widehat{I}_{t+1} - \widehat{I}_t)\big]. \tag{67}$$

4.6 Retail Sector

In order to motivate sticky prices, two additional ingredients are added to the model. First, the retail sector is assumed to be monopolistically competitive. Second, there are costs of adjusting nominal prices.

Optimal price setting. Retailers purchase wholesale goods from entrepreneurs and transform them into differentiated varieties according to

$$Y_t = \frac{Y_t^W}{\nu_t^p},$$

where $Y_t = \big(\int_0^1 Y_t(i)^{1-1/\epsilon_p}\,di\big)^{\epsilon_p/(\epsilon_p-1)}$ and $\nu_t^p = \int_0^1 \big(P_t(i)/P_t\big)^{-\epsilon_p}di$ is price dispersion. Dispersion is second order around a zero-inflation steady state and therefore does not appear in the log-linearized system.

Price setting is subject to Calvo rigidity: each retailer may reoptimize with probability $1-\theta_p$ in any period, and by the law of large numbers a fraction $1-\theta_p$ of retailers resets each period. Given a common real marginal cost MC_t, a retailer resetting at t chooses P_t^* to maximize expected discounted real profits

$$\mathbb{E}_t \sum_{k=0}^{\infty} \theta_p^k\, M_{t,t+k}\, Y_{t+k}(i) \left[\frac{P_t^*}{P_{t+k}} - MC_{t+k}\right], \tag{68}$$

subject to the sequence of demand constraints

$$Y_{t+k}(i) = \left(\frac{P_{t+k}(i)}{P_{t+k}}\right)^{-\epsilon_p} Y_{t+k}, \tag{69}$$

where $M_{t,t+k} = \beta^k\Lambda_{t+k}/\Lambda_t$ is the household's real stochastic discount factor. All resetting retailers face the same problem and therefore choose the same price, so the aggregate price level evolves as

$$P_t = \Big[(1-\theta_p)\big(P_t^*\big)^{1-\epsilon_p} + \theta_p P_{t-1}^{1-\epsilon_p}\Big]^{\frac{1}{1-\epsilon_p}}. \tag{70}$$

Log-linearizing (68) and (70) around a zero-inflation steady state gives the New Keynesian Phillips curve

$$\widehat{\pi}_t = \beta\,\mathbb{E}_t\widehat{\pi}_{t+1} - \kappa_p\,\widehat{X}_t, \qquad \kappa_p = \frac{(1-\theta_p)(1-\beta\theta_p)}{\theta_p}, \tag{71}$$

where $X_t = 1/MC_t$ is the gross price markup, so that $\widehat{X}_t = -\widehat{MC}_t$

4.7 Government and Monetary Policy

4.7.1 Government Budget Constraint

The government issues long-term perpetuity bonds to finance spending net of taxes:

$$G_t + \frac{(1+\kappa q_t^L)}{\Pi_t} B_{t-1}^L = T_t + q_t^L B_t^L, \tag{72}$$

where B_t^L is the government bonds outstanding with its price q_t^L, G_t government spending, T_t total real tax revenue, comprising labor-income taxes $\tau_t^\ell w_t H_t$, consumption taxes $\tau_t^c C_t$, and lump-sum taxes T_t^{LS}; the budget constraint and the fiscal rules are written on the *market value* of debt, $q_t^L B_t^L$, irrespective of the intermediaries' internal HTM accounting. The government's *effective* (book) interest rate—the empirical counterpart of interest outlays over the debt stock—reprices only as debt rolls over, $\widehat{i}_t^{\text{eff}} = (1-\kappa)\widehat{R}_t^L + \kappa\, \widehat{i}_{t-1}^{\text{eff}}$: immediately under $\kappa = 0$, at a rate of about 2% per quarter under ten-year duration. This is the model's *debt-rollover channel* and maps directly to the effective-rate responses in the empirical section. The monetary policy bond is in zero net supply, so it does not appear in the government budget constraint.

4.7.2 Fiscal rules

Both spending and taxes contain automatic-stabilizer components in line with Leeper, Plante, and Traum (2010), and taxes additionally react to the lagged market value of debt to guarantee sustainability. Government purchases obey

$$\widehat{G}_t = -\alpha_G\, \widehat{Y}_t + e_t^g, \tag{73}$$

so that spending is counter-cyclical when $\alpha_G > 0$. The labor-tax *rate* follows

$$\widehat{\tau}_t^\ell = \tau_b^\ell \left(\hat{q}_{t-1}^L + \hat{B}_{t-1}^L\right) + \tau_y^\ell\, \widehat{Y}_{t-1}, \tag{74}$$

and lump-sum taxes follow

$$\widehat{T}_t^{LS} = \tau_b \big(\hat{q}_{t-1}^L + \hat{B}_{t-1}^L\big) + \tau_y\, \widehat{Y}_{t-1}, \tag{75}$$

the consumption-tax rate is constant. The distortionary component implies that debt consolidation after a tightening operates partly through a supply-side wedge, another conduit from fiscal positions to activity.

4.7.3 Monetary Policy Rule

The central bank sets the short-term nominal rate R_t according to a Taylor rule:

$$\frac{R_t}{\bar{R}} = \left(\frac{R_{t-1}}{\bar{R}}\right)^{\rho_r} \left(\left(\frac{\Pi_t}{\bar{\Pi}}\right)^{\theta_\pi} (Y_t^{gap})^{\theta_y}\right)^{(1-\rho_r)} e^{\varepsilon_{mt}} \tag{76}$$

where ε_{mt} is the monetary policy shock- the shock of interest in our analysis. The first term on the right-hand-side represents the smoothing behavior of the central bank in setting the interest rate. The second term denotes the deviation of inflation from its steady-state level $\bar{\Pi}$. The third term is the output gap which is proxied by the inverse price markup, as in Gertler and Karadi (2011).

4.8 Resource Constraint

The market for final goods clears in every period

$$Y_t = C_t + C_{E,t} + I_t + G_t + \mu G(\overline{\psi}_t) R_t^k Q_{t-1} K_{t-1}. \tag{77}$$

5 Model Parametrization and Results

5.1 Model Parametrization

The model is parameterized to the US economy at a quarterly frequency. Table 2 reports the parameter values. The standard parameters are taken from the literature.

Households and production. The discount factor is $\beta = 0.995$, implying a steady-state annual real rate of 2.0 percent. The capital share is $\alpha = 0.35$, quarterly depreciation $\delta = 0.025$, the Frisch elasticity $\eta = 3$ and habit persistence $b = 0.80$. The household labor share of output is 0.64, leaving a one-percent entrepreneurial-labor share. The investment-adjustment curvature is $\phi_s = 0.56$.

Nominal rigidities and monetary policy. Calvo price and wage stickiness are both 0.75, with a wage elasticity of substitution $\epsilon_w = 6$. The gross price markup is $X = 1.10$. The policy rule has smoothing $\rho_r = 0.90$, an inflation coefficient $\theta_\pi = 1.5$ and an output coefficient $\theta_y = 0.125$, the output gap being proxied by the inverse markup.

Financial intermediaries. The block is identified by four targets. (i) An annualized private-credit spread over deposits of 138 basis points, which sets $\bar{R}^P$. (ii) A diversion weight, $\Delta = 1/3$ as in Sims and Wu (2021), implying a government-bond spread of 46 basis points. (iii) Ordinary intermediary leverage of 4, which sets net worth. (iv) A government-securities share of intermediary assets of 28 percent, U.S. average, which sets the bond book. Banker survival is $\sigma_F = 0.972$, following Gertler and Karadi (2013), implying an expected horizon of about nine years. In the baseline, the share of government portfolio carried at amortized cost is set at $\chi = 1$. We also consider different values of $\chi \in \{0, 0.25, 0.5, 0.75\}$.

Financial accelerator. Following standard calibration, three restrictions identify the contract: the quarterly default probability of 0.75 percent, implying the three-percent annual business-failure rate; entrepreneurial leverage $QK/N = 2$; and a monitoring cost $\mu = 0.298$, the value used by Rannenberg (2016). Consequently, the idiosyncratic volatility is $\sigma_\psi = 0.271$ and the elasticity of the premium with respect to leverage is $\nu_{lev} = 0.097$. The survival rate is calibrated $\gamma = 0.966$, based on the net-worth identity, and the share of unexpected surprise absorbed by the lender is $\chi_b = 0.028$.

Bond maturity. Government debt is modeled as a perpetuity with a decaying coupon stream following Woodford (2001). This is the parameter of interest, so we consider two cases. First, the coupon decay rate κ is set to $\kappa = 0$ corresponding to a one-period bond or

short maturity. Second, we set $\kappa = 0.981$ to imply a duration of 40 quarters. The steady-state debt-to-annual-output ratio is set to 70% of GDP. The two maturity calibrations share an identical steady state allocation, in terms of real quantities, market values, rates of return, leverage and net worth.

The debt-supply wedge. Households price government debt subject to a wedge increasing in the market value outstanding, which is what allows debt supply to affect its own yield. We set the elasticity to $\eta_d = 0.012/4$, so that a ten percent increase in the market value of debt relative to GDP raises the expected annualized excess holding return by 12 basis points, in line with Krishnamurthy and Vissing-Jorgensen (2012) as well as in the conservative end of the range surveyed by Mian et al. (2025), which runs from 0.008 to 0.031.

Fiscal policy. Government purchases are $G/Y = 0.20$ and the automatic-stabilizer elasticity of spending is $\alpha_G = 0.18$ (Price, Dang, and Botev, 2015). Steady-state tax rates are $\tau^\ell = 0.22$ and $\tau^c = 0.03$, generating revenues of 12.8 and 1.5 percent of GDP respectively; lump-sum taxes close the budget at 7.4 percent. The labor-tax rule responds to lagged debt and output with coefficients $(0.045, 0.36)$ and the lump-sum rule with $(0.80, 0.11)$.

Table 2: Calibration: primitives, targets and implied objects

Block	Parameter	Symbol	Value
Preferences			
	Discount factor	β	0.995
	Habit persistence	b	0.80
	Frisch elasticity	η	3
Technology			
	Capital share	α	0.35
	Depreciation (quarterly)	δ	0.025
	Investment adjustment cost	ϕ_s	0.56
Nominal rigidities			
	Calvo prices	θ_p	0.75
	Calvo wages	θ_w	0.75
	Wage elasticity of substitution	ϵ_w	6
	Gross price markup	X	1.10
Financial intermediaries			
	Banker survival	σ_F	0.972
	Private spread, annualized	$\bar{R}^P - \bar{R}^D$	138 bp
	Government spread, annualized	$\bar{R}^L - \bar{R}^D$	46 bp
	Diversion weight	Δ	1/3
	Ordinary leverage	$(\bar{X}^P + \bar{A}^{LB})/\bar{N}^F$	4
	Government bond share of FI assets	s_G	0.28
	Adjusted leverage	$\bar{\phi}$	3.253
	Enforcement parameter	θ	0.508
Financial accelerator			
	Idiosyncratic volatility	σ_ψ	0.271
	Default probability, quarterly	$F(\bar{\psi})$	0.75%
	Entrepreneurial leverage	QK/N	2.01
	Monitoring cost	μ	0.298
	Default cutoff	$\bar{\psi}$	0.498
	Survival probability	γ	0.966
	EFP elasticity to leverage	ν_{lev}	0.097
	Loan-loss loading	χ_b	0.028
Monetary policy			
	Inflation coefficient	θ_π	1.5
	Output coefficient	θ_y	0.125
	Smoothing	ρ_r	0.90
Bond maturity and debt			
	Coupon decay: short / long	κ	0 / 0.9810
	Duration, quarters: short / long	$\bar{q}^L \bar{R}^L$	1 / 40
	Debt to annual GDP	$q^L B^L / 4Y$	0.70
	Debt-supply elasticity	η_d	0.012/4

continued on next page

Table 2, continued

Block	Parameter	Symbol	Value
Fiscal policy			
	Government spending share	G/Y	0.20
	Labor / consumption tax rate	τ^{ℓ}, τ^{c}	0.22, 0.03
	Spending elasticity to output	α_G	0.18
	Labor-tax rule (debt, output)	$\tau_b^{\ell}, \tau_y^{\ell}$	0.045, 0.36
	Lump-sum rule (debt, output)	τ_b, τ_y	0.80, 0.11
Implied steady-state ratios			
	Investment share	I/Y	17.2%
	Consumption share	C/Y	50.2%
	Entrepreneur consumption share	C^E/Y	11.9%
	Labor / consumption tax revenue	T^{ℓ}/Y, T^{c}/Y	12.8%, 1.5%
	Lump-sum tax revenue	T^{LS}/Y	7.4%

5.2 Baseline Results

The model is solved by first-order perturbation around the deterministic steady state; all results below are impulse responses of that linear system. Figure 9 reports responses to a contractionary monetary policy shock of 100 basis points under two alternative debt maturity structures: a short maturity, i.e. one-period bond as in BGG, and a long maturity, i.e. a (Macaulay) duration of forty quarters.

The model reproduces all three key empirical facts. Debt and debt-to-GDP increase more under a short maturity structure. Specifically, the par debt-to-GDP ratio rises 3.2 percentage points of annual GDP at peak under short maturity against 2.2 under long. The private lending rate rises above 1.0 annualized percentage points under short maturity against 0.2 under long. Output also falls 2.5 per cent at trough under short maturity against 1.7 under long, a gap of 0.78 percentage points or 31 per cent of the short-maturity response, which is largely driven by the drop in investment.

Three channels generate these gaps.

The rollover channel. A monetary tightening raises the return the government must offer on newly issued debt. With short maturity, the corresponding yield is more sensitive to the policy rate, whereas a long bond's yield is an average of expected short rates over time and therefore moves far less. Also, with a short maturity structure, debt needs to be rolled over more frequently, leading to a faster and fuller pass-through of the policy rate to the effective interest rate on government debt. The effective interest rate rises 0.74 percentage points under short maturity but only 0.02 under long. Under long maturity, the effective rate combines the smaller response of the yield with the slow rollover of the legacy stock, therefore insulating the government's interest bill from the policy rate hike. By contrast, the faster pass-through under short maturity directly raises the debt service burden, contributing mechanically to the accumulation of real debt and the rise in the debt-to-GDP ratio.

The lender's opportunity cost. This is the paper's central mechanism which matters for the real economy and operates through equation (52). The lender to the private sector is a constrained intermediary that also holds government bonds, and the two assets compete for the same scarce pledgeable capacity. A dollar of lending capacity used on Treasuries is a dollar unavailable for firms. So when the government bond becomes a better deal, the intermediary's opportunity cost of extending private credit rises with it, and by more than one-for-one: because a Treasury absorbs only a fraction Δ of the capacity a private loan absorbs, the required return on private credit is tied to the government yield with pass-through $1/\Delta > 1$. Whether duration matters for that opportunity cost in a non-trivial way, however, depends on an institutional feature of bank balance sheets: an intermediary that marks its bond book to market and can sell at will prices the one-period holding return, whose expected value tracks the policy rate whatever the bond's duration, whereas one holding the bonds at amortized cost recognizes the yield locked in at purchase, which does depend on duration. The maturity of public debt therefore shows up directly in the price firms pay for external finance, and Section 5.3.2 shows that adopting the first convention removes most of the maturity gap documented here. The latter is our baseline model. Consistently, the promised nominal return on private credit Z_t rises 1.1 percentage points under short maturity against 0.2 under long.

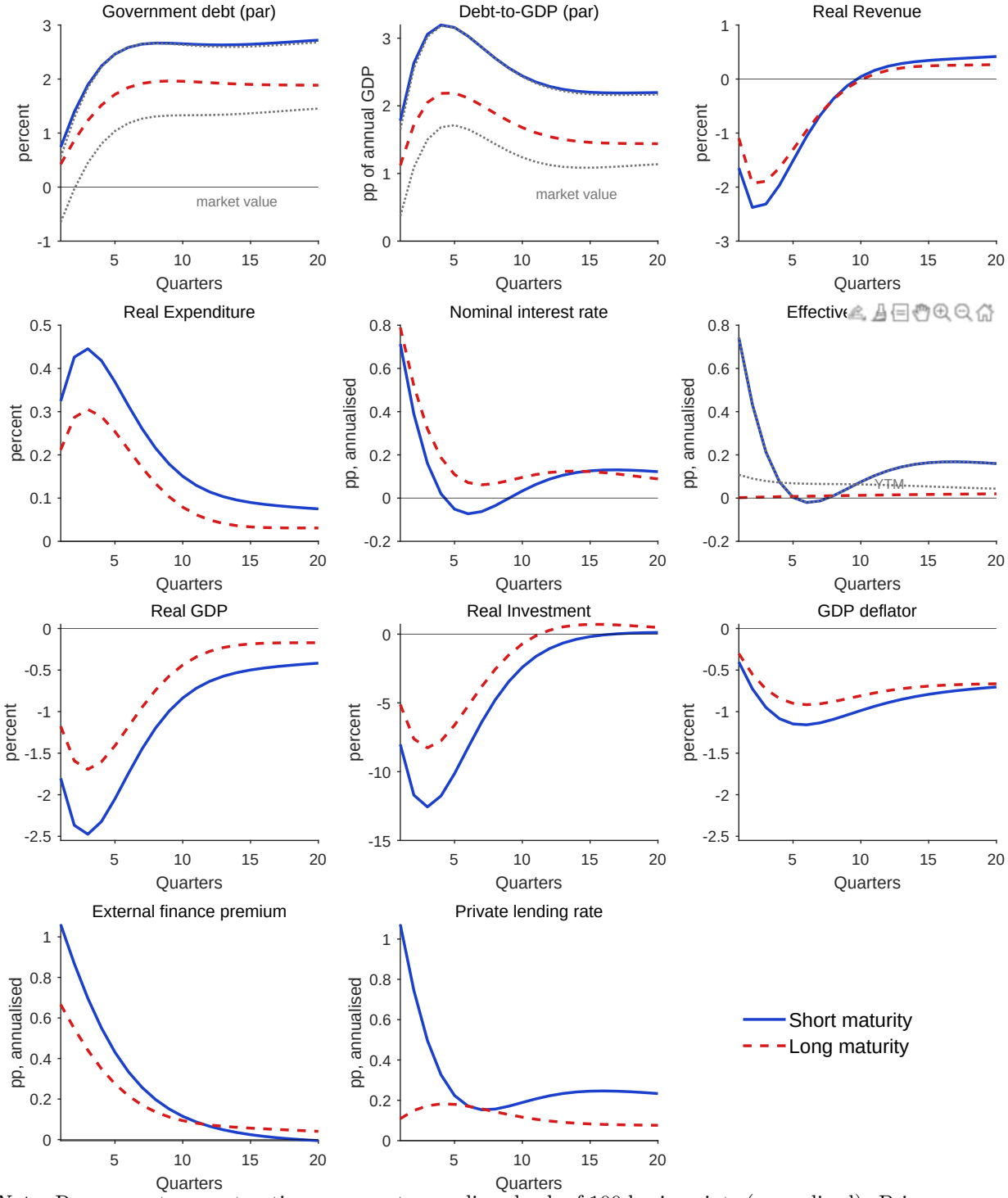
The accelerator. Finally, the higher required return associated with short maturity com-

presses entrepreneurial net worth, raising leverage and feeding back into the external finance premium, particularly under short maturity. The external finance premium rises by 1.1 pp under short maturity compared with 0.7 under long. Consequently, investment and output fall by more, with the collapse in the output denominator further amplifying the rise in the debt-to-GDP ratio. However, Section 5.3.1 shows this channel amplifies but does not generate the maturity gap.

The differential real contraction across maturity structures has direct implications for the automatic stabilizers. Government revenue falls sharply under short maturity—by approximately 2.4 percent on impact—as the collapse in economic activity erodes the tax base, before gradually recovering as output returns to steady state. Under long maturity, the revenue decline is smaller. Meanwhile, government expenditure rises more under the short maturity in response to a larger contraction due to the automatic stabilizer. The combination of lower revenues and higher expenditures contributes to the greater debt accumulation observed under short maturity, reinforcing the role of the roll-over and financial accelerator channels in amplifying the fiscal consequences of monetary tightening.

The valuation effect can be observed in the par and market-value debt divergence under long maturity: par rises 2.0 per cent while market value rises only 1.5, because bond price q^L falls. Under short maturity the two are indistinguishable, around 2.8. Finally, the GDP deflator falls by less across maturity regimes than output does, because price stickiness damps the pass-through, which is consistent with our baseline empirical result.

FIGURE 9: MODEL-BASED EFFECTS OF MONETARY POLICY SHOCKS



Note: Responses to a contractionary monetary policy shock of 100 basis points (annualized). Price response is the cumulated response of inflation. Debt is at par value; the dotted line is market value. YTM: Yield to maturity

5.3 Discussions

5.3.1 The role of two frictions

The model contains two financial frictions: enforcement constraint facing financial intermediaries and costly state verification between FIs and entrepreneurs. We now assess to what extent each friction matters. Table 3 switches each off in turn.

The GK's enforcement constraint. When we switch off the enforcement constraint, so that private credit is priced at the deposit rate, and the intermediary no longer trades the government bond at the margin, the maturity effect to the real economy ceases to matter entirely.[18] Output and investment responses become similar across both maturity regimes (Table 3). Without the enforcement constraint, the external finance premium still rises, so the accelerator remains relevant. However, it rises by the *same* amount of 0.90 percentage points regardless of maturity duration. This indicates that the macro and financial effects of public debt maturity duration work primarily through the intermediary channel. This point echoes the findings of Andreolli (2021) who argues that the maturity structure does not matter to the real economy even when the BGG-typed financial friction exists if the credit cost to entrepreneurs is pinned down by the same (policy) rate across maturity regimes (i.e., the same lender's opportunity cost). Additionally, in the current setting, most of the feedback of debt happens via the endogenous lump-sum tax response and the feedback of debt risk on government bond yield does not alter deposit rate or private lending rate.

In terms of the magnitude, compared to the baseline, output drops more under long maturity (2.03 percent compared to 1.70 percent) because of a sharper increase in credit cost to firms (from 0.18 pp to 0.99 pp). In contrast, shutting down the GK financial friction channel helps mitigate the impacts under the short maturity, leading to a smaller increase in the lending rate to firms (from 1.07 pp to 0.99 pp).

One thing does survive. Government debt at par still rises more under short maturity, by 0.22 percentage points against 0.85 in the baseline. This residue is the rollover channel and the debt-supply wedge: with short debt the Treasury refinances a larger share of its stock at the higher rate. Roughly a quarter of the fiscal maturity effect on debt is therefore mechanical, and the remaining three quarters comes from the contraction in activity that the intermediary channel produces.

The BGG's financial accelerator. Switching off the BGG accelerator instead, reducing external finance premium to zero while leaving the intermediary constrained, does not alter the finding that under short maturity debt increases more and output falls more.[19] That said, the corresponding gap between short and long maturity reduces. For instance, the output gap decreases from 0.78 percent to 0.56 percent and debt-to-GDP from 1 percentage point to 0.8 percentage point. This implies that the accelerator works as an amplifier.

Therefore, the dependence of price of credit on the maturity of government debt comes from the intermediary, because only the intermediary holds both assets and must trade them off against a common constraint. Remove it and the government bond becomes irrelevant to the corporate borrower; maturity then affects the Treasury's interest bill.

[18] See Appendix A137 for the impulse response functions to a tightening monetary policy shock, when the GK's enforcement constraint facing the FIs is turned off.

[19] See Appendix A138 for impulse response functions to a tightening monetary policy shock, when the BGG's costly state verification is turned off.

When shutting both financial frictions, the results reflect both features. First, in line with the no-GK case, the response becomes similar for real variables across maturity regimes, while debt still increases more under the short duration due to rollover exposure. Second, output falls less and debt increases less compared to the case when turning off only the GK friction, therefore confirming the amplification role played by the BGG's financial accelerator for the monetary policy shocks.

Table 3: Peak/Trough Responses by Selected Variables and Model Variants

	Baseline	No BGG	No GK	Neither frictions	MTM	No debt wedge	No fiscal reactions
Real GDP (trough), %							
Short maturity	−2.48	−1.75	−2.03	−1.33	−2.48	−2.03	−2.42
Long maturity	−1.70	−1.19	−2.03	−1.33	−2.34	−1.33	−1.53
Difference	**−0.78**	**−0.56**	**−0.00**	**−0.00**	**−0.13**	**−0.71**	**−0.89**
Investment (trough), %							
Short maturity	−12.56	−8.62	−10.08	−6.22	−12.56	−10.08	−11.74
Long maturity	−8.27	−5.58	−10.07	−6.20	−11.82	−6.22	−7.06
Difference	**−4.29**	**−3.04**	**−0.01**	**−0.01**	**−0.74**	**−3.85**	**−4.68**
Government debt at par (peak), %							
Short maturity	2.81	2.62	2.44	2.17	2.81	2.35	1.27
Long maturity	1.96	1.82	2.21	1.72	2.50	1.64	0.96
Difference	**0.85**	**0.80**	**0.22**	**0.44**	**0.31**	**0.71**	**0.31**
Debt-to-GDP at par (peak), pp							
Short maturity	3.19	2.58	2.73	2.08	3.19	2.71	2.52
Long maturity	2.19	1.76	2.52	1.78	2.88	1.76	1.64
Difference	**1.01**	**0.82**	**0.21**	**0.30**	**0.32**	**0.94**	**0.89**
Private lending rate Z_t (peak), pp							
Short maturity	1.07	0.84	0.99	0.82	1.07	0.99	1.07
Long maturity	0.18	−0.10	0.99	0.82	1.03	0.08	0.14
Difference	**0.89**	**0.94**	**0.00**	**−0.00**	**0.04**	**0.91**	**0.93**
External finance premium (peak), pp							
Short maturity	1.06	0.02	0.90	0.02	1.06	0.90	1.02
Long maturity	0.67	0.01	0.90	0.02	1.01	0.54	0.59
Difference	**0.40**	**0.01**	**0.00**	**0.00**	**0.05**	**0.37**	**0.42**

Note: Peak/trough responses to a contractionary monetary policy shock of 100 basis points (annualized), that is, the largest absolute deviation over the response horizon, reported signed. Short maturity is one-period debt, long maturity a duration of forty quarters, and the difference is short minus long. The friction switches operate on dynamics only, while holding steady-state fixed (except the case with exogenous tax/spending). *No BGG*: no external finance premium or unexpected loan losses. *No GK*: the intermediary is unconstrained and private credit is priced at the deposit rate. *Neither friction*: turning off both BGG and GK frictions. *MTM*: mark-to-market portfolio. *No debt wedge*: shutting down the debt supply wedge impacting the government bond yield. *No fiscal reactions*: the output-stabilization terms in the spending and tax rules are removed, the steady-state income and consumption tax rates are set to zero, and the income tax no longer responds to debt; only the lump-sum response to debt is retained, which keeps the government solvent.

5.3.2 The carrying convention: Amortized cost versus mark-to-market

Having established that the mechanism runs through the intermediary's balance sheet, we now ask which characteristic of that balance sheet admits a role for duration. The answer is the basis on which the government portfolio is carried. To establish it, we replace the

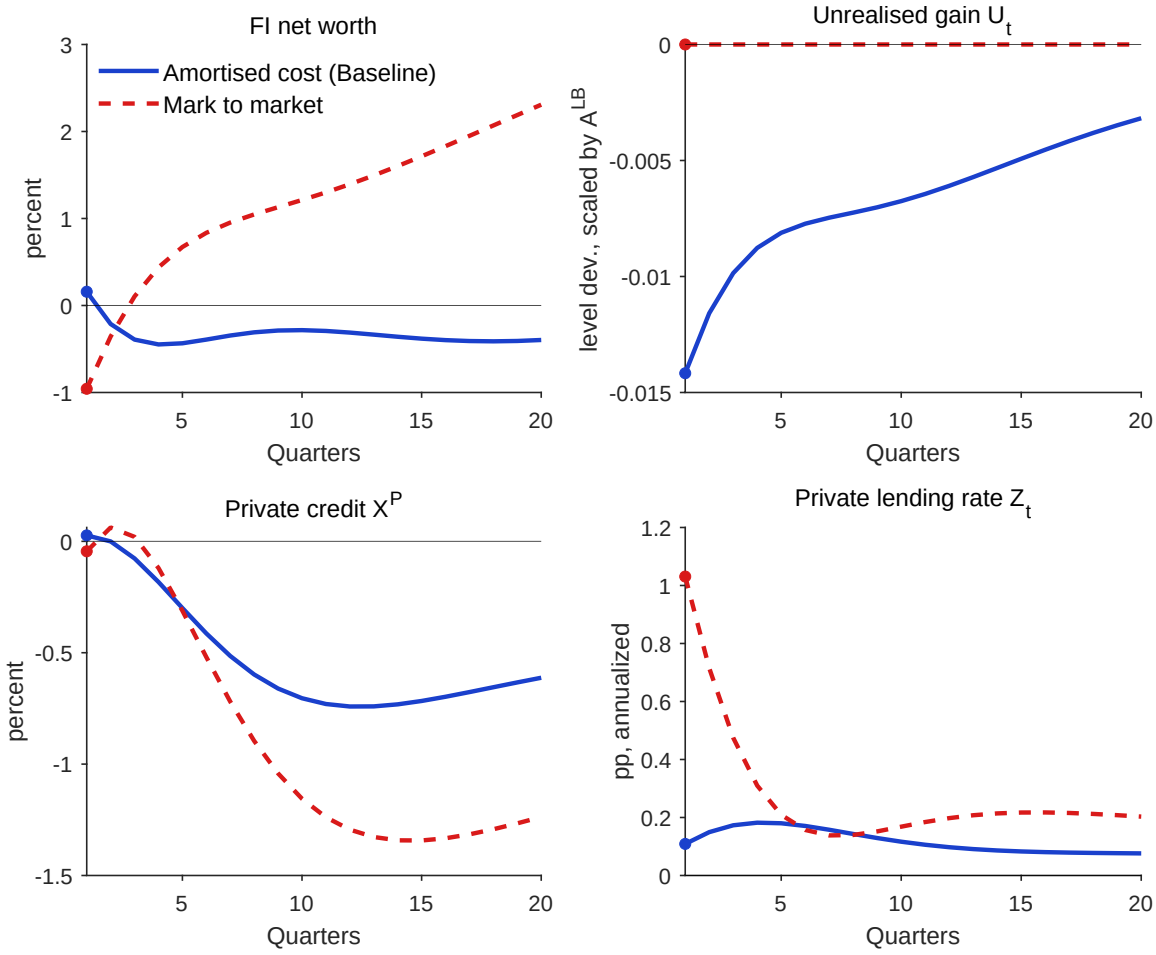
baseline amortized-cost convention with mark-to-market: the financial intermediary carries its bonds at market value, so valuation changes pass directly into book net worth, and its portfolio condition prices the one-period holding return rather than the yield locked in at acquisition.

With one-period debt the carrying value equals the market value in every period, so the two conventions coincide exactly, the accounting choice does not matter, and the responses are identical to the baseline (Table 3, column 5). Under long maturity two things change together (Figure 10).

First, the financial intermediary recognizes the fall in the bond price immediately. With the carrying value equal to the market value, the unrealized gain/loss U_t is identically zero. Hence, the valuation changes pass into book net worth in full. Net worth drops sharply on impact instead of declining gradually as it does under amortized cost (Panel 1, Figure 10).

Second, and more consequentially, the return relevant to the intermediary's bond choice is now the one-period holding return, whose expected value tracks the policy rate whatever the bond's duration. It is this second change that removes the reason the response should differ across maturity regimes at all. This higher return is what raises the lending rate on private credit. Consequently, output falls 2.34 percent at long maturity against 1.70 in the baseline, so the maturity gap narrows from 0.78 to 0.13 percentage points, and the long-maturity economy comes to behave much like the short-maturity one (see also Appendix A140). The mechanism is therefore an interaction between duration and the carrying basis rather than two separate effects.

FIGURE 10: FINANCIAL INTERMEDIARIES UNDER LONG MATURITY



Note: Responses to a contractionary monetary policy shock of 100 basis points (annualized).

5.3.3 Partial held-to-maturity

Banks do not classify their entire securities book one way or the other. We therefore let a share χ be classified held-to-maturity and carried at amortized cost, with the remaining $1-\chi$ carried at market, and trace the economy as χ moves between the two poles. In other words, χ is a valuation parameter, setting the carrying basis for a share of the book.

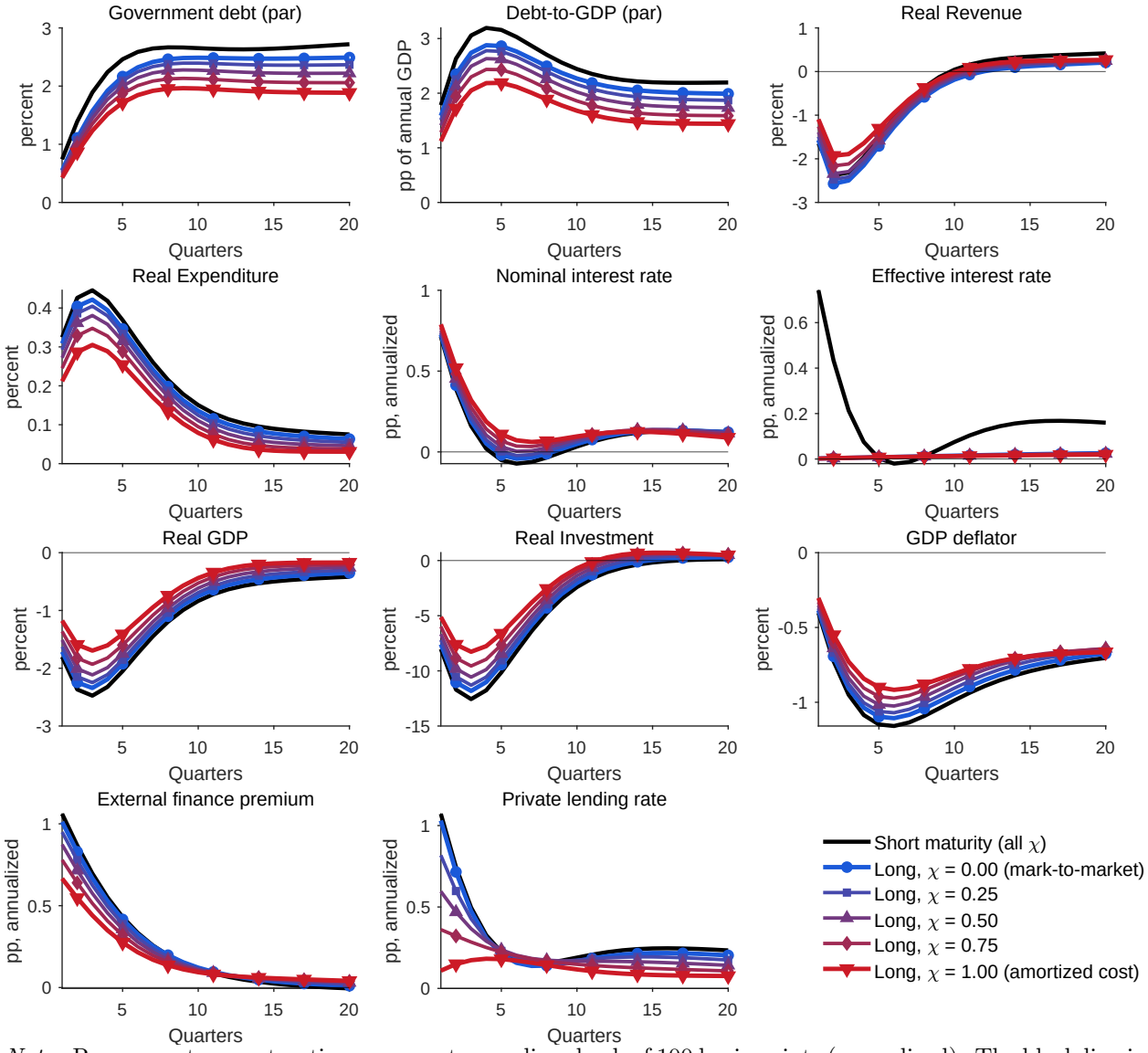
The responses are monotone in χ and close to linear. The accounting composition of the banking system's securities book therefore maps smoothly into the strength of monetary transmission. The more of the book is insulated from valuation losses, the more the lender's opportunity cost is governed by the locked-in yield rather than the policy rate, and the more the maturity of public debt matters.

Figure 11 makes the interaction visible. For one-period government bonds, the carrying value equals the market value in every period, so every χ delivers the same response, which is the black line in the figure. Under long-maturity government bonds, the responses separate, and they order monotonically in χ: the more of the book is carried at amortized cost, the smaller the contraction in output and investment, the smaller the rise in the cost of credit,

and the smaller the increase in debt and the debt ratio. The fiscal aggregates follow: with a milder contraction, revenue falls by less and the automatic stabilizers raise expenditure by less.

The smaller is χ, the closer the long-maturity responses move to the short-maturity path. The effective interest rate is the exception: the short-maturity economy's financing cost rises roughly in parallel with the policy rate while the long-maturity economy's barely moves. This is as expected, since the effective rate is governed by the rollover of the outstanding stock, which is a feature of the Treasury's balance sheet rather than of how banks carry their securities. This in turn feeds back into debt dynamics through rollover exposure. This exercise confirms that the accounting treatment of securities governs the credit channel, and with it much of the role that debt maturity plays in the macro-fiscal response to a monetary policy shock.

FIGURE 11: EFFECTS OF MONETARY POLICY SHOCKS UNDER PARTIAL HELD-TO-MATURITY



Note: Responses to a contractionary monetary policy shock of 100 basis points (annualized). The black line is the short-maturity economy; the colored lines are the long-maturity economy for values of χ, the share of the intermediary's securities book carried at amortized cost, from $\chi = 0$ (mark-to-market) to $\chi = 1$ (amortized cost throughout). Only one short-maturity line is shown because at $\kappa = 0$ the carrying value equals the market value in every period, so every χ delivers the same response. Price response is the cumulated response of inflation. Debt is at par value.

5.3.4 Fiscal Reactions and Debt-Supply Wedge

We further consider the role played by two other ingredients. First, the household's convenience-yield wedge makes the bond yield respond to the stock of debt outstanding. Removing it collapses the household's bond Euler to the pure expectations hypothesis and eliminates the time variation in the term premium by construction.[20] As noted in Table 3 (column 6),

[20]See Appendix A141 for the impulse responses to a monetary policy shock.

shutting this would reduce the borrowing cost of government, contributing to a reduction in the private credit cost and external finance premium. Consequently, output and investment fall less compared to the baseline, while debt also increases less across both maturity regimes.

Second, we switch off endogenous tax responses by setting steady-state tax rates to zero, removing the output-stabilization (automatic stabilizer) from both the spending and the tax instruments, shutting down the response of income tax to debt (which is already small), while allowing only lump-sum response to debt to keep the government solvent.[21] Doing so, the revenue drop is significantly mitigated compared to the baseline and reverses quickly to positive due to lump-sum tax in response to debt. At the same time, government expenditure remains unchanged. This improves the primary surplus relative to the baseline, contributing to improve the debt dynamics, as shown in Table 3 (column 7).[22] In both cases the ranking across maturity regimes survives, so neither is the source of the mechanism, but instead operates as a contributing factor.

[21] See Appendix A142 for the impulse responses to a monetary policy shock.

[22] Only in this exercise, steady states also change due to setting the consumption and income tax rate to zero at the steady state. Then differences reflect not only the dynamics, but also to some extent changes in the steady states.

6 Conclusions

This paper shows that the maturity structure of sovereign debt is a key determinant of monetary policy transmission. Using U.S. data and a nonlinear local-projections framework, the analysis finds that contractionary monetary policy shocks have substantially stronger effects on public debt dynamics, fiscal balances, financial spreads, and real activity when the government relies more heavily on short-term debt. When sovereign rollover exposure is high, monetary tightening produces persistent increases in public debt and the debt-to-GDP ratio, larger declines in output and private investment, and sharper deteriorations in fiscal balances and financial conditions. These effects are considerably weaker when debt maturity is longer. The findings reveal a strong two-way interaction between monetary policy and sovereign debt management. On the one hand, monetary policy has more pronounced fiscal consequences when sovereign rollover exposure is high. On the other hand, debt maturity shapes the strength of monetary transmission by influencing how aggregate demand, financial conditions, and public debt accumulation respond to policy-rate changes. Debt maturity therefore determines both how monetary policy affects fiscal outcomes and how fiscal conditions feed back into monetary transmission. These results imply that debt-management choices can materially influence both the macroeconomic effects and the fiscal costs of monetary policy. More broadly, the paper shows that high public debt combined with shorter maturities can intensify tensions between macroeconomic stabilization and fiscal sustainability, speaking directly to recent debates on fiscal dominance and central bank independence (Bajaro, Galimberti, and Qureshi, 2025; Schnabel, 2026).

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Appendix

Table A1: Descriptive statistics for the dataset used in the quarterly-frequency analysis.

Variable	Mean	St. Dev.	Min	Max	Sample available	
T-Bills share	0.24	0.08	0.11	0.43	1952q3	2025q3
Real debt (log)	15.46	0.95	14.41	17.19	1952q3	2025q3
Debt-to-GDP	63.73	29.06	30.60	132.66	1966q1	2025q3
Real GDP (log)	9.02	0.70	7.68	10.09	1947q1	2025q3
GDP deflator (log)	3.71	0.78	2.41	4.86	1947q1	2025q3
Real Investment (log)	5.03	1.39	2.35	7.11	1947q1	2025q3
EBP	0.06	0.52	-0.78	3.31	1973q1	2025q3
Real revenues (log)	7.71	0.78	6.10	8.83	1947q1	2025q3
Real primary expenditure (log)	7.98	0.62	6.71	9.18	1960q1	2025q3
Effective rate	5.21	2.62	1.57	11.91	1952q3	2025q3
RR (2004) shock	0.01	0.26	-2.07	1.61	1969q1	2019q4
BS (2023) shock	0.00	0.03	-0.12	0.07	1988q2	2023q4
Privately held short-term debt share	37.06	6.70	28.01	58.67	1975q1	2025q3
Aaa Corporate Bond Yield	6.19	2.79	2.26	15.49	1947q1	2025q4
Baa Corporate Bond Yield	7.13	3.04	3.15	16.92	1947q1	2025q4

Table A2: Descriptive statistics for the dataset used in the monthly-frequency analysis.

Variable	Mean	St. Dev.	Min	Max	Sample available	
T-Bills share	0.24	0.08	0.10	0.43	1952m7	2026m1
Real debt (log)	15.56	0.93	14.43	17.19	1959m1	2025m10
Debt-to-GDP	61.95	27.71	29.89	131.24	1959m1	2025m10
Real GDP (log)	9.22	0.56	8.12	10.09	1959m1	2025m10
GDP deflator (log)	3.91	0.67	2.71	4.87	1959m1	2025m10
Real Investment (log)	5.09	0.45	4.14	5.93	1950m1	2025m3
EBP	0.06	0.53	-1.06	3.43	1973m1	2025m12
Real revenues (log)	4.95	0.61	3.59	6.53	1959m1	2025m10
Real primary expenditure (log)	4.95	0.67	3.51	6.81	1959m1	2025m10
Effective rate	5.20	2.62	1.56	11.91	1952m7	2026m1
RR (2004) shock	0.00	0.27	-3.30	1.84	1969m3	2019m12
BS (2023) shock	0.00	0.05	-0.38	0.19	1988m2	2023m12
Aaa Corporate Bond Yield	6.32	2.14	2.14	15.49	1950m1	2026m1
Baa Corporate Bond Yield	7.27	3.16	3.16	17.18	1950m1	2026m1

Table A3: Kleibergen-Paap rk Wald F Statistics for IV Estimation. LP in Levels.

Variable	**First-stage F**
Real debt	10.42
Debt-to-GDP	13.18
Real revenues	8.89
Real primary expenditure	8.57
Effective rate	11.89
Real GDP	7.81
GDP deflator	6.67
Real Private Fixed Investment	9.55
EBP	7.15
Aaa Corporate Bond Yield	12.02
Baa Corporate Bond Yield	12.20
Stock-Yogo 10% Critical Value	**7.03**

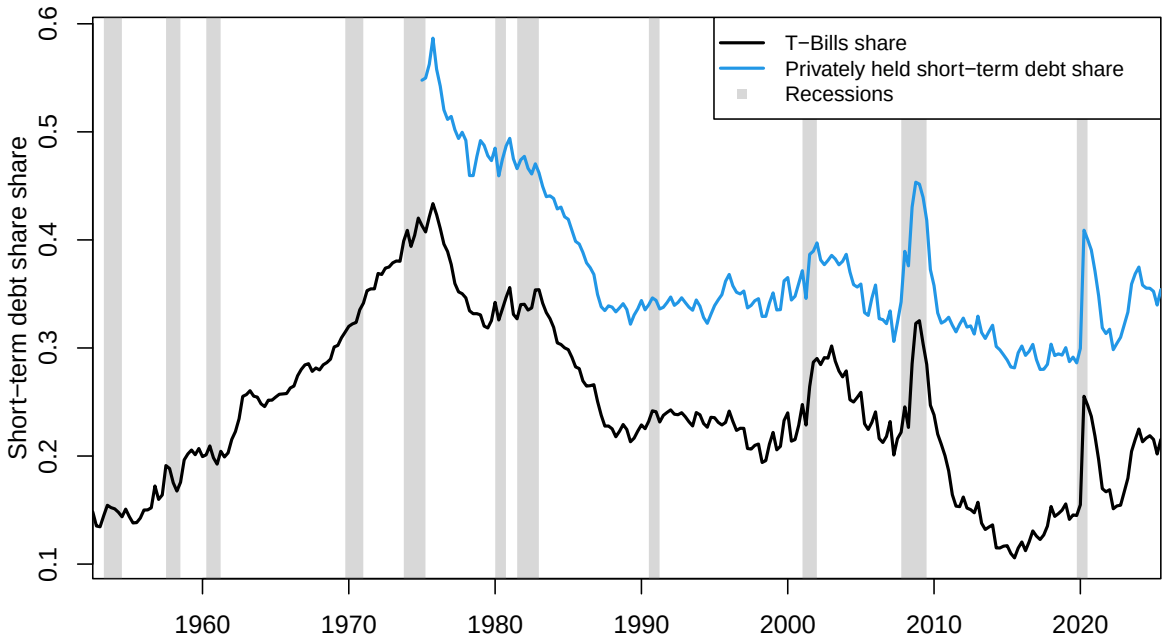
Note: This table shows the average across all horizons of the first-stage Kleibergen-Paap rk Wald F statistic for all variables. Estimates are obtained via 2SLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta^{(h)}_{\text{high}}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta^{(h)}_{\ell,\text{high}}W_{t-\ell}) + (1 - F(z_{t-1}))(\beta^{(h)}_{\text{low}}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta^{(h)}_{\ell,\text{low}}W_{t-\ell}) + \delta^{(h)}F(z_{t-1}) + \varepsilon^{(h)}_{t+h}$, where $\widehat{\Delta FFR}_t$ is instrumented with the narrative exogenous monetary policy shock series described in the main text.

Table A4: Kleibergen-Paap rk Wald F Statistics for IV Estimation. LP in Long Differences.

Variable	**First-stage F**
Real debt	9.42
Debt-to-GDP	7.68
Real revenues	6.48
Real primary expenditure	5.44
Effective rate	3.93
Real GDP	4.90
GDP deflator	9.60
Real Private Fixed Investment	8.16
EBP	5.99
Aaa Corporate Bond Yield	5.00
Baa Corporate Bond Yield	4.53
Stock-Yogo 10% Critical Value	**7.03**

Note: This table shows the average across all horizons of the first-stage Kleibergen-Paap rk Wald F statistic for all variables. Estimates are obtained via 2SLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + F(z_{t-1})(\beta^{(h)}_{\text{high}}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta^{(h)}_{\ell,\text{high}}W_{t-\ell}) + (1 - F(z_{t-1}))(\beta^{(h)}_{\text{low}}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta^{(h)}_{\ell,\text{low}}W_{t-\ell}) + \delta^{(h)}F(z_{t-1}) + \varepsilon^{(h)}_{t+h}$, where $\widehat{\Delta FFR}_t$ is instrumented with the narrative exogenous monetary policy shock series described in the main text.

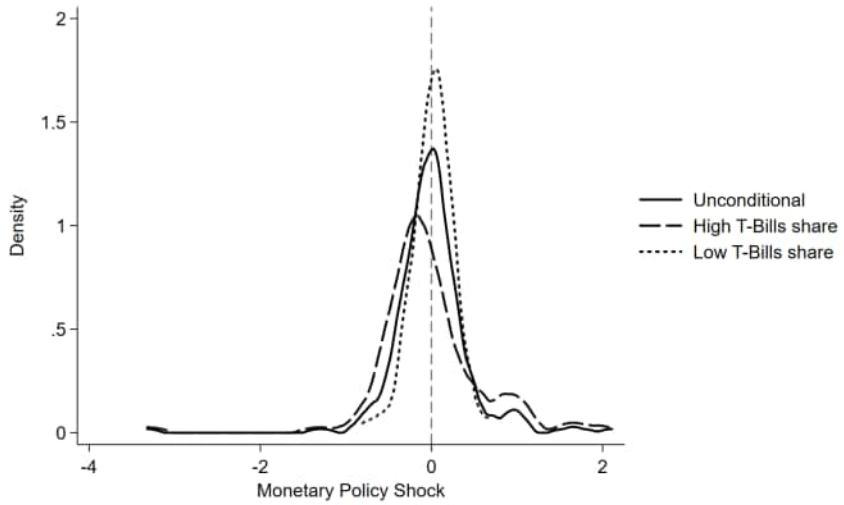
Figure A1: T-Bills share and privately held short-term debt share



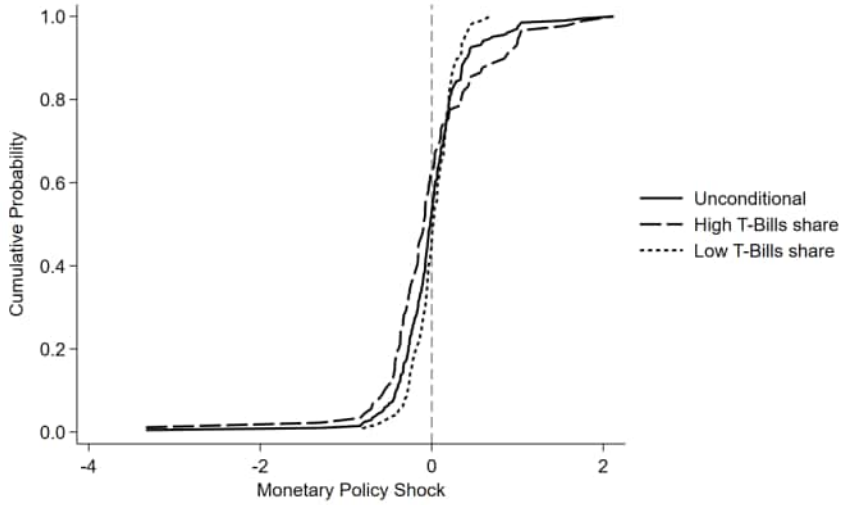
Note: Shaded areas indicate NBER recessions.

Figure A2: PDFs and CDFs of the Regime-Specific Shocks.

Panel A. PDF

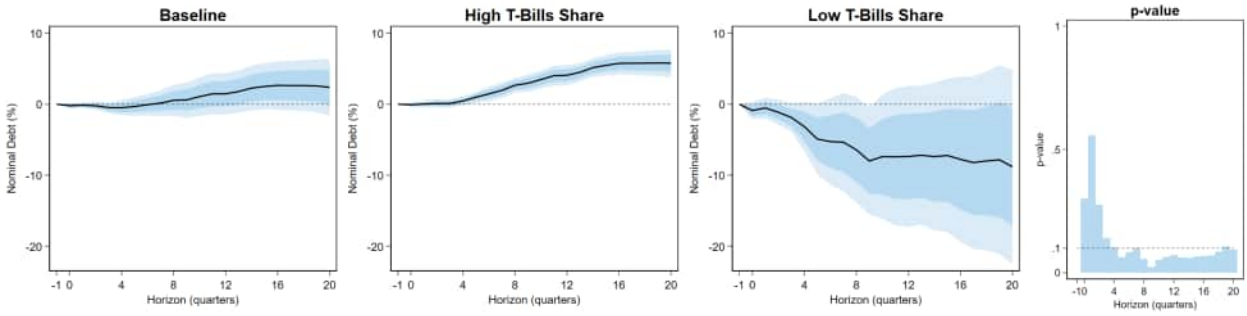


Panel B. CDF



Note: The figure plots the Probability Density Functions (Panel A) and Cumulative Density Functions (Panel B) of the updated Romer-Romer monetary policy shock (purged from its predictable component) in the different regimes. The solid lines show the unconditional distribution, the long-dashed lines in a High T-Bills share regime, and the short-dashed lines in a Low T-Bills share regime.

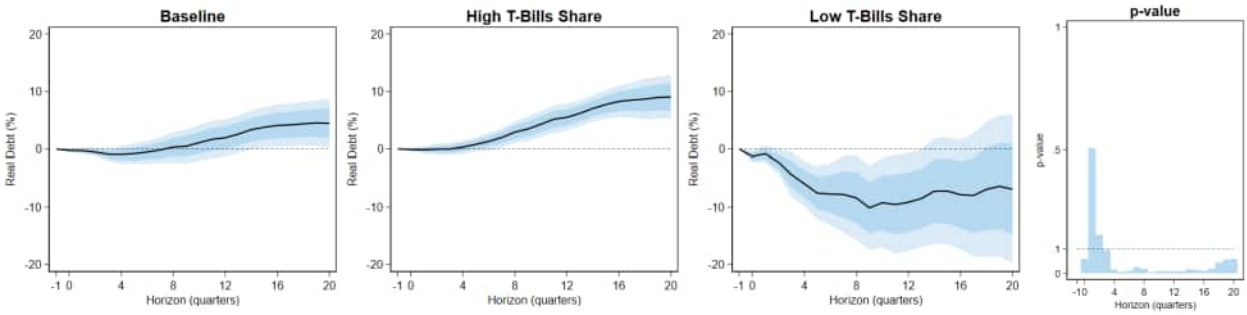
FIGURE A3: THE EFFECTS OF MONETARY POLICY SHOCKS ON NOMINAL DEBT DYNAMICS



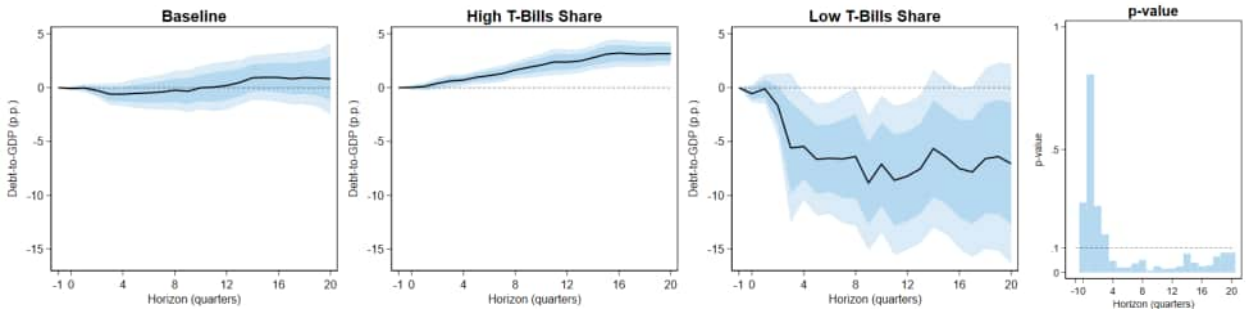
Note: The figure plots the impulse response of Nominal Debt (Panel A) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A4: ROBUSTNESS CHECK - LP IN LONG DIFFERENCES.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



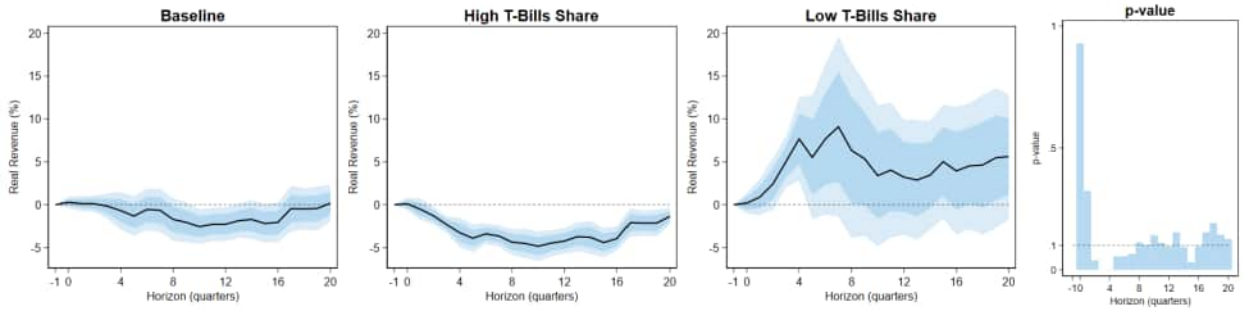
Panel B. Debt-to-GDP



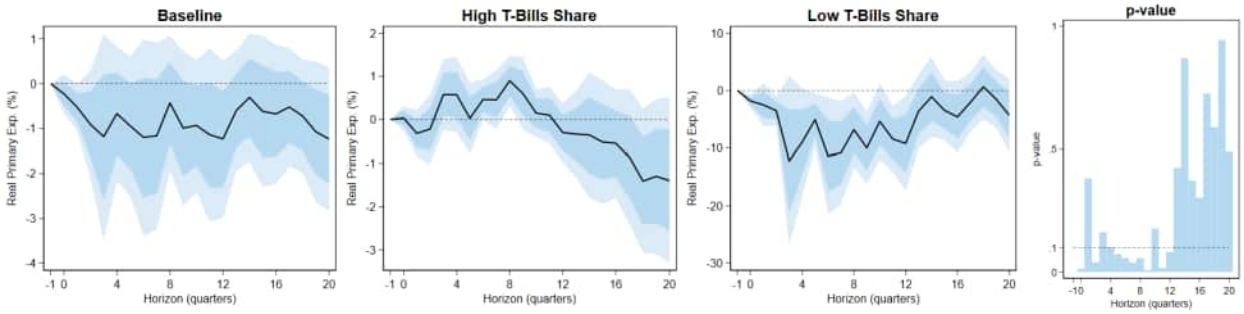
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A5: ROBUSTNESS CHECK - LP IN LONG DIFFERENCES.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

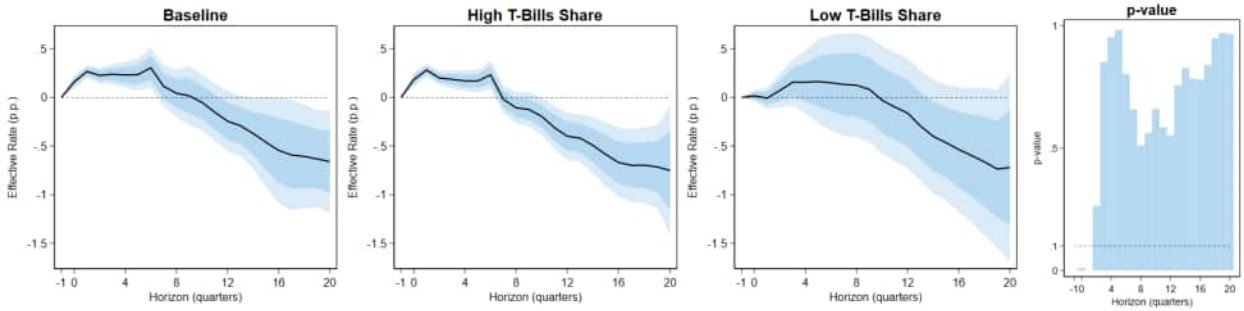
Panel A. Real Revenue



Panel B. Real Primary Expenditure



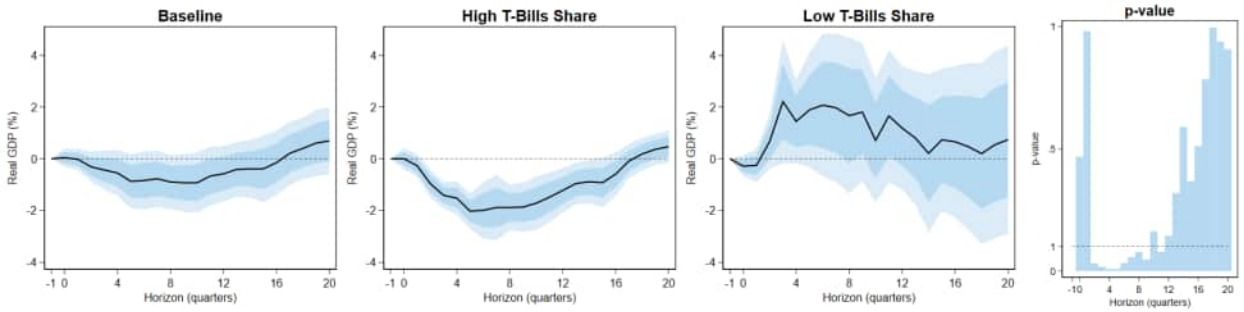
Panel C. Effective interest rate



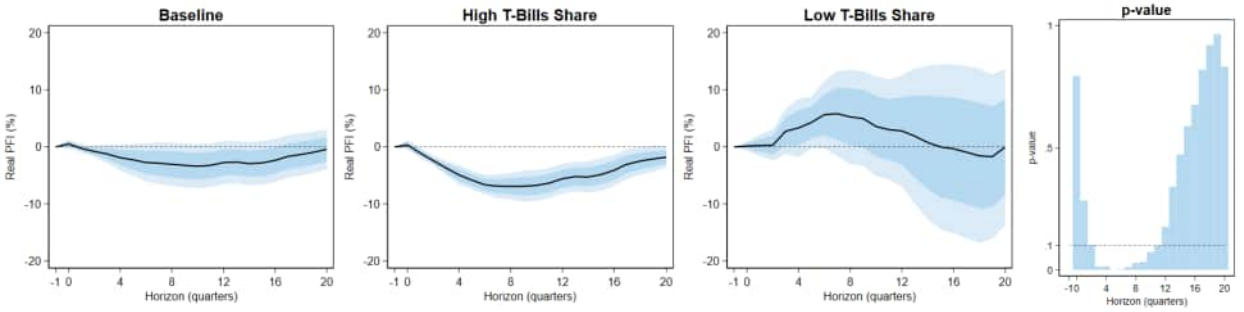
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A6: ROBUSTNESS CHECK - LP IN LONG DIFFERENCES.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

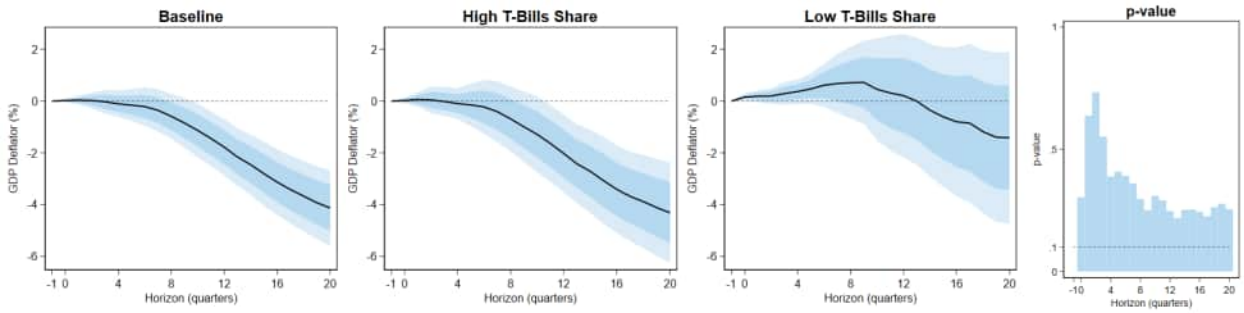
Panel A. Real GDP



Panel B. Real Private Fixed Investments



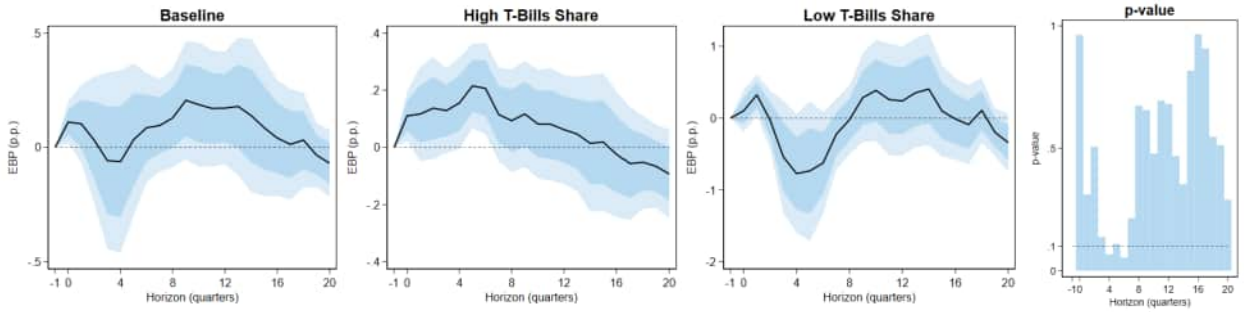
Panel C. GDP Deflator



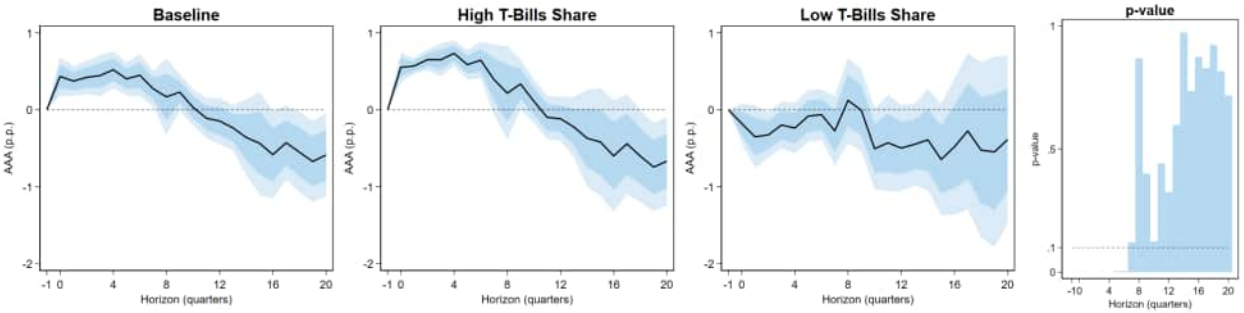
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and the GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A7: ROBUSTNESS CHECK - LP IN LONG DIFFERENCES.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

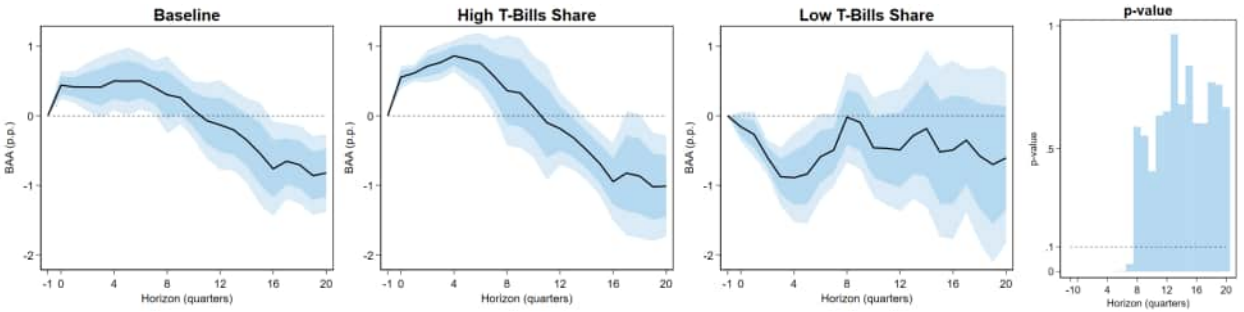
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



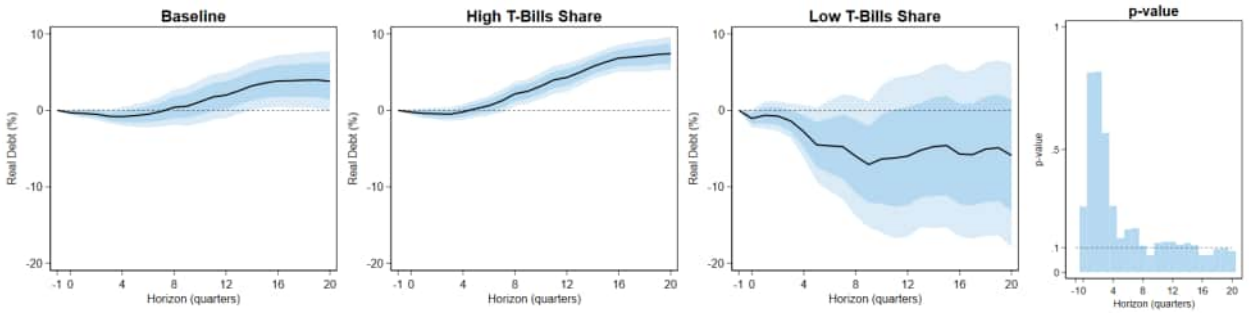
Panel C. Baa Corporate Bond Yield



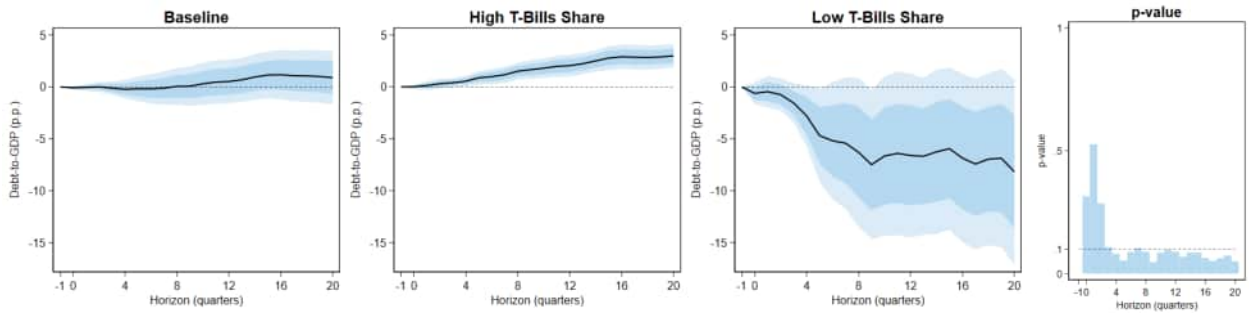
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

Figure A8: Robustness Check - Alternative Lag Structure (2 Lags).
The Effects of Monetary Policy Shocks on Debt Dynamics

Panel A. Real Debt



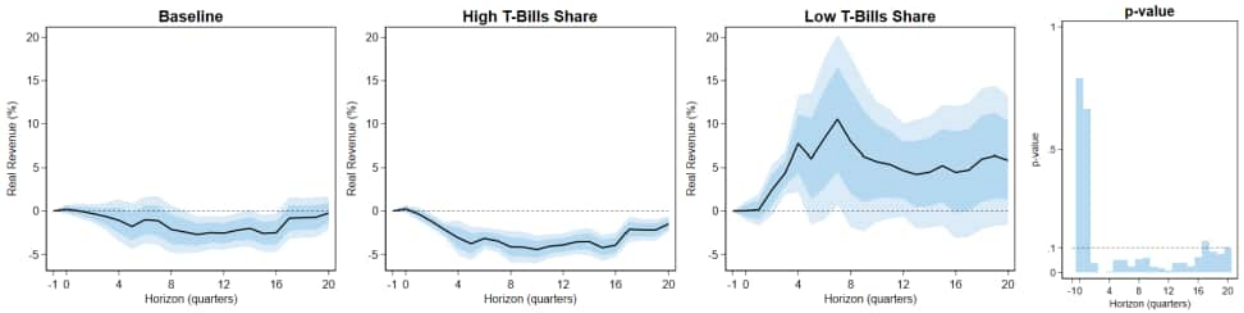
Panel B. Debt-to-GDP



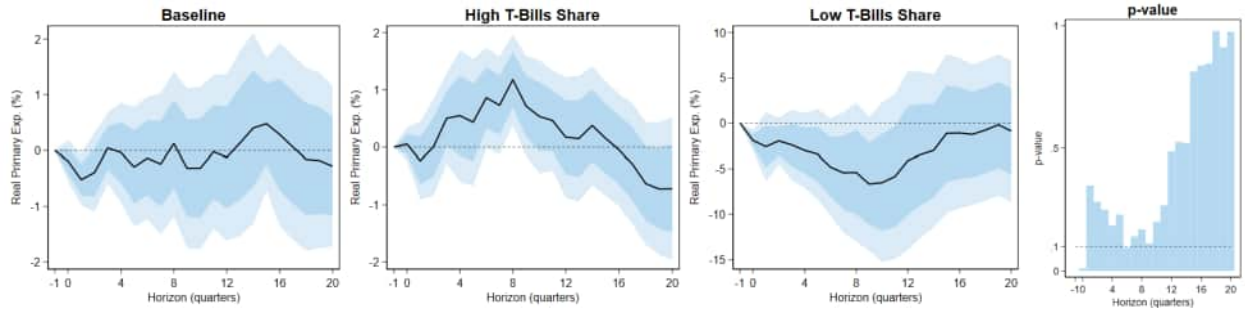
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with two lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

Figure A9: Robustness Check - Alternative Lag Structure (2 Lags).
The Effects of Monetary Policy Shocks on Fiscal Dynamics

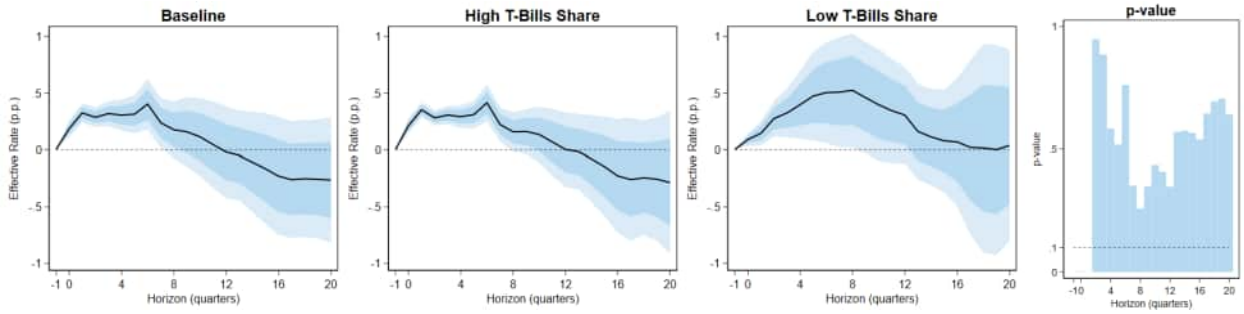
Panel A. Real Revenue



Panel B. Real Primary Expenditure



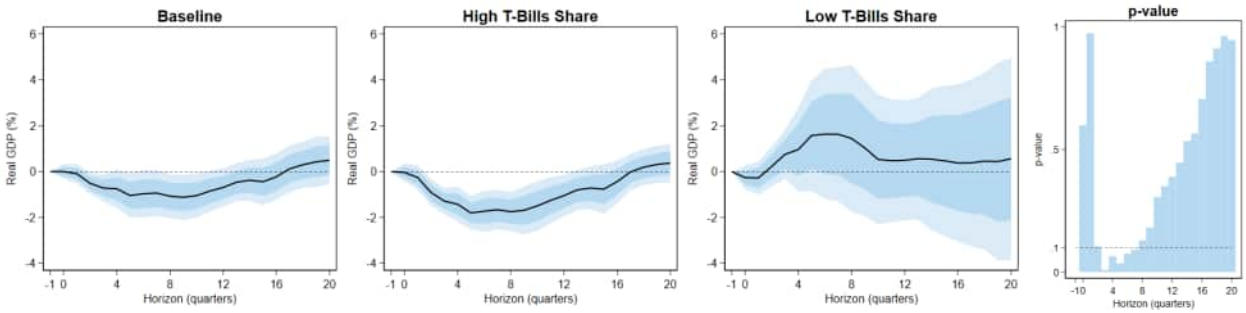
Panel C. Effective interest rate



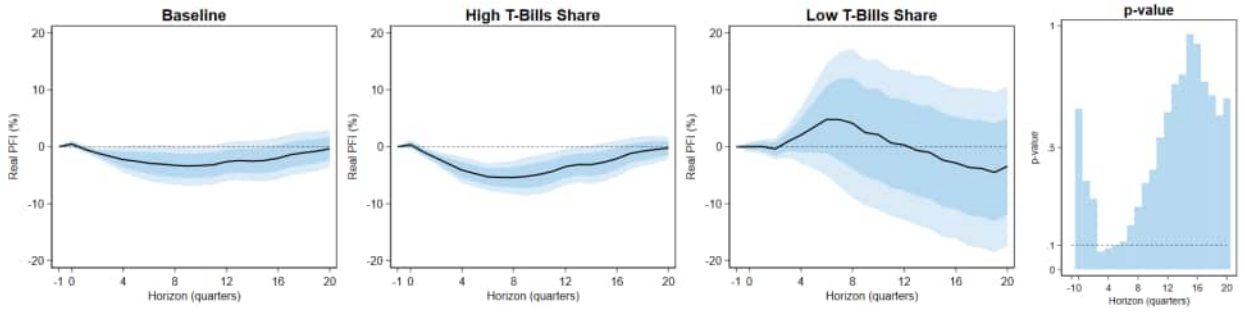
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with two lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A10: ROBUSTNESS CHECK - ALTERNATIVE LAG STRUCTURE (2 LAGS).
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

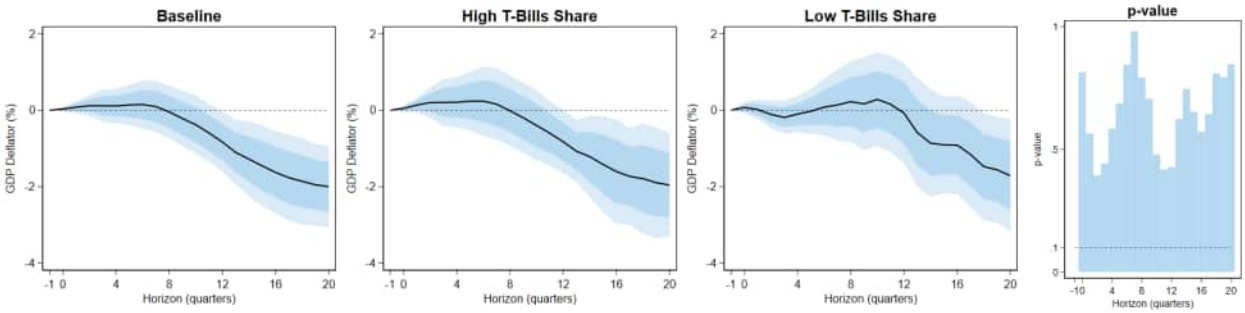
Panel A. Real GDP



Panel B. Real Private Fixed Investments



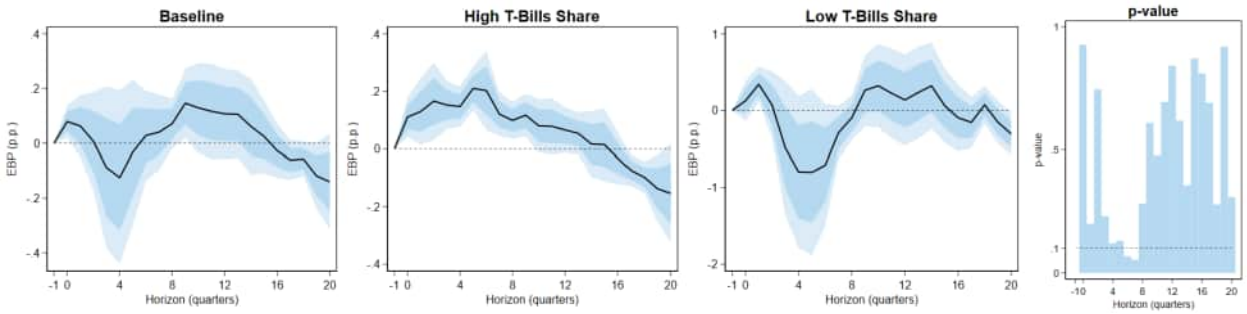
Panel C. GDP Deflator



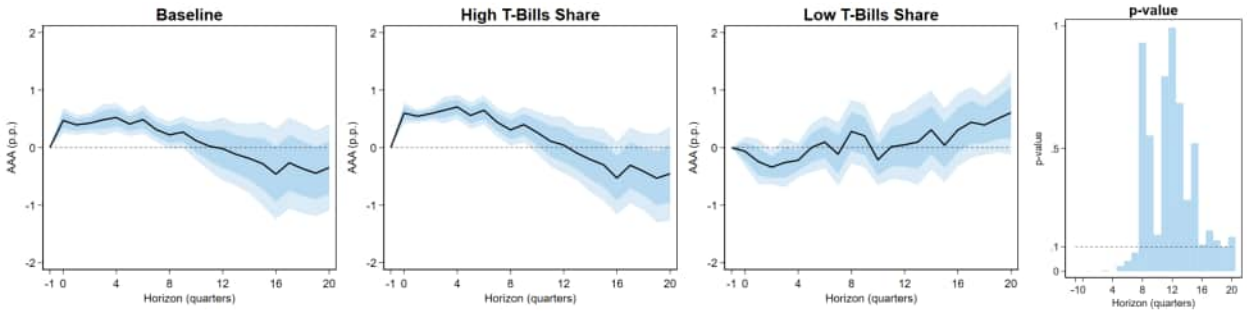
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with two lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A11: ROBUSTNESS CHECK - ALTERNATIVE LAG STRUCTURE (2 LAGS).
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

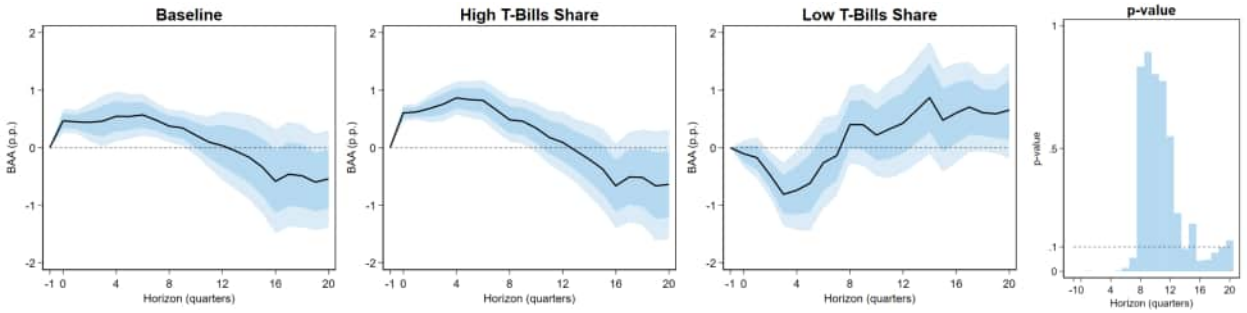
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



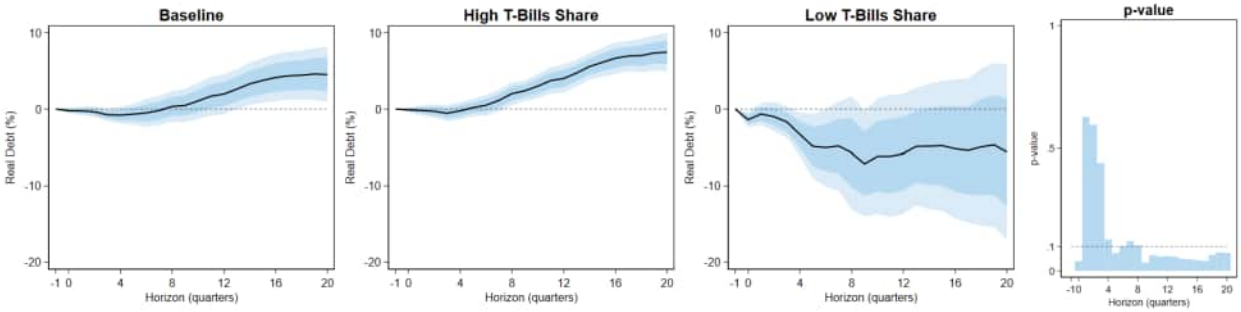
Panel C. Baa Corporate Bond Yield



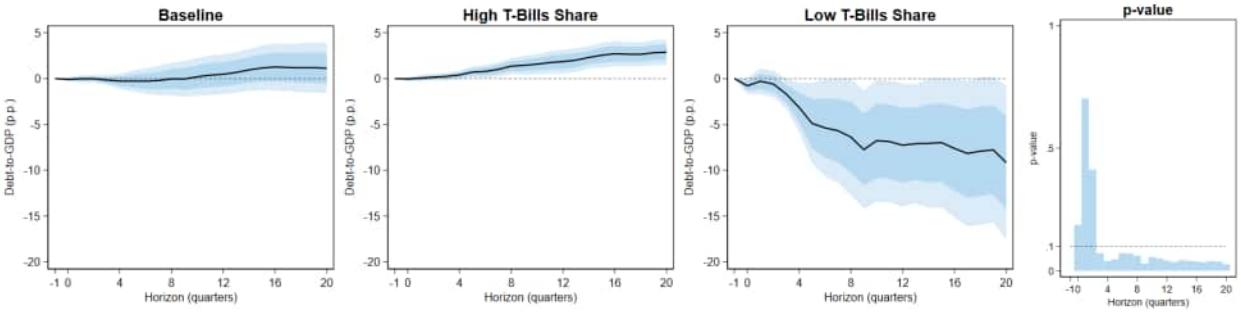
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with two lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A12: ROBUSTNESS CHECK - ALTERNATIVE LAG STRUCTURE (6 LAGS).
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



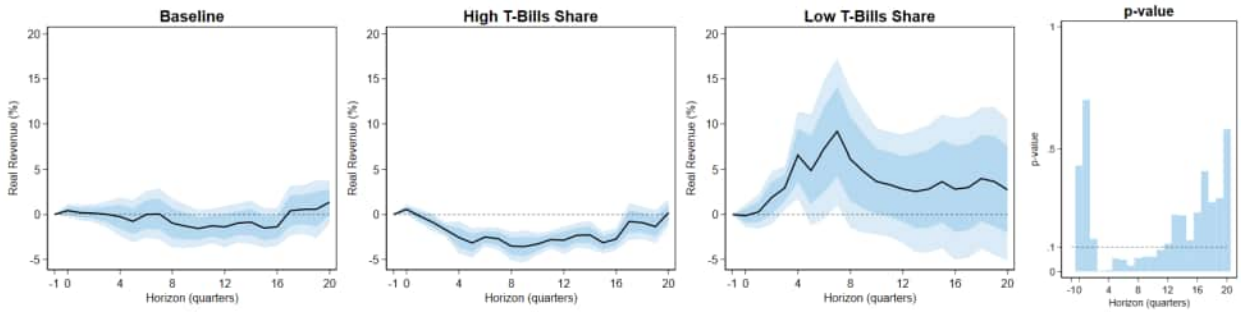
Panel B. Debt-to-GDP



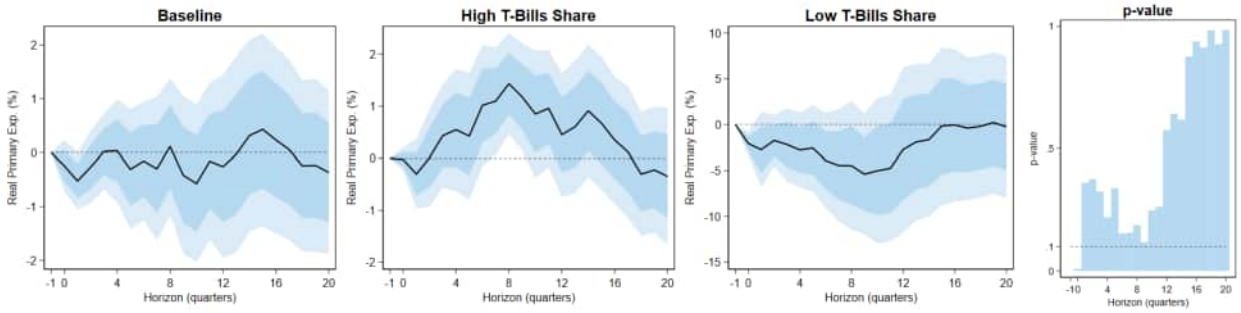
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{6} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with six lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{6} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{6} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A13: ROBUSTNESS CHECK - ALTERNATIVE LAG STRUCTURE (6 LAGS).
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

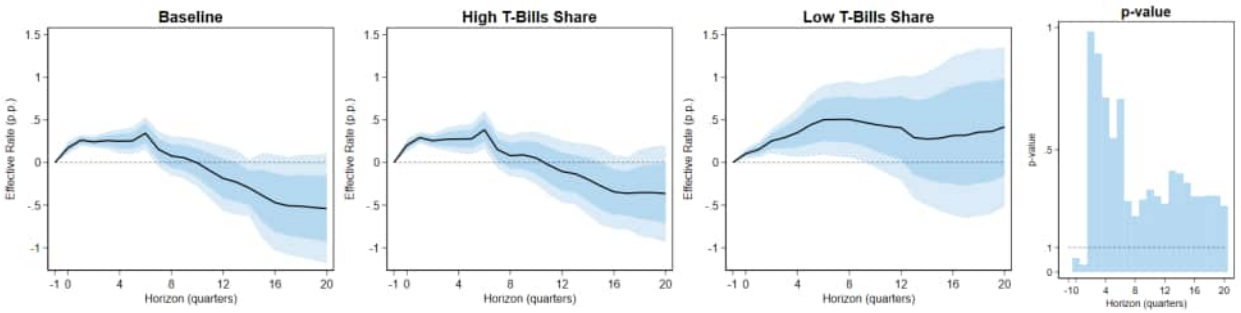
Panel A. Real Revenue



Panel B. Real Primary Expenditure



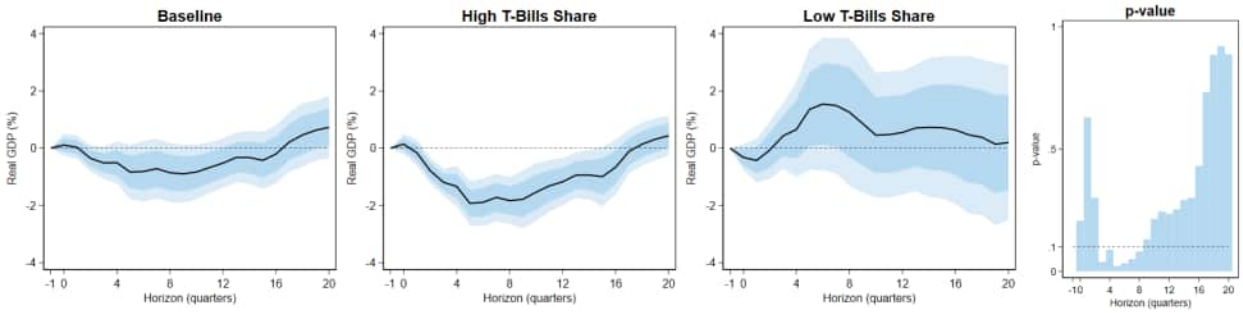
Panel C. Effective interest rate



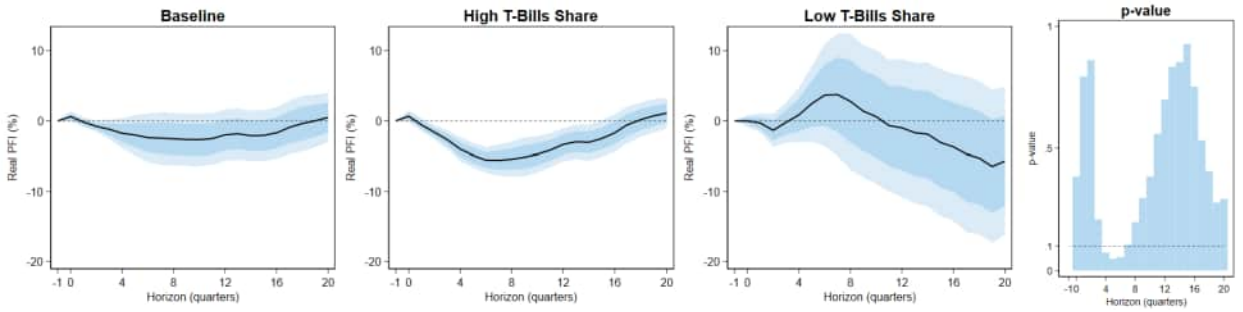
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{6} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with six lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{6} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{6} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A14: ROBUSTNESS CHECK - ALTERNATIVE LAG STRUCTURE (6 LAGS).
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

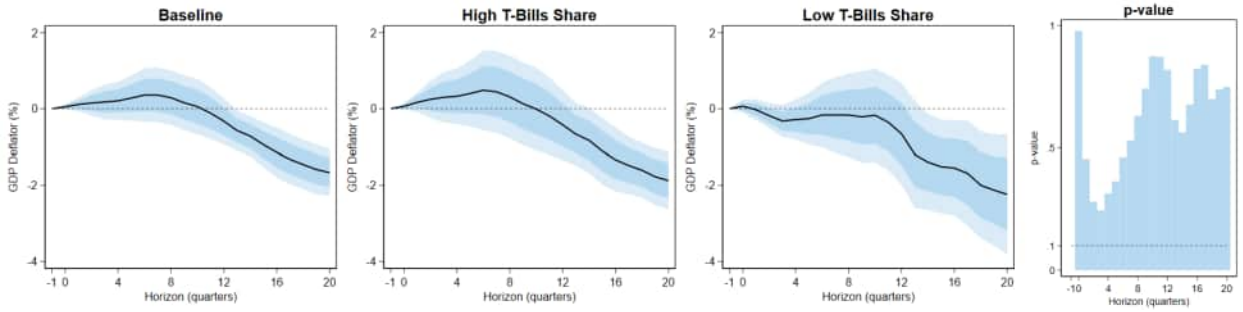
Panel A. Real GDP



Panel B. Real Private Fixed Investments



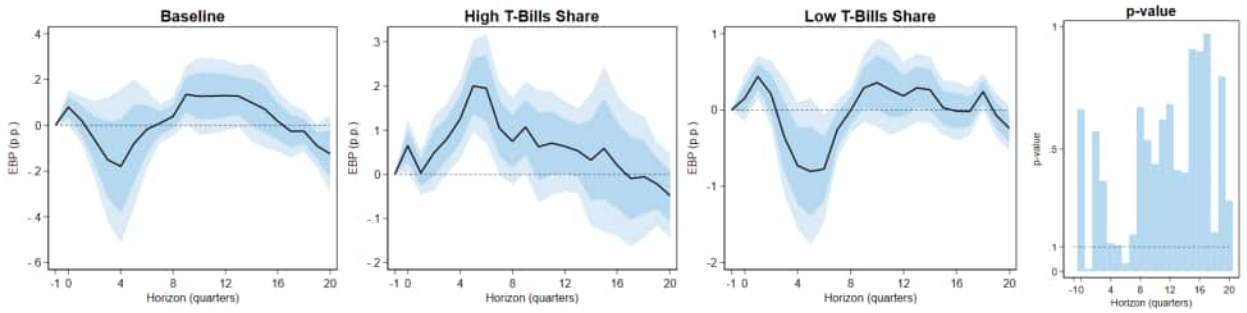
Panel C. GDP Deflator



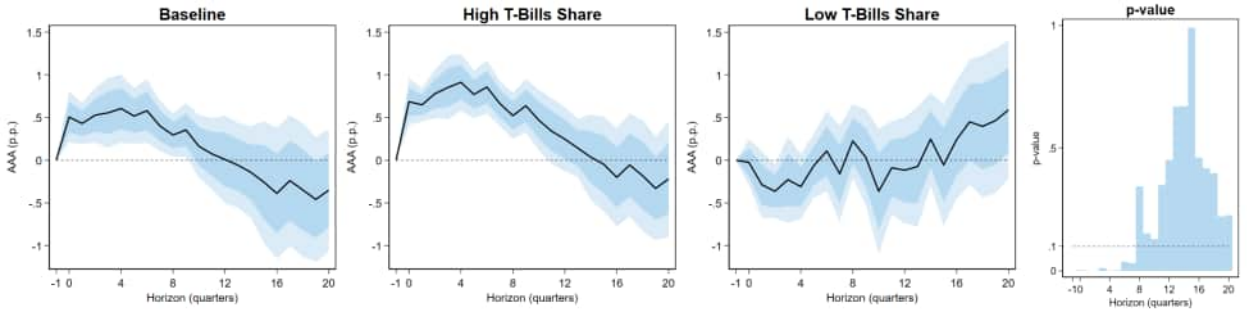
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{6} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with six lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{6} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{6} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A15: ROBUSTNESS CHECK - ALTERNATIVE LAG STRUCTURE (6 LAGS).
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



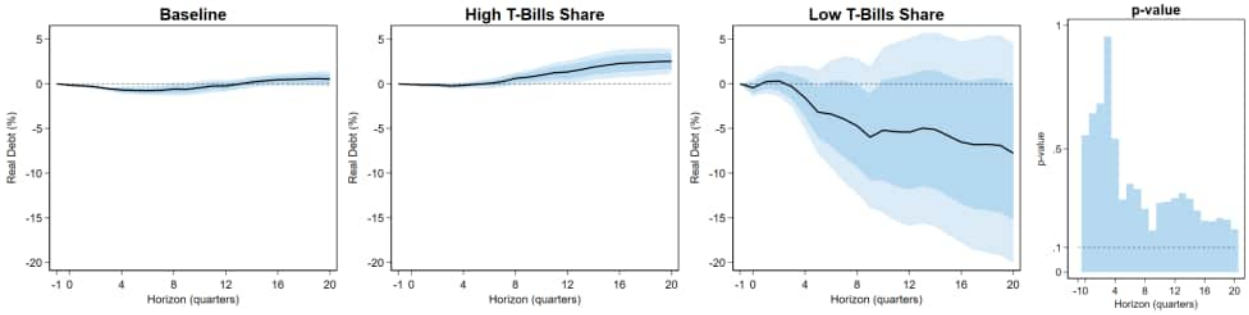
Panel C. Baa Corporate Bond Yield



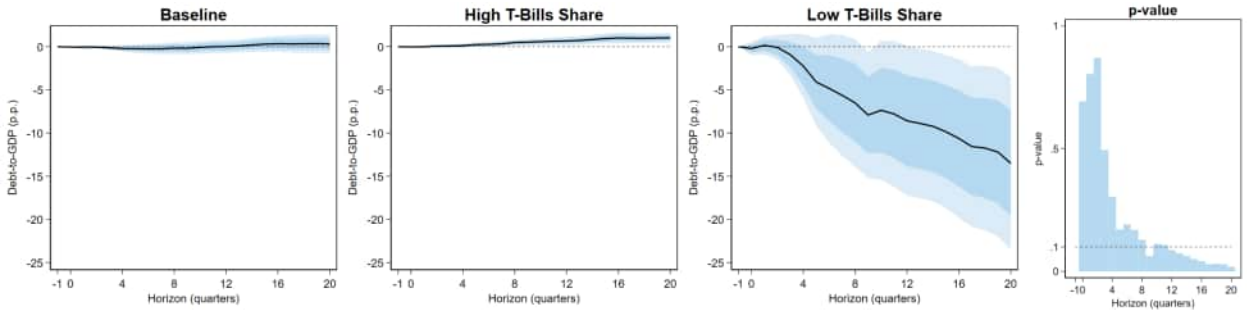
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa corporate bond yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{6} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with six lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{6} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{6} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A16: ROBUSTNESS CHECK, INSTRUMENTAL VARIABLE.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



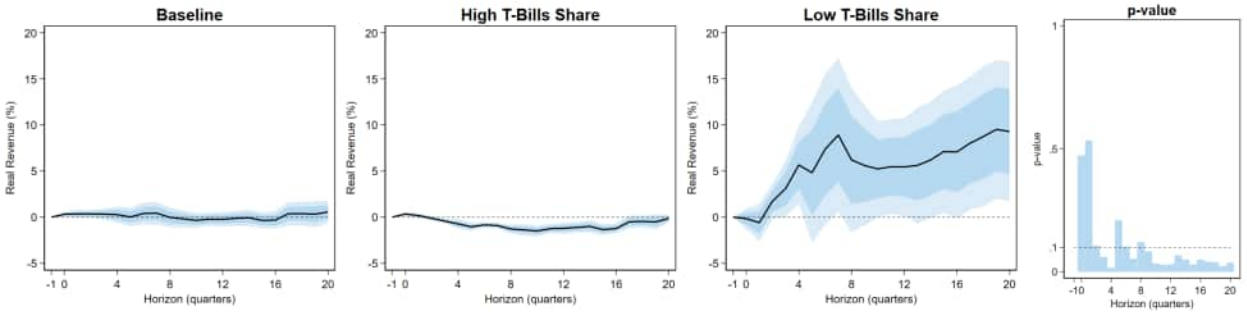
Panel B. Debt-to-GDP



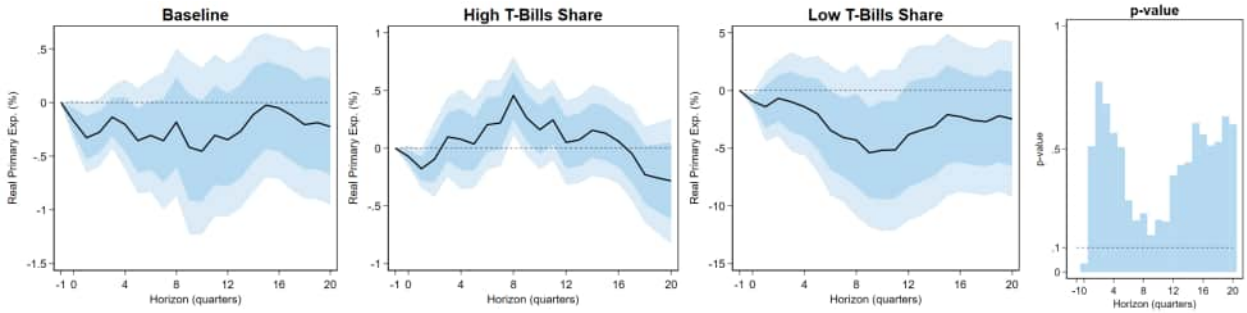
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via 2SLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} \widehat{\Delta FFR}_t + \sum_{\ell=1}^{4} \theta_\ell^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where $\widehat{\Delta FFR}_t = \gamma + \delta MPS_t + \sum_{\ell=1}^{4} \iota_\ell' W_{t-\ell} + \nu_t$. Here y_{t+h} is the variable of interest h quarters after the shock, FFR_t is the federal funds rate, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via 2SLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} \widehat{\Delta FFR}_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} \widehat{\Delta FFR}_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A17: ROBUSTNESS CHECK, INSTRUMENTAL VARIABLE.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

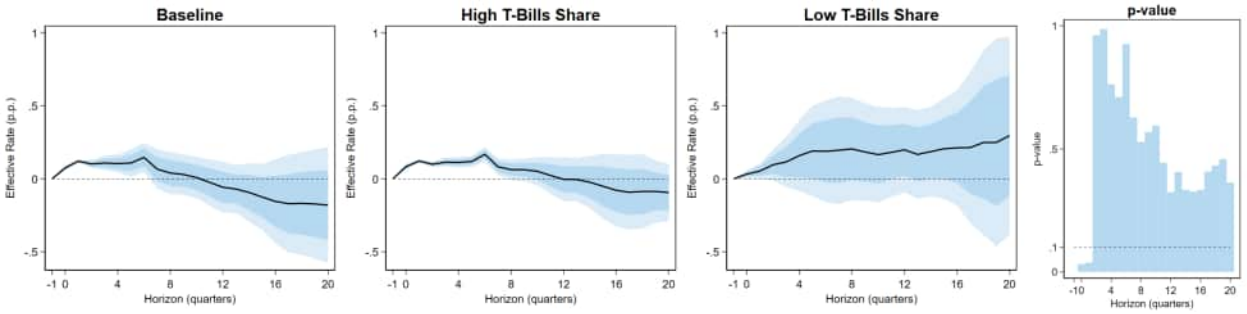
Panel A. Real Revenue



Panel B. Real Primary Expenditure



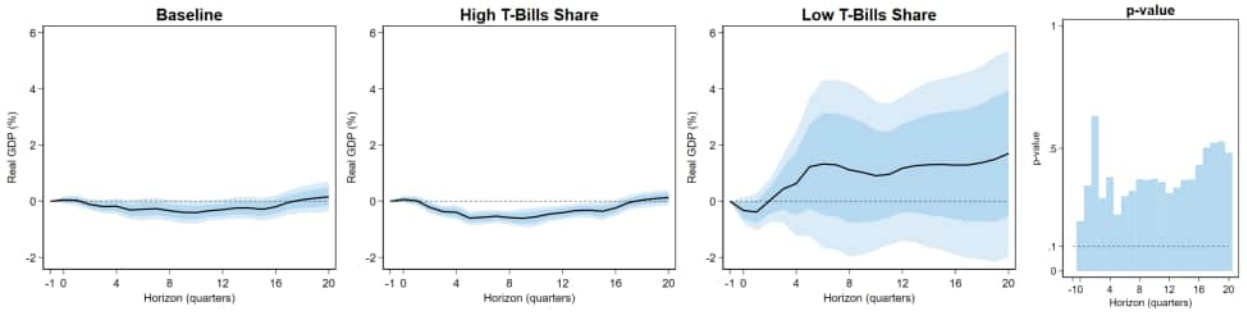
Panel C. Effective Interest Rate



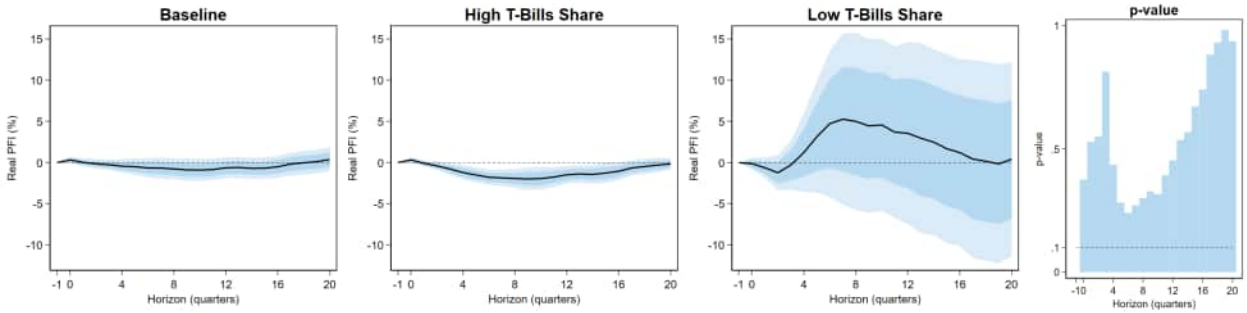
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via 2SLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_\ell^{(h)}W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where $\widehat{\Delta FFR}_t = \gamma + \delta MPS_t + \sum_{\ell=1}^{4}\iota_\ell' W_{t-\ell} + \nu_t$. Here y_{t+h} is the variable of interest h quarters after the shock, FFR_t is the federal funds rate, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via 2SLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{high}}^{(h)}W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{low}}^{(h)}W_{t-\ell}) + \delta^{(h)}F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A18: ROBUSTNESS CHECK, INSTRUMENTAL VARIABLE.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

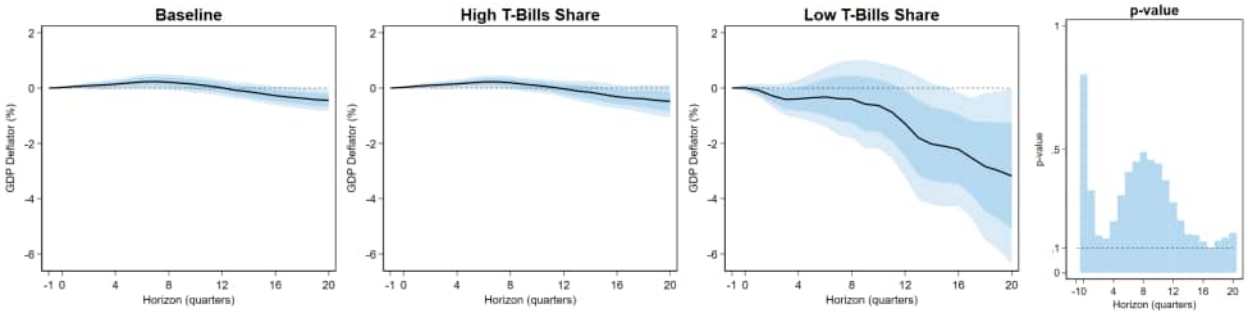
Panel A. Real GDP



Panel B. Real Private Fixed Investments



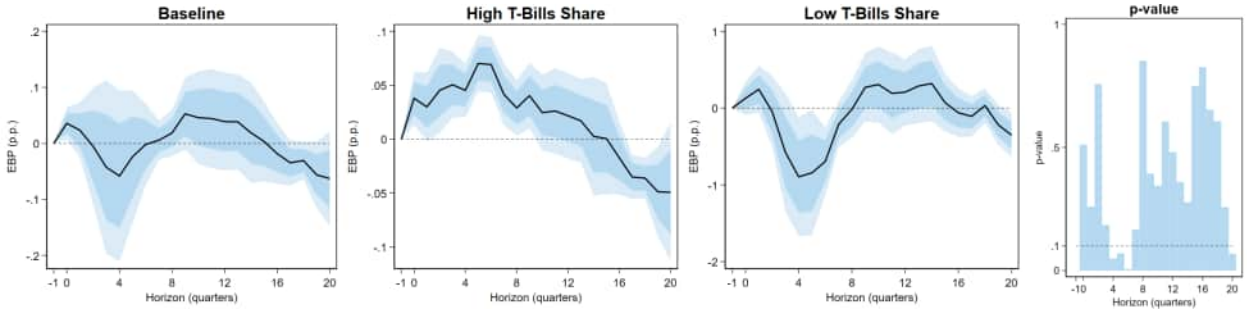
Panel C. GDP Deflator



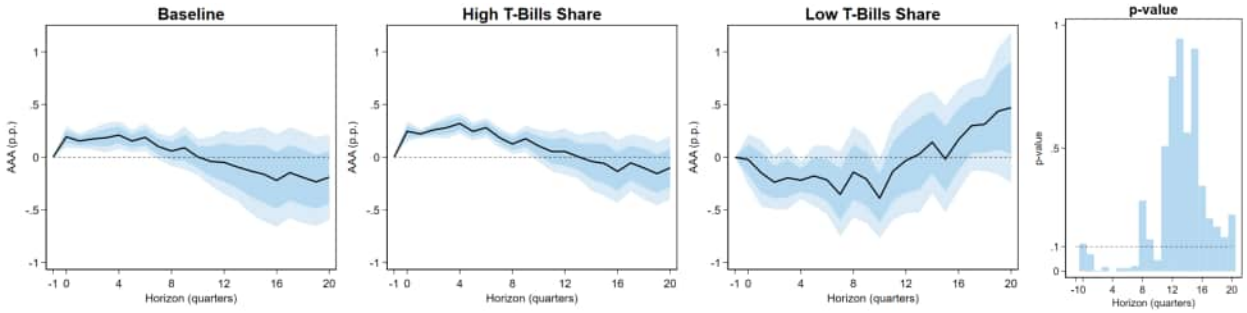
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via 2SLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_\ell^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where $\widehat{\Delta FFR}_t = \gamma + \delta MPS_t + \sum_{\ell=1}^{4}\iota_\ell' W_{t-\ell} + \nu_t$. Here y_{t+h} is the variable of interest h quarters after the shock, FFR_t is the federal funds rate, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via 2SLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A19: ROBUSTNESS CHECK, INSTRUMENTAL VARIABLE.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

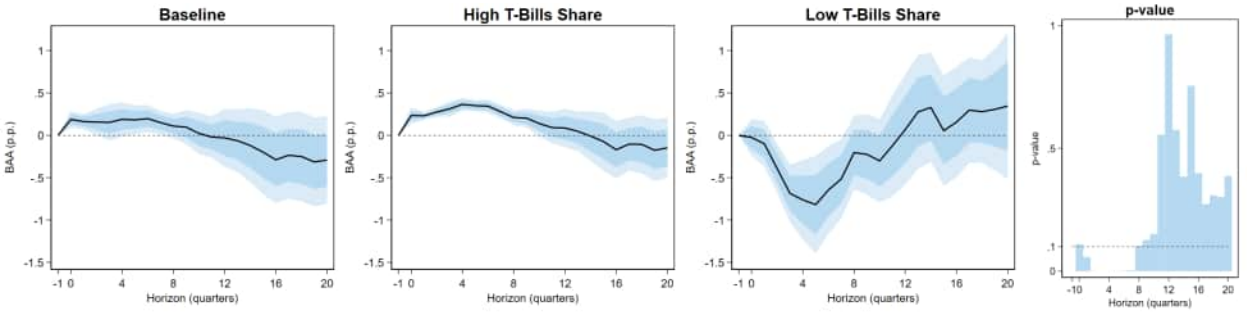
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



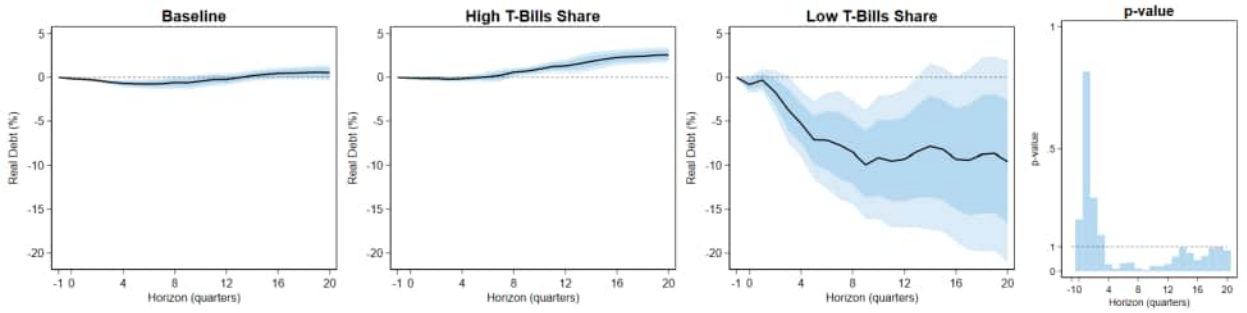
Panel C. Baa Corporate Bond Yield



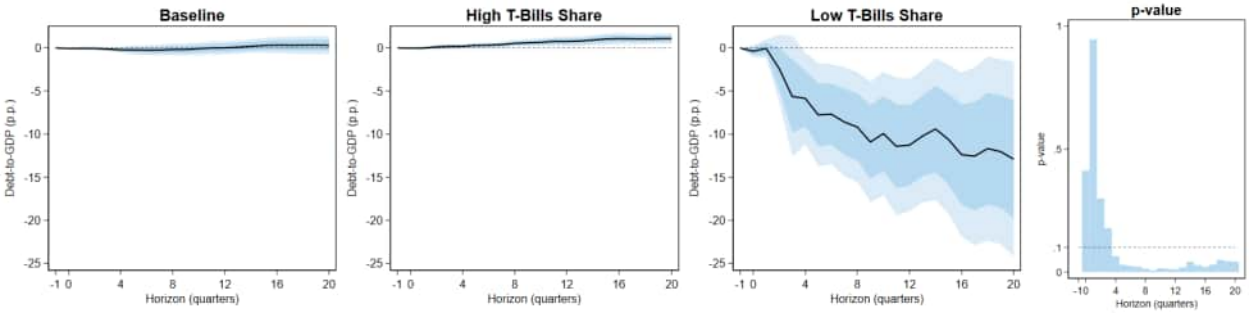
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via 2SLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_\ell^{(h)}W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where $\widehat{\Delta FFR}_t = \gamma + \delta MPS_t + \sum_{\ell=1}^{4}\iota_\ell' W_{t-\ell} + \nu_t$. Here y_{t+h} is the variable of interest h quarters after the shock, FFR_t is the federal funds rate, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via 2SLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{high}}^{(h)}W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{low}}^{(h)}W_{t-\ell}) + \delta^{(h)}F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A20: ROBUSTNESS CHECK, INSTRUMENTAL VARIABLE, LP IN LONG DIFFERENCES. THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



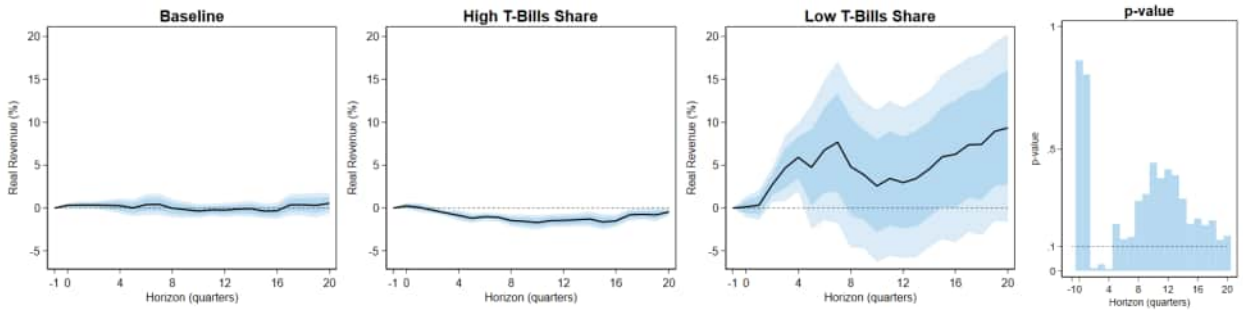
Panel B. Debt-to-GDP



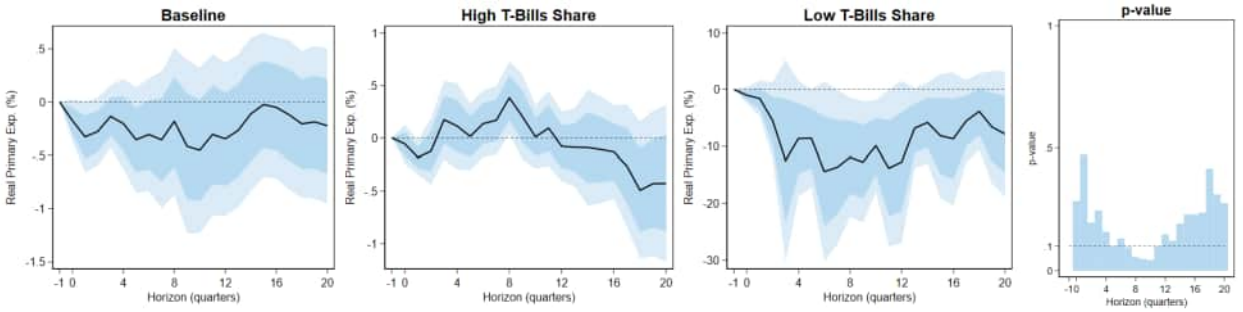
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via 2SLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + \beta^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_\ell^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where $\widehat{\Delta FFR}_t = \gamma + \delta MPS_t + \sum_{\ell=1}^{4}\iota_\ell' W_{t-\ell} + \nu_t$. Here y_{t+h} is the variable of interest h quarters after the shock, FFR_t is the federal funds rate, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via 2SLS on the same sample, using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A21: ROBUSTNESS CHECK, INSTRUMENTAL VARIABLE, LP IN LONG DIFFERENCES. THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

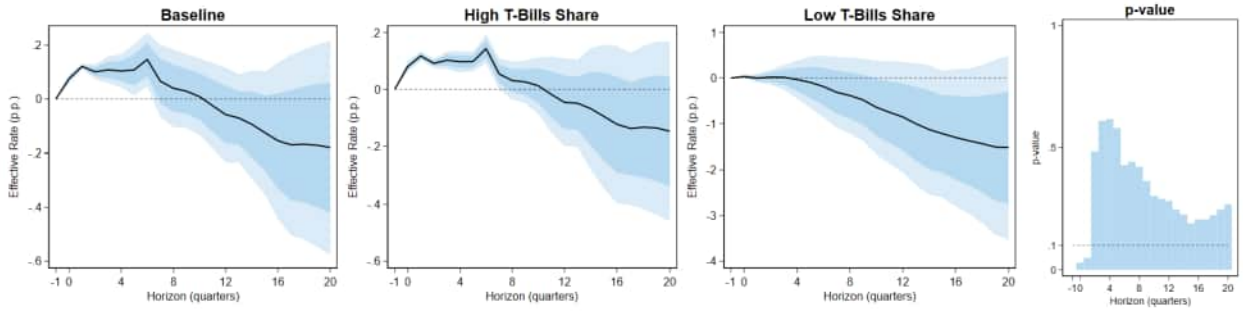
Panel A. Real Revenue



Panel B. Real Primary Expenditure



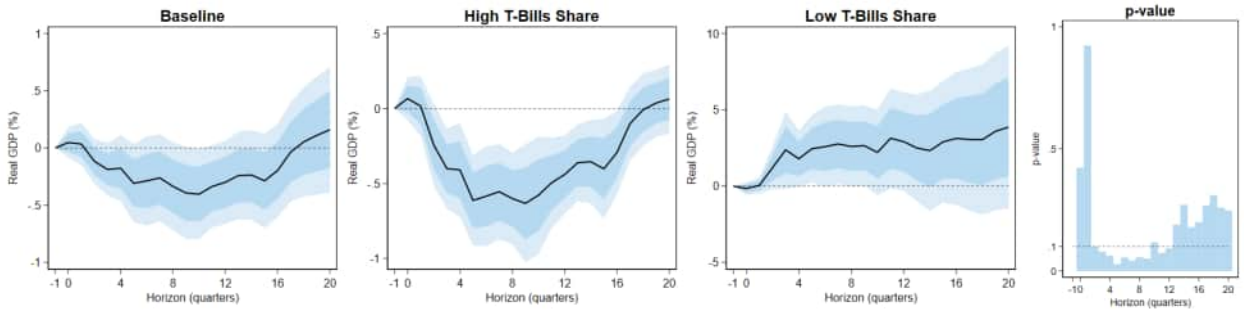
Panel C. Effective Interest Rate



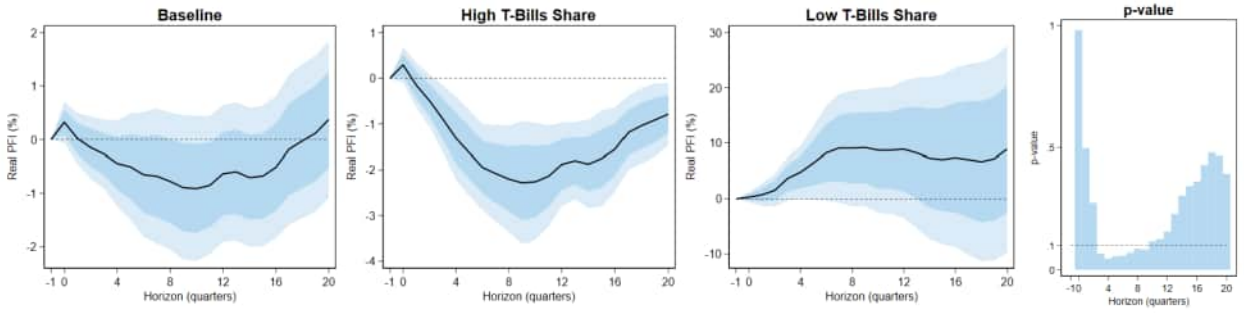
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via 2SLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + \beta^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_\ell^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where $\widehat{\Delta FFR}_t = \gamma + \delta MPS_t + \sum_{\ell=1}^{4}\iota_\ell' W_{t-\ell} + \nu_t$. Here y_{t+h} is the variable of interest h quarters after the shock, FFR_t is the federal funds rate, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via 2SLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A22: ROBUSTNESS CHECK, INSTRUMENTAL VARIABLE, LP IN LONG DIFFERENCES. THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

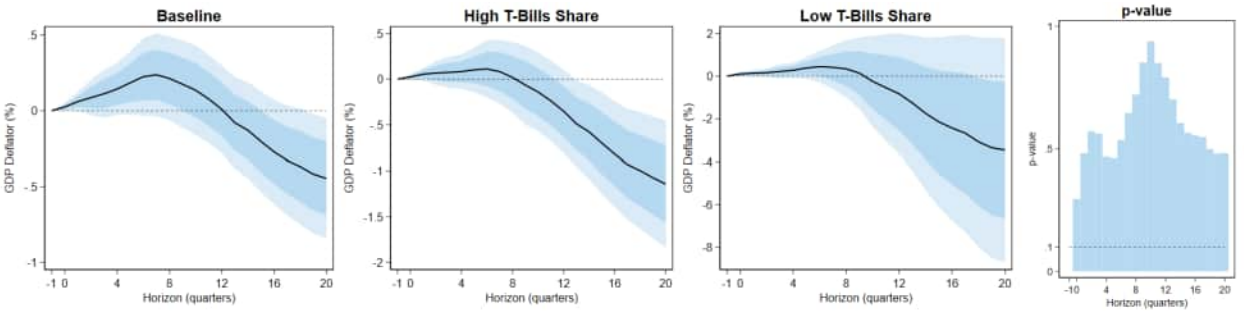
Panel A. Real GDP



Panel B. Real Private Fixed Investments



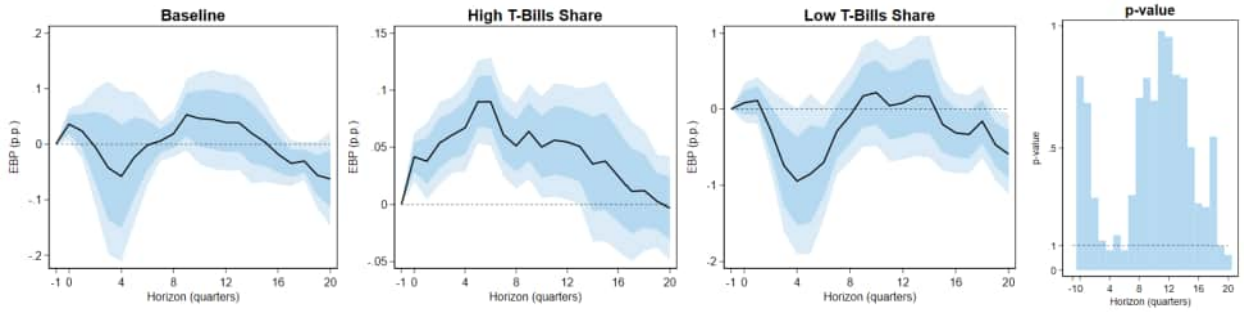
Panel C. GDP Deflator



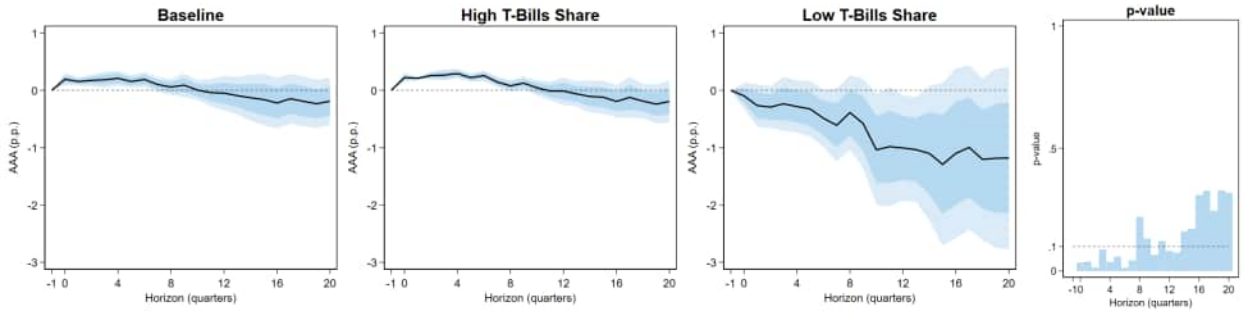
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via 2SLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + \beta^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_\ell^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where $\widehat{\Delta FFR}_t = \gamma + \delta MPS_t + \sum_{\ell=1}^{4}\iota_\ell' W_{t-\ell} + \nu_t$. Here y_{t+h} is the variable of interest h quarters after the shock, FFR_t is the federal funds rate, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via 2SLS on the same sample, using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A23: ROBUSTNESS CHECK, INSTRUMENTAL VARIABLE, LP IN LONG DIFFERENCES. THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

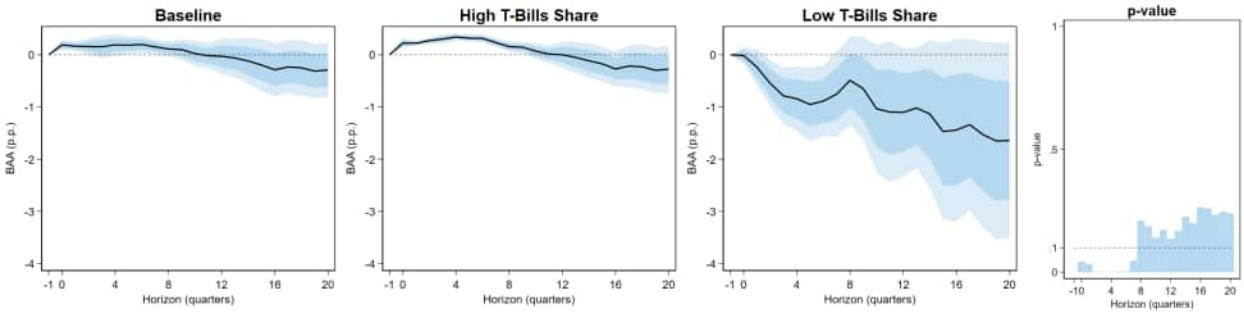
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



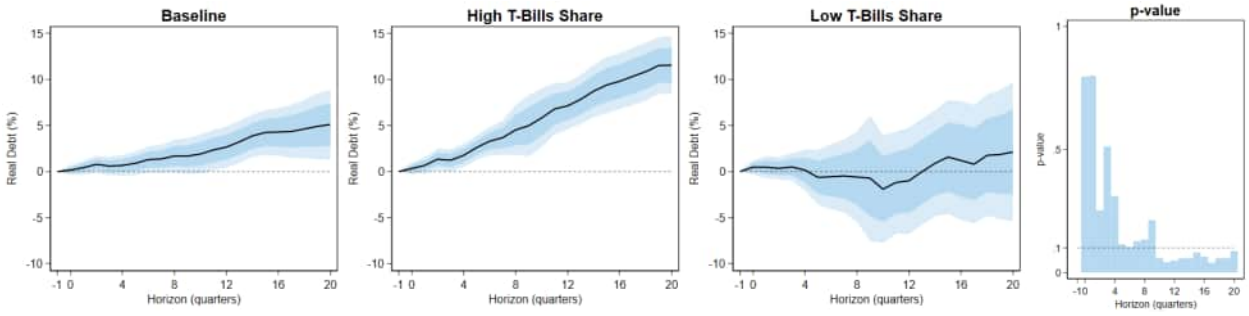
Panel C. Baa Corporate Bond Yield



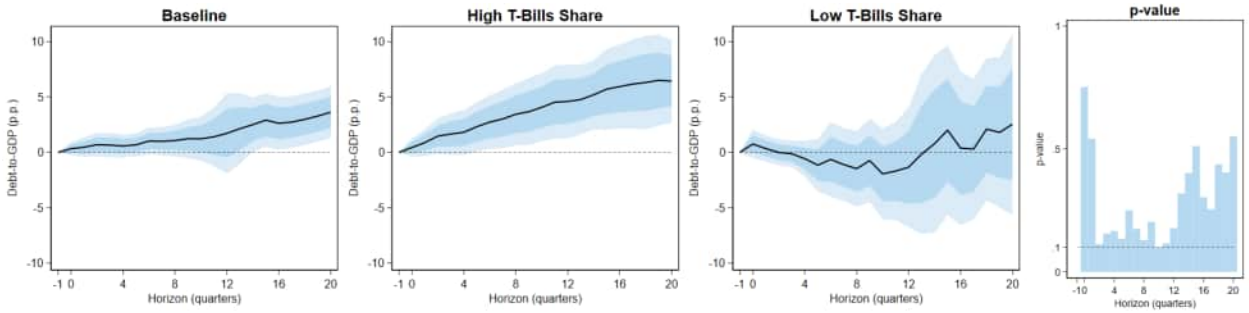
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via 2SLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + \beta^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_\ell^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where $\widehat{\Delta FFR}_t = \gamma + \delta MPS_t + \sum_{\ell=1}^{4}\iota_\ell' W_{t-\ell} + \nu_t$. Here y_{t+h} is the variable of interest h quarters after the shock, FFR_t is the federal funds rate, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via 2SLS on the same sample, using local projections: $y_{t+h} - y_{t-1} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)}\widehat{\Delta FFR}_t + \sum_{\ell=1}^{4}\theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A24: ROBUSTNESS CHECK, HIGH-FREQUENCY SHOCK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



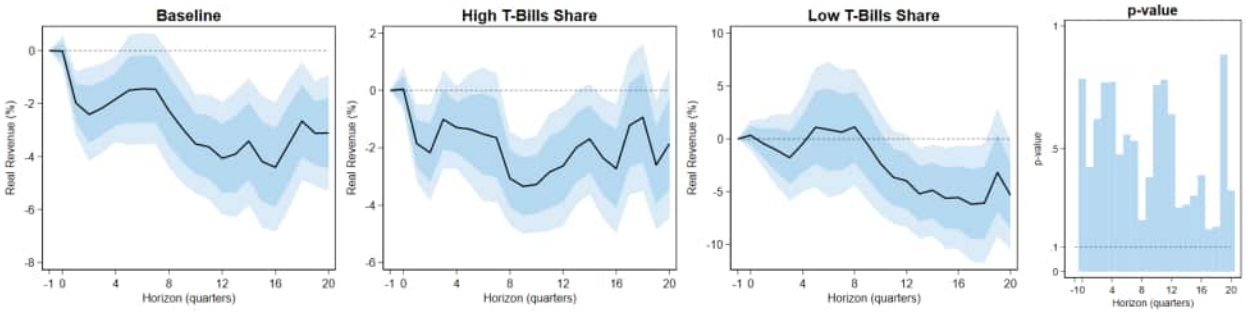
Panel B. Debt-to-GDP



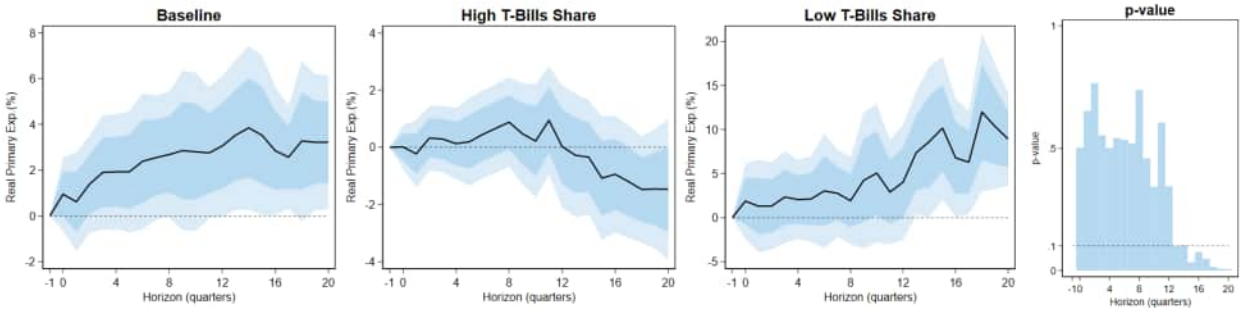
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. Structural shock series identified through high frequency identification in a proxy SVAR. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1963q1-2025q3) using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell'^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3) using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A25: ROBUSTNESS CHECK, HIGH-FREQUENCY SHOCK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

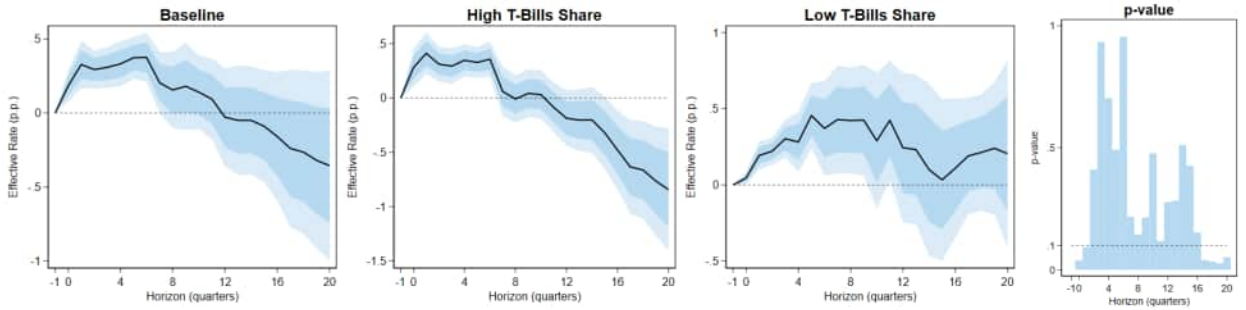
Panel A. Real Revenue



Panel B. Real Primary Expenditure



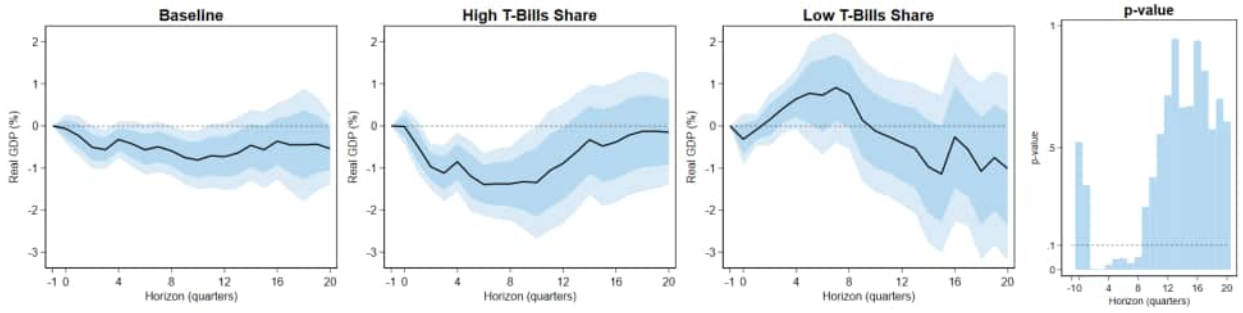
Panel C. Effective interest rate



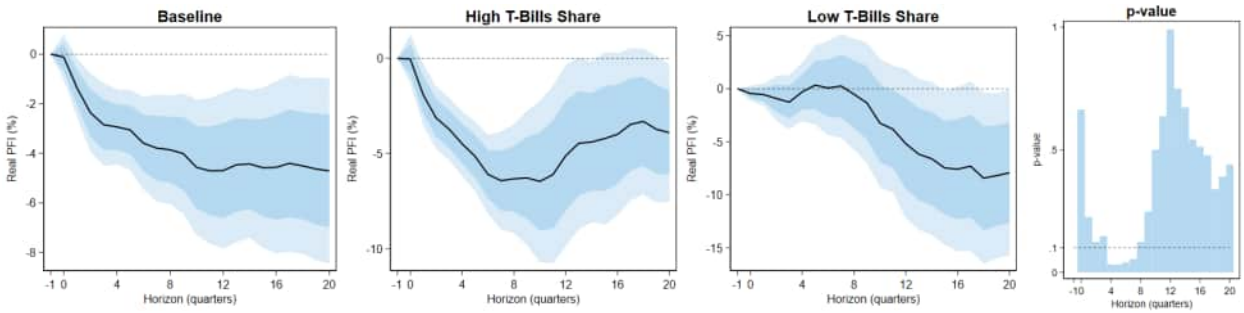
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. Structural shock series identified through high frequency identification in a proxy SVAR. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1963q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3) using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A26: ROBUSTNESS CHECK, HIGH-FREQUENCY SHOCK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

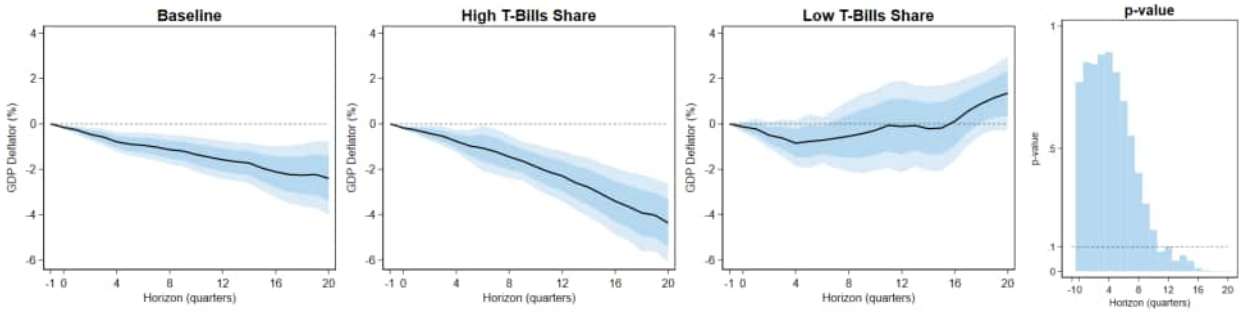
Panel A. Real GDP



Panel B. Real Private Fixed Investments



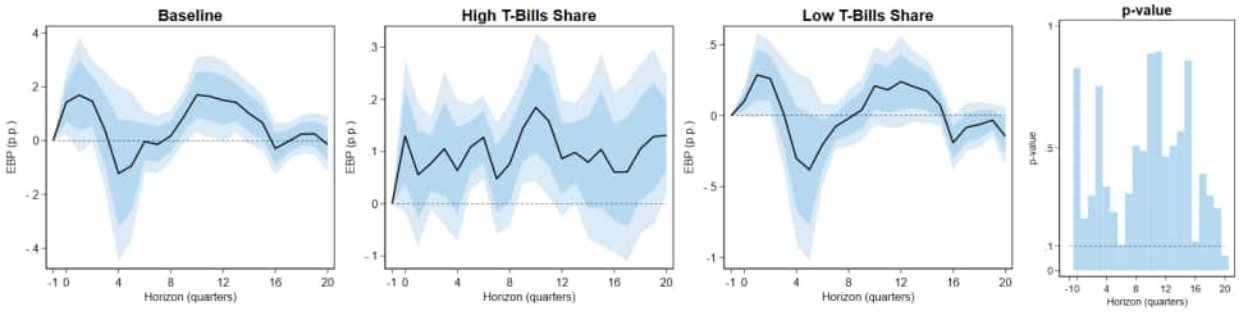
Panel C. GDP Deflator



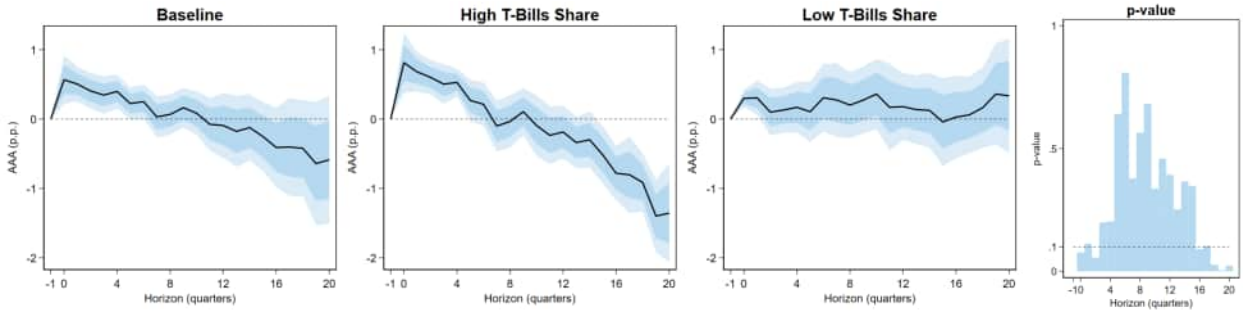
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. Structural shock series identified through high frequency identification in a proxy SVAR. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3) using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A27: ROBUSTNESS CHECK, HIGH-FREQUENCY SHOCK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

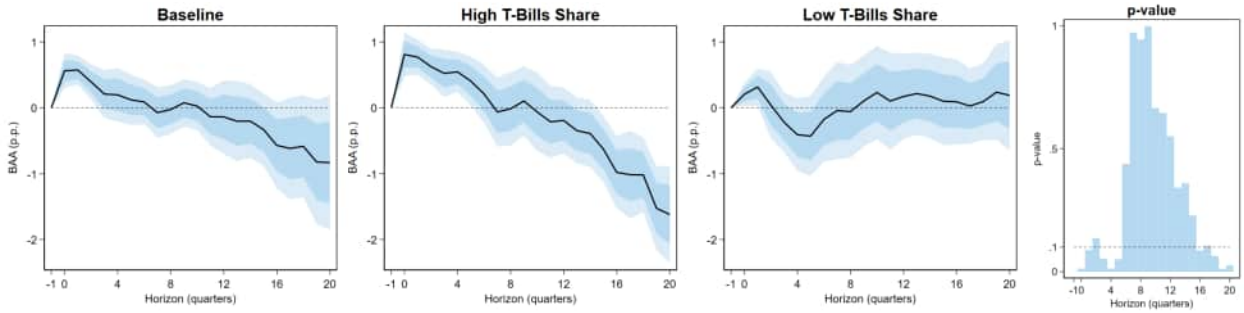
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



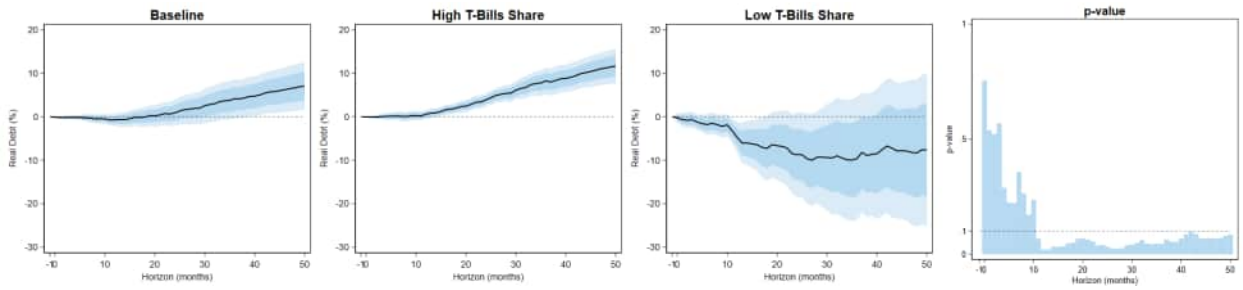
Panel C. Baa Corporate Bond Yield



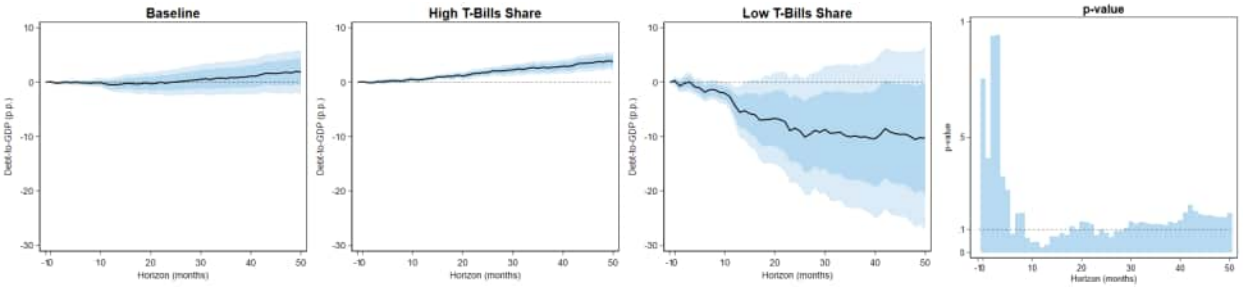
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. Structural shock series identified through high frequency identification in a proxy SVAR. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3) using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A28: ROBUSTNESS CHECK, MONTHLY NARRATIVE SHOCK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



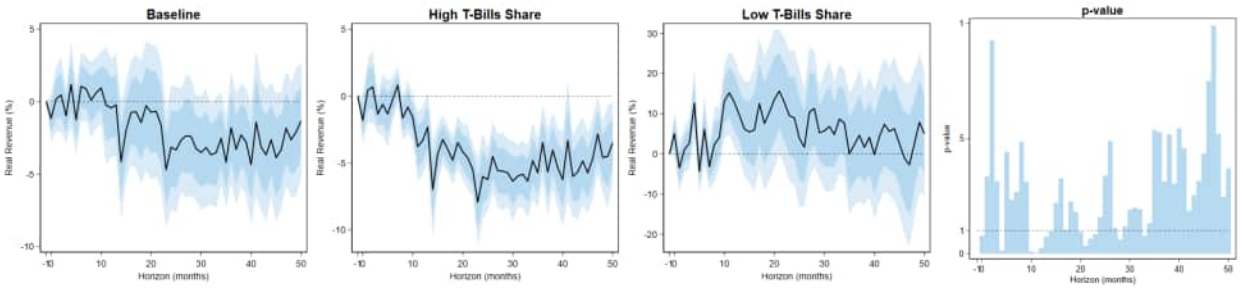
Panel B. Debt-to-GDP



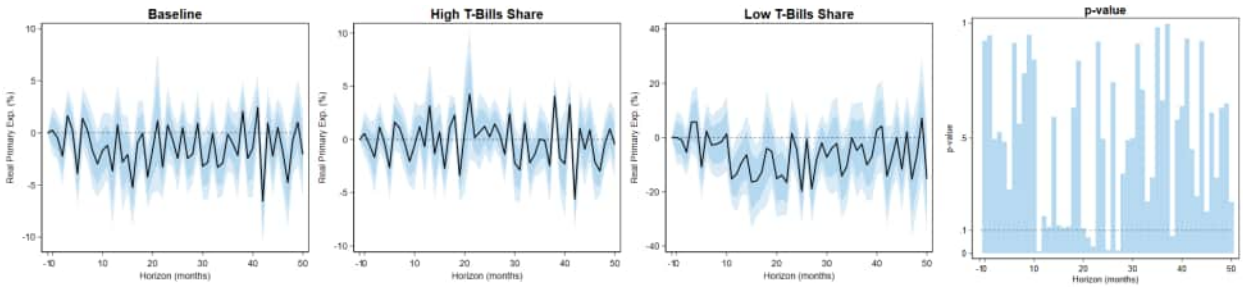
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969m3-2019m12), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with twelve lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A29: ROBUSTNESS CHECK, MONTHLY NARRATIVE SHOCK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

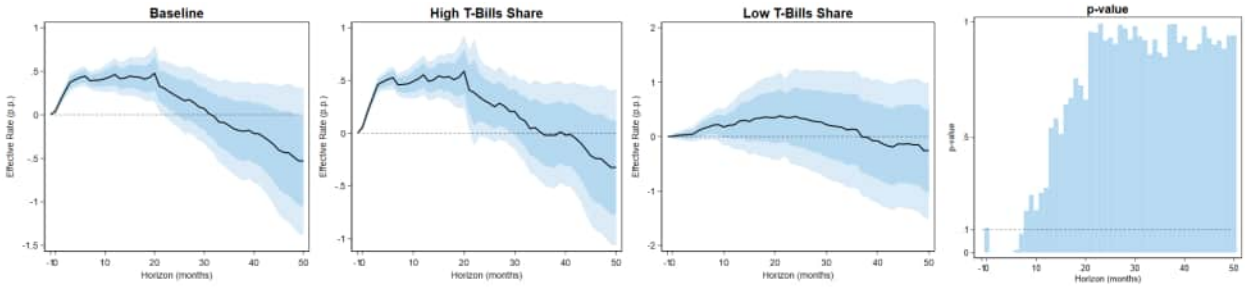
Panel A. Real Revenue



Panel B. Real Primary Expenditure



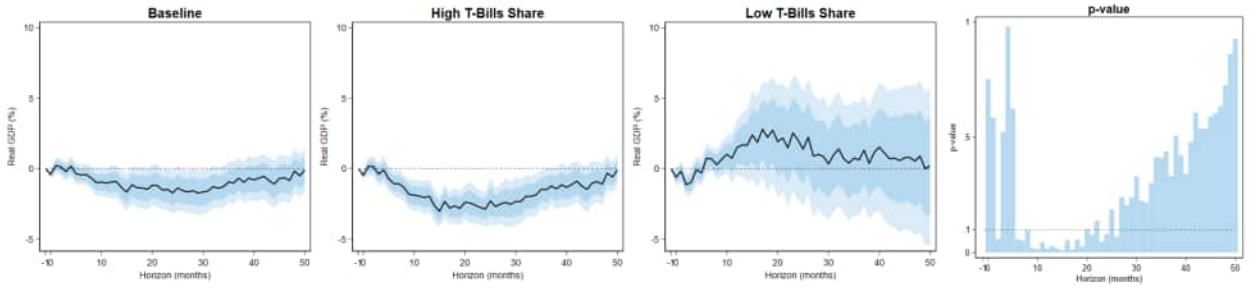
Panel C. Effective interest rate



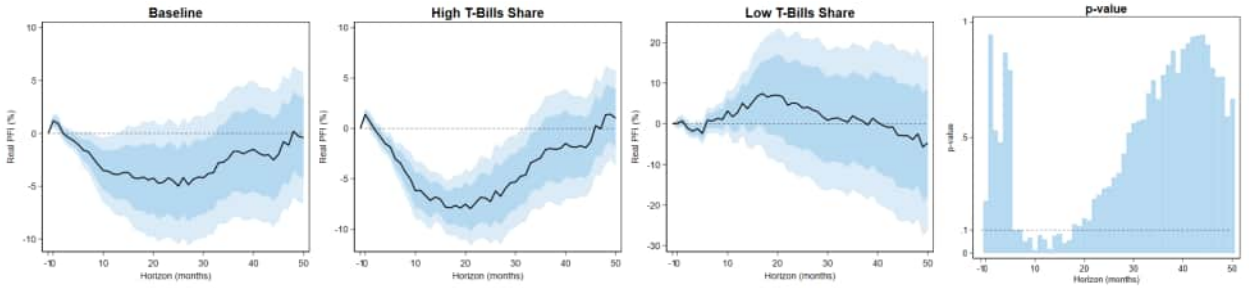
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969m3-2019m12), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with twelve lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A30: ROBUSTNESS CHECK, MONTHLY NARRATIVE SHOCK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

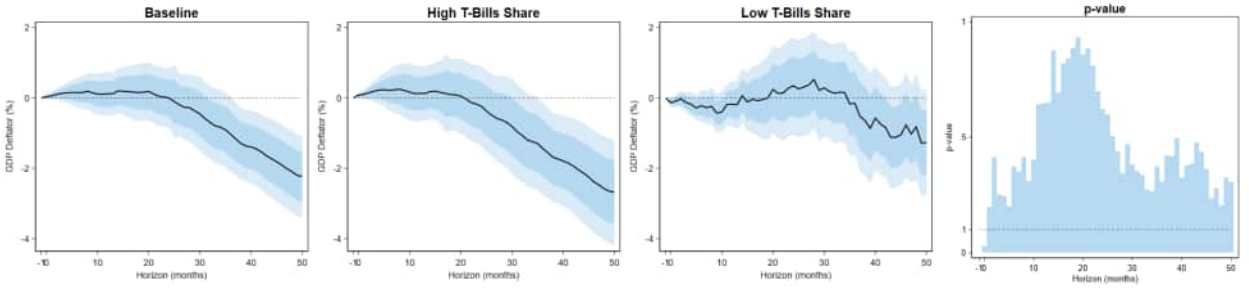
Panel A. Real GDP



Panel B. Real Private Fixed Investments



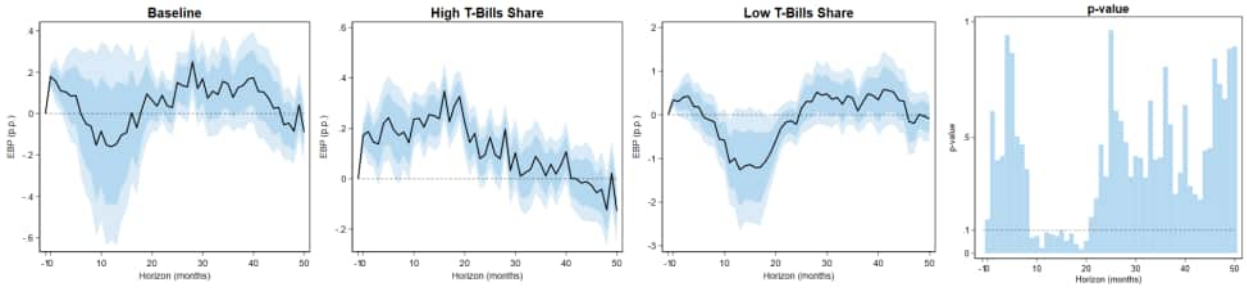
Panel C. GDP Deflator



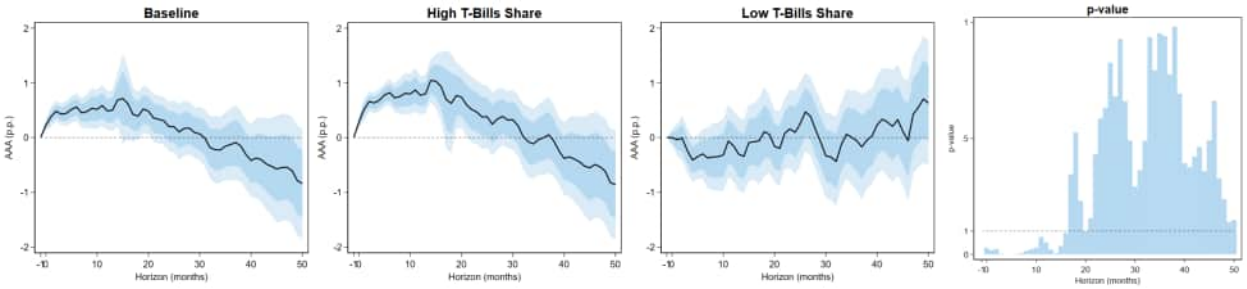
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969m3-2019m12), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with twelve lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A31: ROBUSTNESS CHECK, MONTHLY NARRATIVE SHOCK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

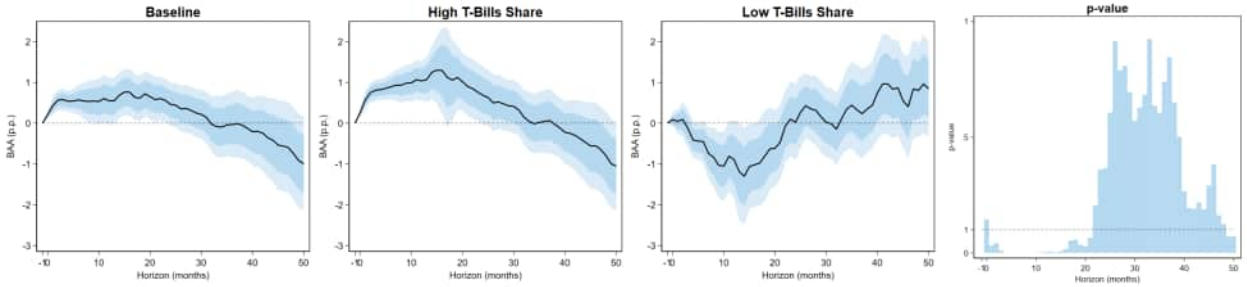
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



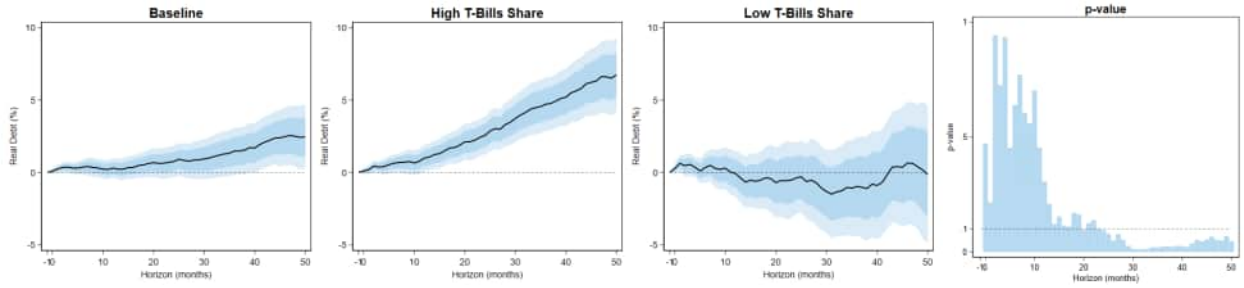
Panel C. Baa Corporate Bond Yield



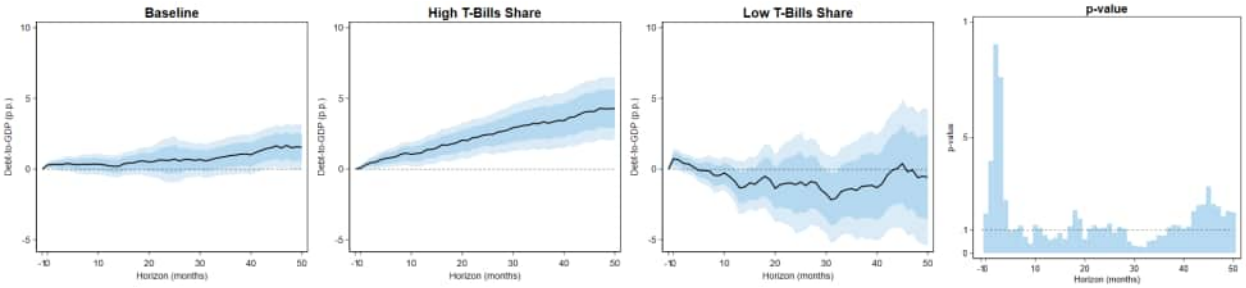
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969m3-2019m12), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with twelve lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A32: ROBUSTNESS CHECK, MONTHLY HIGH-FREQUENCY SHOCK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



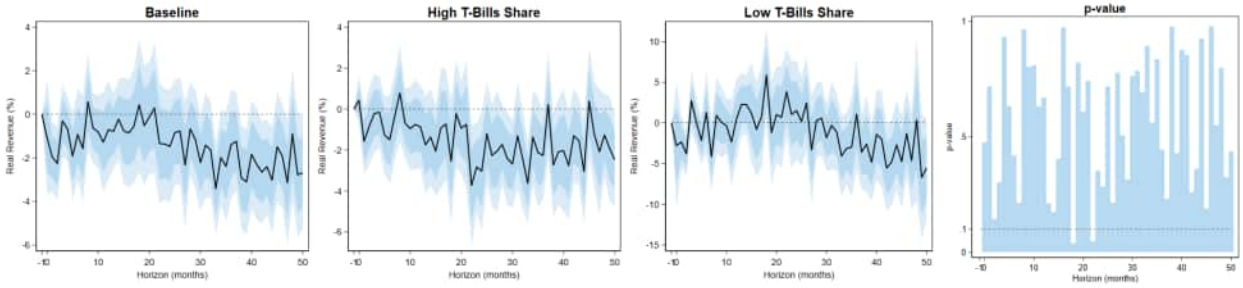
Panel B. Debt-to-GDP



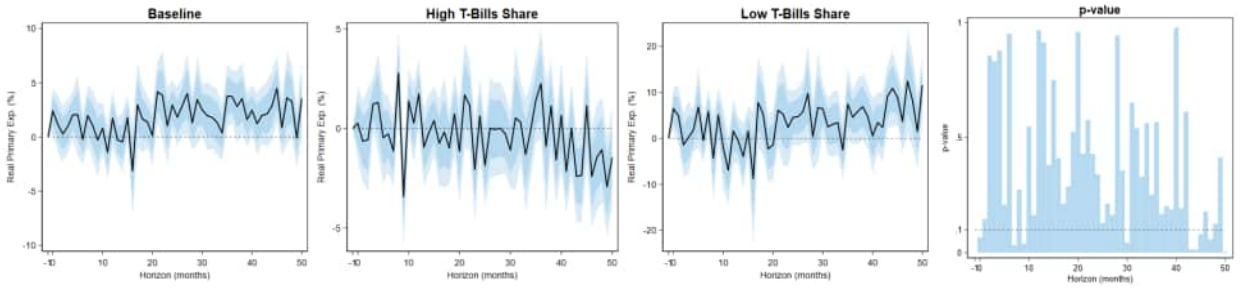
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. Structural shock series identified through high frequency identification in a proxy SVAR. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1963m1-2025m9) using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with twelve lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969m1-2025m9) using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A33: ROBUSTNESS CHECK, MONTHLY HIGH-FREQUENCY SHOCK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

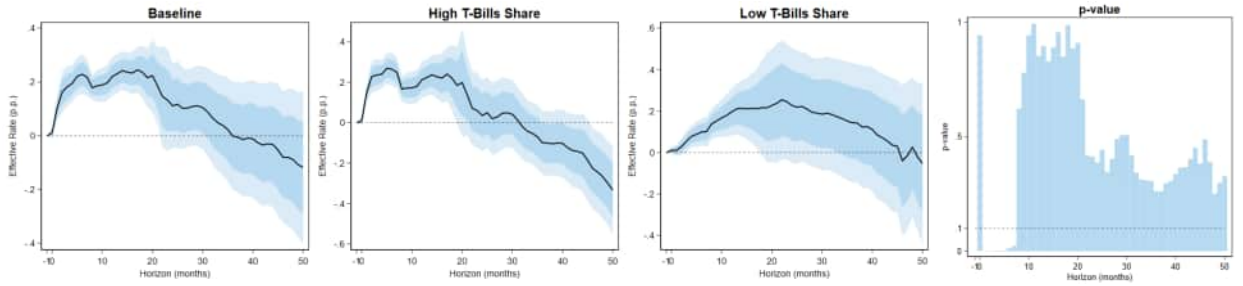
Panel A. Real Revenue



Panel B. Real Primary Expenditure



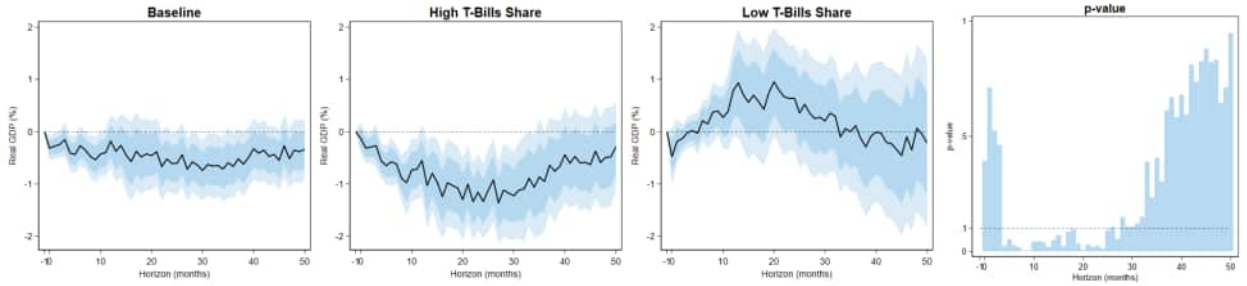
Panel C. Effective interest rate



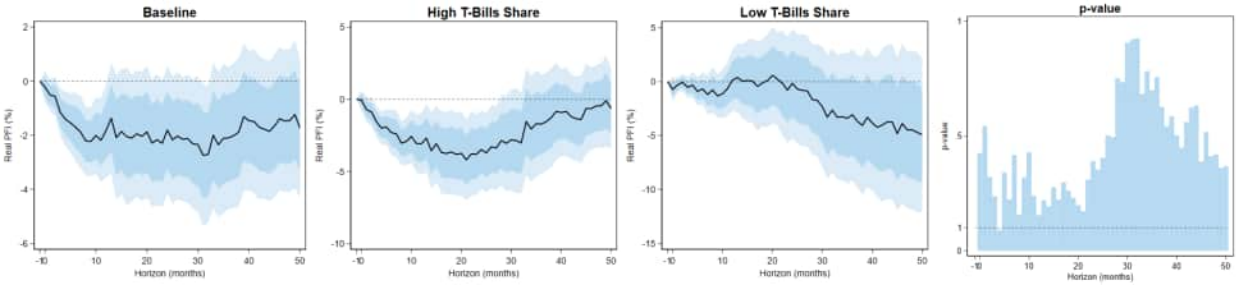
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the 10-year rate (Panel C) to a 100 bps monetary policy tightening shock. Structural shock series identified through high frequency identification in a proxy SVAR. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1963m1-2025m9), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with twelve lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969m1-2025m9) using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A34: ROBUSTNESS CHECK, MONTHLY HIGH-FREQUENCY SHOCK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

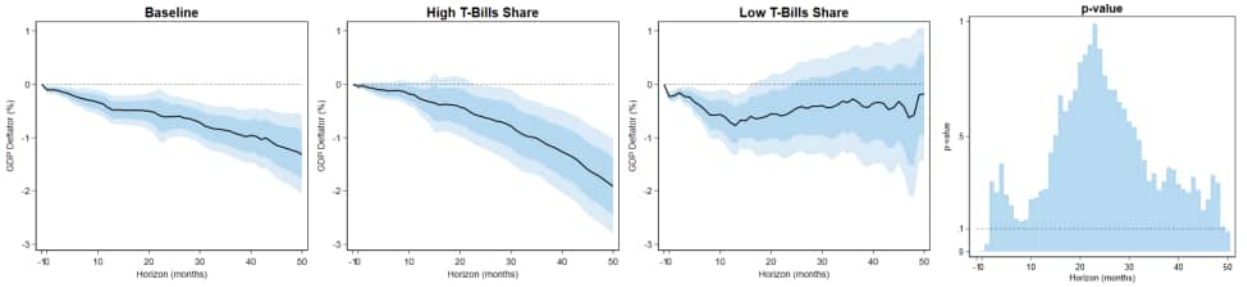
Panel A. Real GDP



Panel B. Real Private Fixed Investments



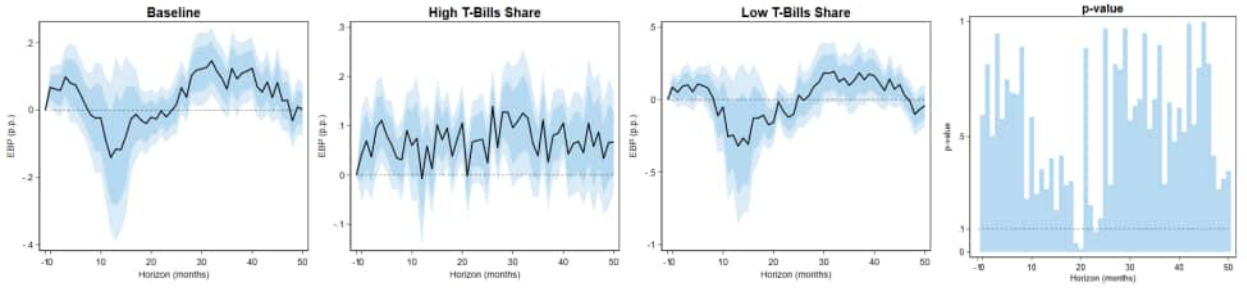
Panel C. GDP Deflator



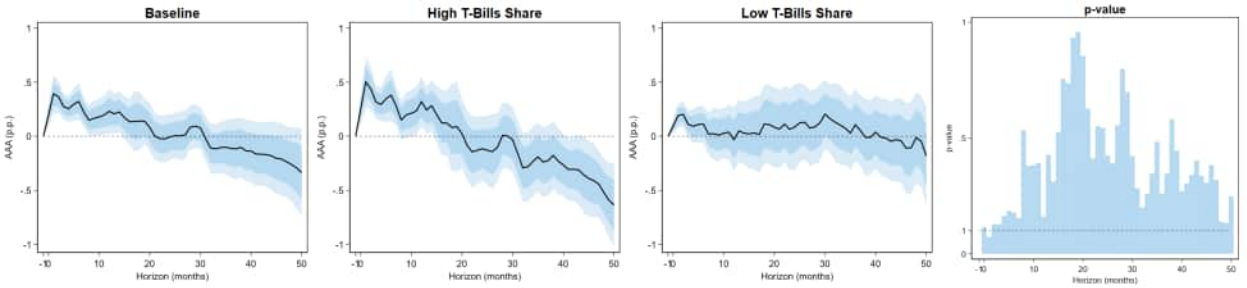
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. Structural shock series identified through high frequency identification in a proxy SVAR. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969m1-2025m9), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with twelve lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969m1-2025m9) using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A35: ROBUSTNESS CHECK, MONTHLY HIGH-FREQUENCY SHOCK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

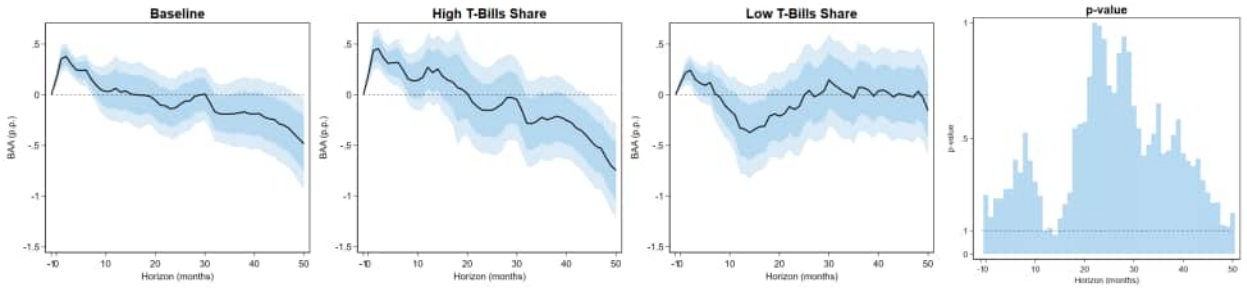
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



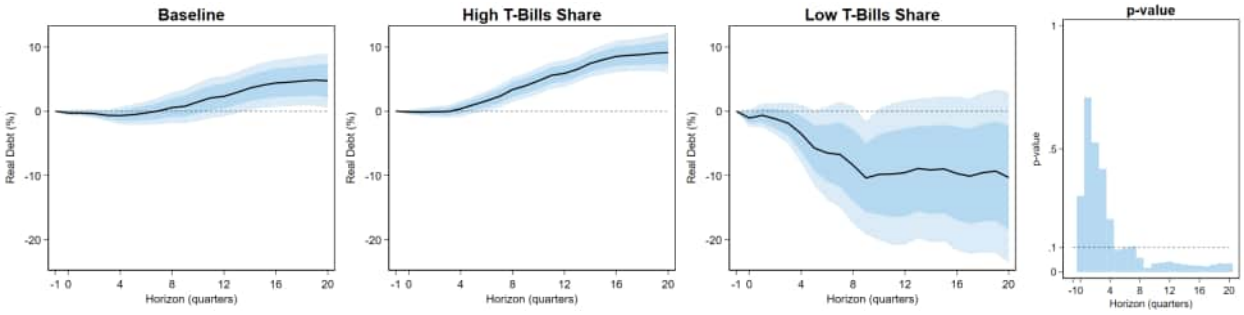
Panel C. Baa Corporate Bond Yield



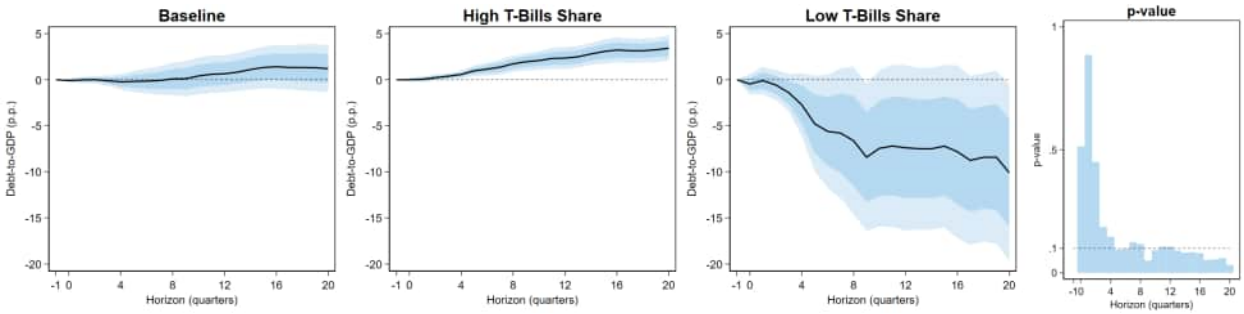
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. Structural shock series identified through high frequency identification in a proxy SVAR. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1963m1-2025m9), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell'^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with twelve lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969m1-2025m9) using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A36: ROBUSTNESS CHECK, ALTERNATIVE STF PARAMETER $\theta = 3$.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



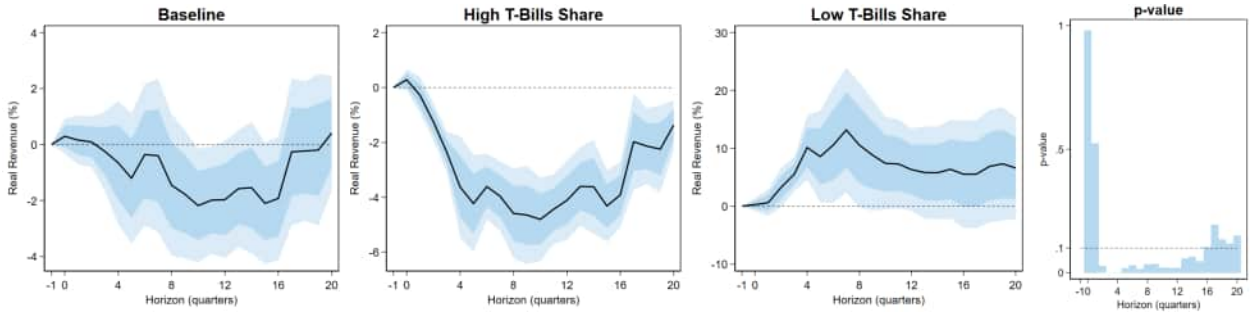
Panel B. Debt-to-GDP



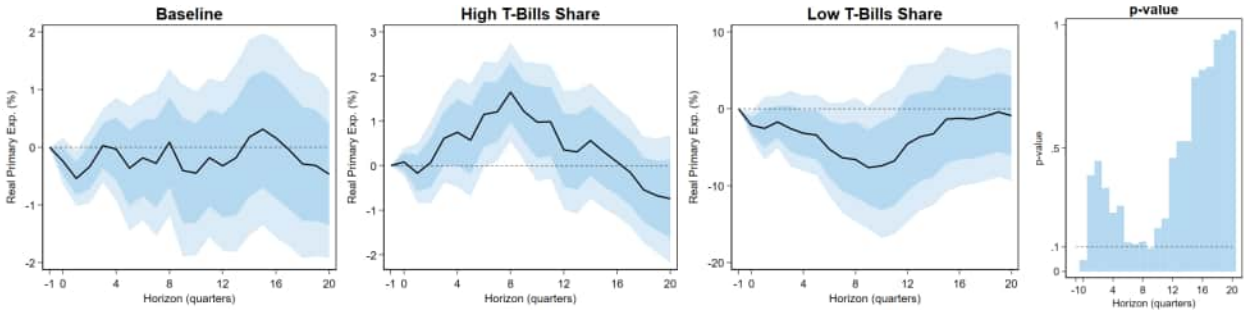
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1975q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A37: ROBUSTNESS CHECK, ALTERNATIVE STF PARAMETER $\theta = 3$.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

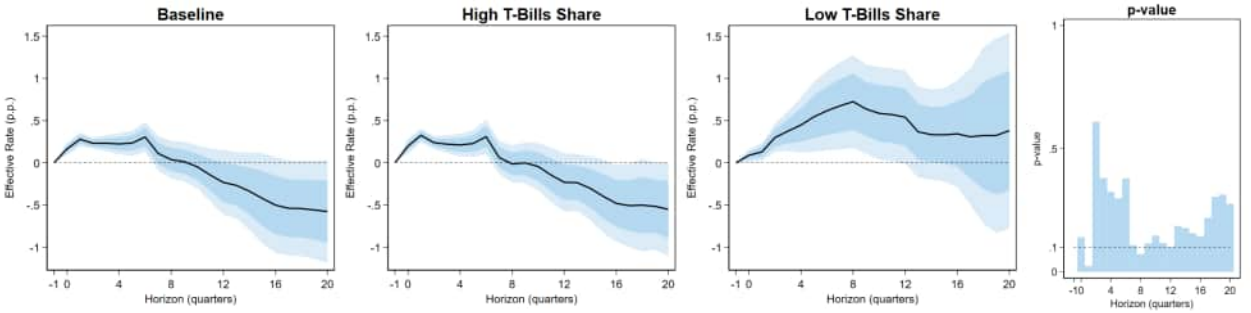
Panel A. Real Revenue



Panel B. Real Primary Expenditure



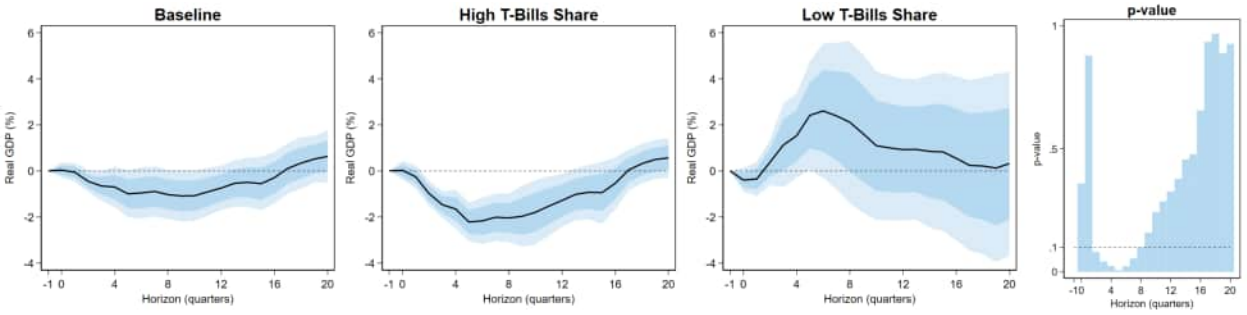
Panel C. Effective Interest Rate



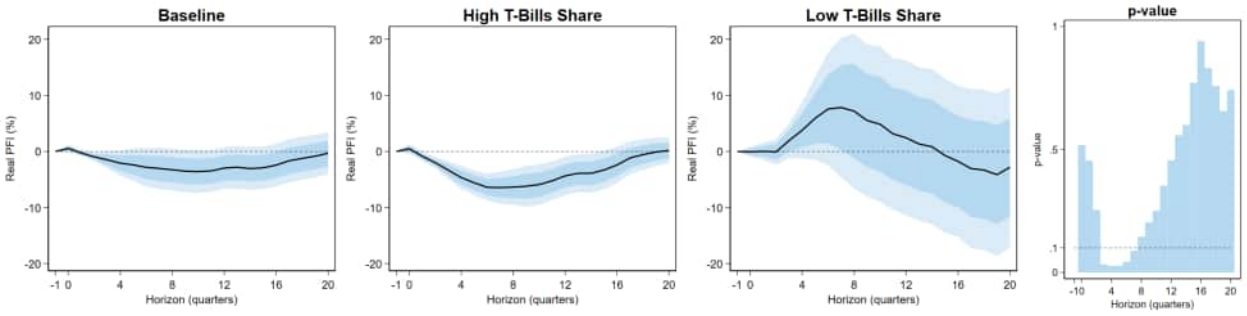
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1975q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A38: ROBUSTNESS CHECK, ALTERNATIVE STF PARAMETER $\theta = 3$.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

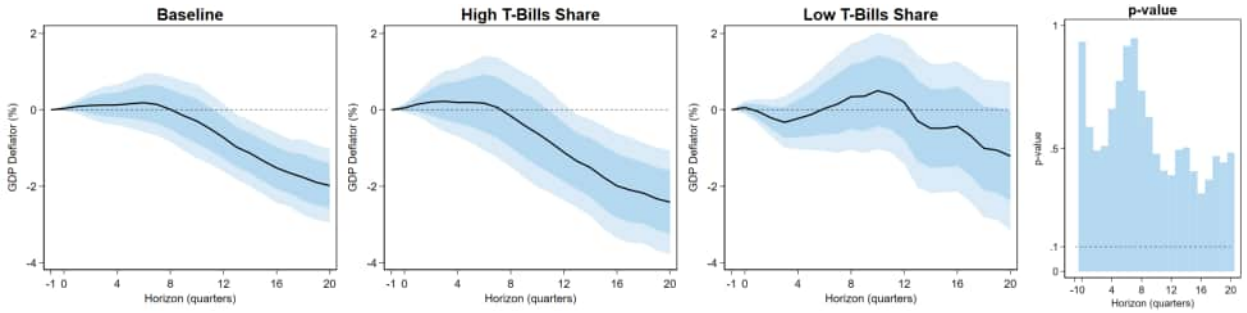
Panel A. Real GDP



Panel B. Real Private Fixed Investments



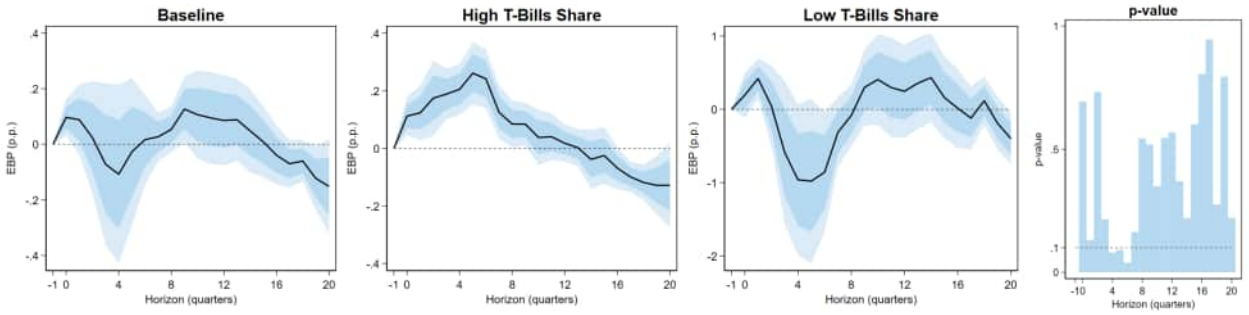
Panel C. GDP Deflator



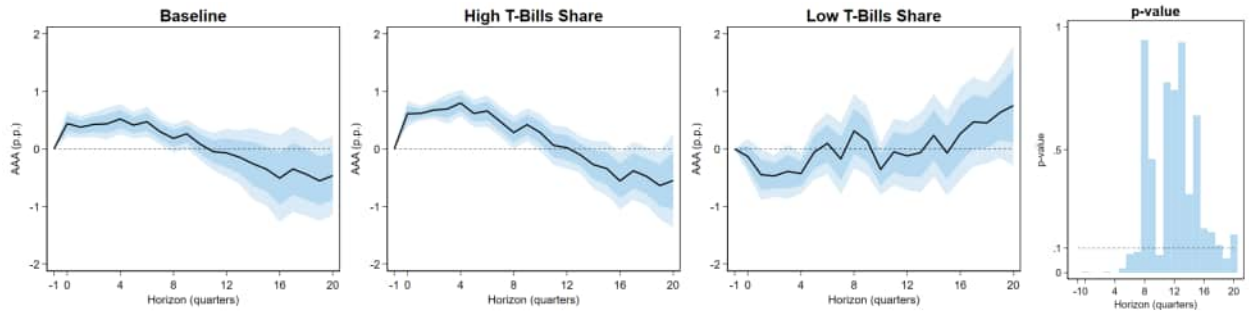
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1975q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A39: ROBUSTNESS CHECK, ALTERNATIVE STF PARAMETER $\theta = 3$.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

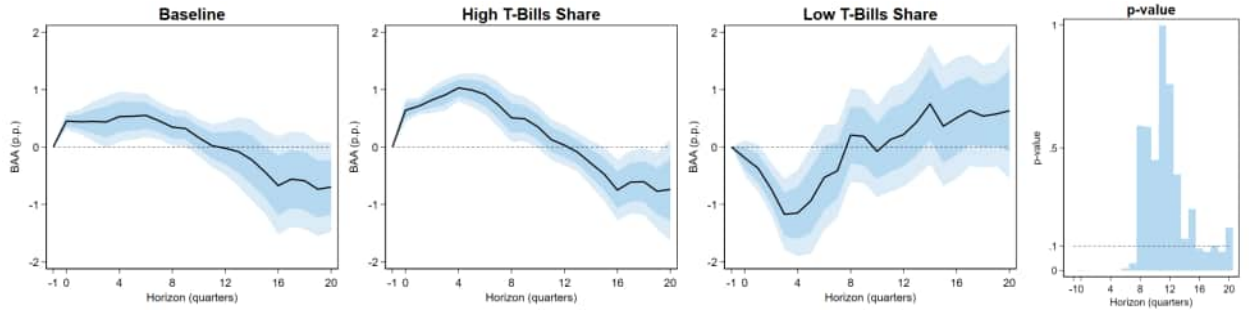
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



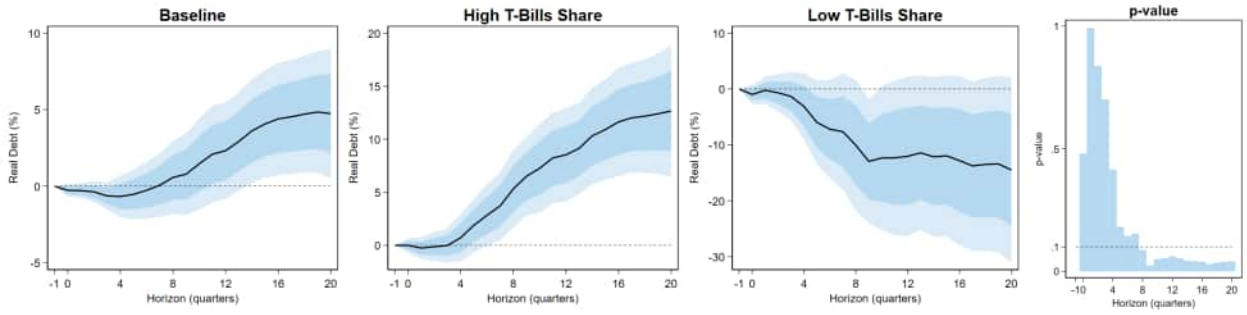
Panel B. Baa Corporate Bond Yield



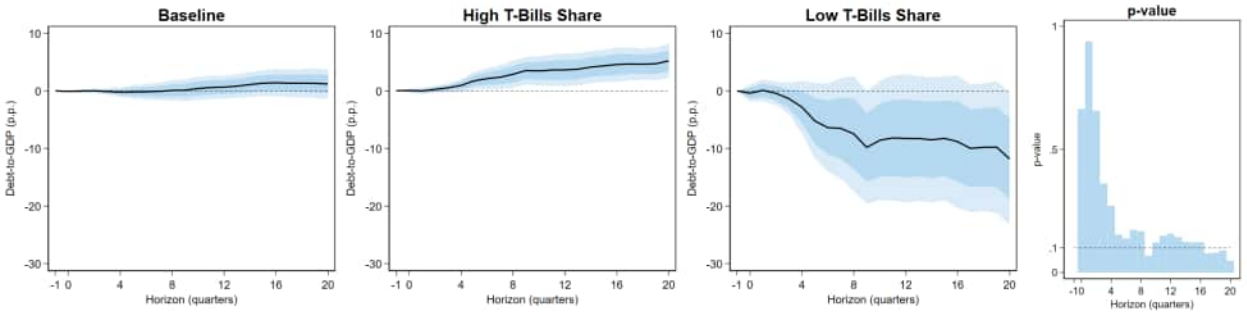
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1975q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A40: ROBUSTNESS CHECK, ALTERNATIVE STF PARAMETER $\theta = 1.5$.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



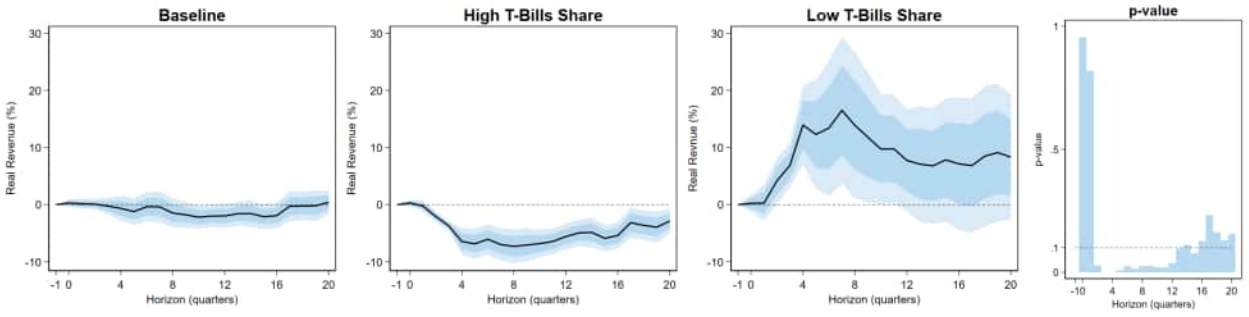
Panel B. Debt-to-GDP



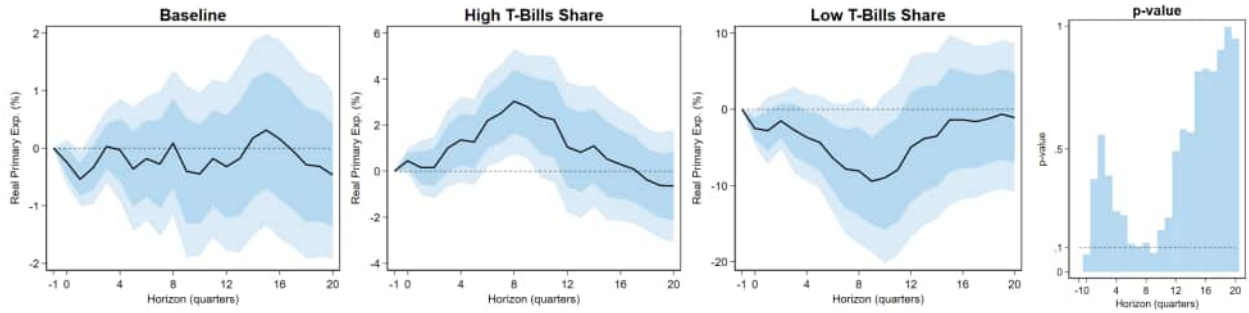
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1975q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A41: ROBUSTNESS CHECK, ALTERNATIVE STF PARAMETER $\theta = 1.5$.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

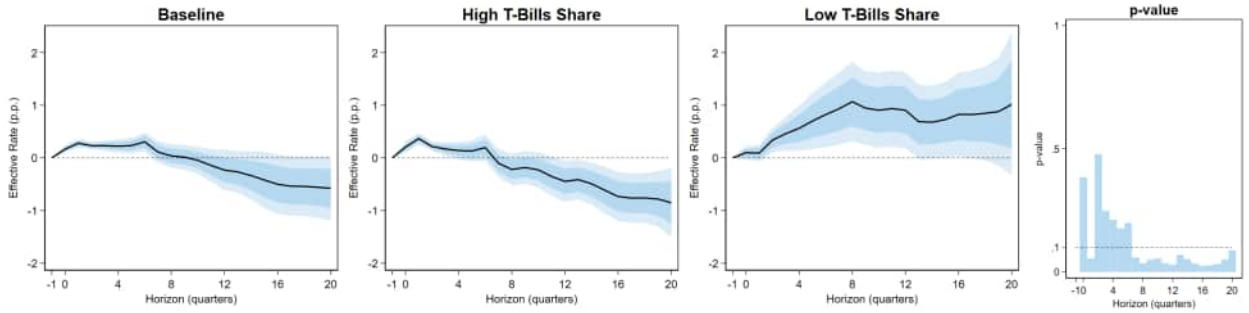
Panel A. Real Revenue



Panel B. Real Primary Expenditure



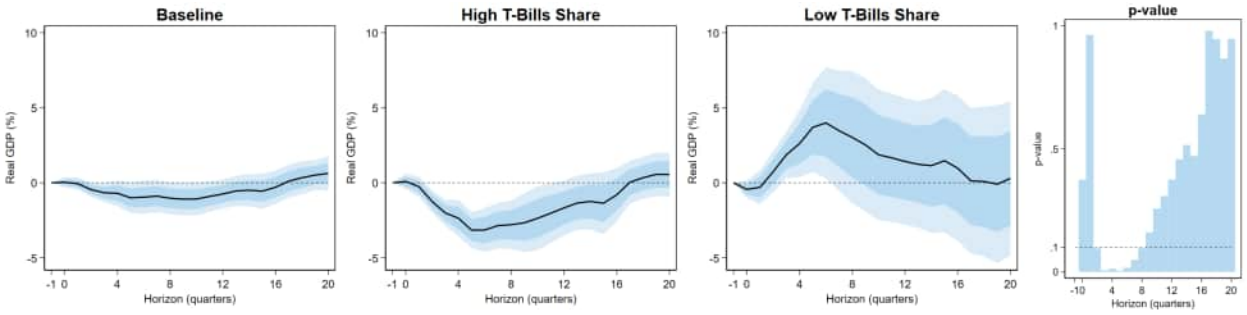
Panel C. Effective Interest Rate



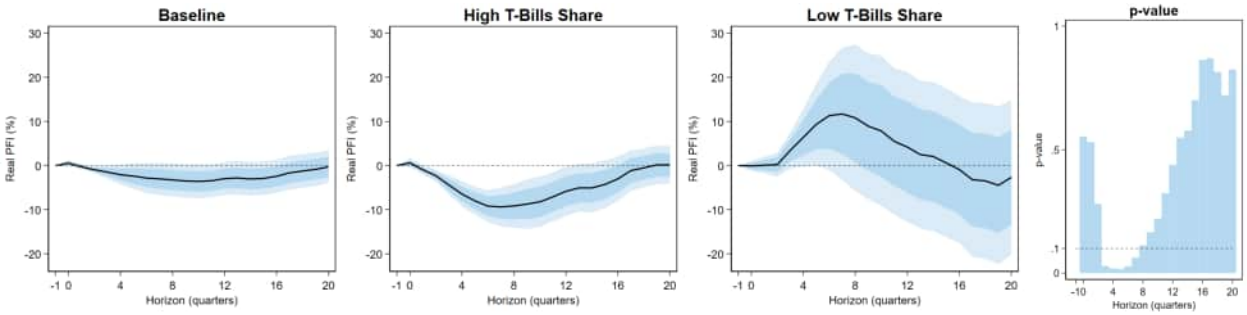
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1975q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A42: ROBUSTNESS CHECK, ALTERNATIVE STF PARAMETER $\theta = 1.5$.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

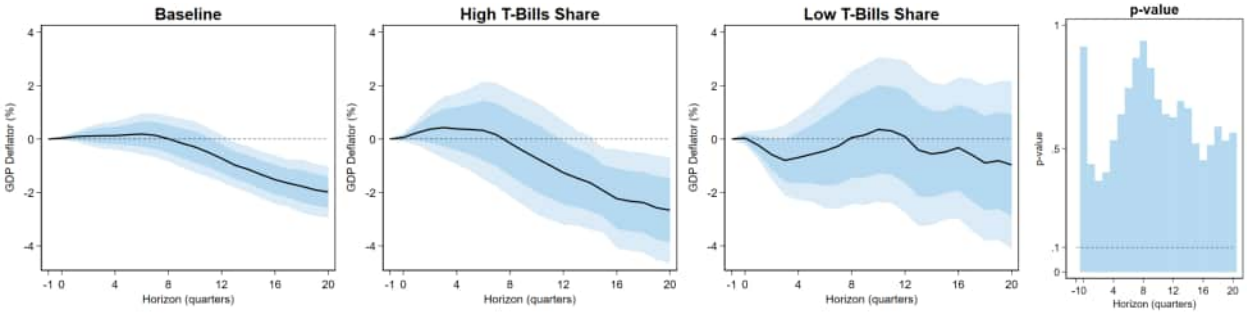
Panel A. Real GDP



Panel B. Real Private Fixed Investments



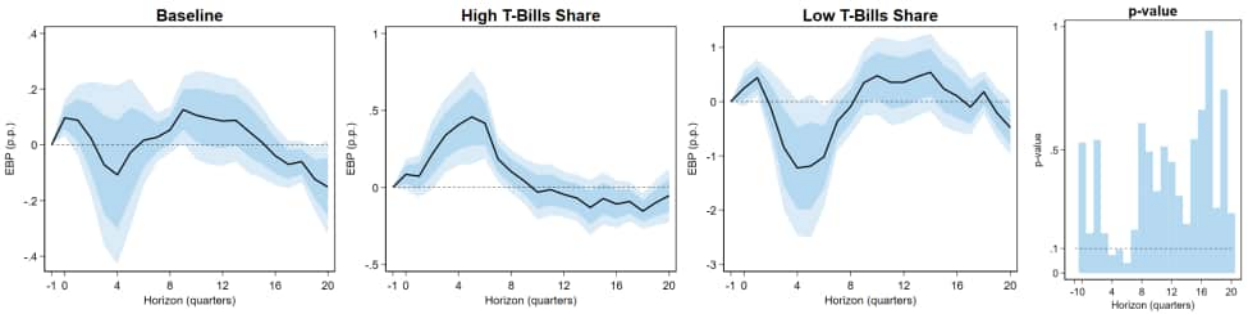
Panel C. GDP Deflator



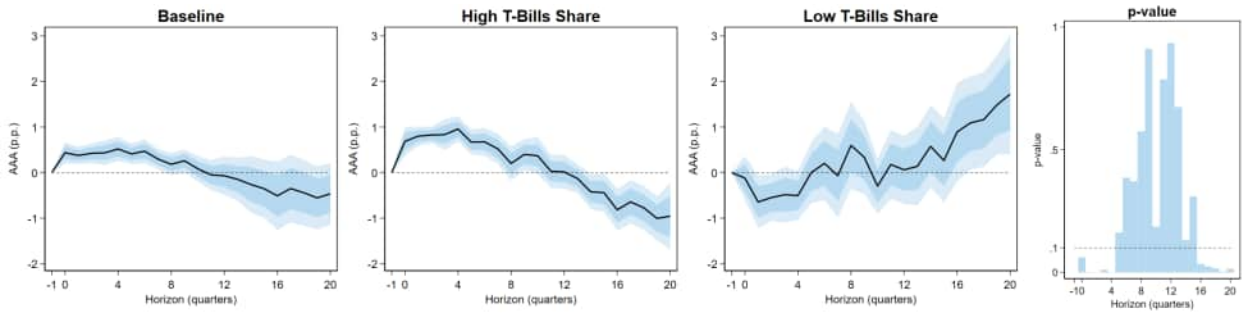
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1975q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A43: ROBUSTNESS CHECK, ALTERNATIVE STF PARAMETER $\theta = 1.5$.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

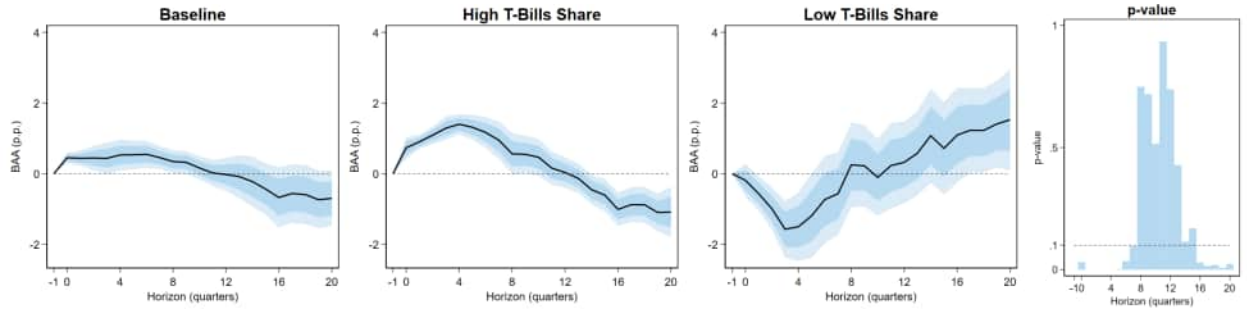
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



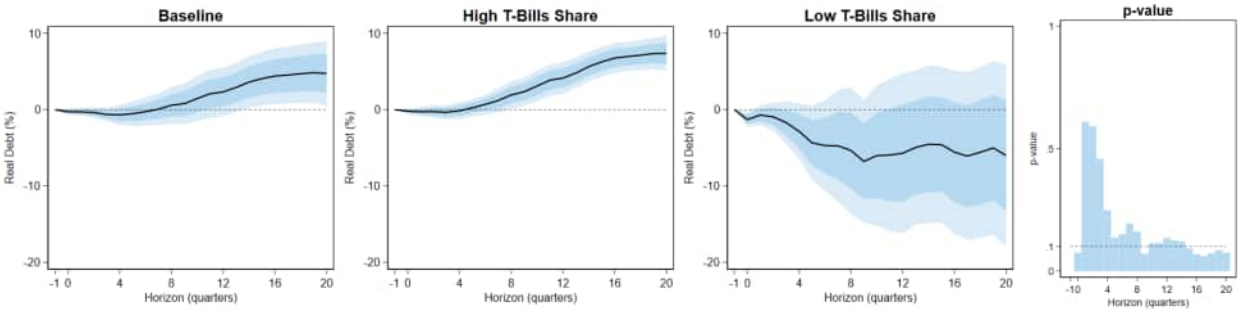
Panel C. Baa Corporate Bond Yield



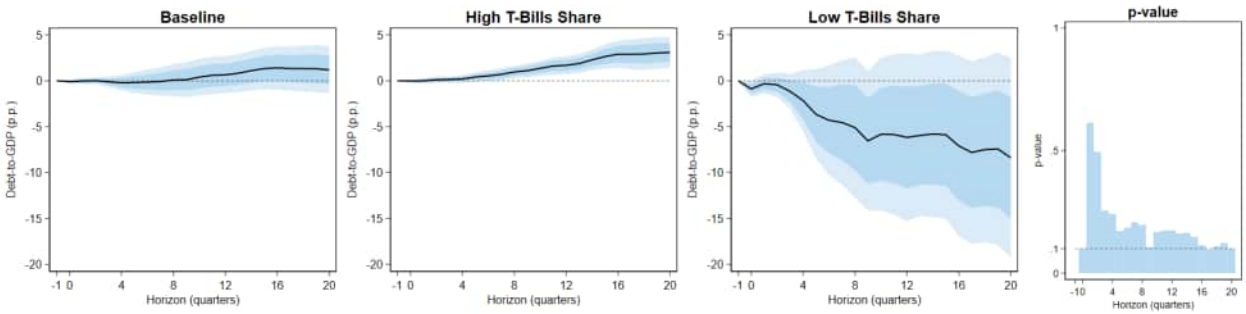
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1975q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

Figure A44: Robustness Check, T-Bills Share Above Median Interaction.
The Effects of Monetary Policy Shocks on Debt Dynamics

Panel A. Real Debt



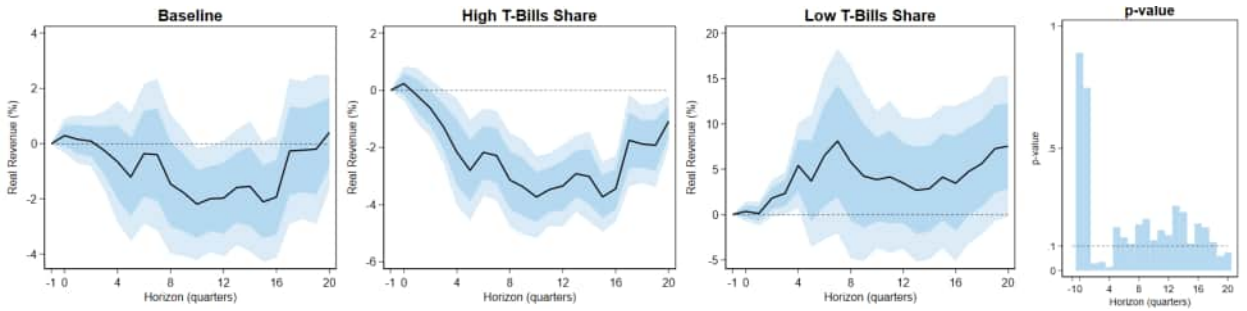
Panel B. Debt-to-GDP



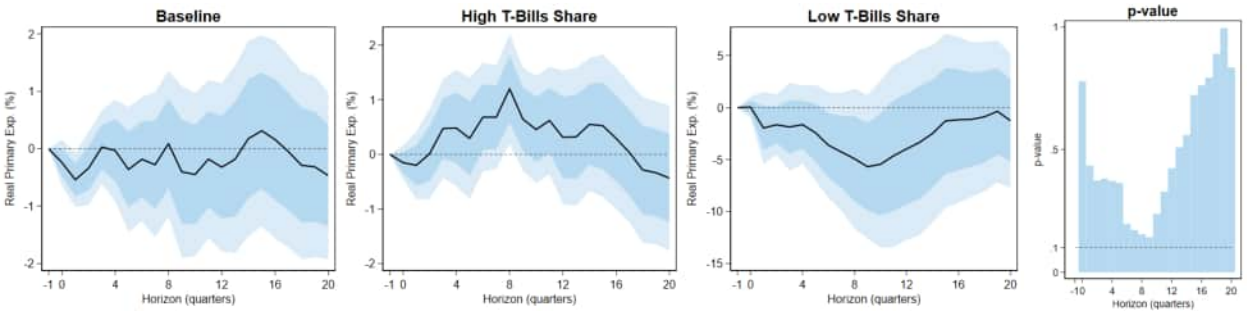
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + D_{t-1}(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - D_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} D_{t-1} + \varepsilon_{t+h}^{(h)}$, where D_{t-1} is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A45: ROBUSTNESS CHECK, T-BILLS SHARE ABOVE MEDIAN INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

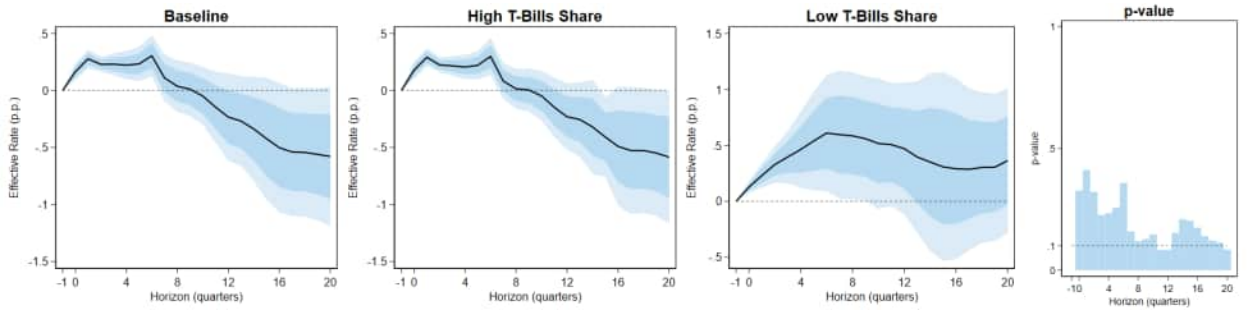
Panel A. Real Revenue



Panel B. Real Primary Expenditure



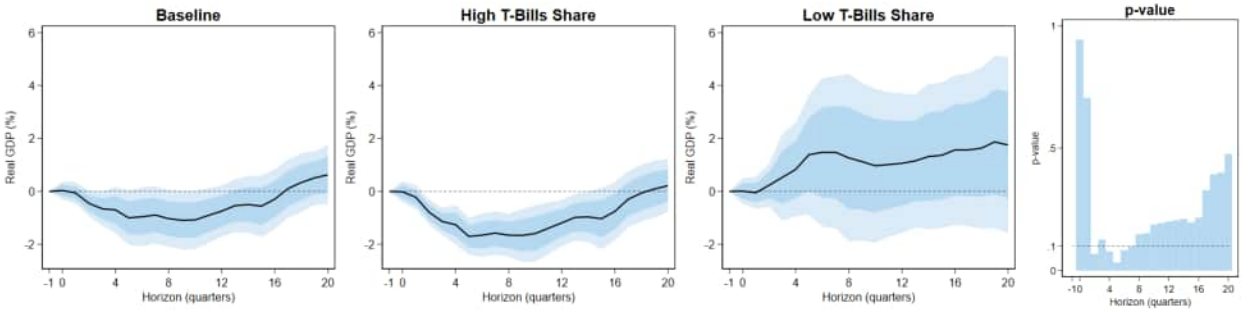
Panel C. Effective Interest Rate



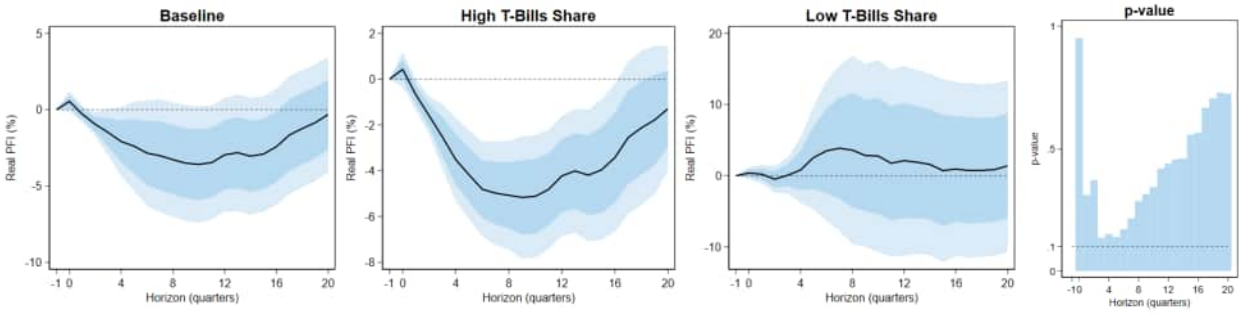
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell'^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + D_{t-1}(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - D_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} D_{t-1} + \varepsilon_{t+h}^{(h)}$, where D_{t-1} is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A46: ROBUSTNESS CHECK, T-BILLS SHARE ABOVE MEDIAN INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

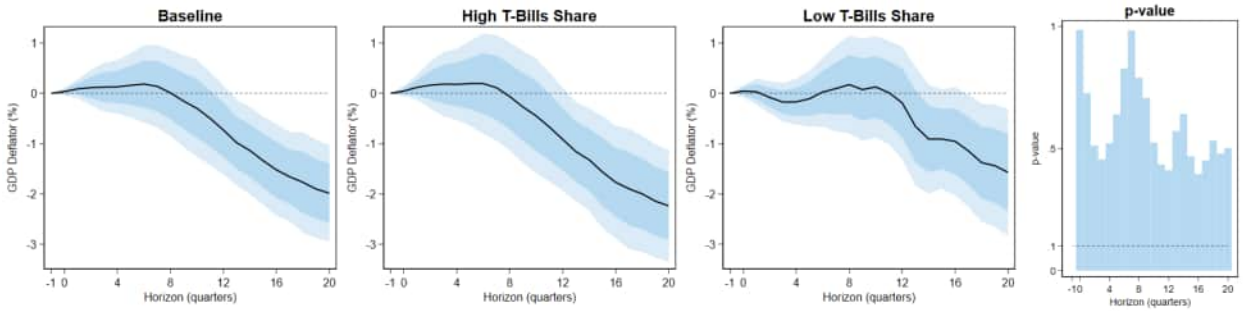
Panel A. Real GDP



Panel B. Real Private Fixed Investments



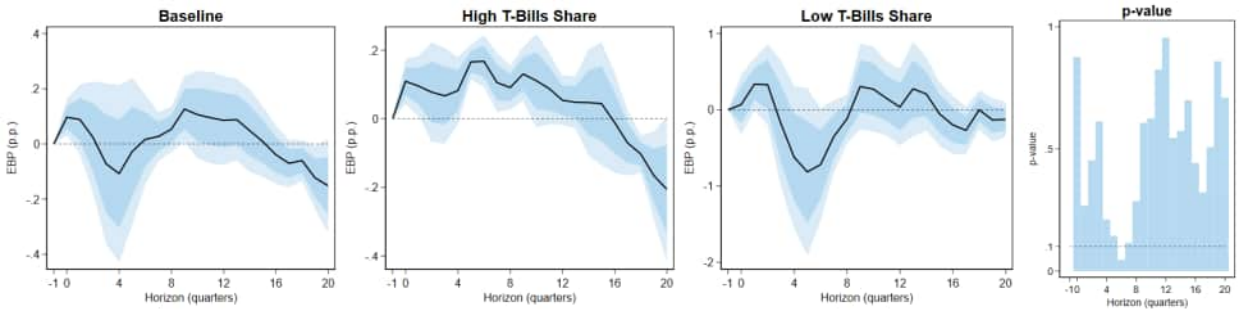
Panel C. GDP Deflator



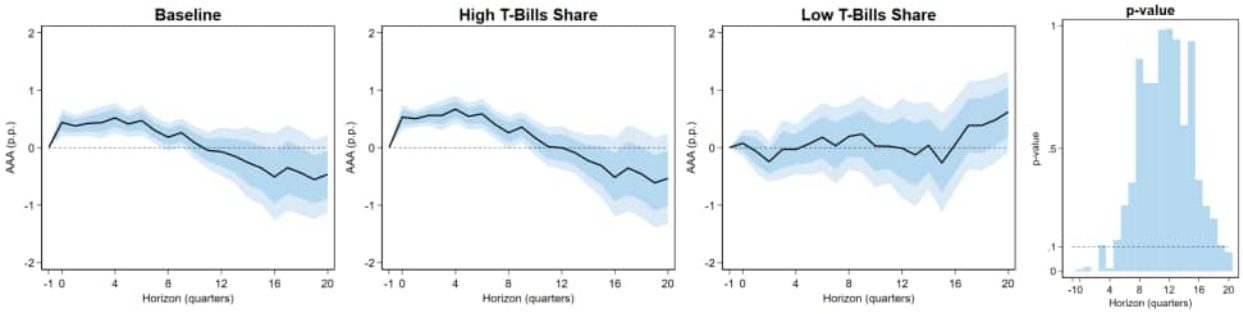
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + D_{t-1}(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - D_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} D_{t-1} + \varepsilon_{t+h}^{(h)}$, where D_{t-1} is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A47: ROBUSTNESS CHECK, T-BILLS SHARE ABOVE MEDIAN INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

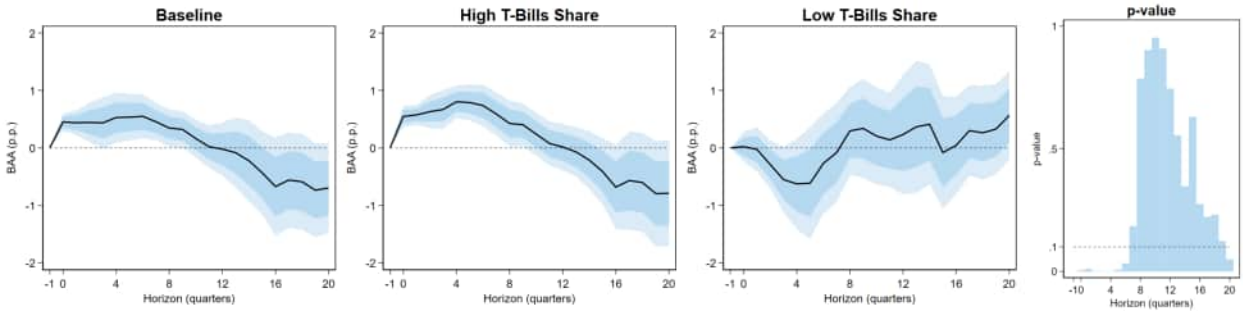
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



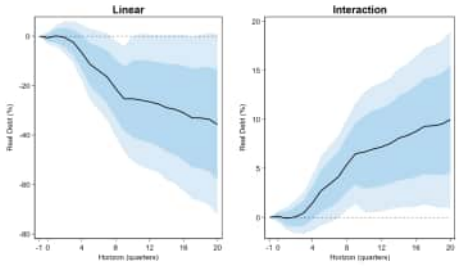
Panel C. Baa Corporate Bond Yield



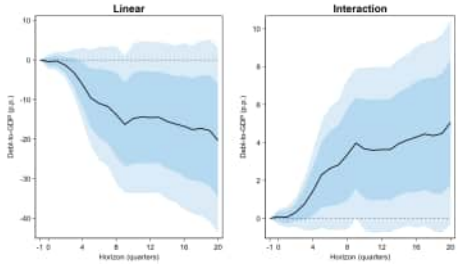
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + D_{t-1}(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - D_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} D_{t-1} + \varepsilon_{t+h}^{(h)}$, where D_{t-1} is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A48: ROBUSTNESS CHECK, T-BILLS SHARE LINEAR INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



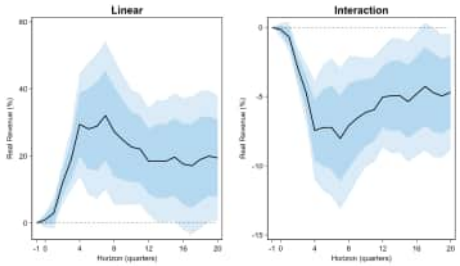
Panel B. Debt-to-GDP



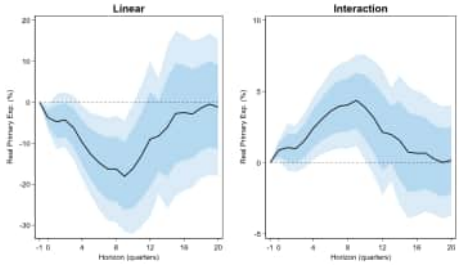
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. The solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \gamma^{(h)} MPS_t T_{t-1} + \delta^{(h)} T_{t-1} + \sum_{\ell=1}^{4} \theta_\ell'^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$ where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, T_{t-1} is the share of T-Bills over total public debt, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Column 1 shows $\beta^{(h)}$ (Column 1) and thus the effects of a monetary policy tightening shock in an hypothetical state in which the share of T-Bills is zero, while Column 2 shows $\gamma^{(h)}$, which measures how this effect varies as the share of T-Bills increases.

FIGURE A49: ROBUSTNESS CHECK, T-BILLS SHARE LINEAR INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

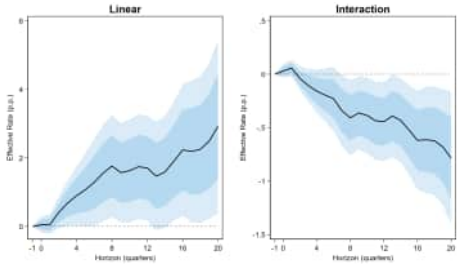
Panel A. Real Revenue



Panel B. Real Primary Expenditure



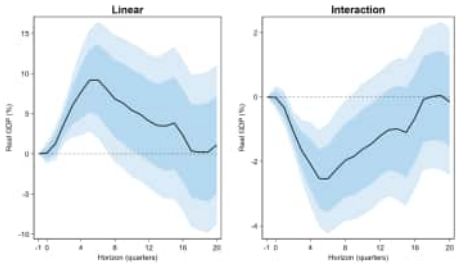
Panel C. Effective Interest Rate



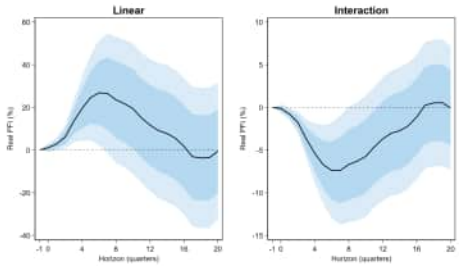
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. The solid black line shows the response, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \gamma^{(h)} MPS_t T_{t-1} + \delta^{(h)} T_{t-1} + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$ where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, T_{t-1} is the share of T-Bills over total public debt, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Column 1 shows $\beta^{(h)}$ (Column 1) and thus the effects of a monetary policy tightening shock in an hypothetical state in which the share of T-Bills is zero, while Column 2 shows $\gamma^{(h)}$, which measures how this effect varies as the share of T-Bills increases.

Figure A50: Robustness Check, T-Bills Share Linear Interaction.
The Effects of Monetary Policy Shocks on Economic Activity

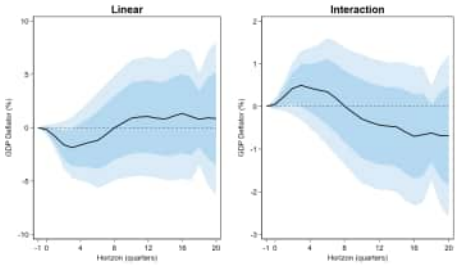
Panel A. Real GDP



Panel B. Real Private Fixed Investments



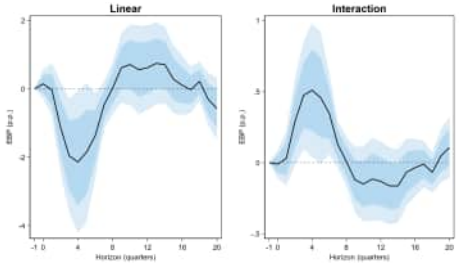
Panel C. GDP Deflator



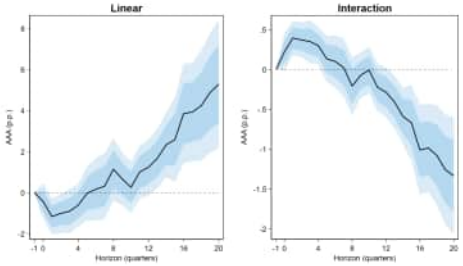
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. The solid black line shows the response, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \gamma^{(h)} MPS_t T_{t-1} + \delta^{(h)} T_{t-1} + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$ where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, T_{t-1} is the share of T-Bills over total public debt, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Column 1 shows $\beta^{(h)}$ (Column 1) and thus the effects of a monetary policy tightening shock in an hypothetical state in which the share of T-Bills is zero, while Column 2 shows $\gamma^{(h)}$, which measures how this effect varies as the share of T-Bills increases.

Figure A51: Robustness Check, T-Bills Share Linear Interaction.
The Effects of Monetary Policy Shocks on Financial Conditions

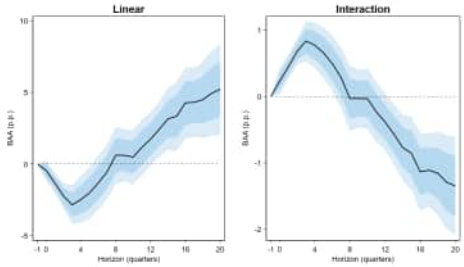
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



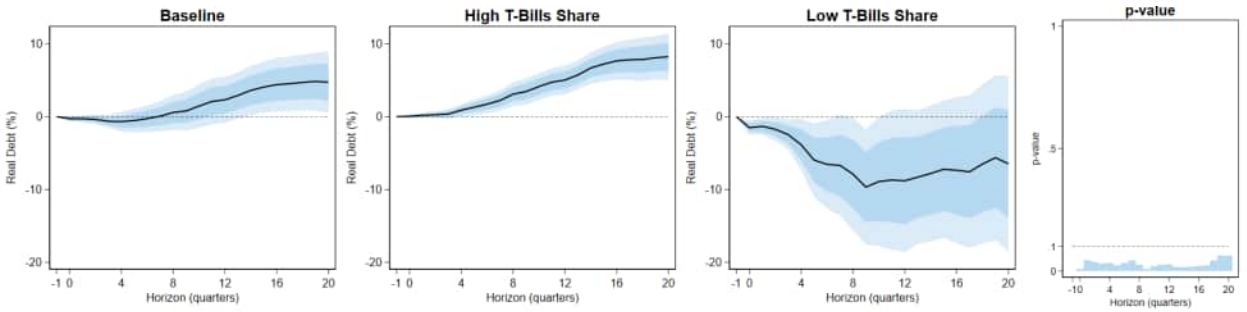
Panel C. Baa Corporate Bond Yield



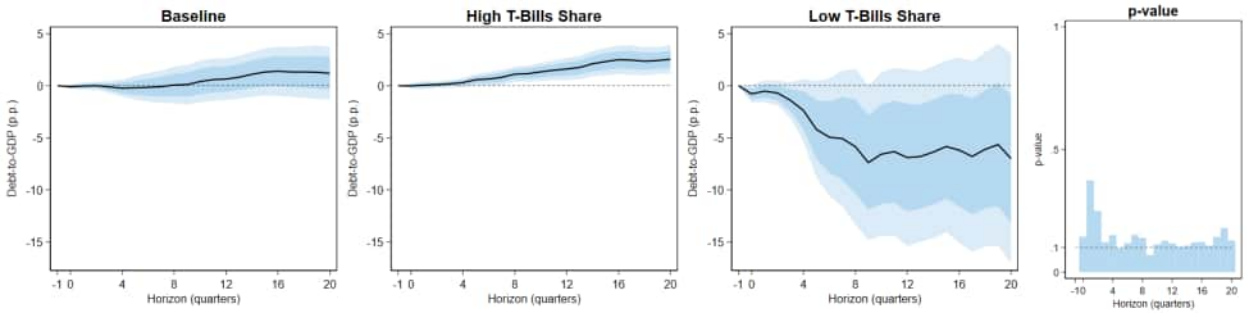
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. The solid black line shows the response, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \gamma^{(h)} MPS_t T_{t-1} + \delta^{(h)} T_{t-1} + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$ where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, T_{t-1} is the share of T-Bills over total public debt, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Column 1 shows $\beta^{(h)}$ (Column 1) and thus the effects of a monetary policy tightening shock in an hypothetical state in which the share of T-Bills is zero, while Column 2 shows $\gamma^{(h)}$, which measures how this effect varies as the share of T-Bills increases.

FIGURE A52: ROBUSTNESS CHECK, ALTERNATIVE MEASURE OF T-BILLS SHARE.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



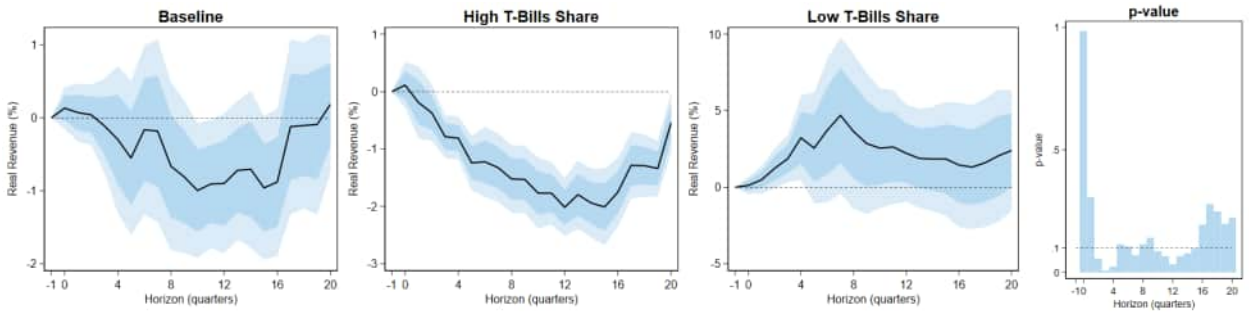
Panel B. Debt-to-GDP



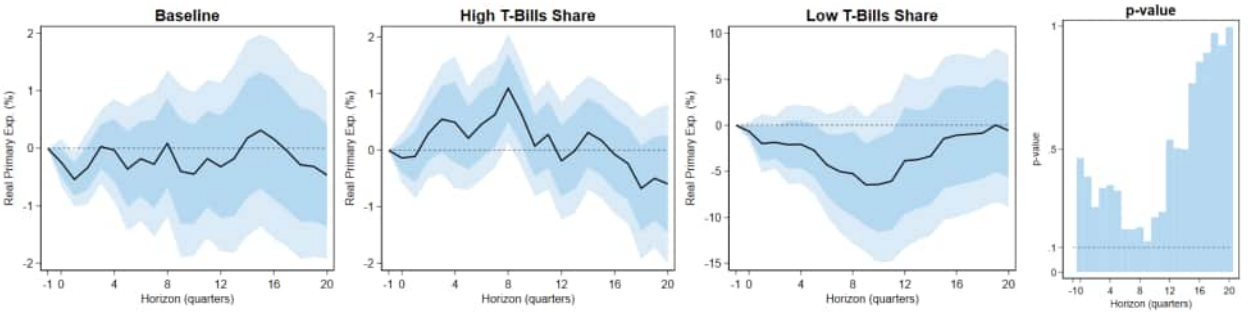
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1975q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A53: ROBUSTNESS CHECK, ALTERNATIVE MEASURE OF T-BILLS SHARE.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

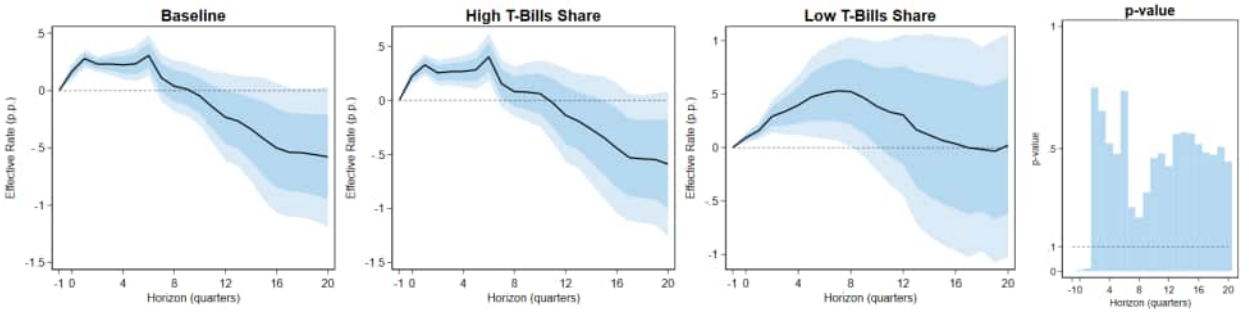
Panel A. Real Revenue



Panel B. Real Primary Expenditure



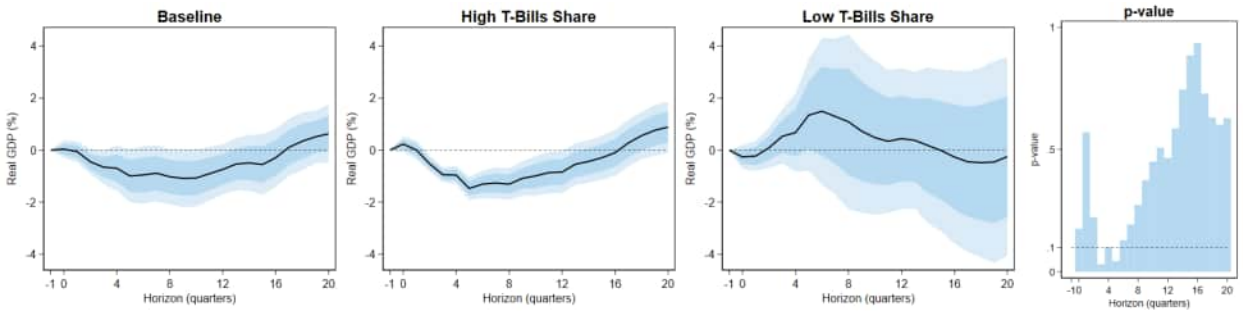
Panel C. Effective Interest Rate



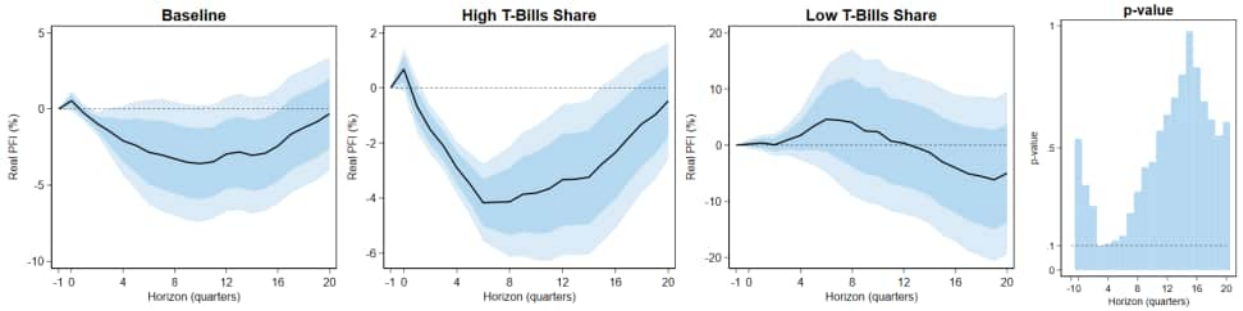
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1975q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A54: ROBUSTNESS CHECK, ALTERNATIVE MEASURE OF T-BILLS SHARE.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

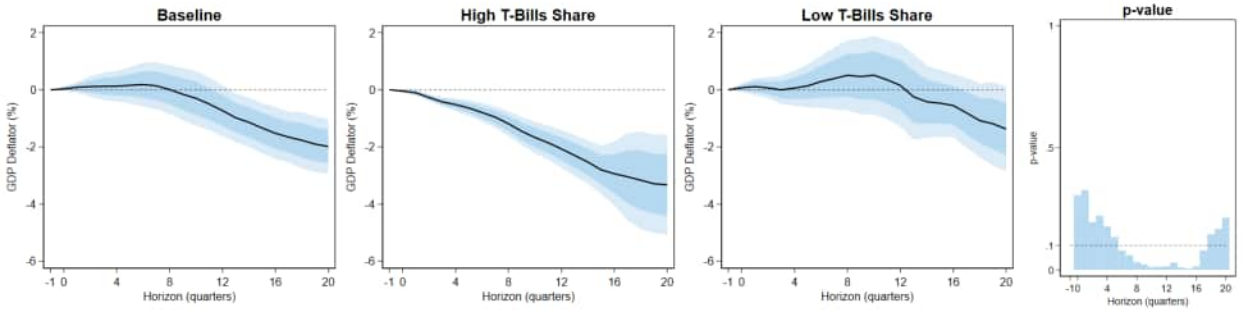
Panel A. Real GDP



Panel B. Real Private Fixed Investments



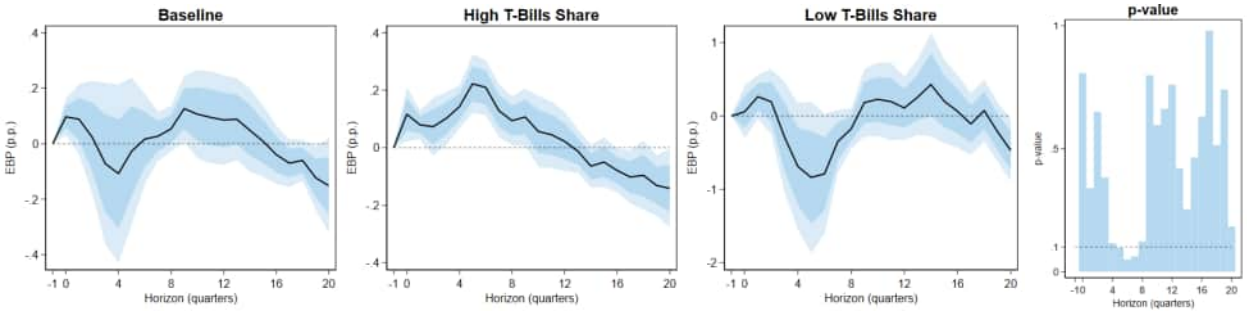
Panel C. GDP Deflator



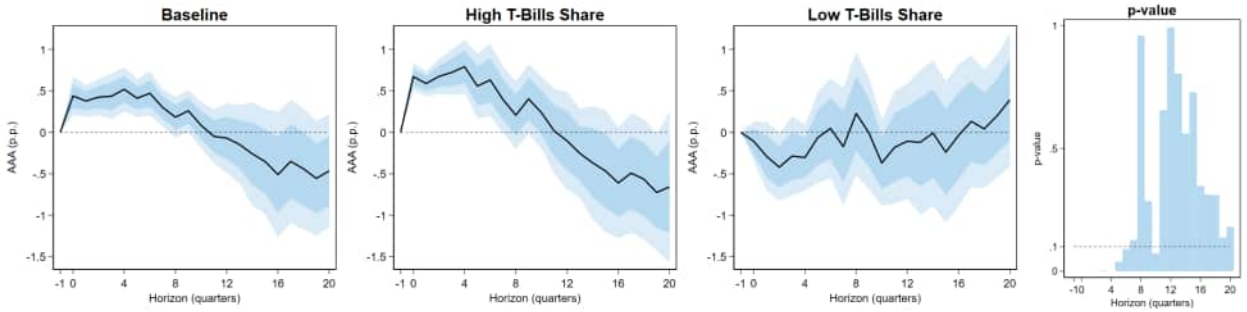
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1975q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A55: ROBUSTNESS CHECK, ALTERNATIVE MEASURE OF T-BILLS SHARE.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



Panel C. Baa Corporate Bond Yield

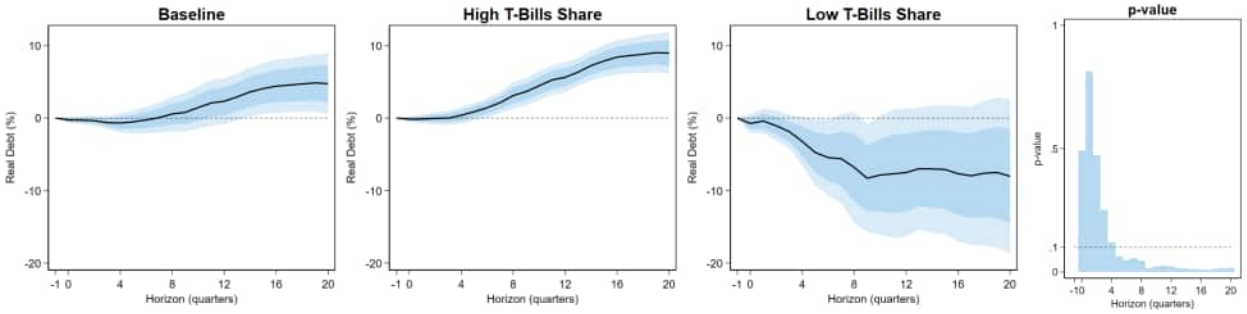


Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1975q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-1}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

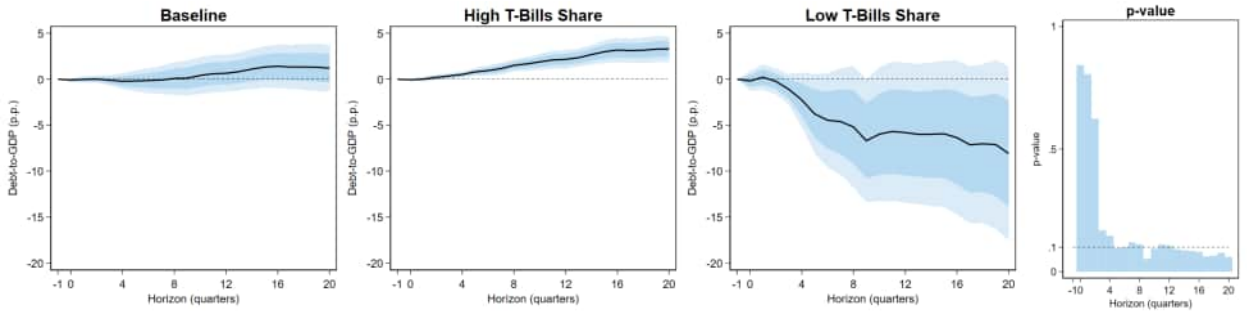
FIGURE A56: ROBUSTNESS CHECK, SMOOTH TRANSITION FUNCTION LAGGED BY TWO QUARTERS.

THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



Panel B. Debt-to-GDP

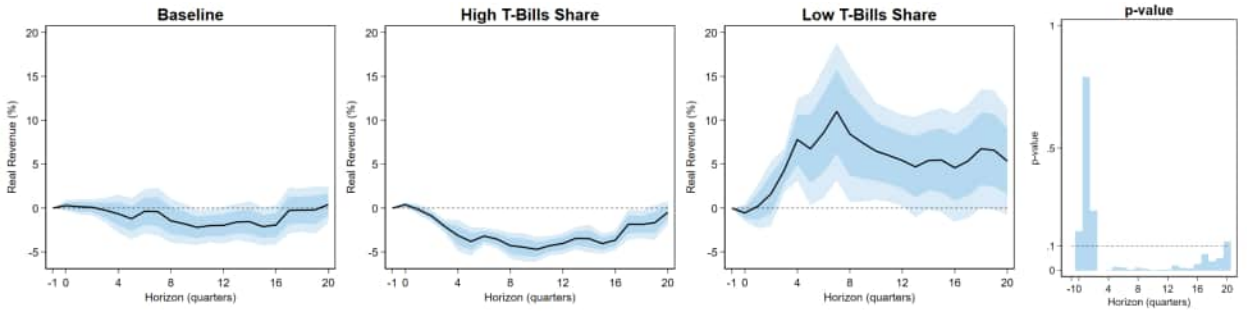


Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-2})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-2}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-2}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-2})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

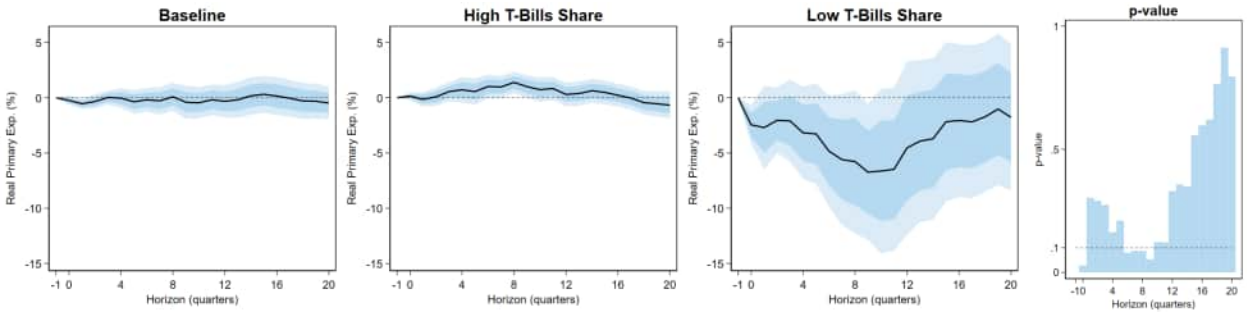
FIGURE A57: ROBUSTNESS CHECK, SMOOTH TRANSITION FUNCTION LAGGED BY TWO QUARTERS.

THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

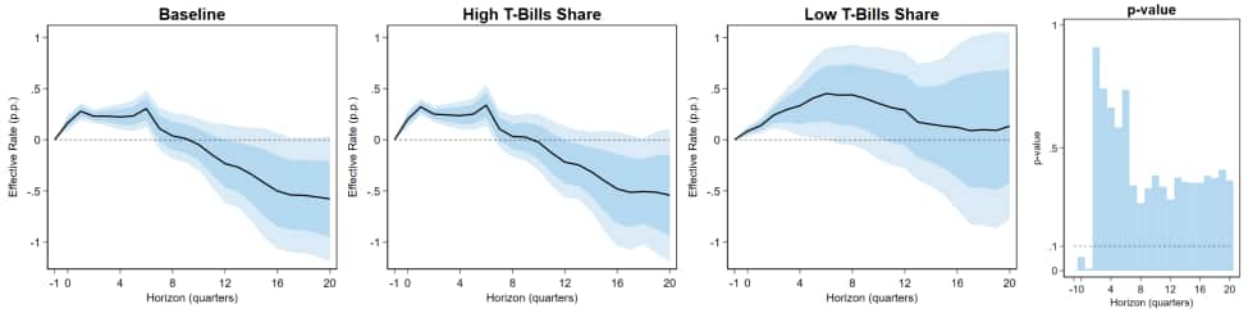
Panel A. Real Revenue



Panel B. Real Primary Expenditure



Panel C. Effective interest rate

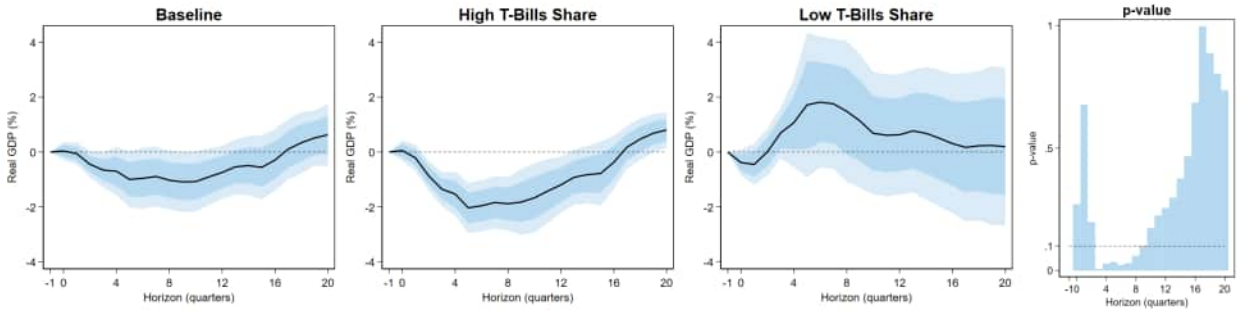


Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-2})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-2}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-2}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-2})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

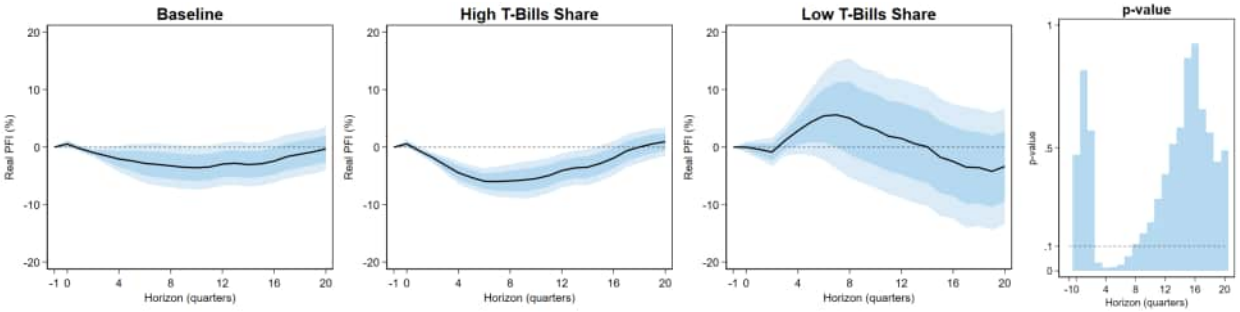
FIGURE A58: ROBUSTNESS CHECK, SMOOTH TRANSITION FUNCTION LAGGED BY TWO QUARTERS.

THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

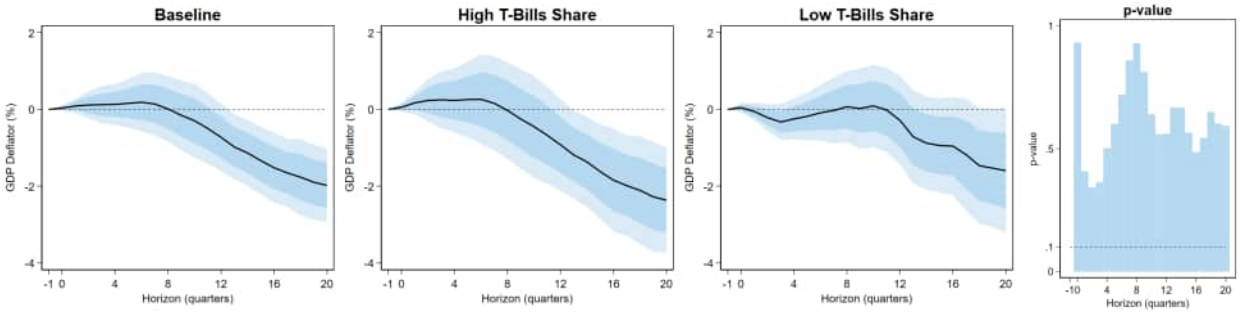
Panel A. Real GDP



Panel B. Real Private Fixed Investments



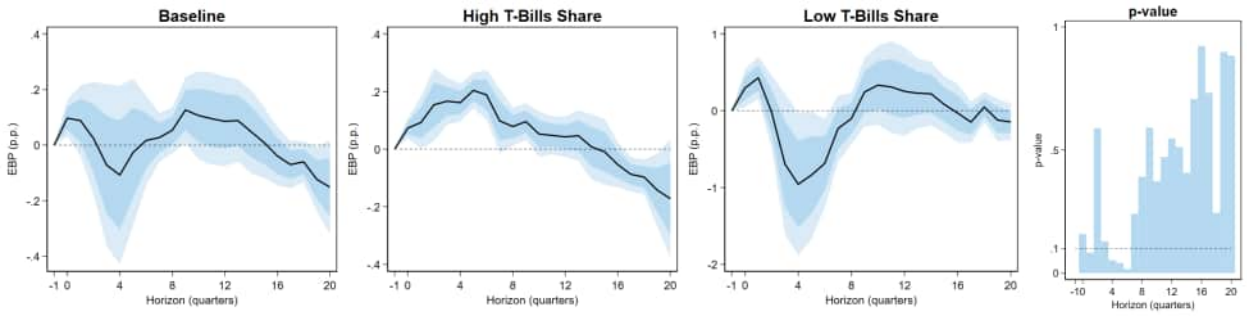
Panel C. GDP Deflator



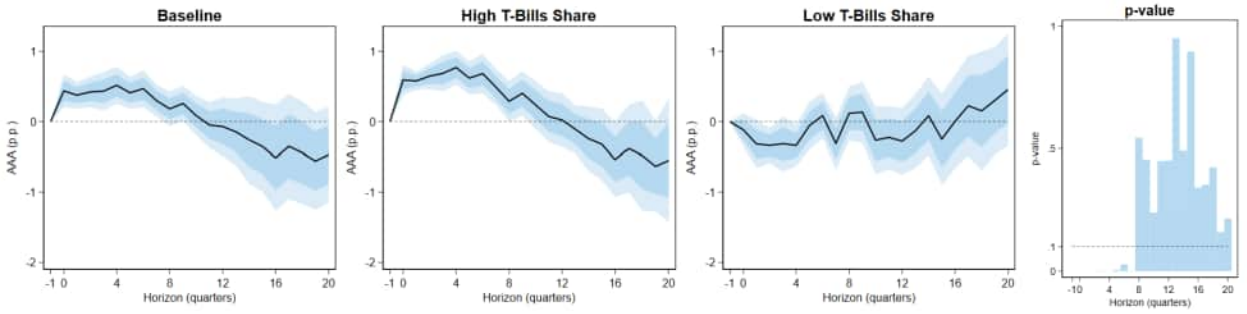
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-2})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-2}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-2}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-2})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A59: ROBUSTNESS CHECK, SMOOTH TRANSITION FUNCTION LAGGED BY TWO QUARTERS.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

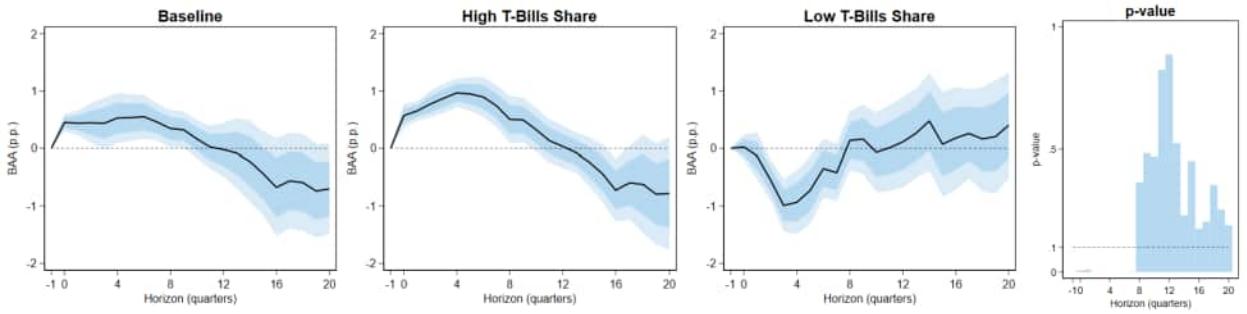
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



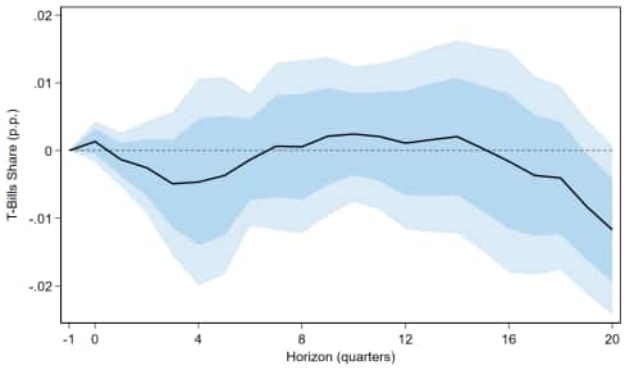
Panel C. Baa Corporate Bond Yield



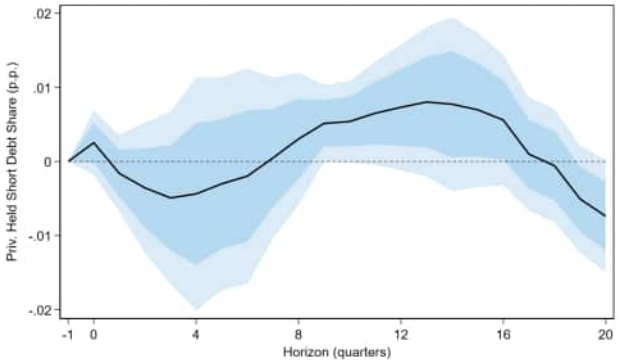
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-2})(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}) + (1 - F(z_{t-2}))(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}) + \delta^{(h)} F(z_{t-2}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-2})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A60: ROBUSTNESS CHECK.
THE EFFECTS OF MONETARY POLICY SHOCKS ON PUBLIC DEBT MATURITY

Panel A. T-Bills Share



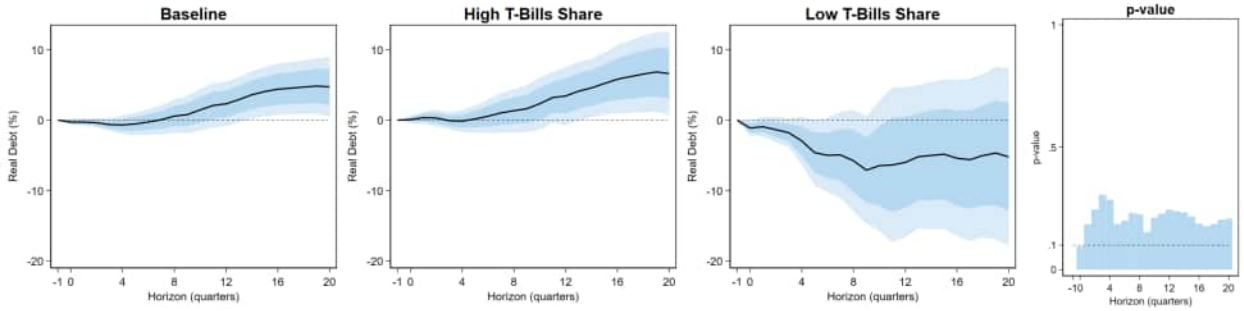
Panel B. Privately Held Short-Term Debt Share



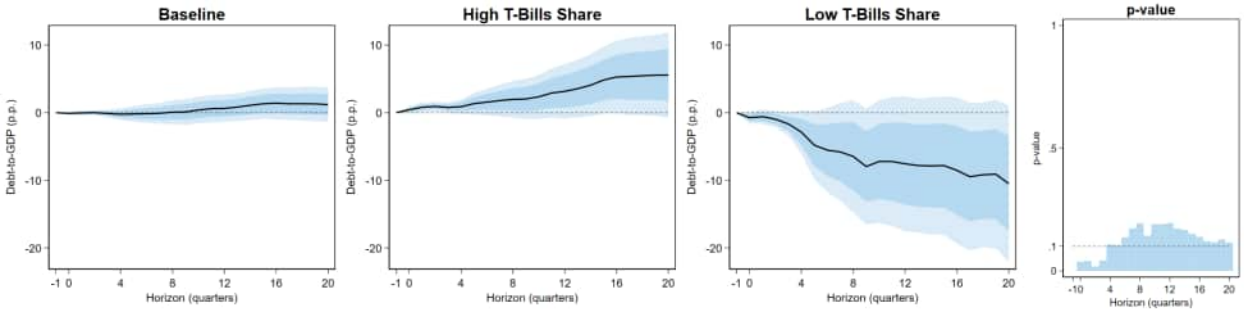
Note: The figure plots the impulse responses of the T-Bills share (Panel A) and the privately held short-term debt share (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{2} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with two lags ℓ of dependent variable and shock.

Figure A61: Robustness Check, Controlling for Interest Rate Interaction. The Effects of Monetary Policy Shocks on Debt Dynamics

Panel A. Real Debt



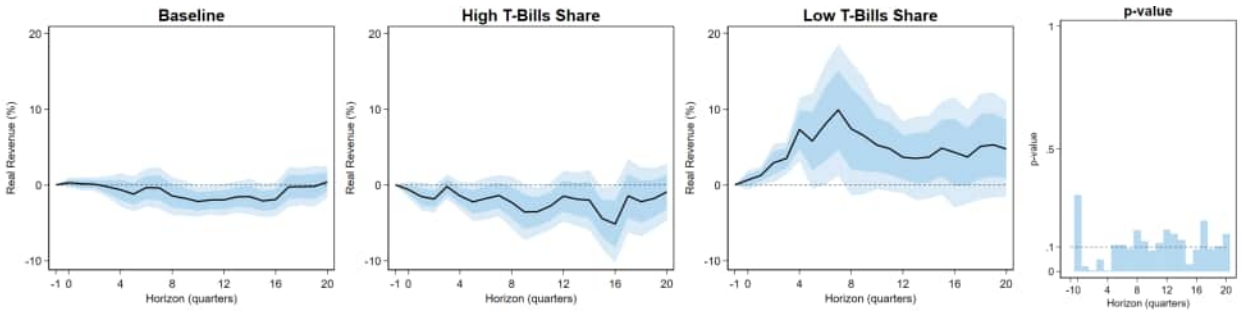
Panel B. Debt-to-GDP



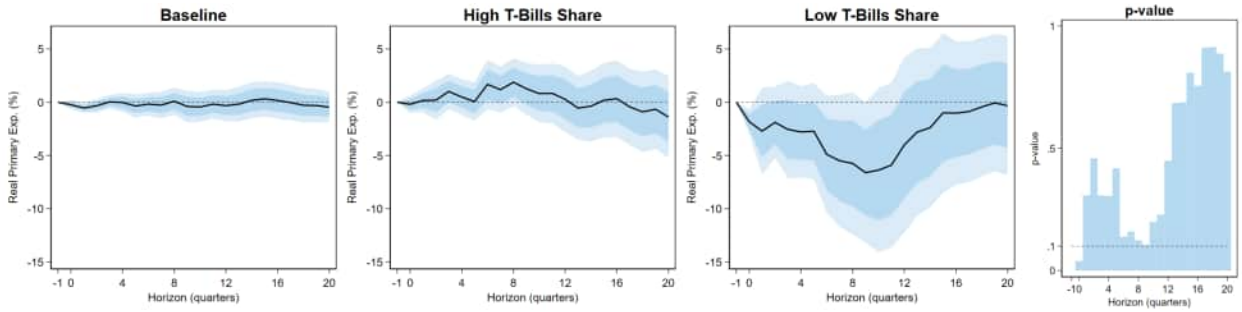
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial federal funds rate level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A62: ROBUSTNESS CHECK, CONTROLLING FOR INTEREST RATE INTERACTION. THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

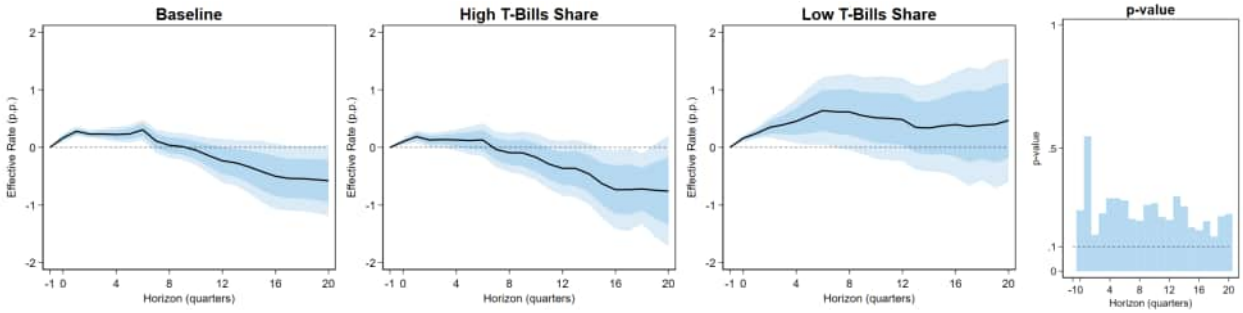
Panel A. Real Revenue



Panel B. Real Primary Expenditure



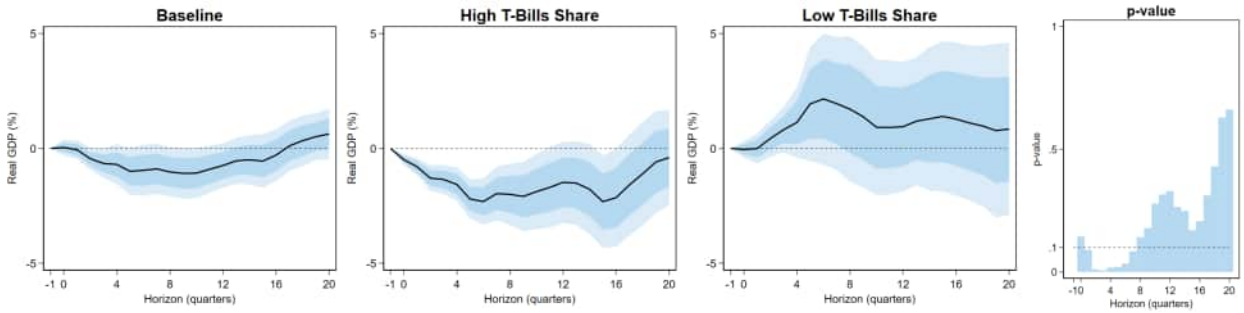
Panel C. Effective Interest Rate



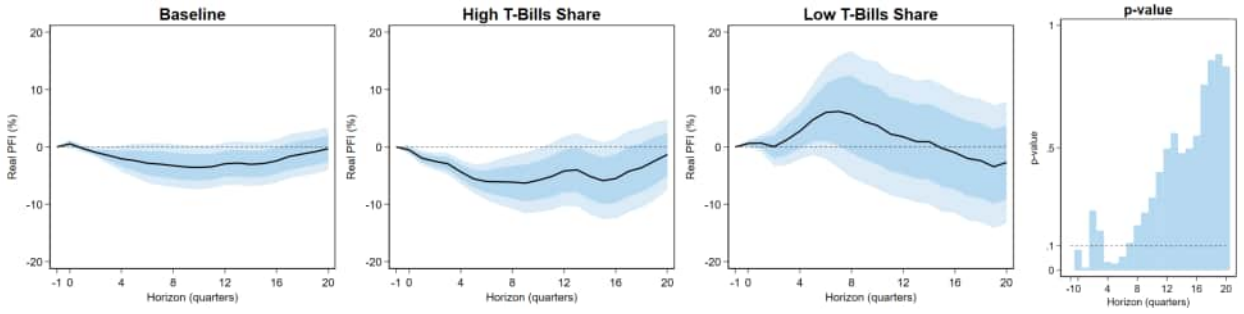
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial federal funds rate level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A63: ROBUSTNESS CHECK, CONTROLLING FOR INTEREST RATE INTERACTION. THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

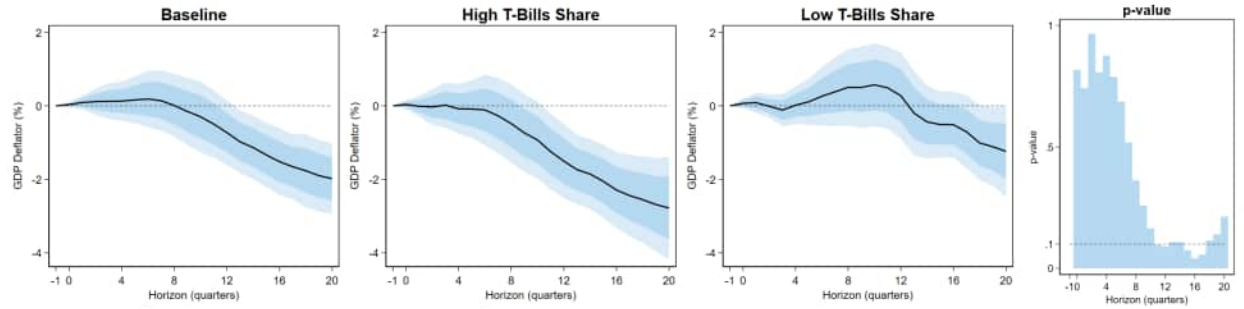
Panel A. Real GDP



Panel B. Real Private Fixed Investments



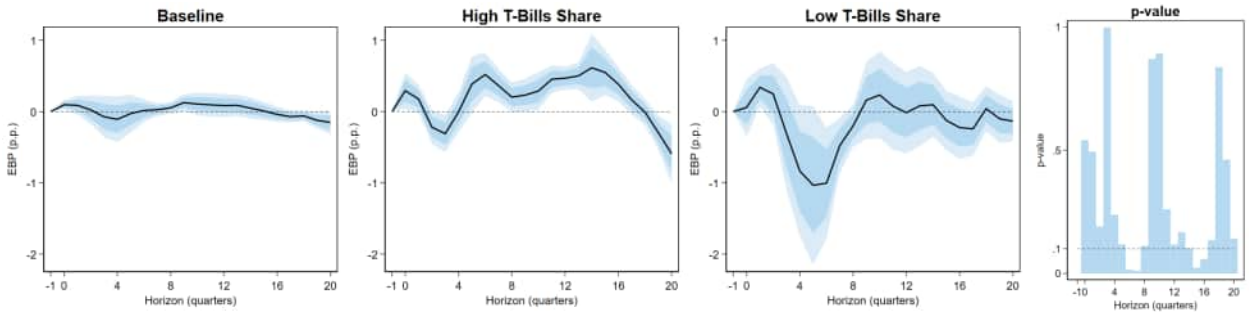
Panel C. GDP Deflator



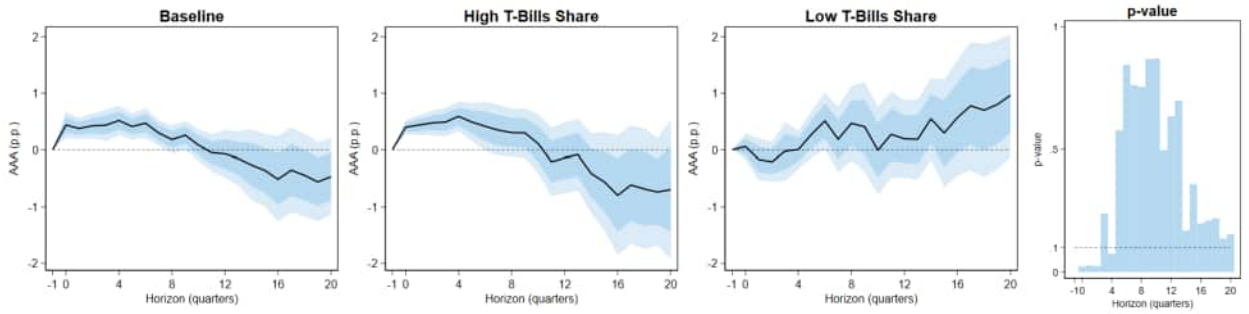
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t\left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)}\left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial federal funds rate level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A64: ROBUSTNESS CHECK, CONTROLLING FOR INTEREST RATE INTERACTION. THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

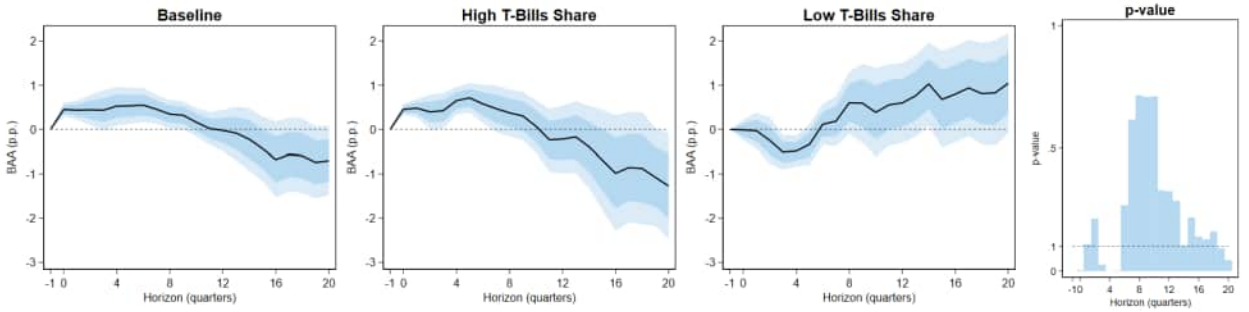
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



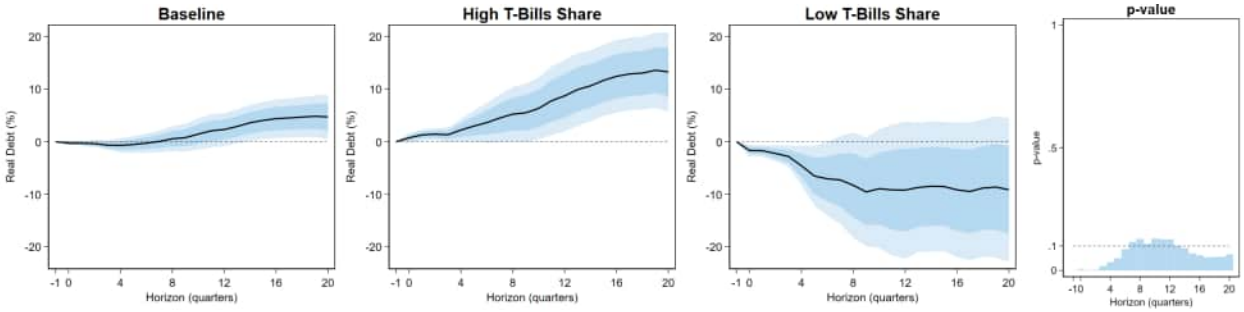
Panel C. Baa Corporate Bond Yield



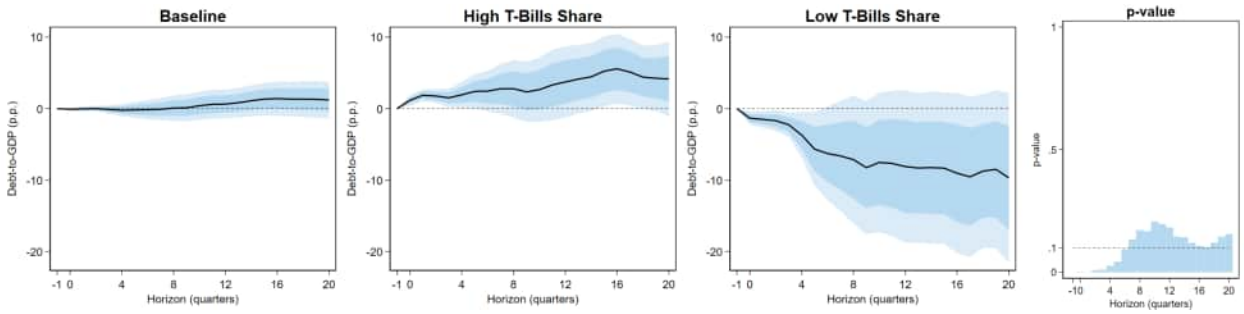
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)}\left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial federal funds rate level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A65: ROBUSTNESS CHECK, CONTROLLING FOR DEBT INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



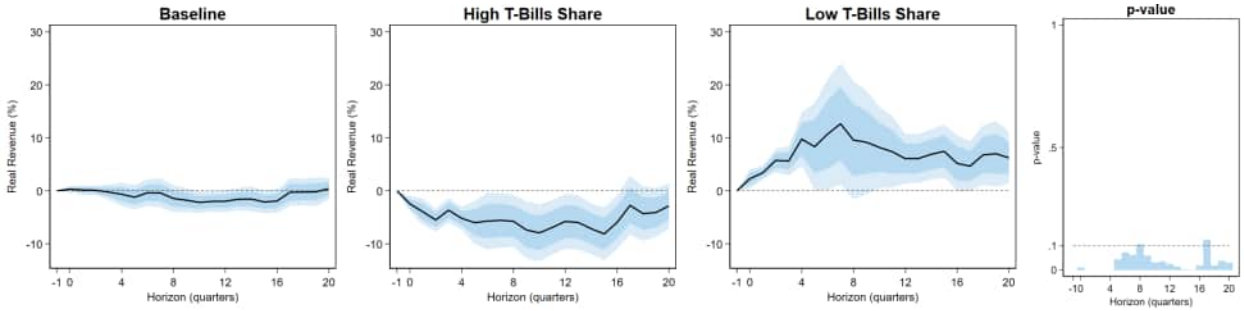
Panel B. Debt-to-GDP



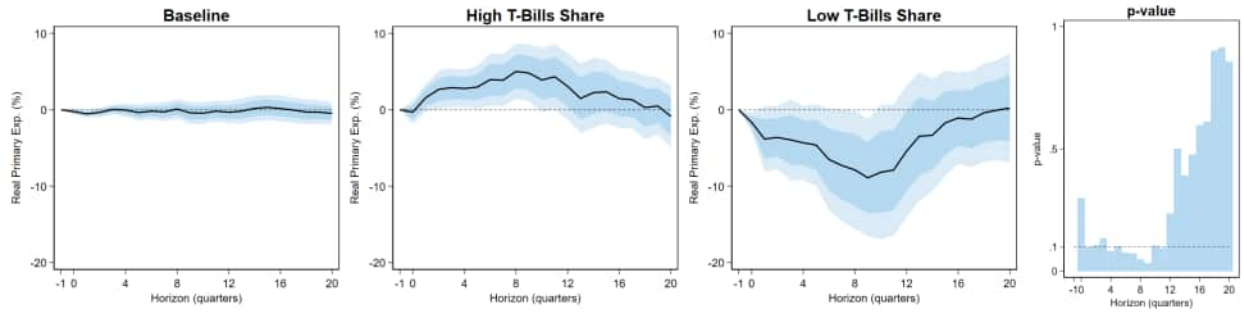
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial debt-to-GDP level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A66: ROBUSTNESS CHECK, CONTROLLING FOR DEBT INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

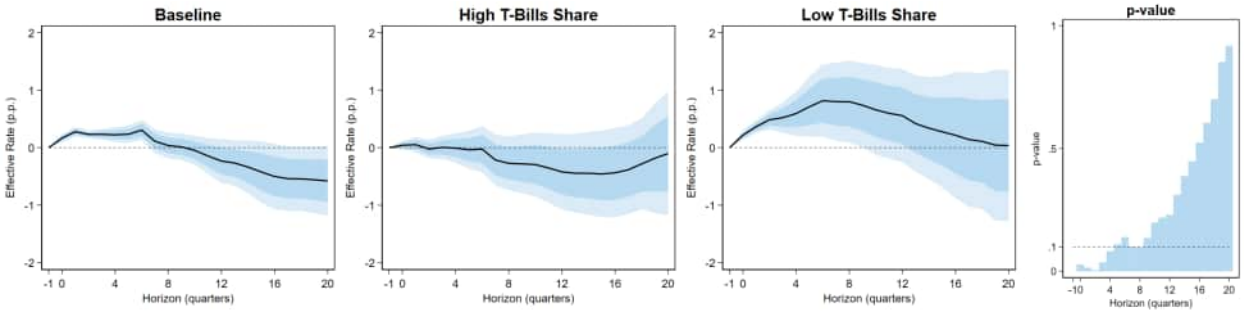
Panel A. Real Revenue



Panel B. Real Primary Expenditure



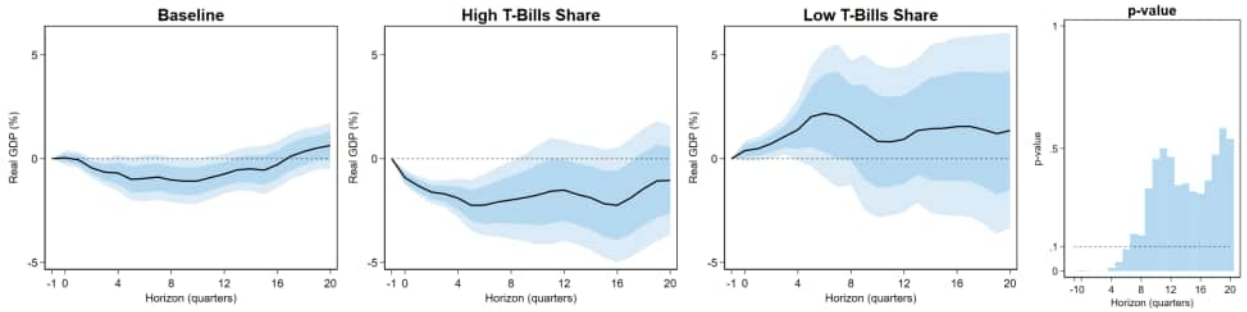
Panel C. Effective Interest Rate



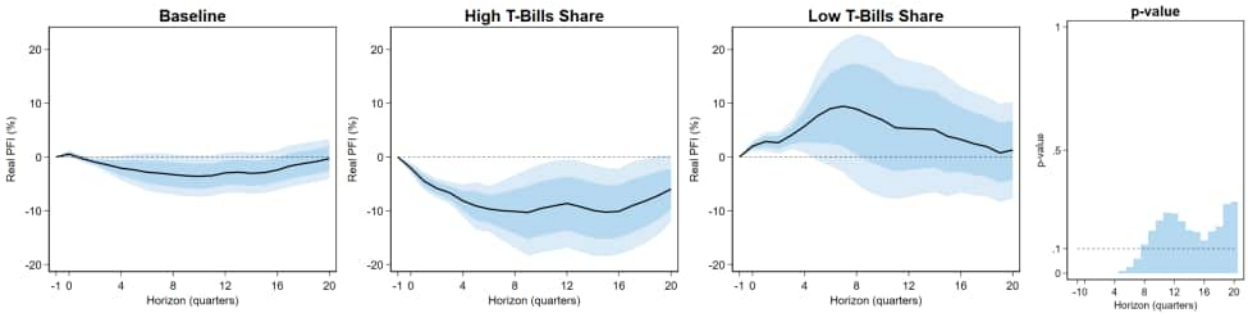
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t\left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)}\left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial debt-to-GDP level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A67: ROBUSTNESS CHECK, CONTROLLING FOR DEBT INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

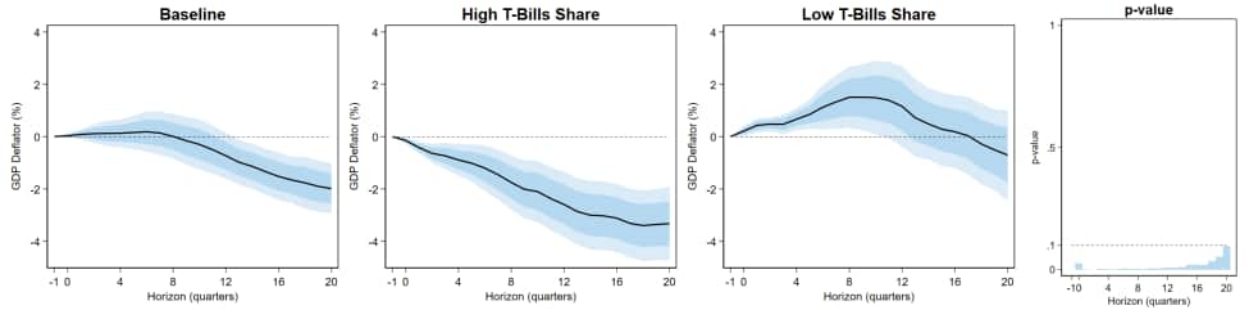
Panel A. Real GDP



Panel B. Real Private Fixed Investments



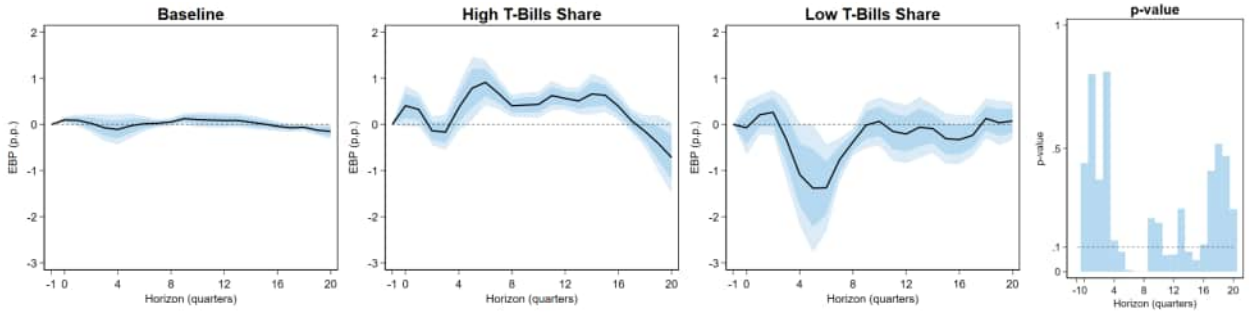
Panel C. GDP Deflator



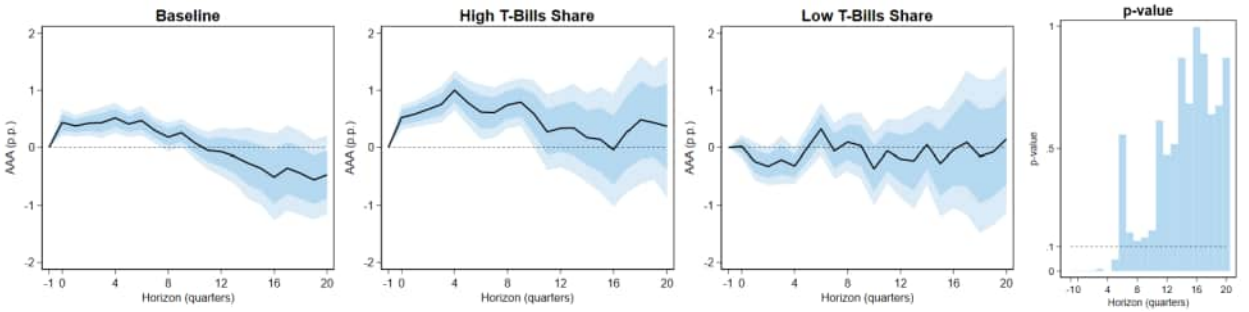
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t\left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)}\left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial debt-to-GDP level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A68: ROBUSTNESS CHECK, CONTROLLING FOR DEBT INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

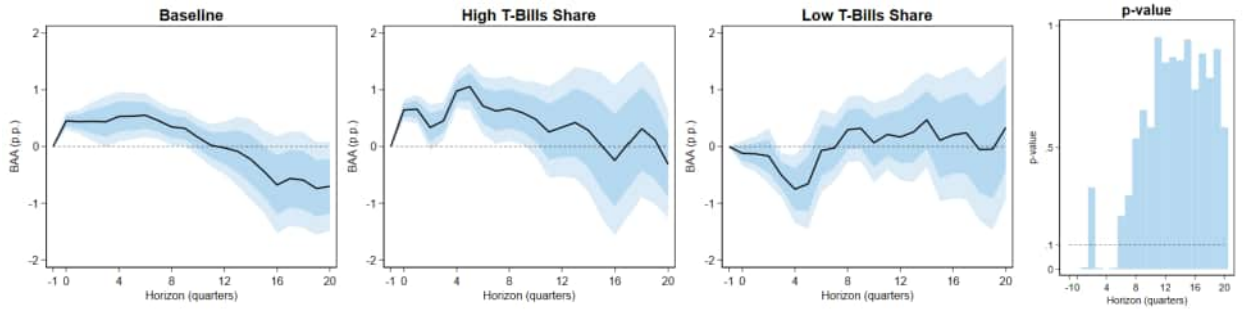
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



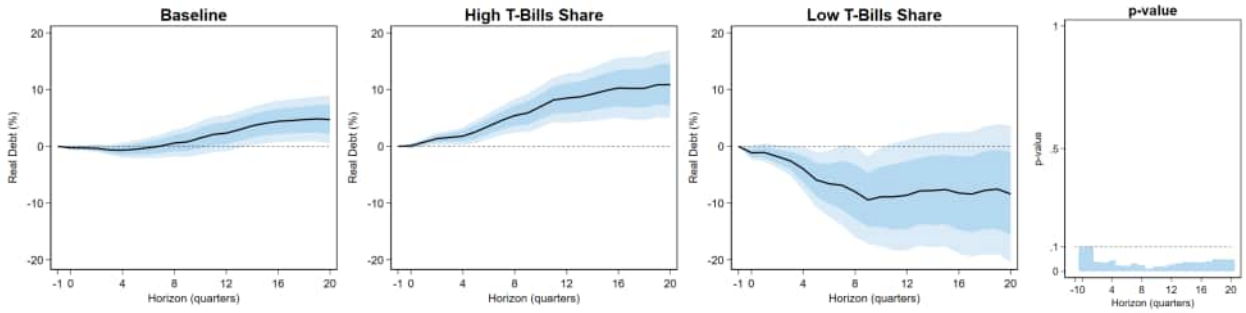
Panel C. Baa Corporate Bond Yield



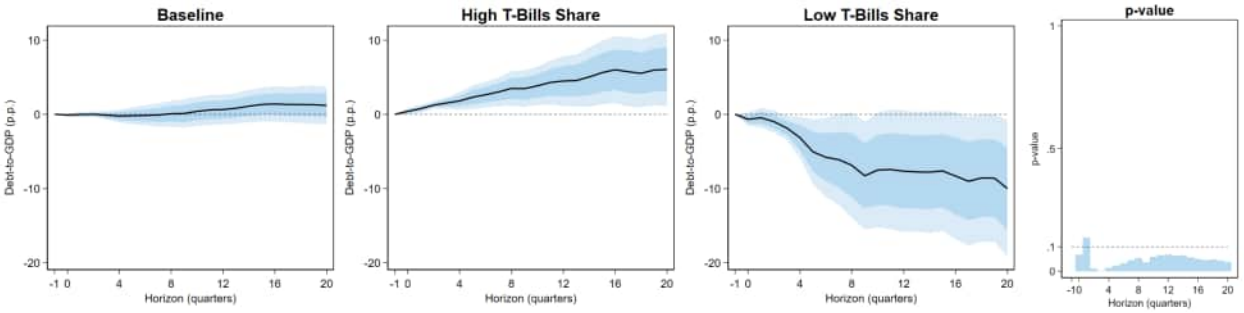
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)}\left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial debt-to-GDP level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A69: ROBUSTNESS CHECK, CONTROLLING FOR INFLATION INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



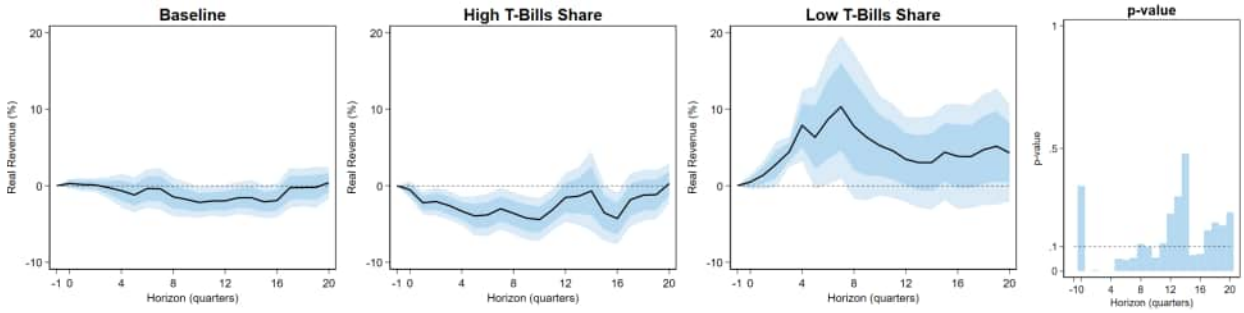
Panel B. Debt-to-GDP



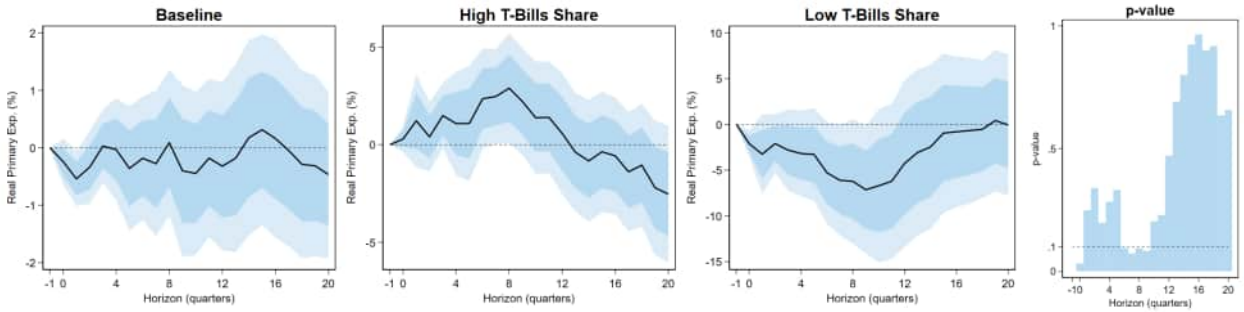
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial inflation level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A70: ROBUSTNESS CHECK, CONTROLLING FOR INFLATION INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

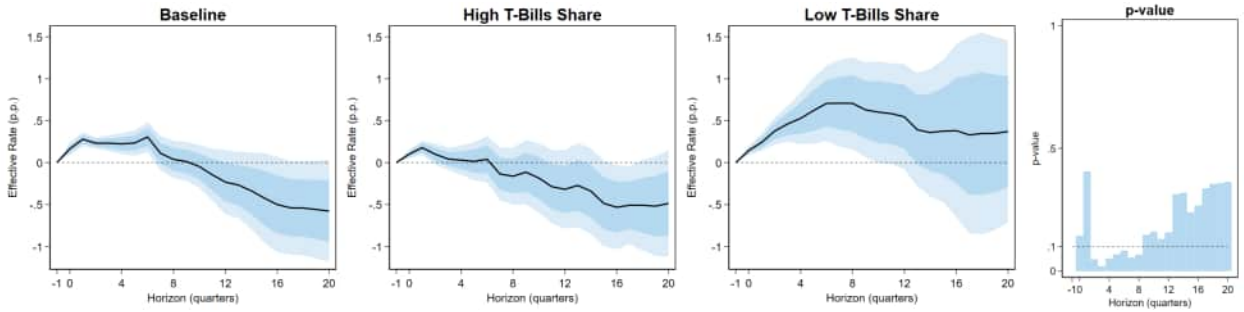
Panel A. Real Revenue



Panel B. Real Primary Expenditure



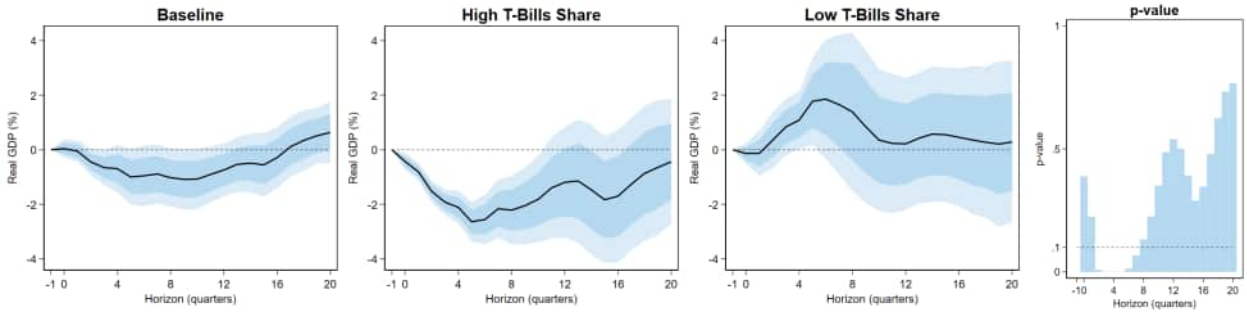
Panel C. Effective Interest Rate



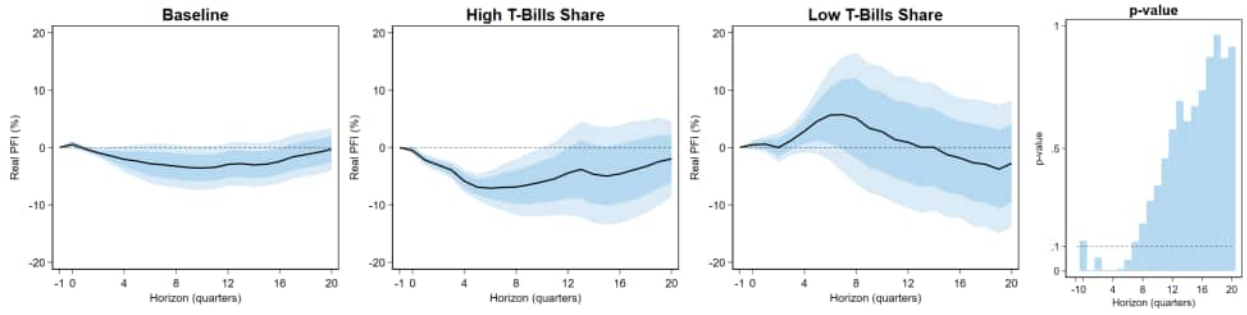
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)}\left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial inflation level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A71: ROBUSTNESS CHECK, CONTROLLING FOR INFLATION INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

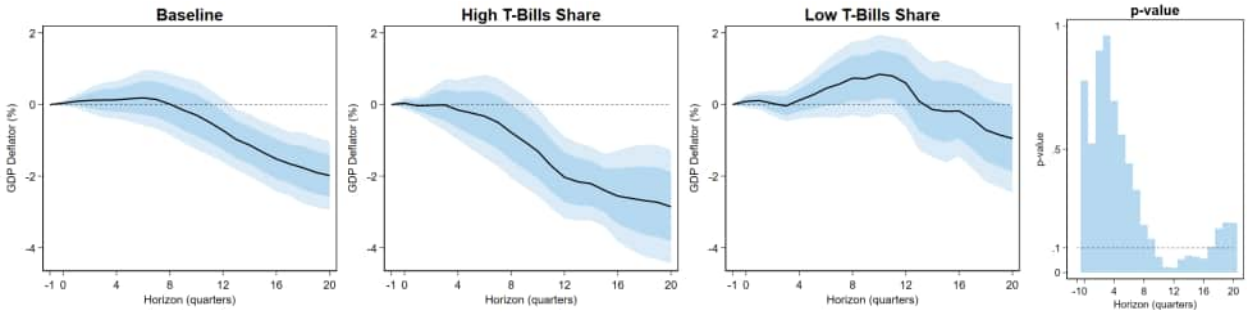
Panel A. Real GDP



Panel B. Real Private Fixed Investments



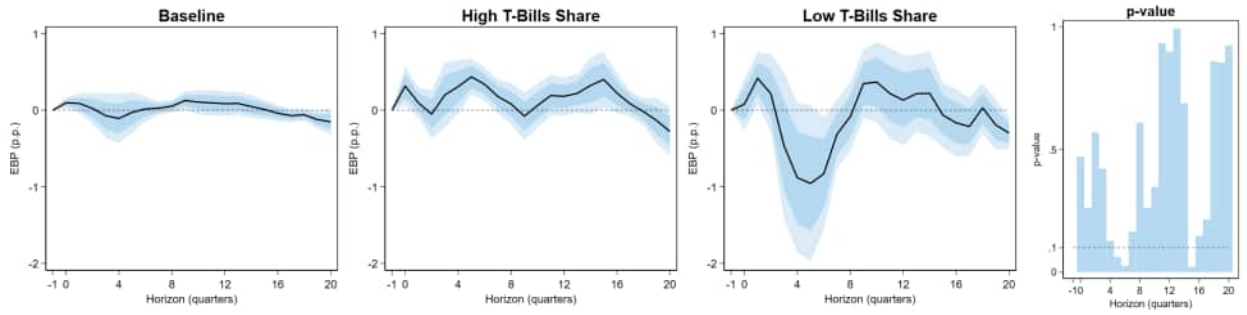
Panel C. GDP Deflator



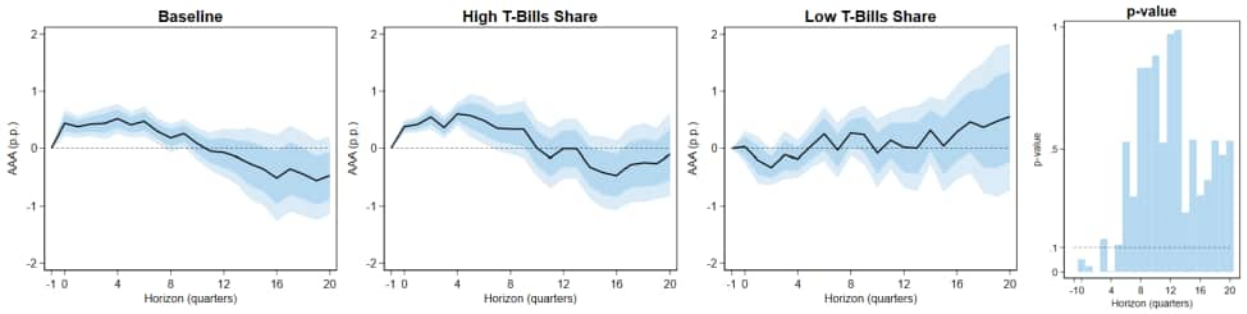
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)} \left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial inflation level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A72: ROBUSTNESS CHECK, CONTROLLING FOR INFLATION INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

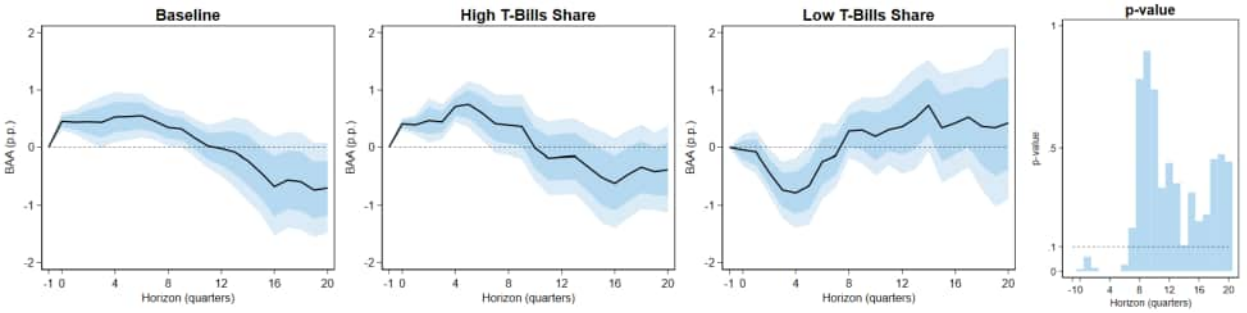
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



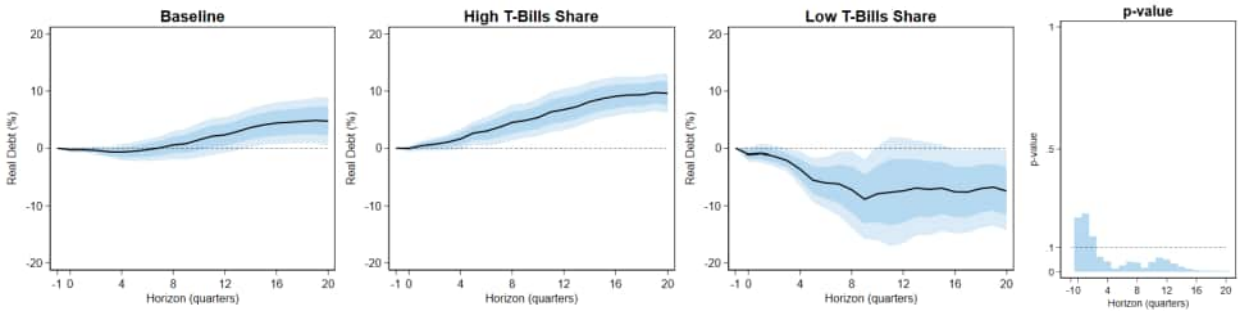
Panel C. Baa Corporate Bond Yield



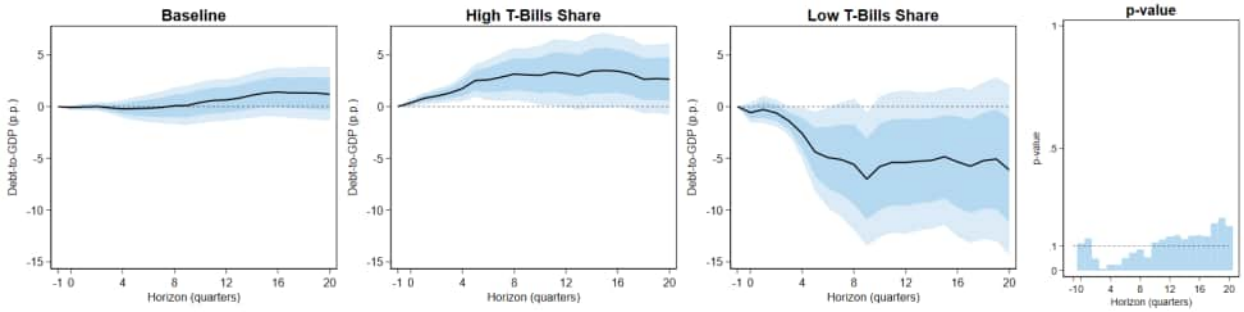
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial inflation level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A73: ROBUSTNESS CHECK, CONTROLLING FOR T-BILLS-TO-GDP INTERACTION. THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



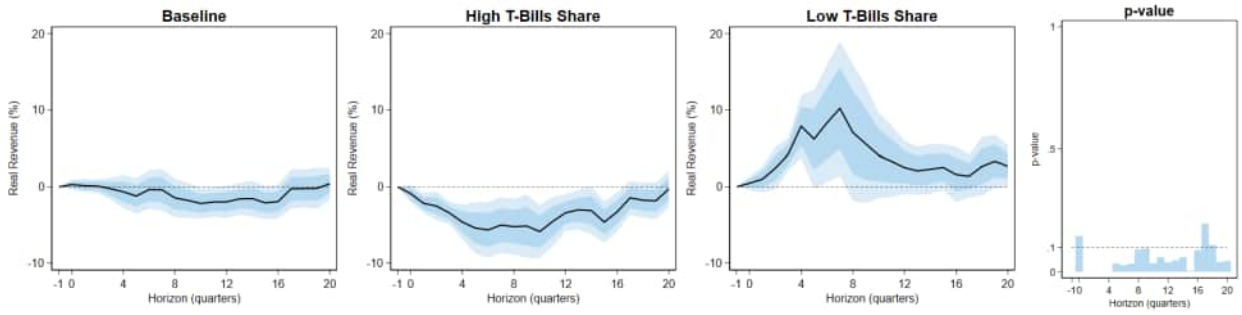
Panel B. Debt-to-GDP



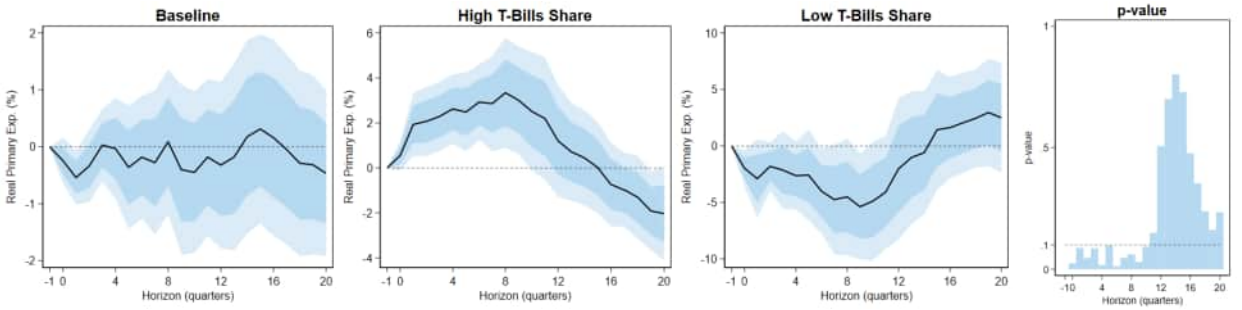
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial T-Bills-to-GDP level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A74: ROBUSTNESS CHECK, CONTROLLING FOR T-BILLS-TO-GDP INTERACTION. THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

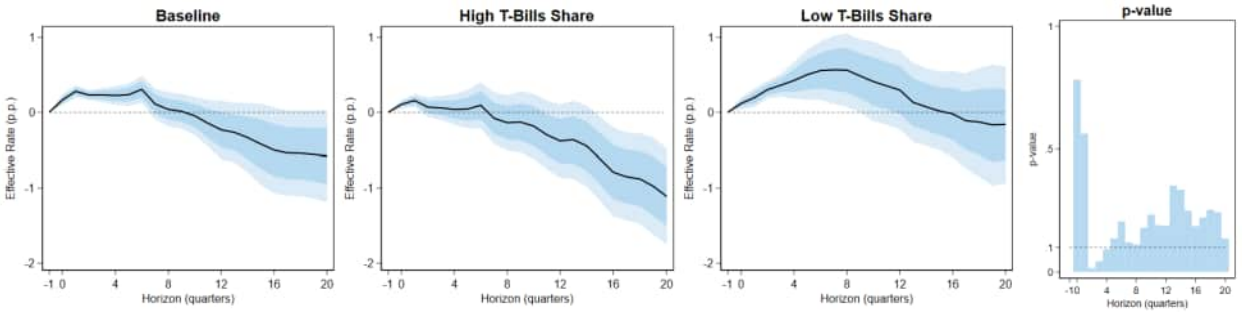
Panel A. Real Revenue



Panel B. Real Primary Expenditure



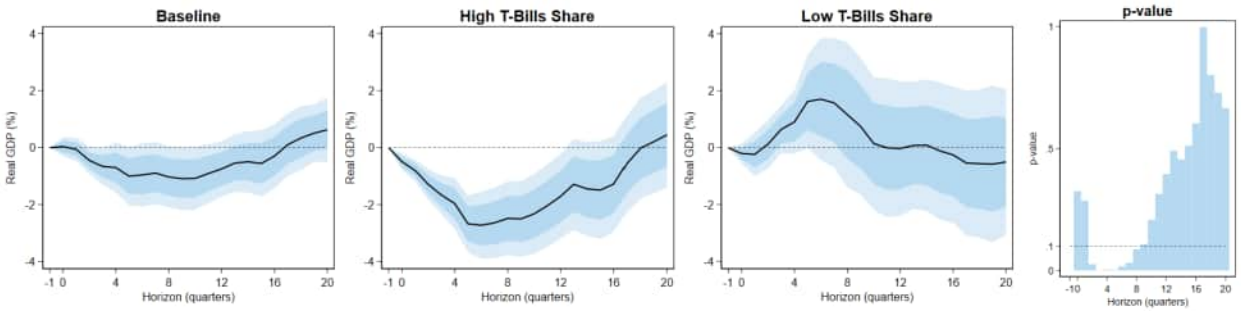
Panel C. Effective Interest Rate



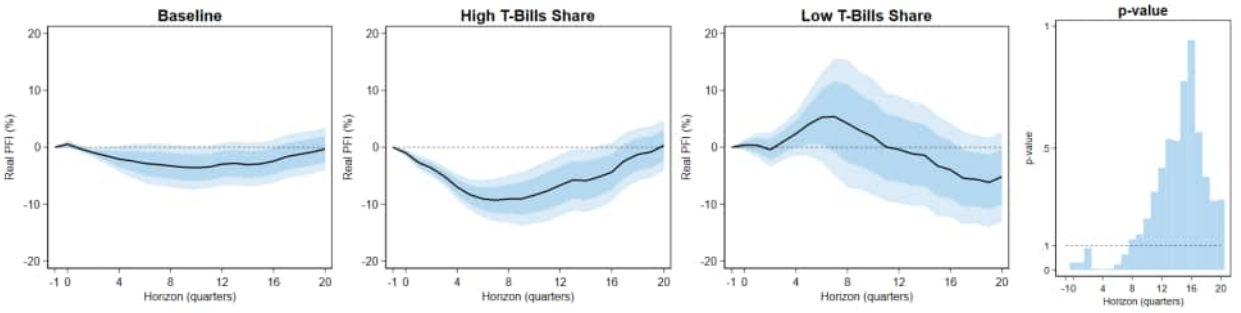
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)}\left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial T-Bills-to-GDP level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A75: ROBUSTNESS CHECK, CONTROLLING FOR T-BILLS-TO-GDP INTERACTION. THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

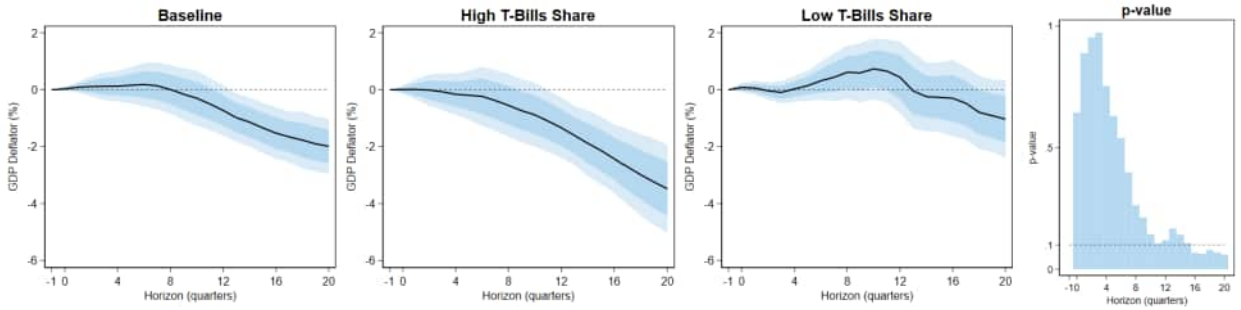
Panel A. Real GDP



Panel B. Real Private Fixed Investments



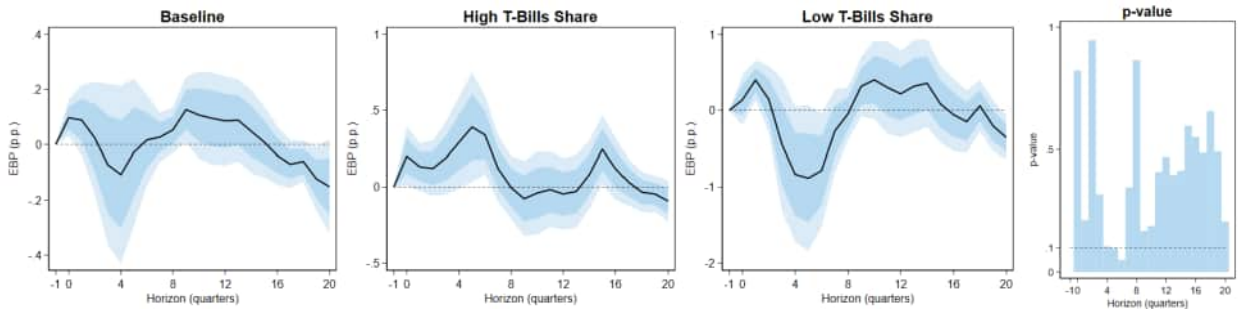
Panel C. GDP Deflator



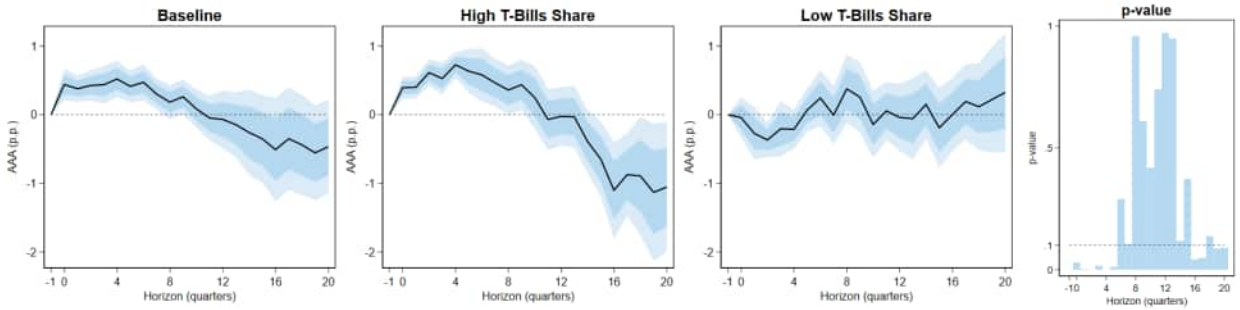
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)} \left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial T-Bills-to-GDP level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A76: ROBUSTNESS CHECK, CONTROLLING FOR T-BILLS-TO-GDP INTERACTION. THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

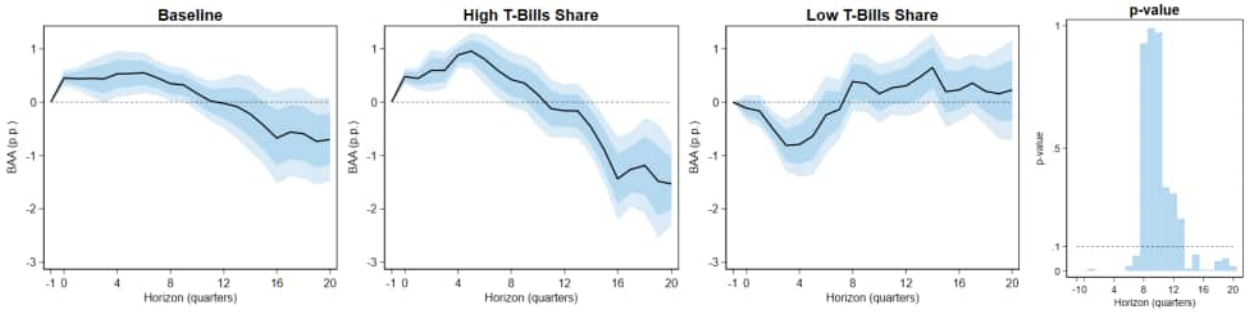
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



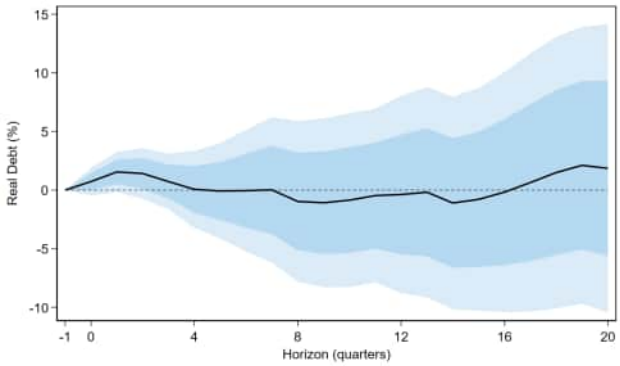
Panel C. Baa Corporate Bond Yield



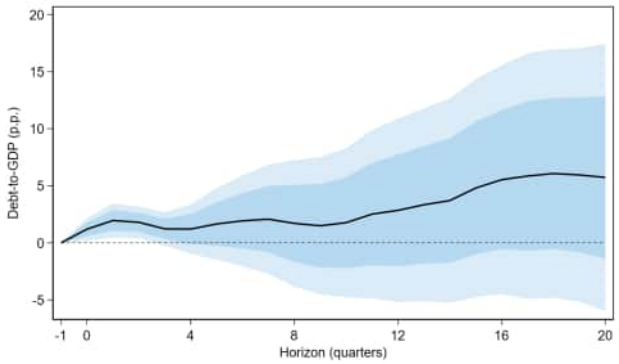
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial T-Bills-to-GDP level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

Figure A77: Fiscal-Monetary Interactions, The Role of Initial FFR Levels. The Effects of Monetary Policy Shocks on Debt Dynamics

Panel A. Real Debt



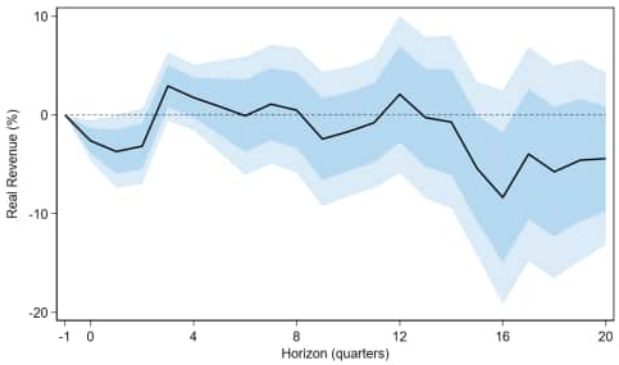
Panel B. Debt-to-GDP



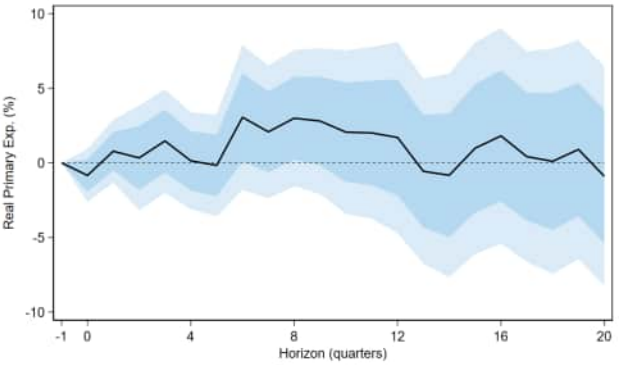
Note: The figure plots how the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock vary with the initial interest rate level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial interest rate level.

Figure A78: Fiscal-Monetary Interactions, The Role of Initial FFR Levels. The Effects of Monetary Policy Shocks on Fiscal Dynamics

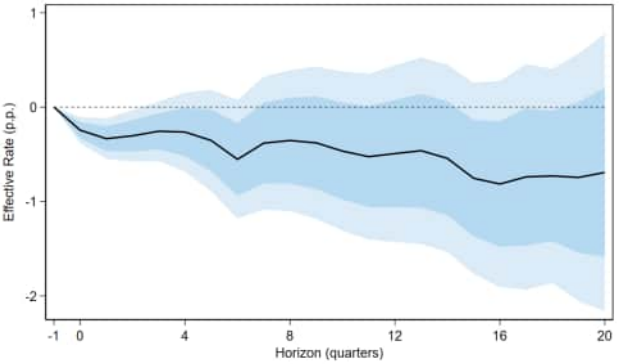
Panel A. Real Revenue



Panel B. Real Primary Expenditure



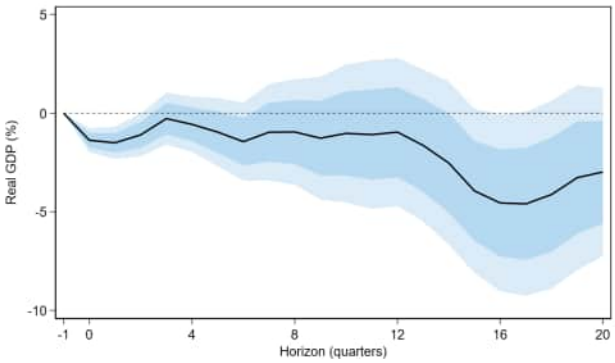
Panel C. Effective Interest Rate



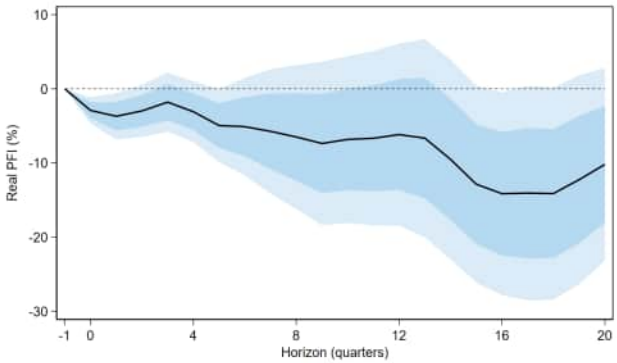
Note: The figure plots how the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock vary with the initial interest rate level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial interest rate level.

Figure A79: Fiscal-Monetary Interactions, The Role of Initial FFR Levels. The Effects of Monetary Policy Shocks on Economic Activity

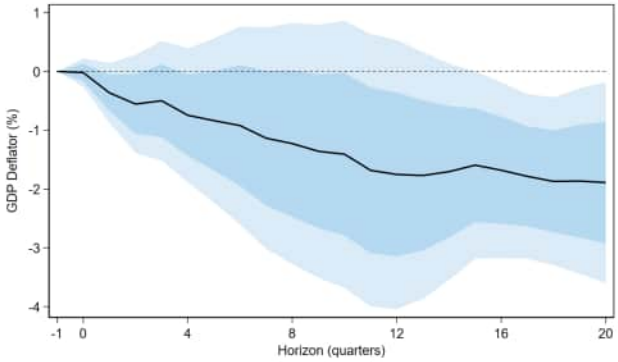
Panel A. Real GDP



Panel B. Real Private Fixed Investments



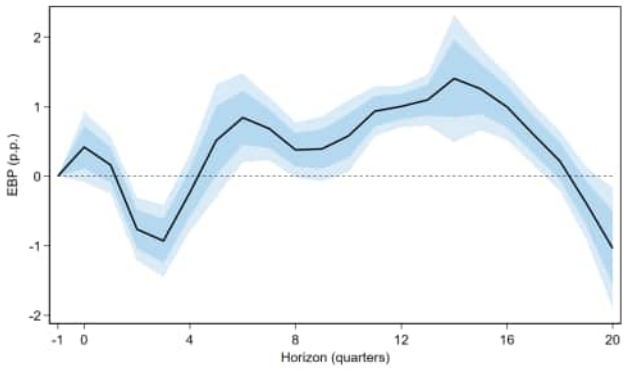
Panel C. GDP Deflator



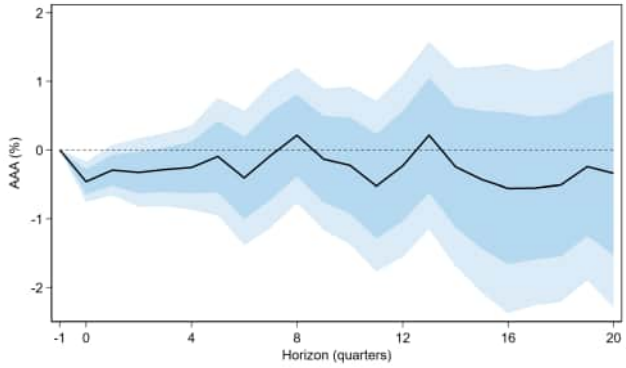
Note: The figure plots how the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and the GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock vary with the initial interest rate level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial interest rate level.

Figure A80: Fiscal-Monetary Interactions, The Role of Initial FFR Levels. The Effects of Monetary Policy Shocks on Financial Conditions

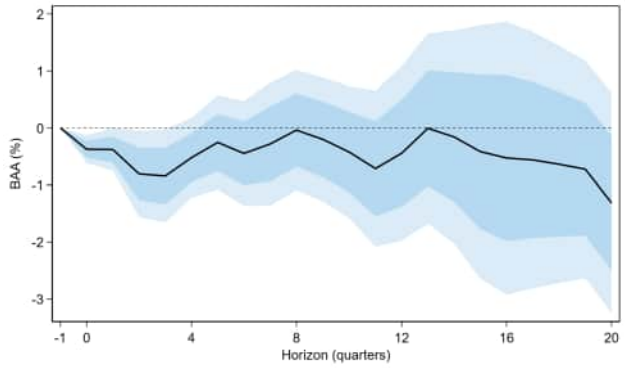
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



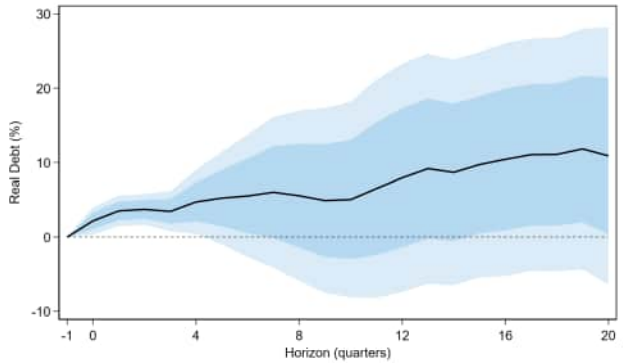
Panel C. Baa Corporate Bond Yield



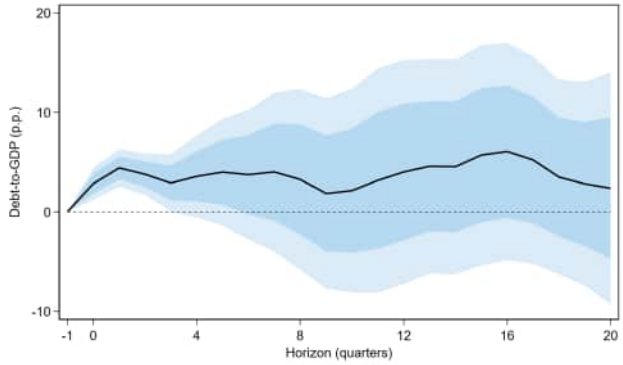
Note: The figure plots how the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock vary with the initial interest rate level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial interest rate level.

FIGURE A81: FISCAL-MONETARY INTERACTIONS, THE ROLE OF INITIAL DEBT LEVELS. THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



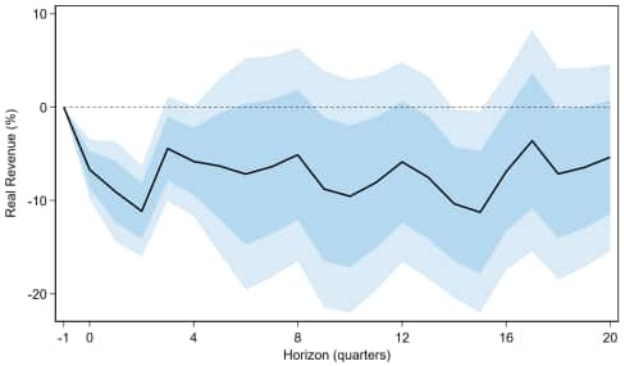
Panel B. Debt-to-GDP



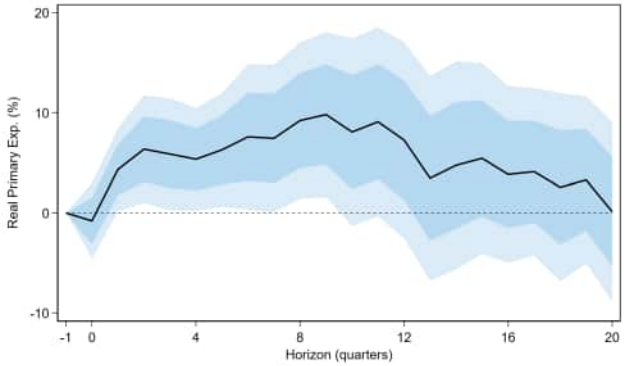
Note: The figure plots how the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock vary with the initial debt level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)} \big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial debt-to-GDP level.

Figure A82: Fiscal-Monetary Interactions, The Role of Initial Debt Levels. The Effects of Monetary Policy Shocks on Fiscal Dynamics

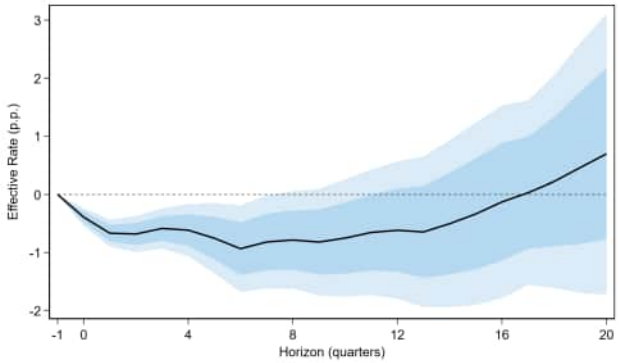
Panel A. Real Revenue



Panel B. Real Primary Expenditure



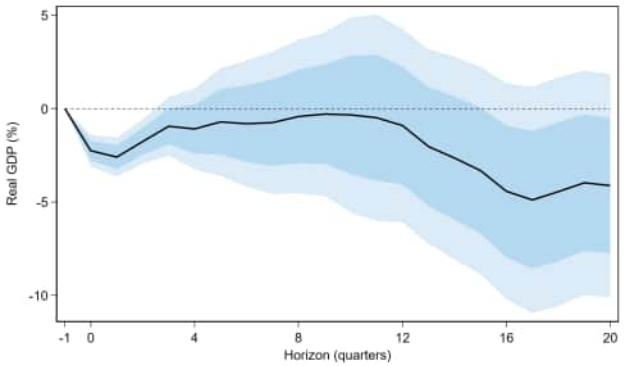
Panel C. Effective Interest Rate



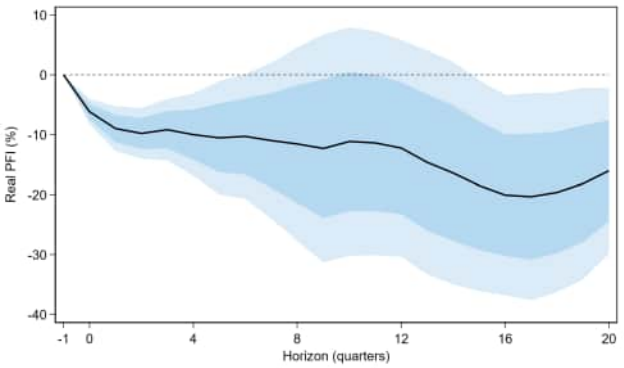
Note: The figure plots how the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the Effective Interest Rate (Panel C) to a 100 bps monetary policy tightening shock vary with the initial debt level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial debt-to-GDP level.

FIGURE A83: FISCAL-MONETARY INTERACTIONS, THE ROLE OF INITIAL DEBT LEVELS. THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

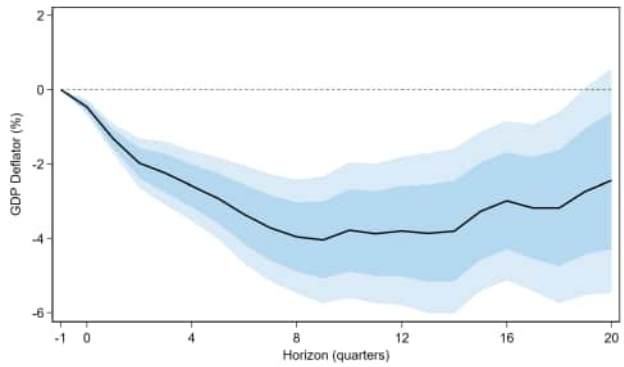
Panel A. Real GDP



Panel B. Real Private Fixed Investments



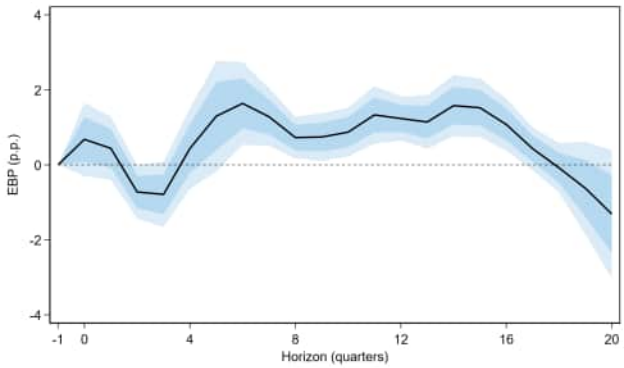
Panel C. GDP Deflator



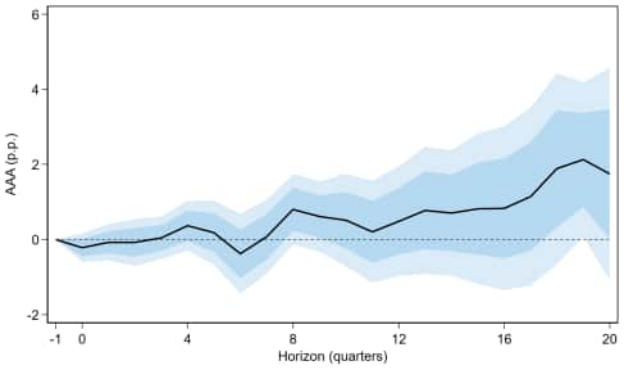
Note: The figure plots how the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and the GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock vary with the initial debt level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial debt-to-GDP level.

FIGURE A84: FISCAL-MONETARY INTERACTIONS, THE ROLE OF INITIAL DEBT LEVELS. THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

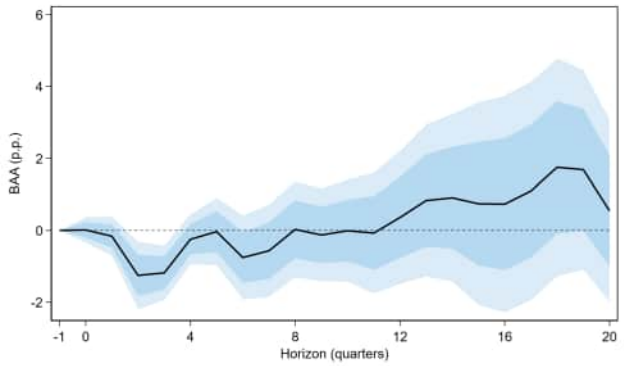
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



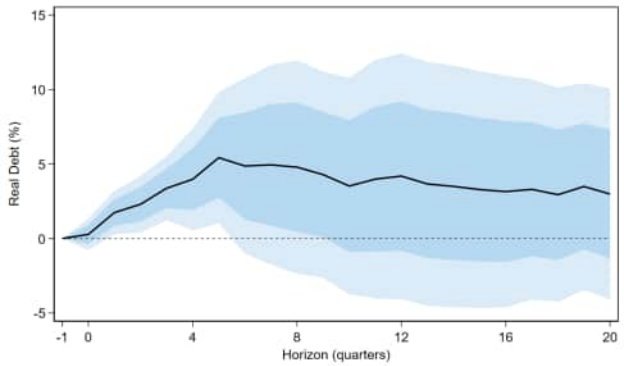
Panel C. Baa Corporate Bond Yield



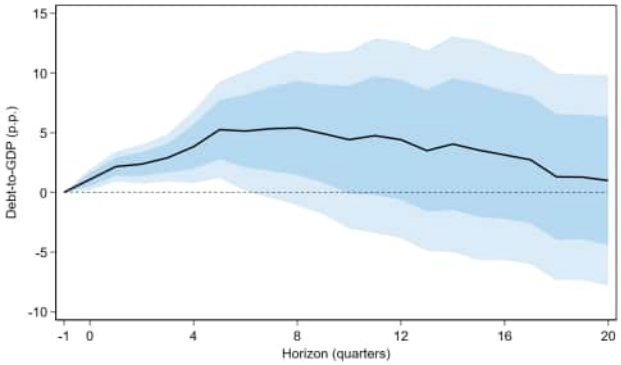
Note: The figure plots how the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock vary with the initial debt level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1-F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial debt-to-GDP level.

Figure A85: Fiscal-Monetary Interactions, The Role of Initial T-Bills Levels. The Effects of Monetary Policy Shocks on Debt Dynamics

Panel A. Real Debt



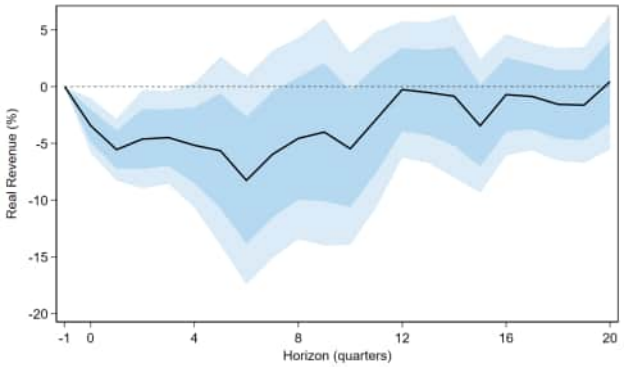
Panel B. Debt-to-GDP



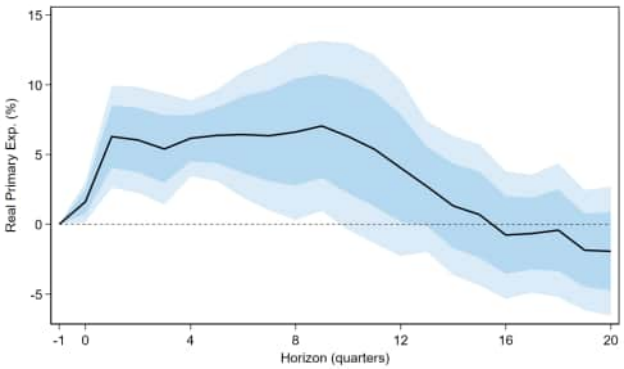
Note: The figure plots how the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock vary with the initial debt level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t\big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial T-Bills-to-GDP level.

Figure A86: Fiscal-Monetary Interactions, The Role of Initial T-Bills Levels. The Effects of Monetary Policy Shocks on Fiscal Dynamics

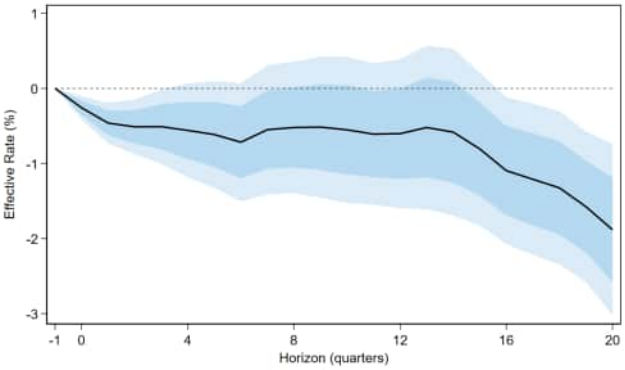
Panel A. Real Revenue



Panel B. Real Primary Expenditure



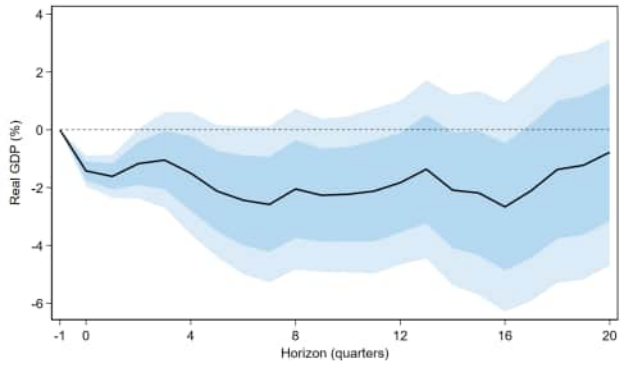
Panel C. Effective Interest Rate



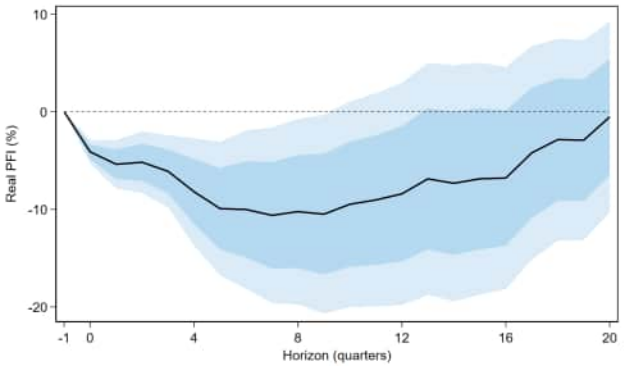
Note: The figure plots how the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the Effective Interest Rate (Panel C) to a 100 bps monetary policy tightening shock vary with the initial debt level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t\big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial T-Bills-to-GDP level.

FIGURE A87: FISCAL-MONETARY INTERACTIONS, THE ROLE OF INITIAL T-BILLS LEVELS. THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

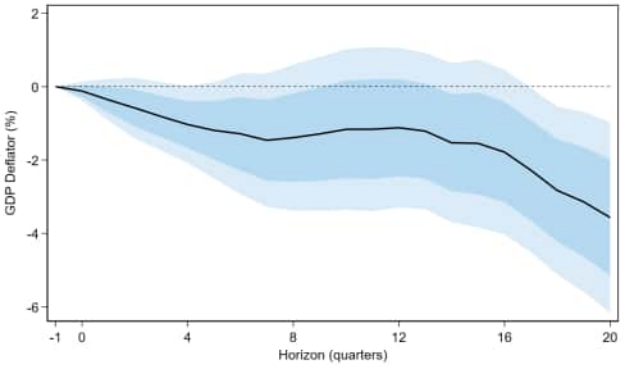
Panel A. Real GDP



Panel B. Real Private Fixed Investments



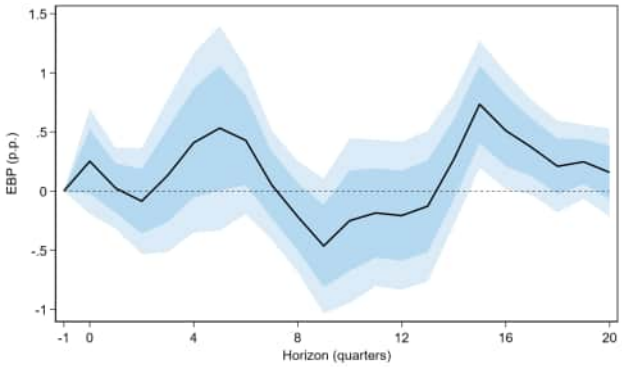
Panel C. GDP Deflator



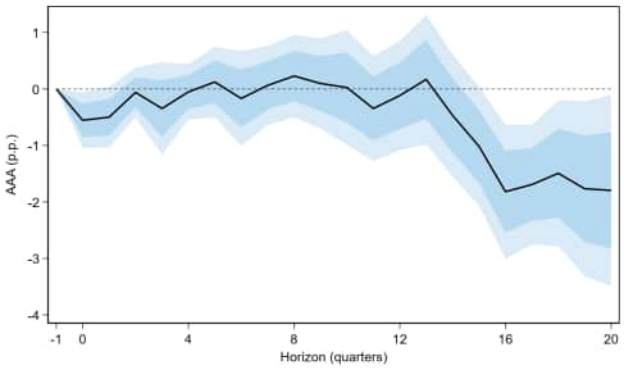
Note: The figure plots how the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and the GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock vary with the initial debt level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial T-Bills-to-GDP level.

FIGURE A88: FISCAL-MONETARY INTERACTIONS, THE ROLE OF INITIAL T-BILLS LEVELS. THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

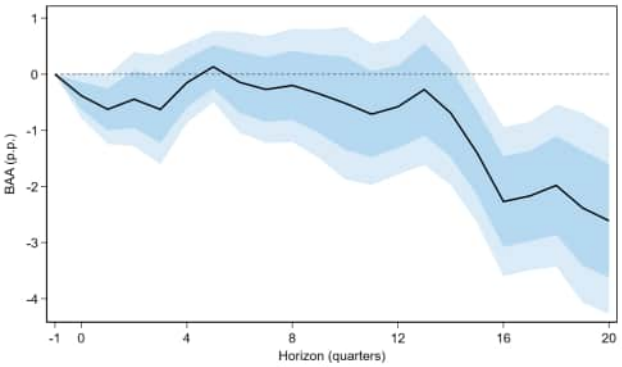
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



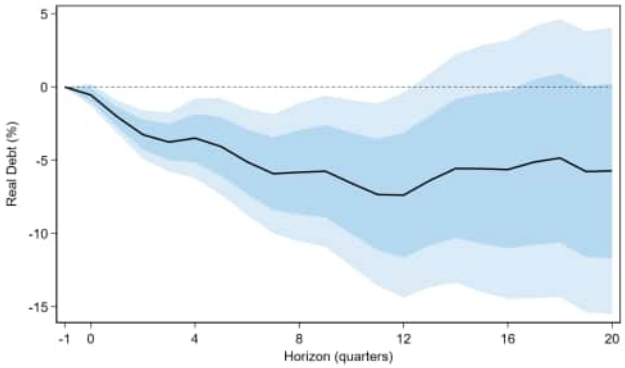
Panel C. Baa Corporate Bond Yield



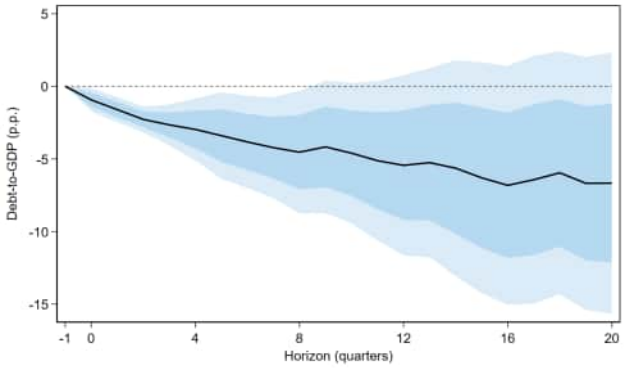
Note: The figure plots how the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock vary with the initial debt level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1-F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial T-Bills-to-GDP level.

FIGURE A89: FISCAL-MONETARY INTERACTIONS, THE ROLE OF INITIAL INFLATION LEVELS. THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



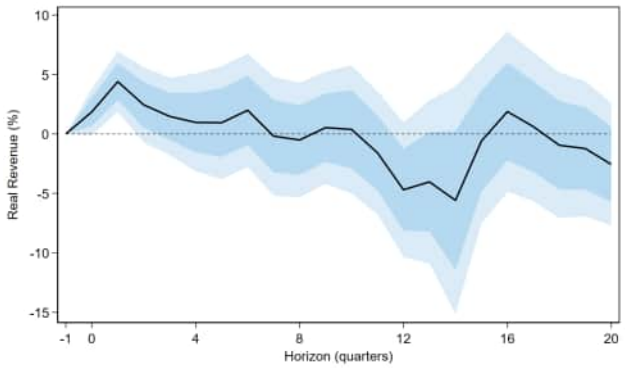
Panel B. Debt-to-GDP



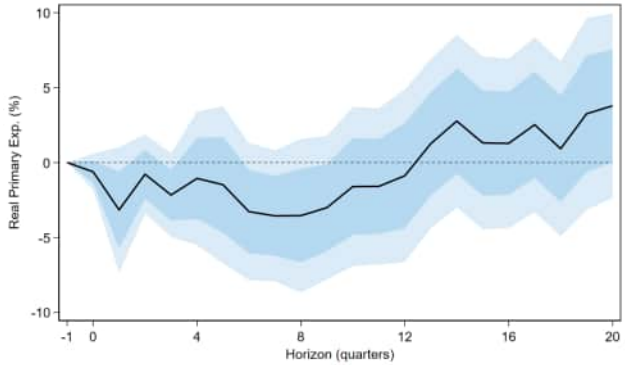
Note: The figure plots how the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock vary with the initial debt level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)} \big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial debt-to-GDP level.

Figure A90: Fiscal-Monetary Interactions, The Role of Initial Inflation Levels. The Effects of Monetary Policy Shocks on Fiscal Dynamics

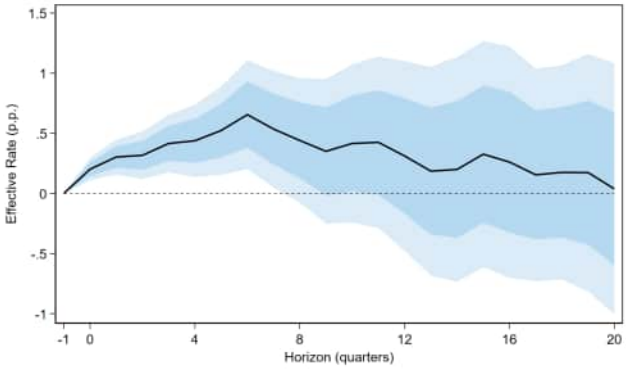
Panel A. Real Revenue



Panel B. Real Primary Expenditure



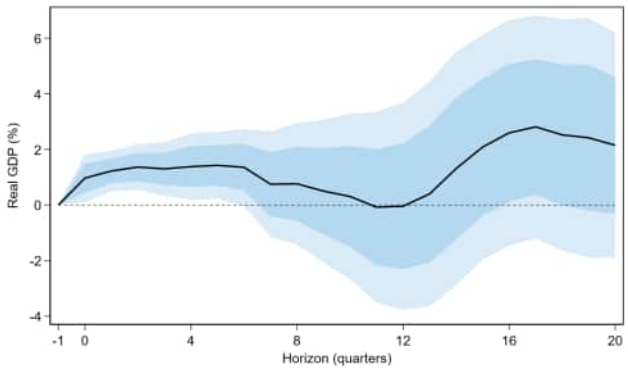
Panel C. Effective Interest Rate



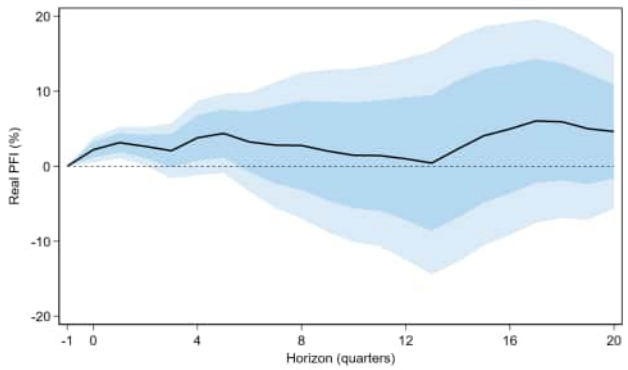
Note: The figure plots how the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the Effective Interest Rate (Panel C) to a 100 bps monetary policy tightening shock vary with the initial debt level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial inflation level.

Figure A91: Fiscal-Monetary Interactions, The Role of Initial Inflation Levels. The Effects of Monetary Policy Shocks on Economic Activity

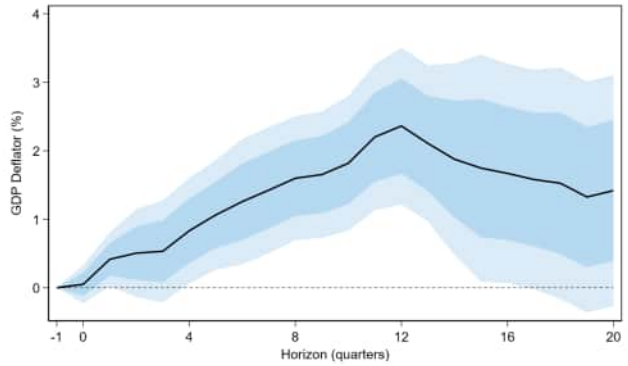
Panel A. Real GDP



Panel B. Real Private Fixed Investments



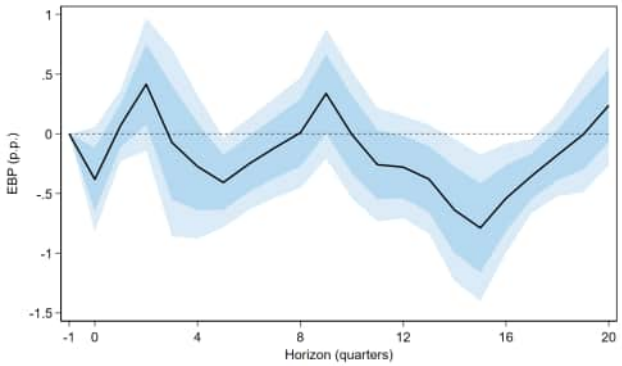
Panel C. GDP Deflator



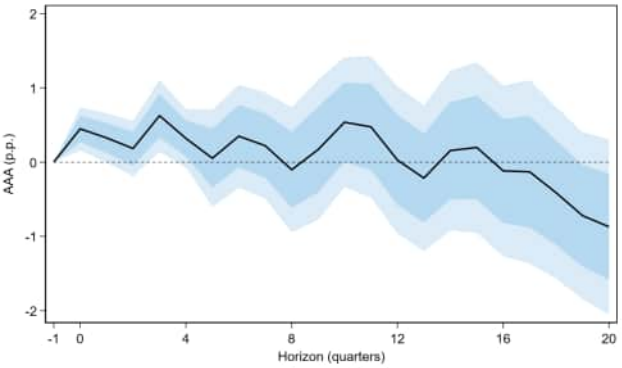
Note: The figure plots how the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and the GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock vary with the initial debt level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial inflation level.

FIGURE A92: FISCAL-MONETARY INTERACTIONS, THE ROLE OF INITIAL INFLATION LEVELS. THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

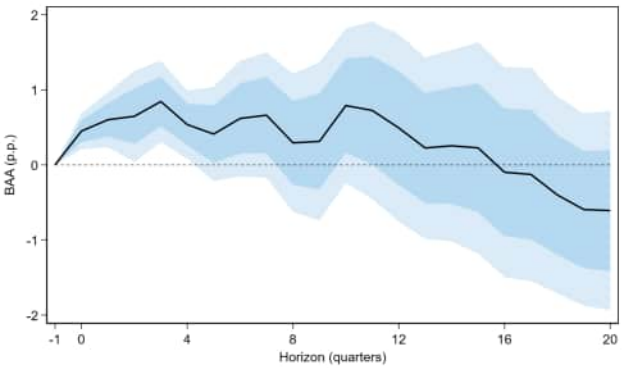
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



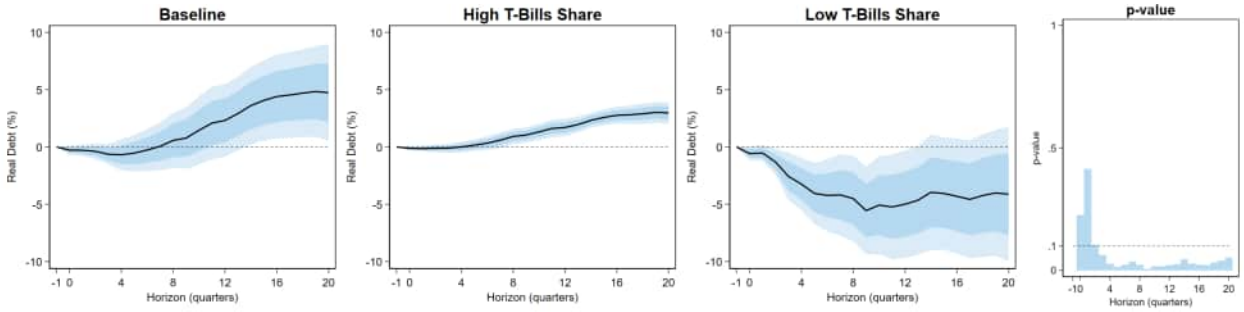
Panel C. Baa Corporate Bond Yield



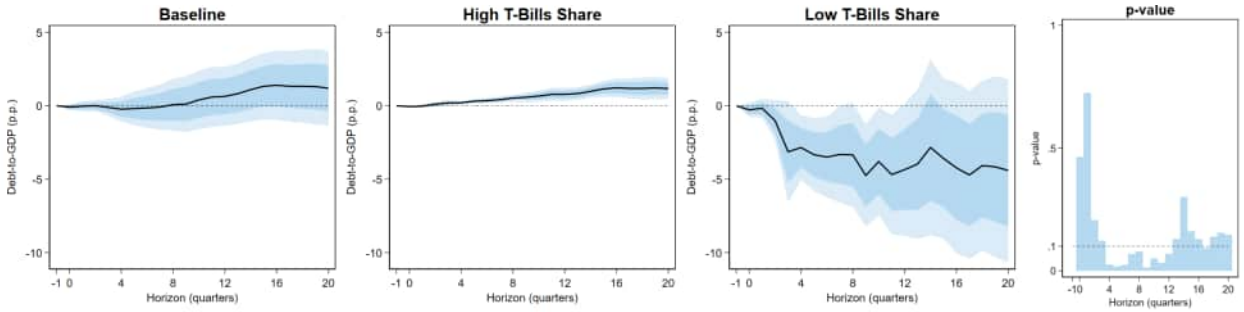
Note: The figure plots how the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock vary with the initial debt level. The solid black line shows the response in a state-dependent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta^{(h)}_{\text{high}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{high}} W_{t-\ell}\Big) + \big(1-F(z_{t-1})\big)\Big(\beta^{(h)}_{\text{low}} MPS_t + \sum_{\ell=1}^{4} \theta^{(h)}_{\ell,\text{low}} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon^{(h)}_{t+h}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock, and $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial inflation level.

FIGURE A93: ROBUSTNESS CHECK, CONTROLLING FOR RECESSION INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



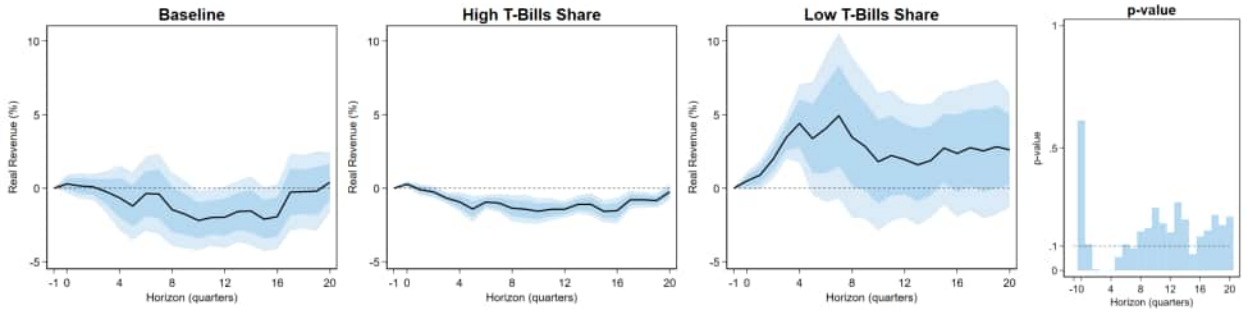
Panel B. Debt-to-GDP



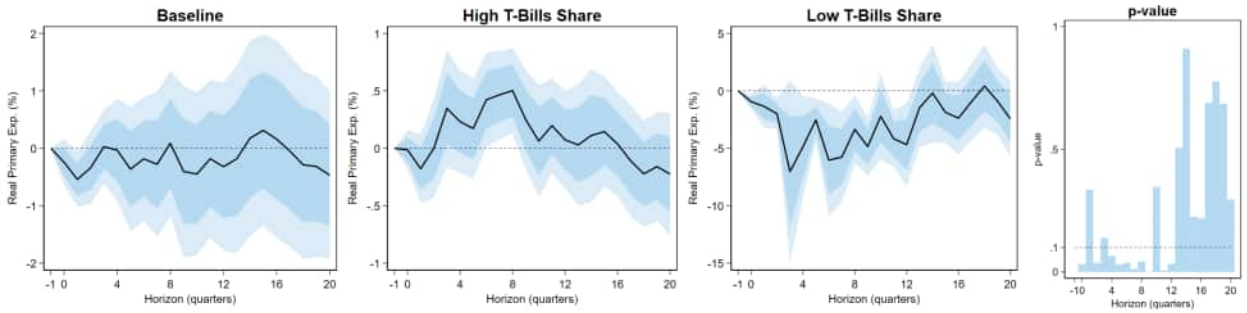
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t(D_{t-1} - \bar{D}) + \iota^{(h)}(D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the recession dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A94: ROBUSTNESS CHECK, CONTROLLING FOR RECESSION INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

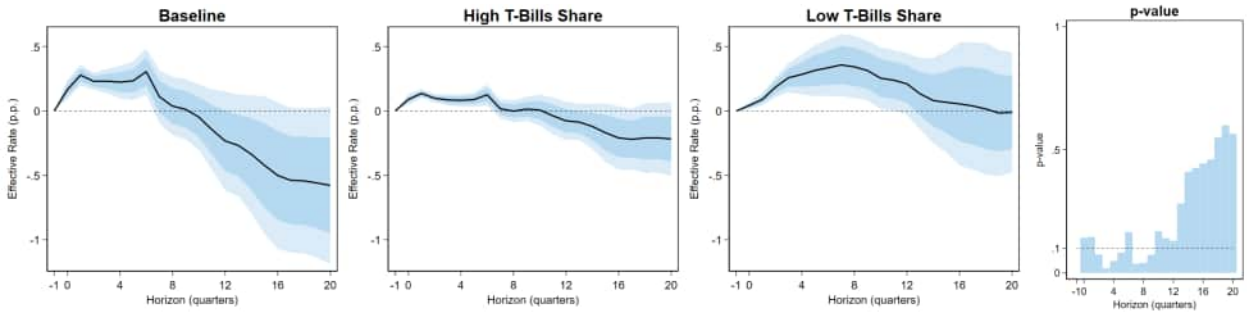
Panel A. Real Revenue



Panel B. Real Primary Expenditure



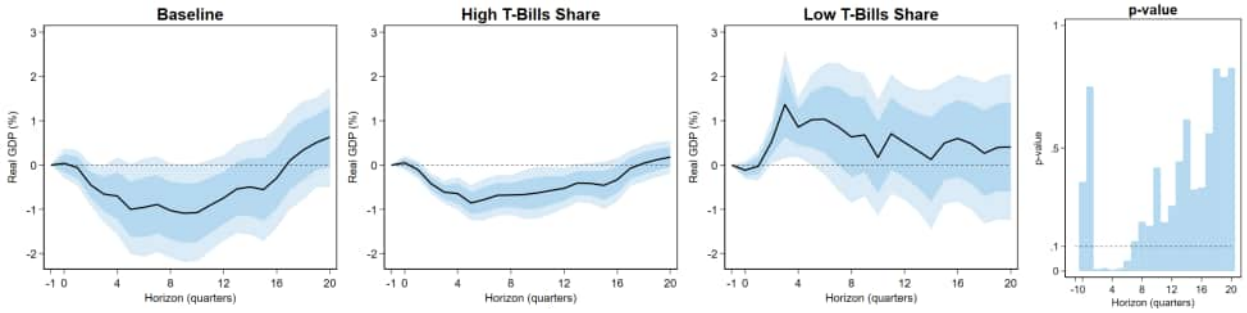
Panel C. Effective Interest Rate



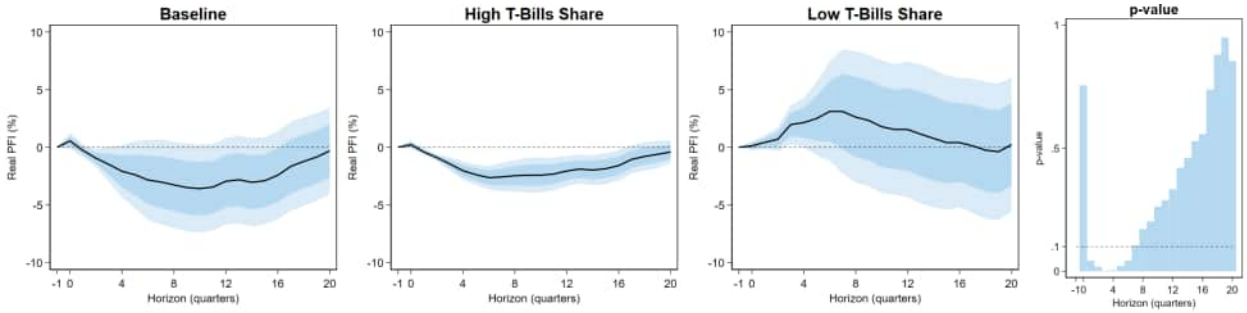
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the recession dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A95: ROBUSTNESS CHECK, CONTROLLING FOR RECESSION INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

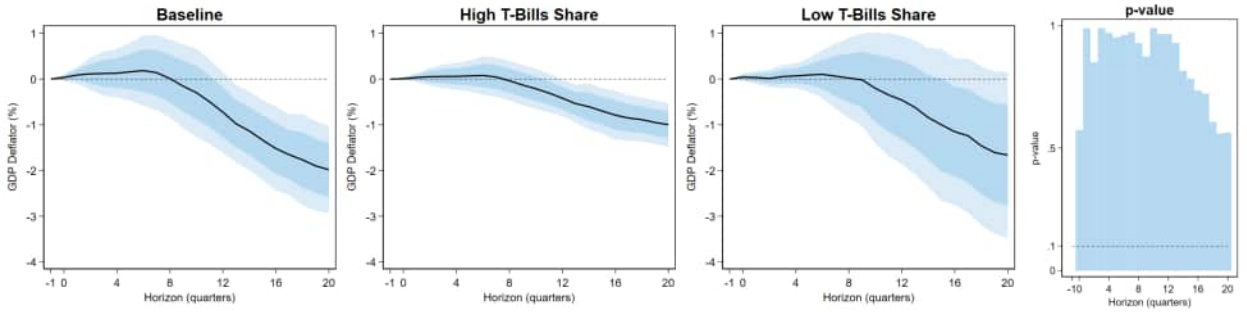
Panel A. Real GDP



Panel B. Real Private Fixed Investments



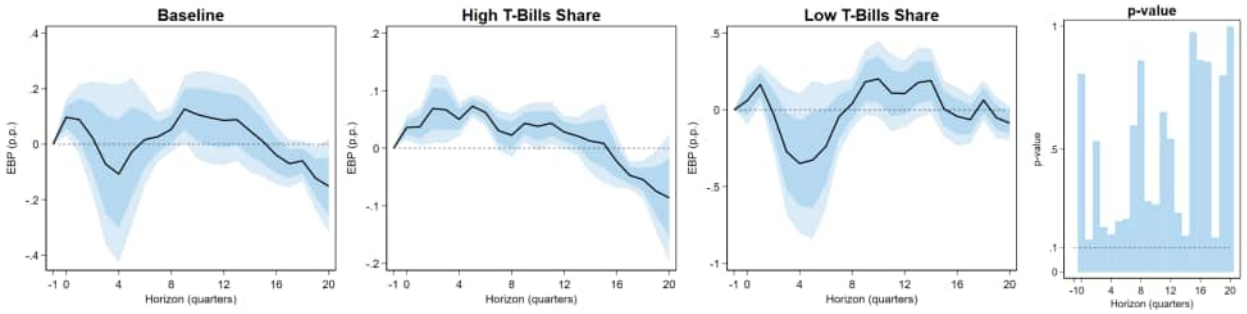
Panel C. GDP Deflator



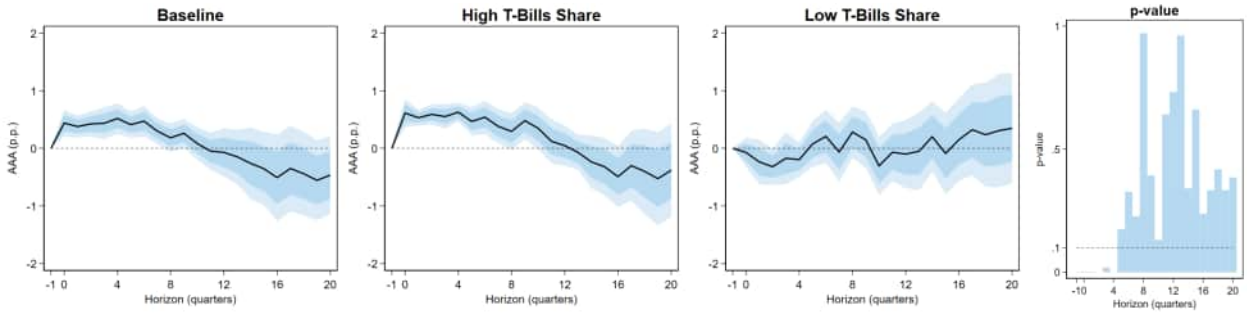
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the recession dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

Figure A96: Robustness Check, Controlling for Recessions Interaction.
The Effects of Monetary Policy Shocks on Financial Conditions

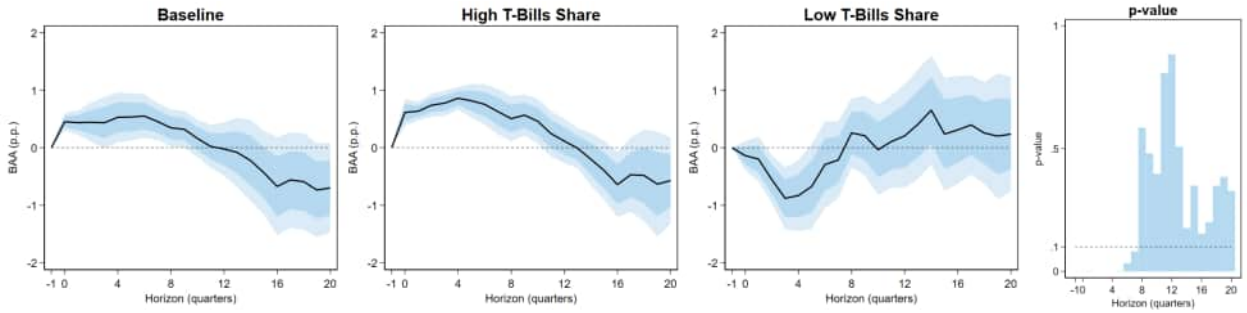
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



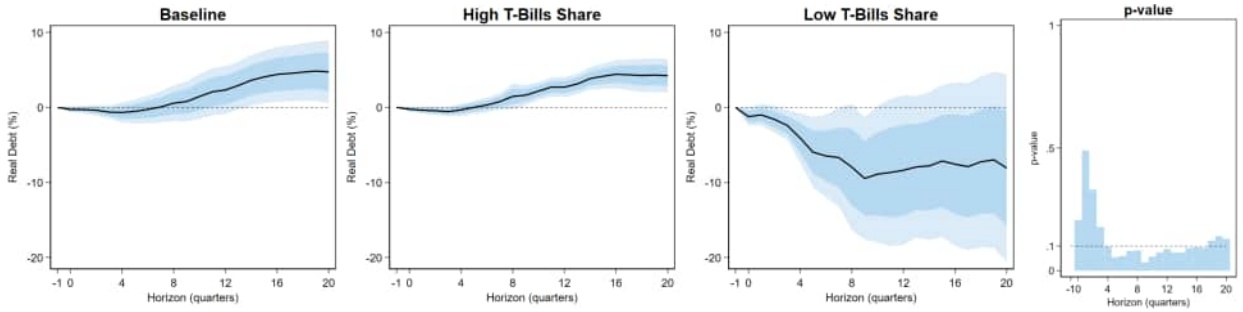
Panel C. Baa Corporate Bond Yield



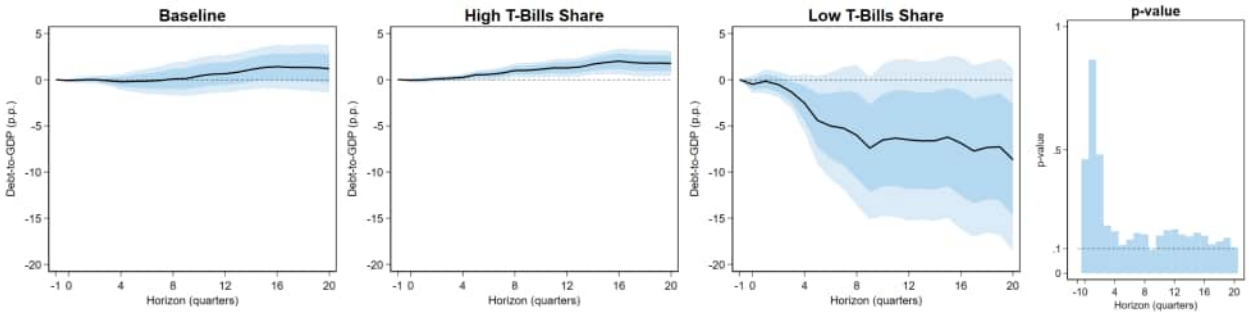
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the recession dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A97: ROBUSTNESS CHECK, CONTROLLING FOR 1970S INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



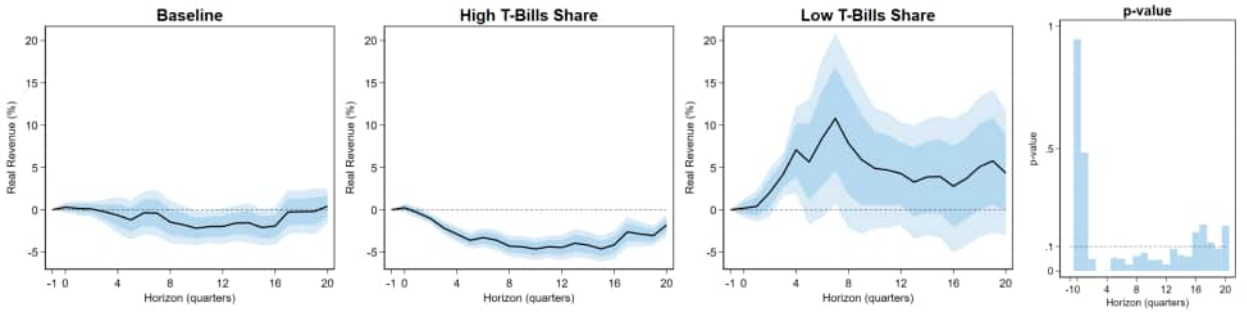
Panel B. Debt-to-GDP



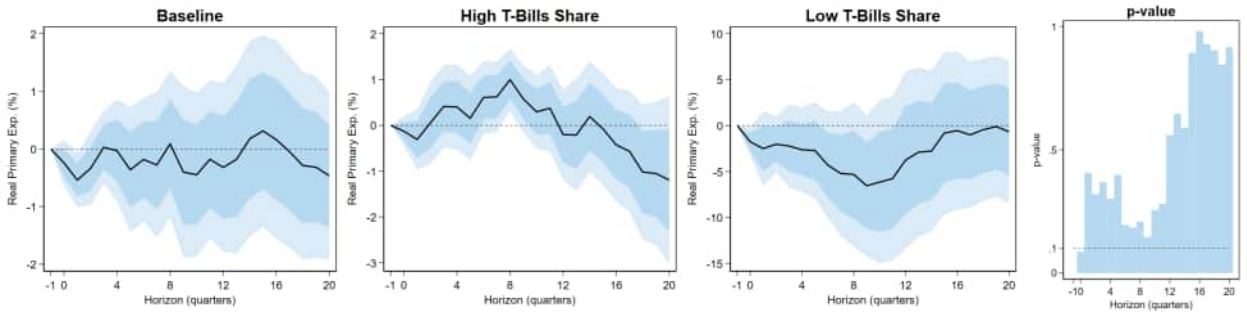
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \big(1 - F(z_{t-1})\big)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the 1970s dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A98: ROBUSTNESS CHECK, CONTROLLING FOR 1970S INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

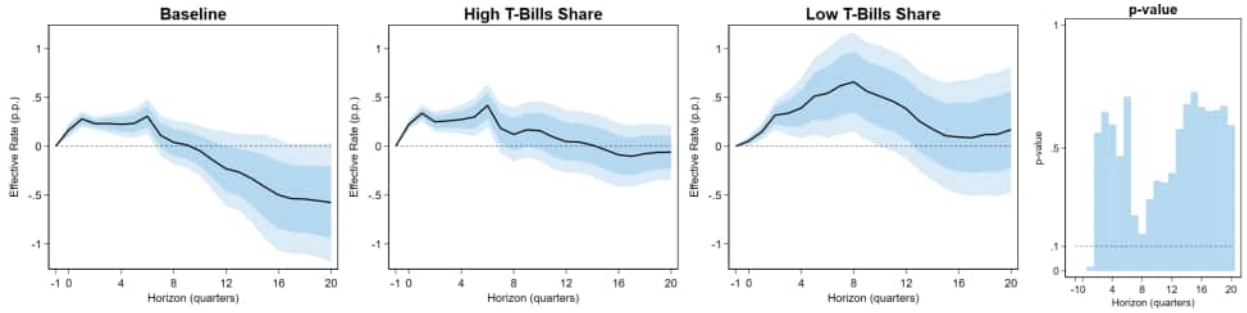
Panel A. Real Revenue



Panel B. Real Primary Expenditure



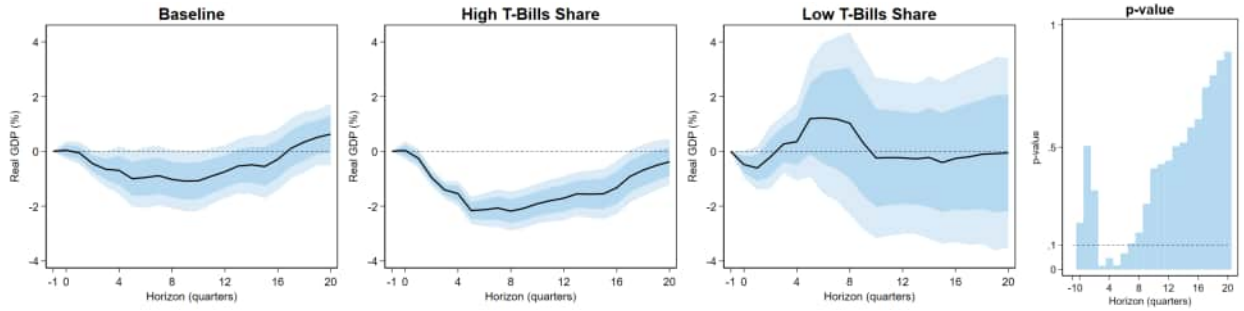
Panel C. Effective Interest Rate



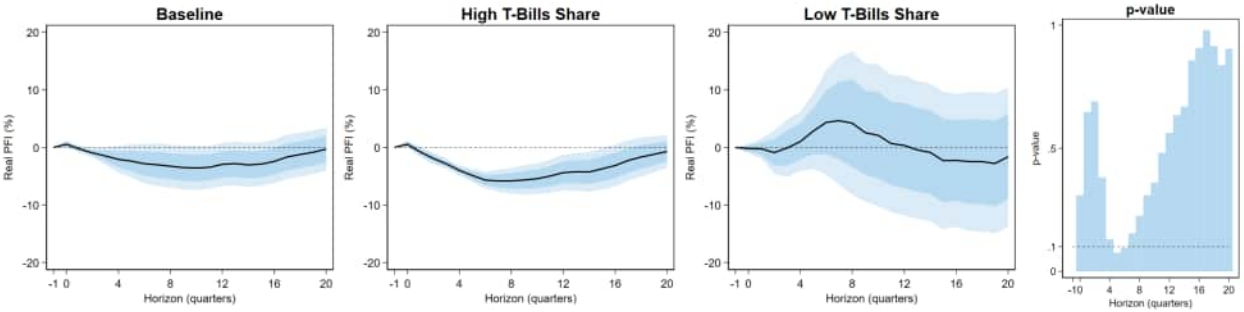
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the 1970s dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A99: ROBUSTNESS CHECK, CONTROLLING FOR 1970S INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

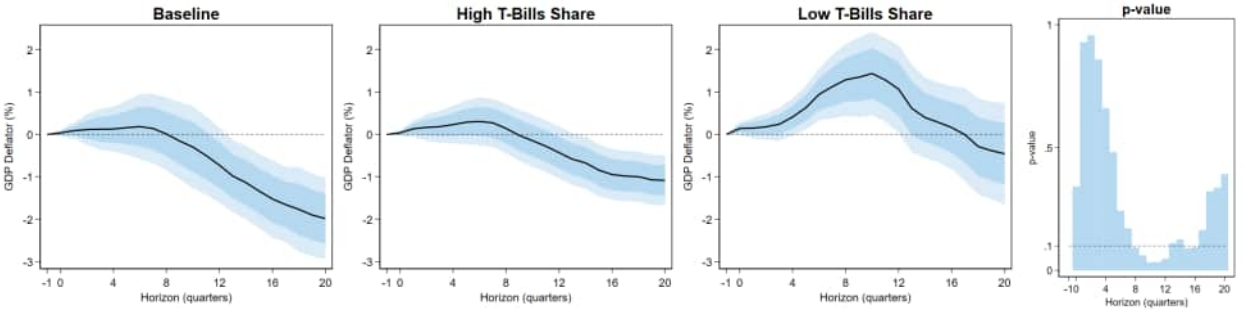
Panel A. Real GDP



Panel B. Real Private Fixed Investments



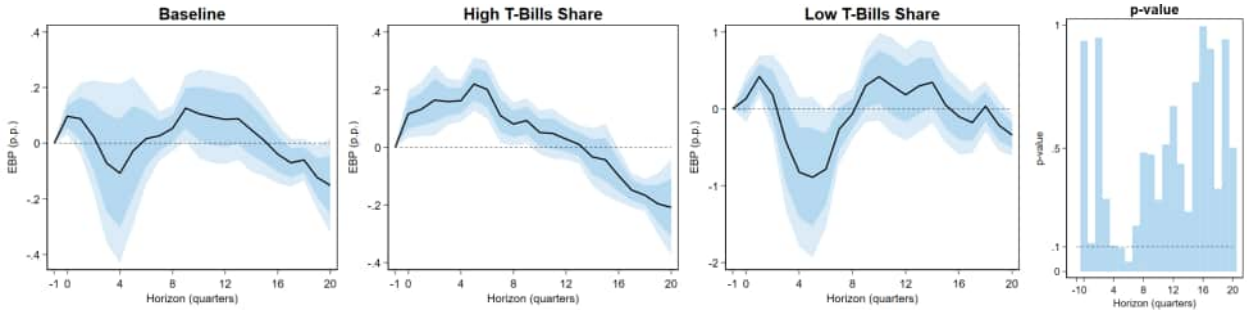
Panel C. GDP Deflator



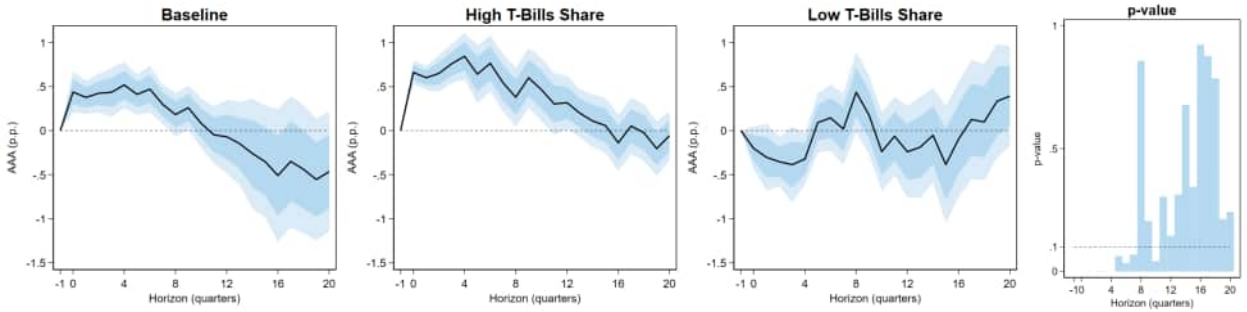
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the 1970s dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A100: ROBUSTNESS CHECK, CONTROLLING FOR 1970S INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

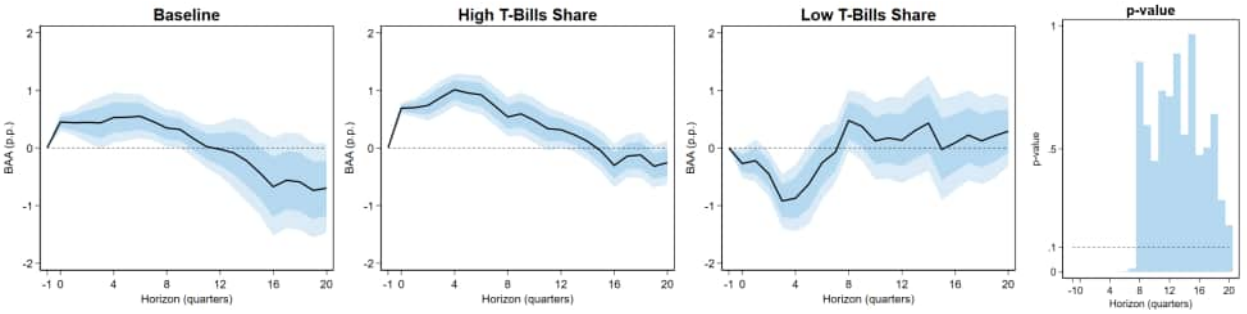
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



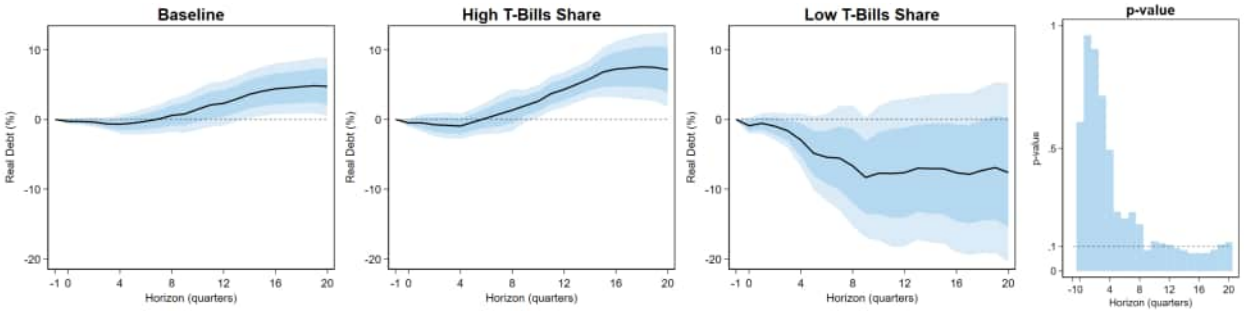
Panel C. Baa Corporate Bond Yield



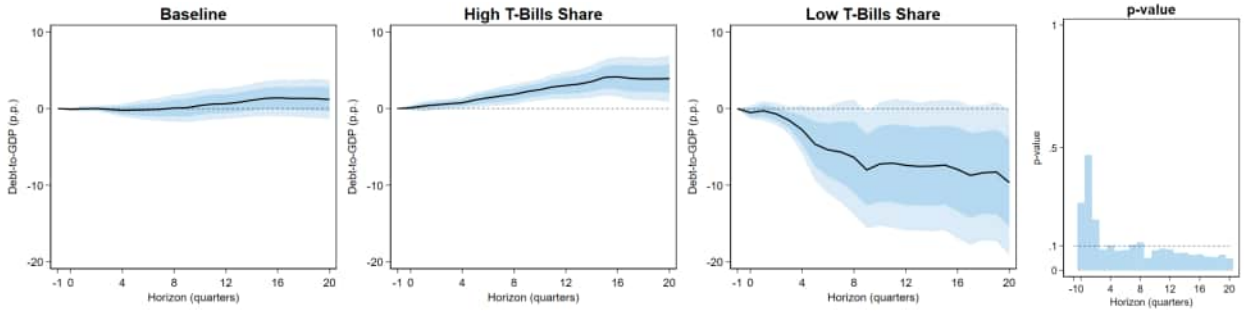
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the 1970s dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A101: ROBUSTNESS CHECK, CONTROLLING FOR VOLCKER INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



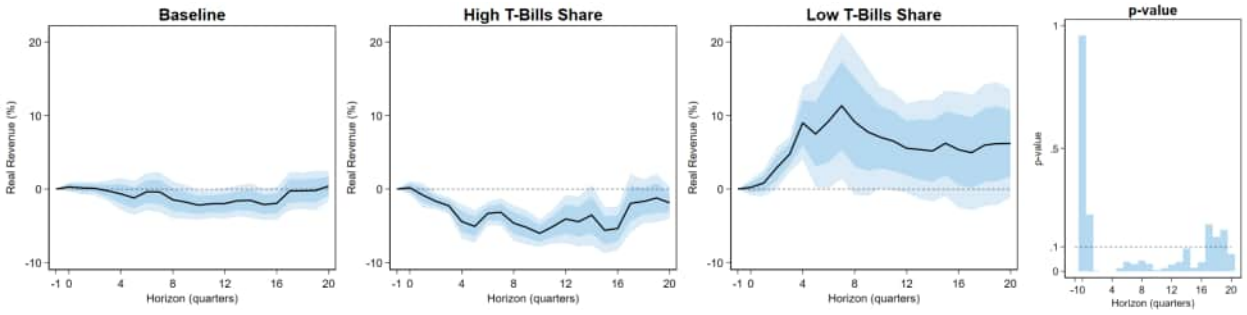
Panel B. Debt-to-GDP



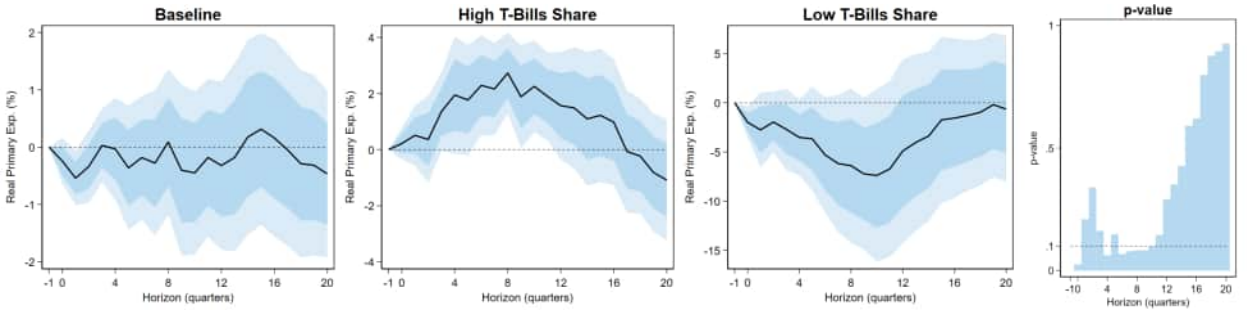
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the Volcker dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A102: ROBUSTNESS CHECK, CONTROLLING FOR VOLCKER INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

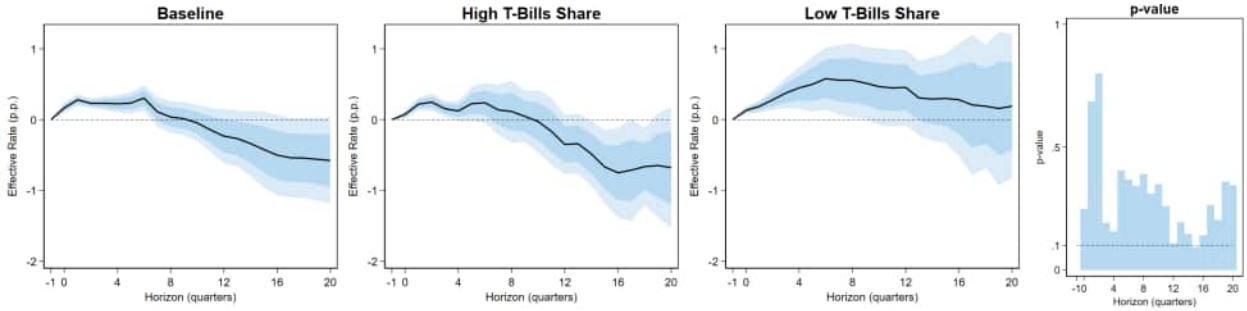
Panel A. Real Revenue



Panel B. Real Primary Expenditure



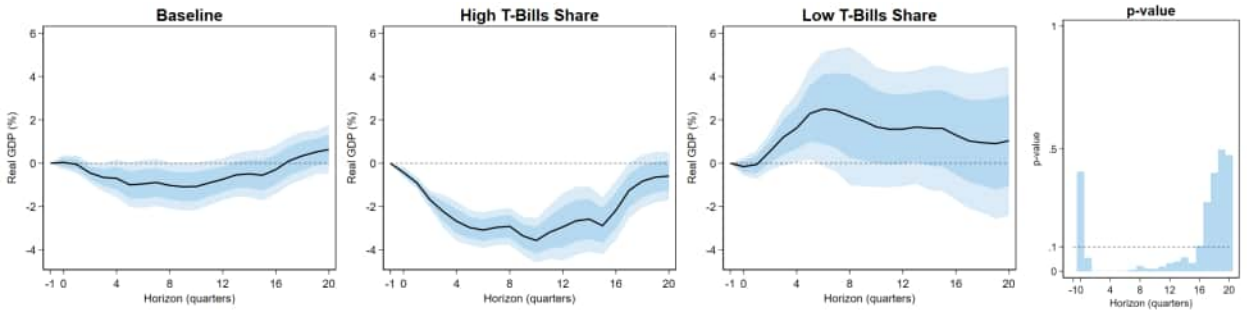
Panel C. Effective Interest Rate



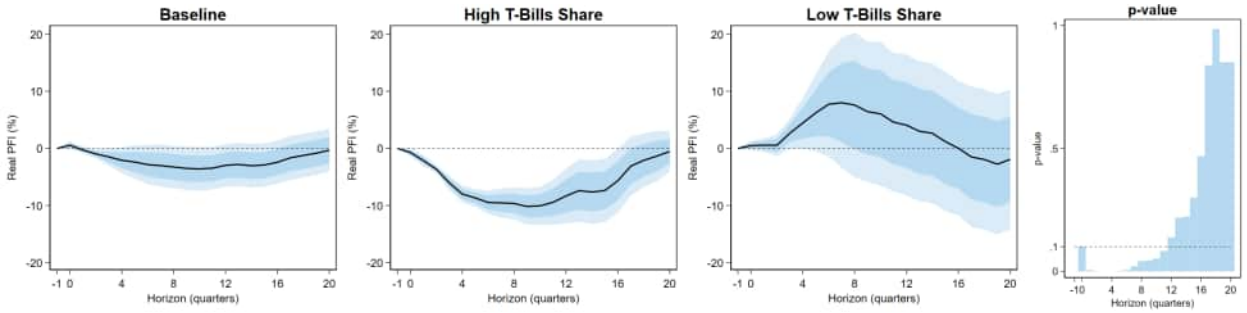
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the Volcker dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A103: ROBUSTNESS CHECK, CONTROLLING FOR VOLCKER INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

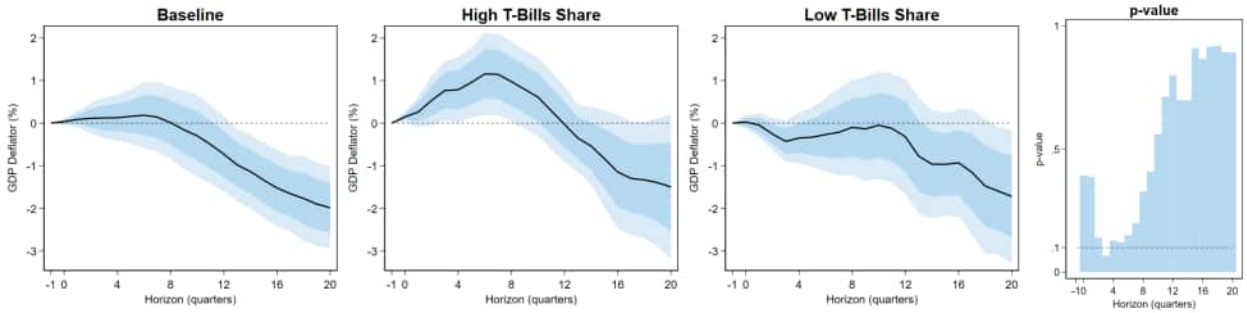
Panel A. Real GDP



Panel B. Real Private Fixed Investments



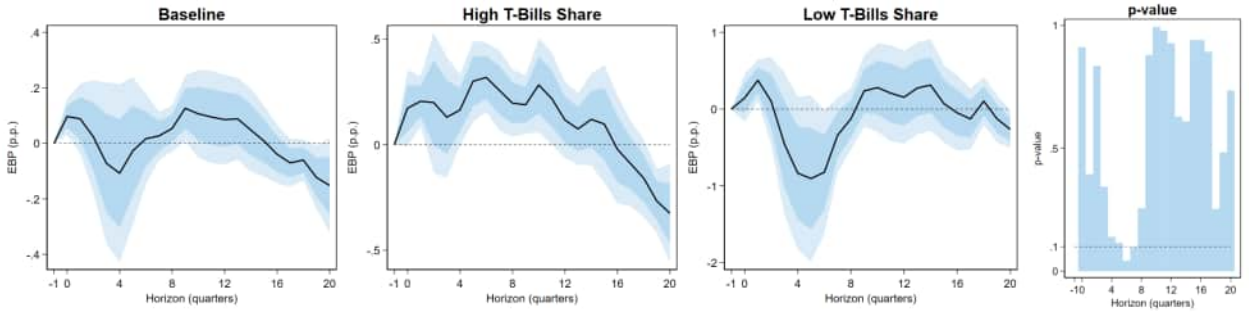
Panel C. GDP Deflator



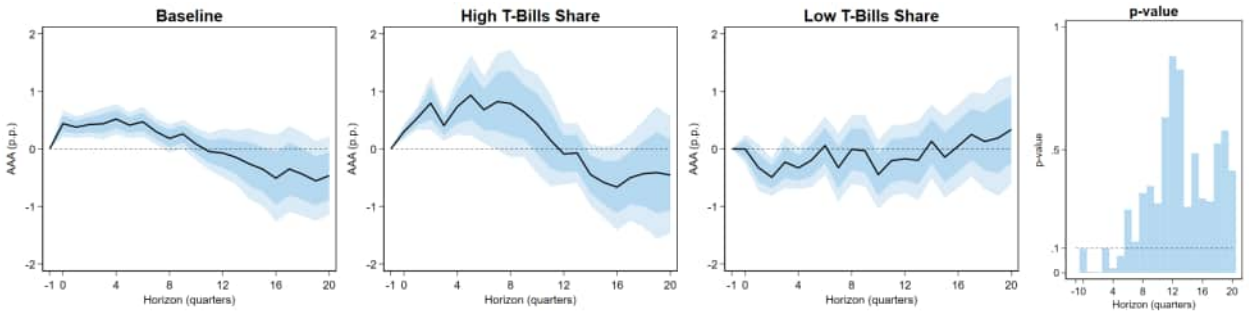
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the Volcker dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A104: ROBUSTNESS CHECK, CONTROLLING FOR VOLCKER INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

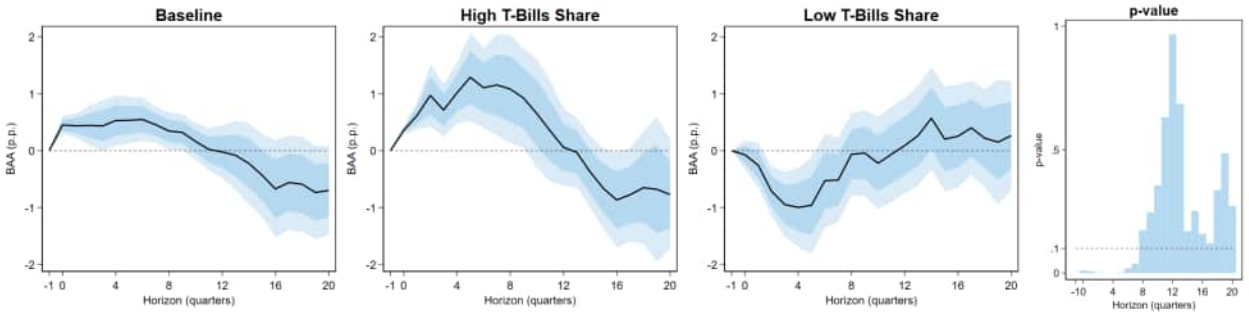
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



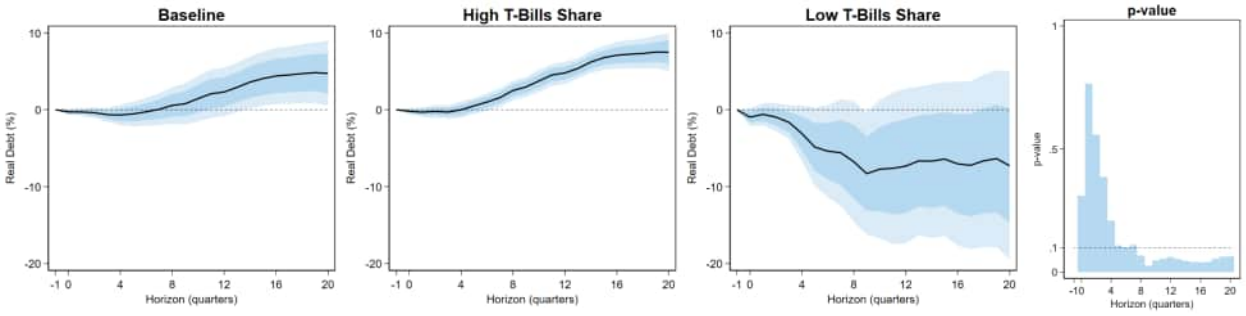
Panel C. Baa Corporate Bond Yield



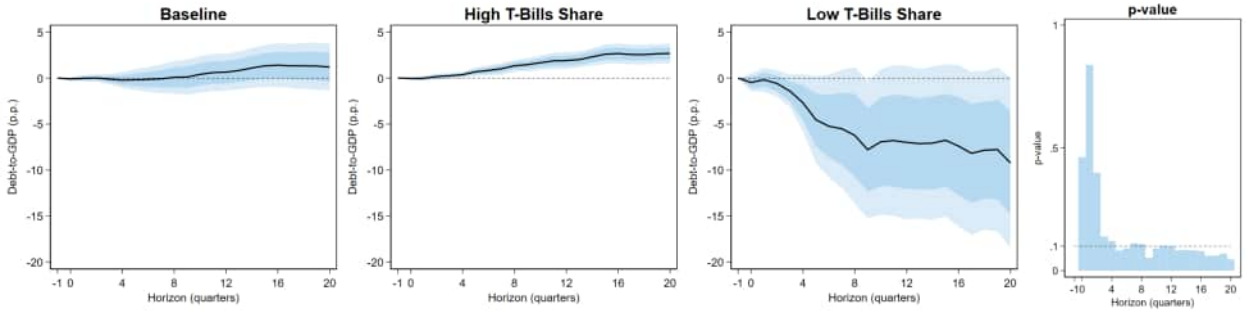
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the Volcker dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A105: ROBUSTNESS CHECK, CONTROLLING FOR 1981-82 CRISIS INTERACTION. THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



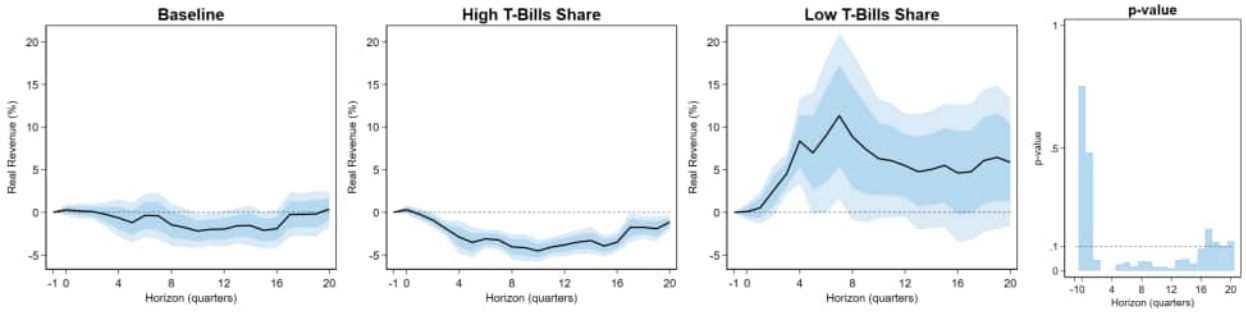
Panel B. Debt-to-GDP



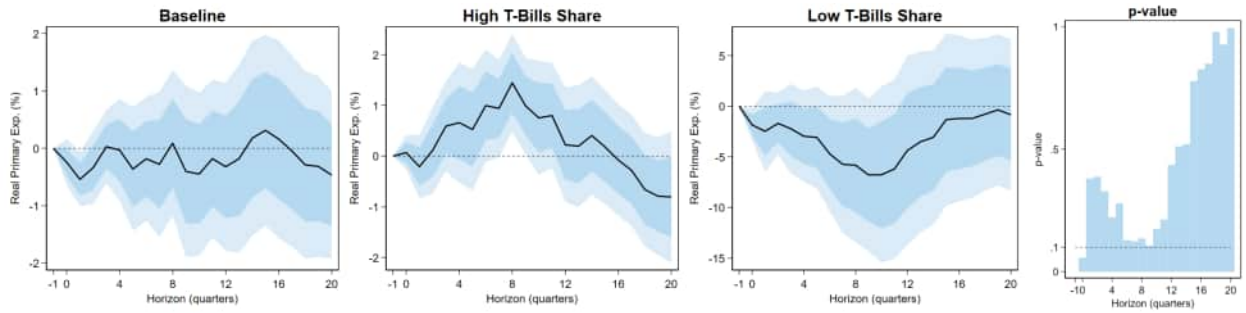
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell'^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + (1 - F(z_{t-1}))\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the 1981-1982 crisis dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A106: ROBUSTNESS CHECK, CONTROLLING FOR 1981-1982 CRISIS INTERACTION. THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

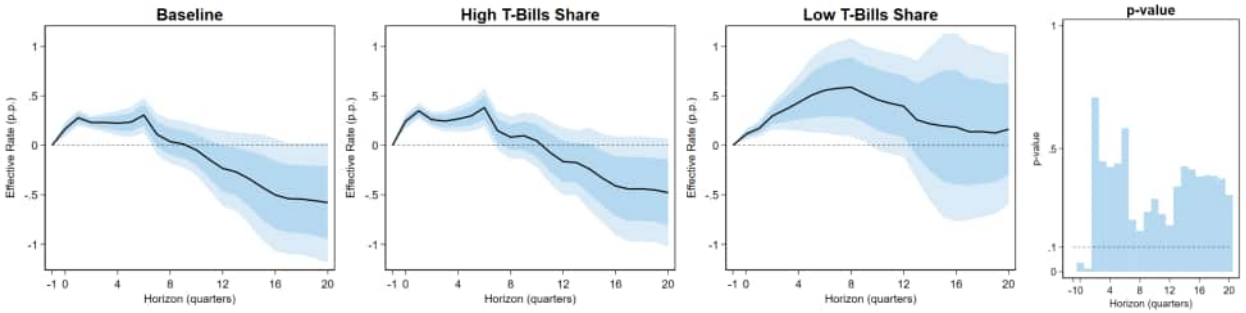
Panel A. Real Revenue



Panel B. Real Primary Expenditure



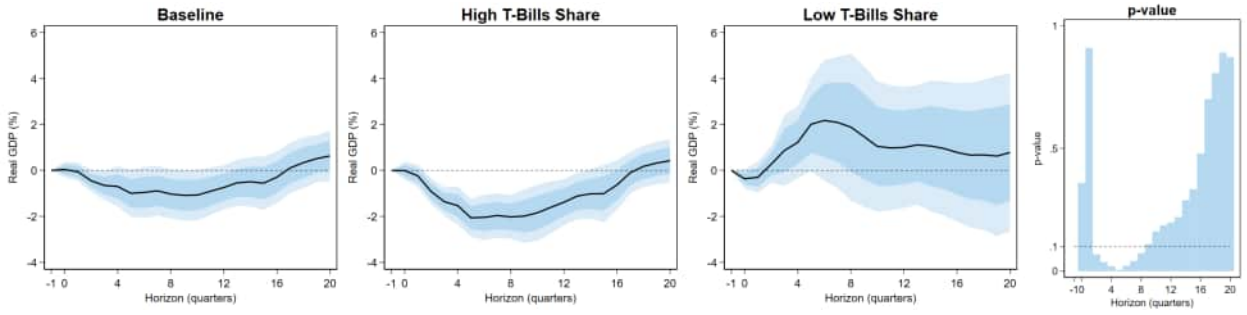
Panel C. Effective Interest Rate



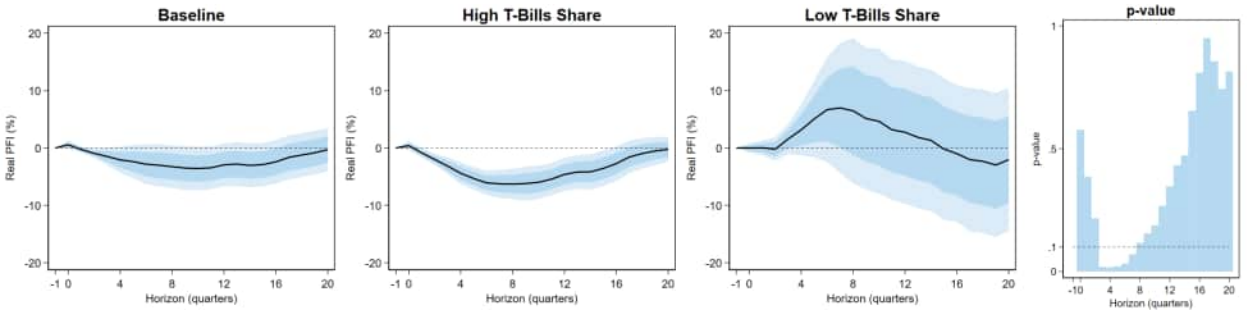
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the 1981-1982 crisis dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A107: ROBUSTNESS CHECK, CONTROLLING FOR 1981-1982 CRISIS INTERACTION. THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

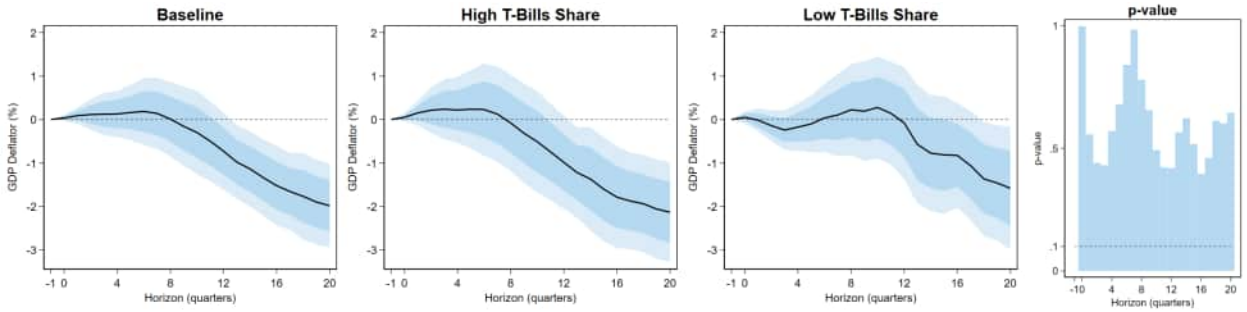
Panel A. Real GDP



Panel B. Real Private Fixed Investments



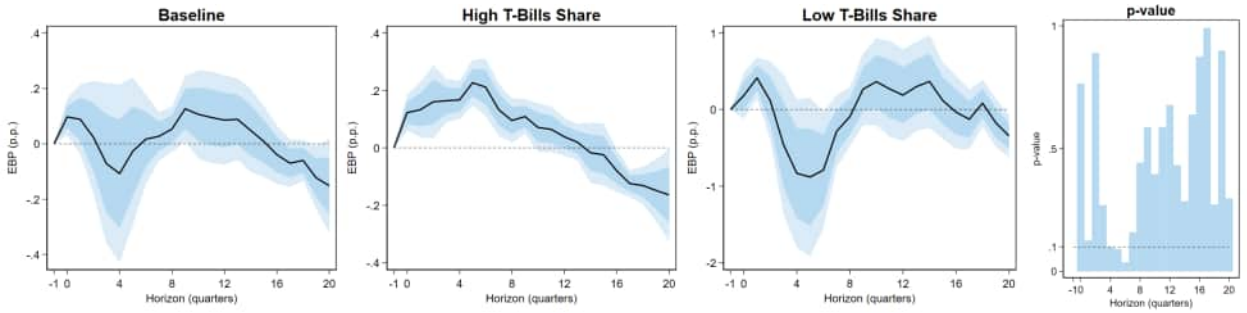
Panel C. GDP Deflator



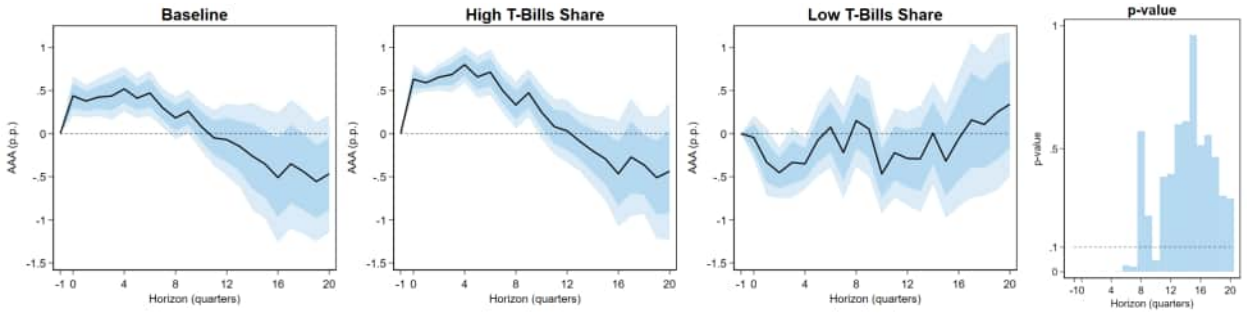
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the 1981-1982 crisis dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A108: ROBUSTNESS CHECK, CONTROLLING FOR 1981-1982 CRISIS INTERACTION. THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

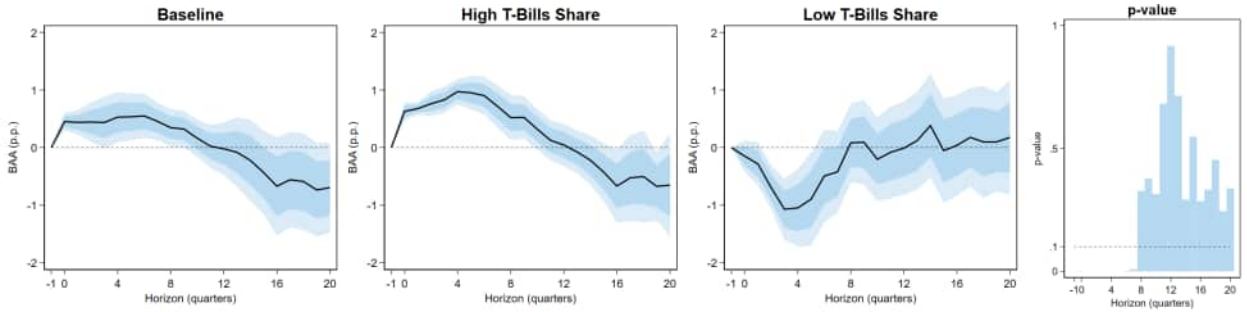
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



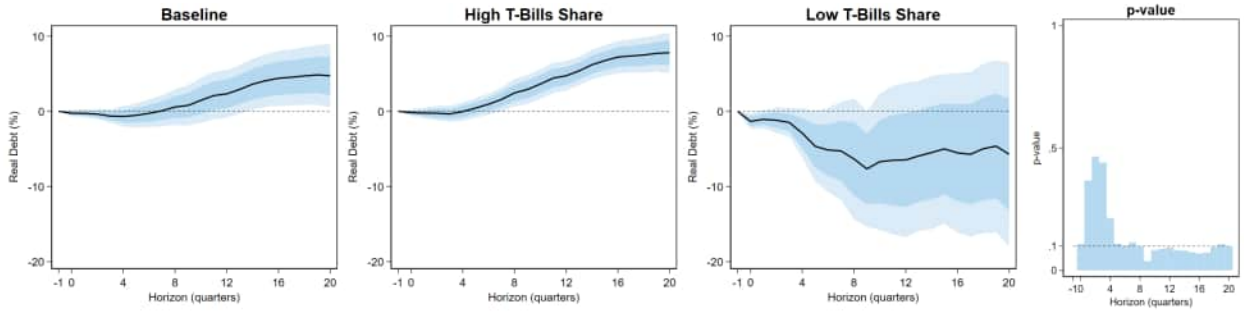
Panel C. Baa Corporate Bond Yield



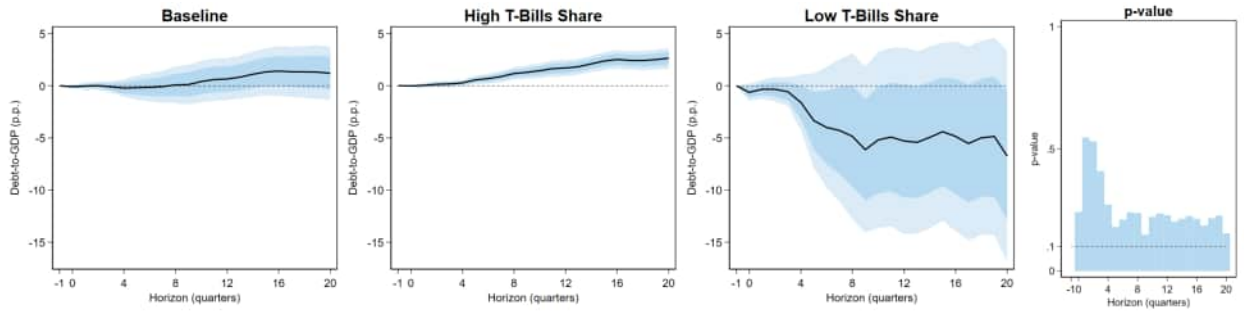
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the 1981-1982 crisis dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A109: ROBUSTNESS CHECK, CONTROLLING FOR GFC INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



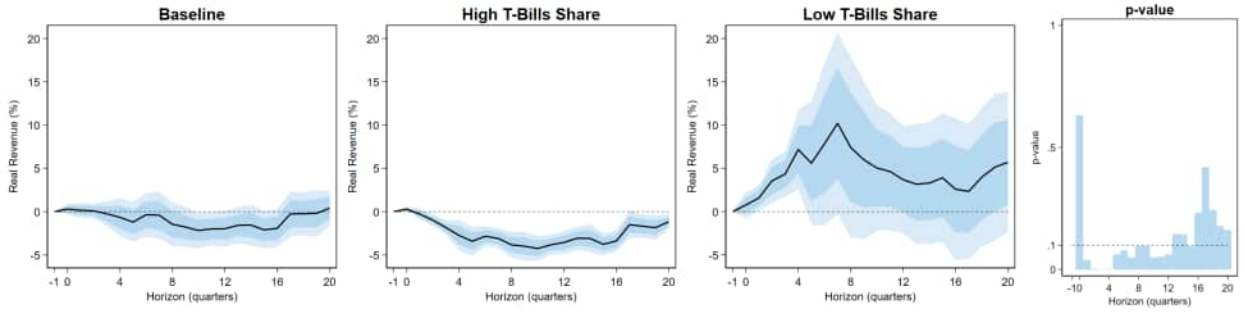
Panel B. Debt-to-GDP



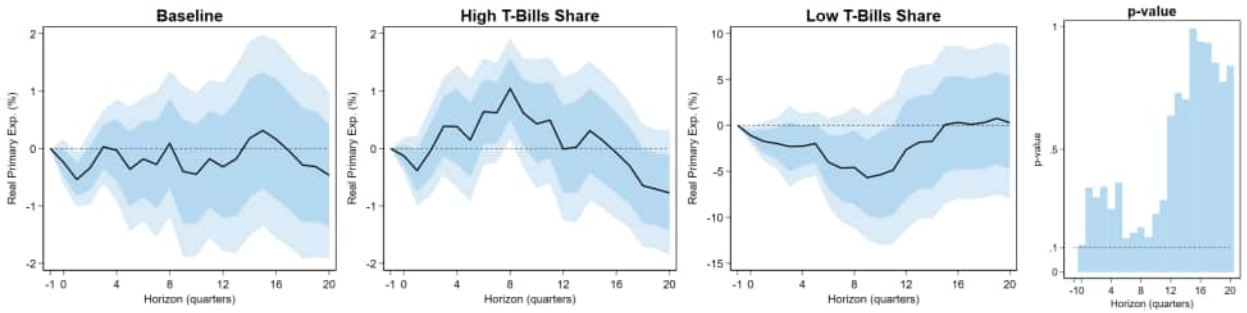
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell'^{(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + (1 - F(z_{t-1}))\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the GFC dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A110: ROBUSTNESS CHECK, CONTROLLING FOR GFC INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

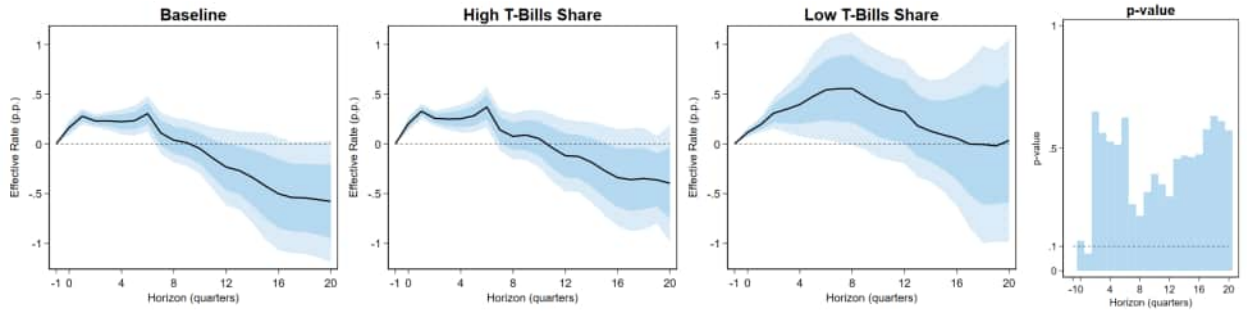
Panel A. Real Revenue



Panel B. Real Primary Expenditure



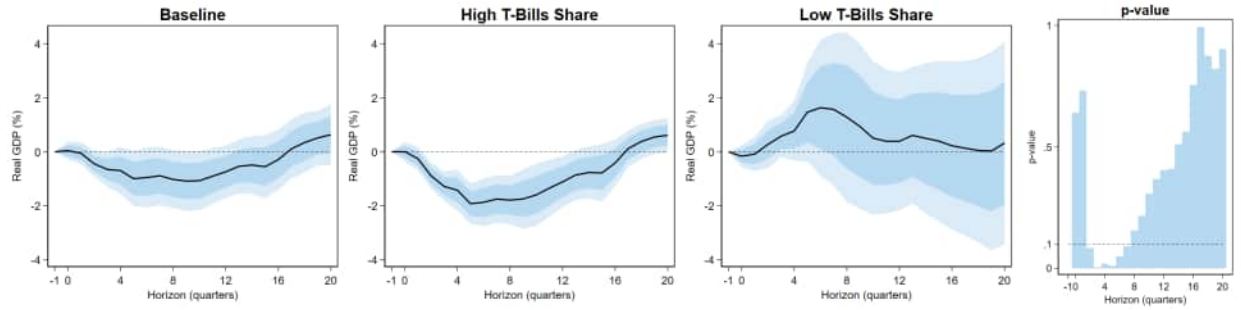
Panel C. Effective Interest Rate



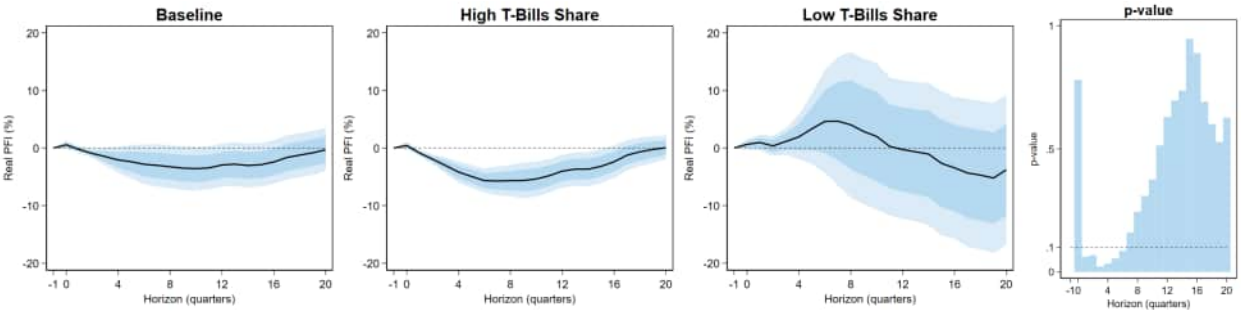
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the GFC dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A111: ROBUSTNESS CHECK, CONTROLLING FOR GFC INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

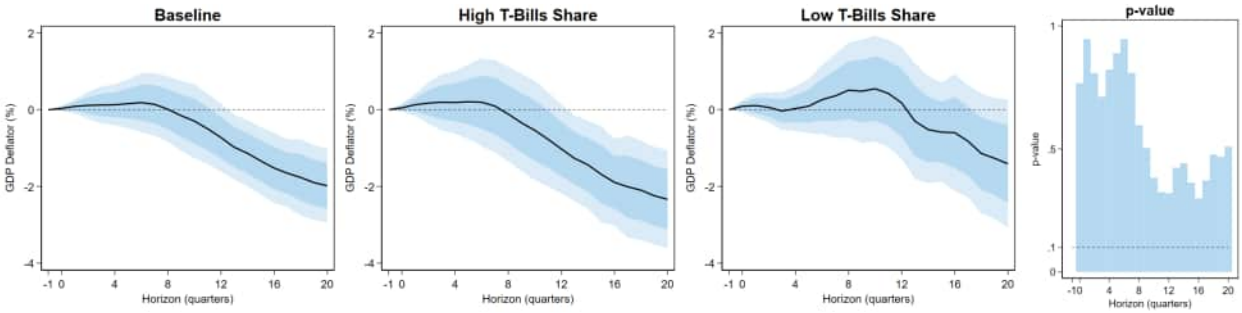
Panel A. Real GDP



Panel B. Real Private Fixed Investments



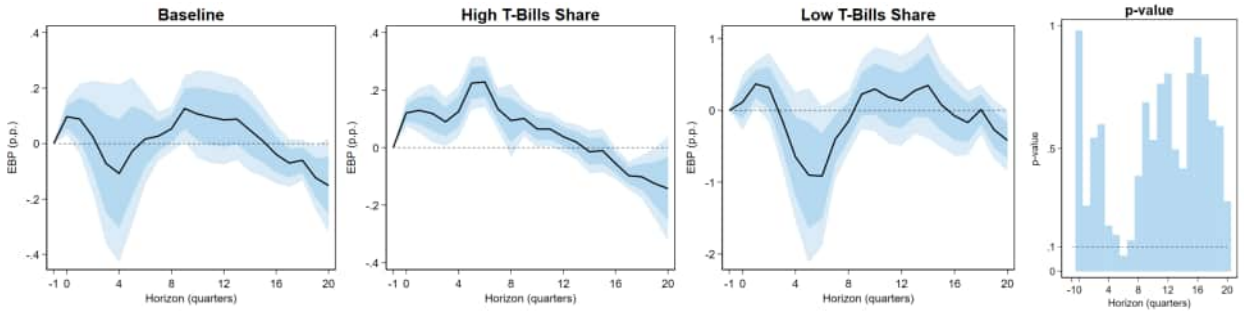
Panel C. GDP Deflator



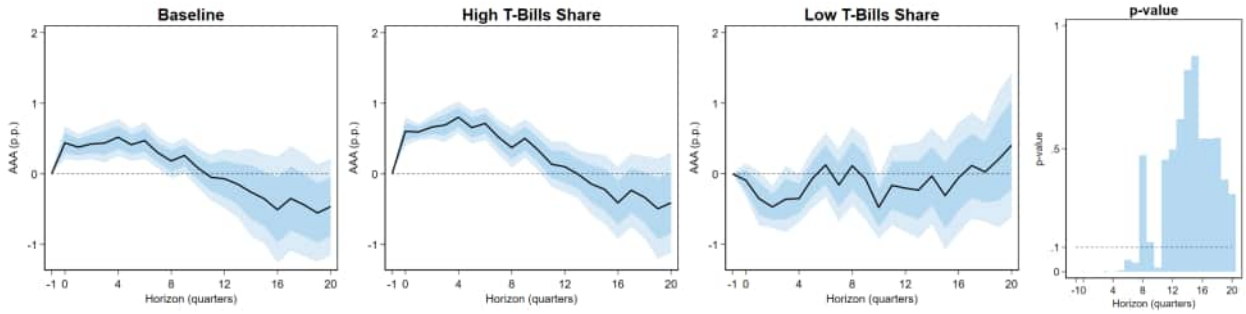
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the GFC dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A112: ROBUSTNESS CHECK, CONTROLLING FOR GFC INTERACTION.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

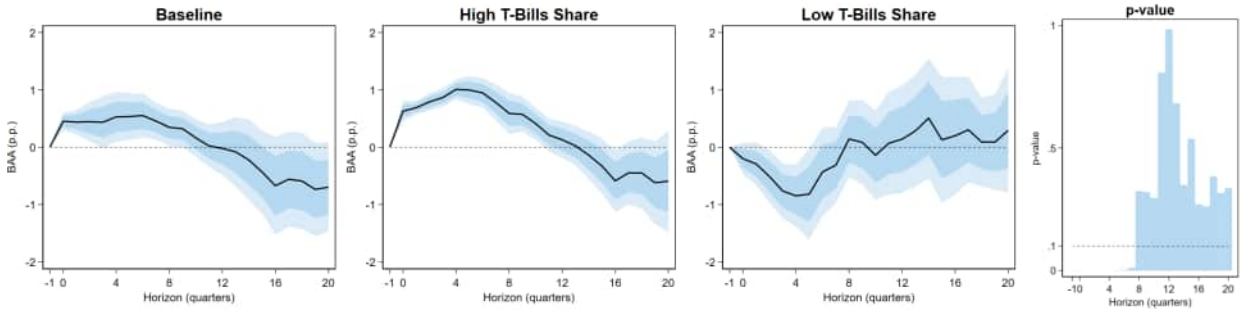
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



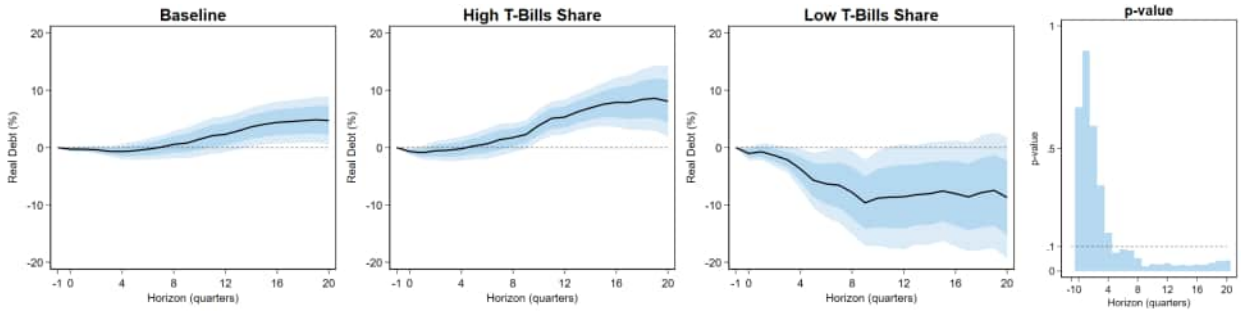
Panel C. Baa Corporate Bond Yield



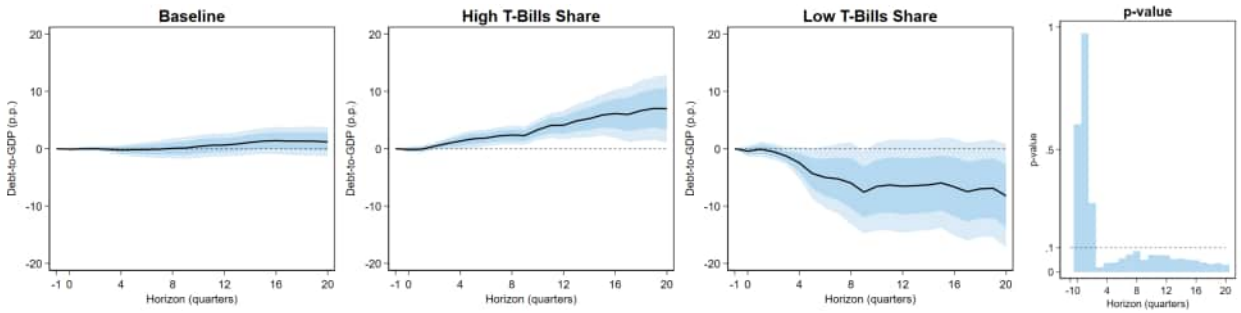
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the GFC dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A113: ROBUSTNESS CHECK, CONTROLLING FOR QE (STF).
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



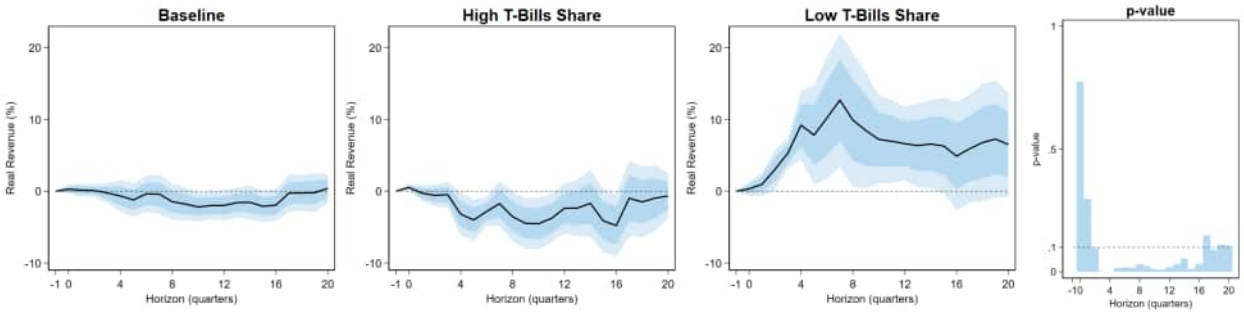
Panel B. Debt-to-GDP



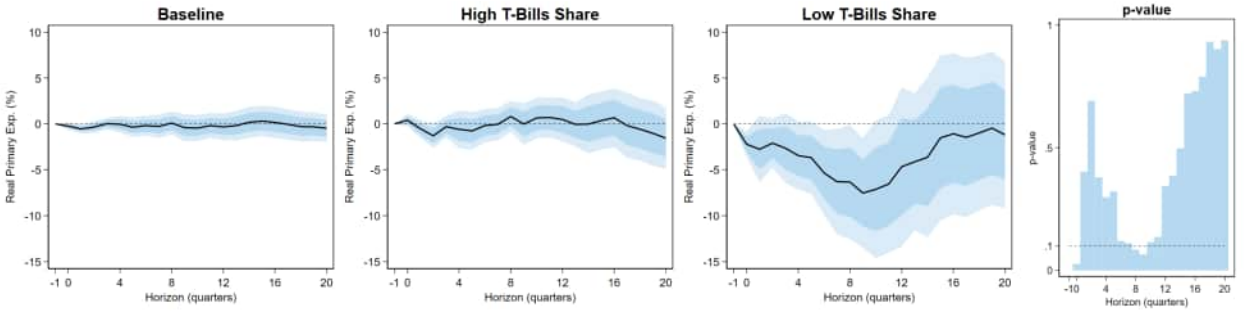
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial level of debt held by the FED. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A114: ROBUSTNESS CHECK, CONTROLLING FOR QE (STF).
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

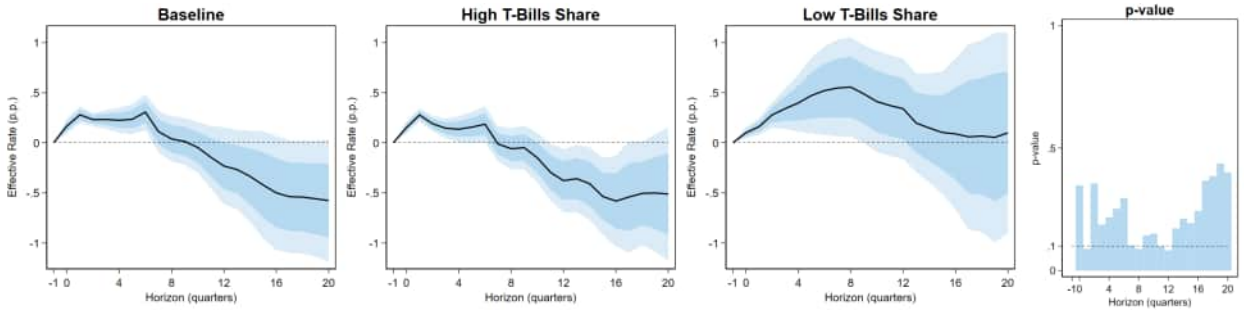
Panel A. Real Revenue



Panel B. Real Primary Expenditure



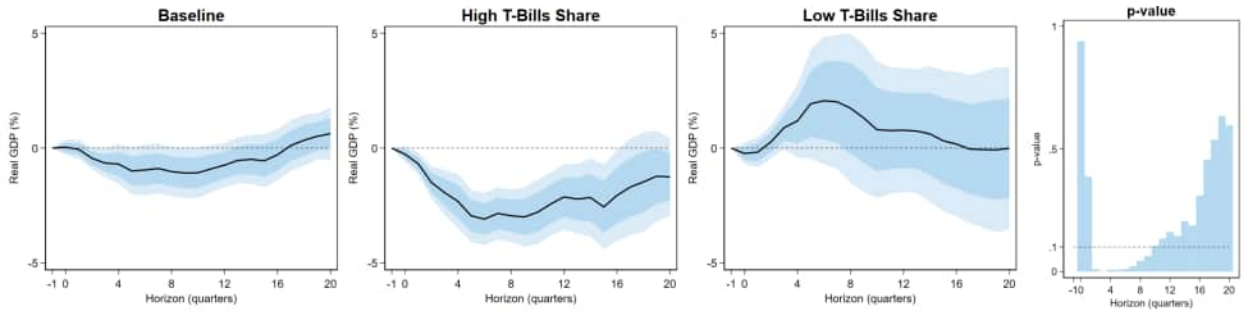
Panel C. Effective Interest Rate



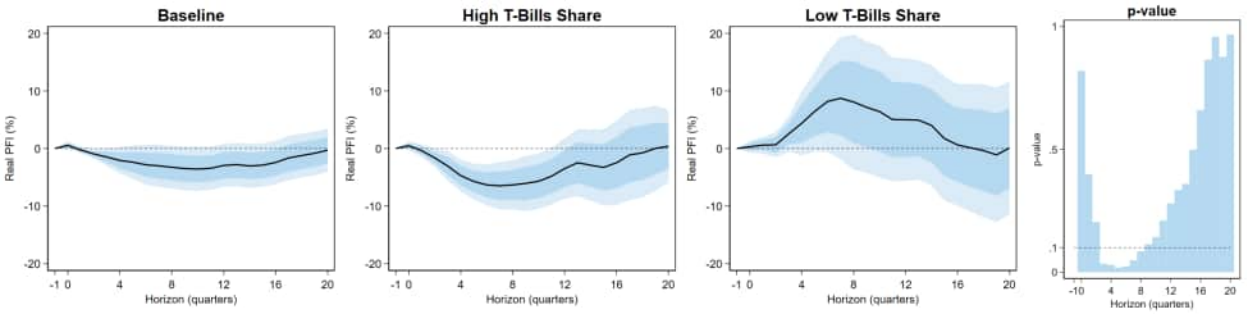
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)}\left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial level of debt held by the FED. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A115: ROBUSTNESS CHECK, CONTROLLING FOR QE (STF).
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

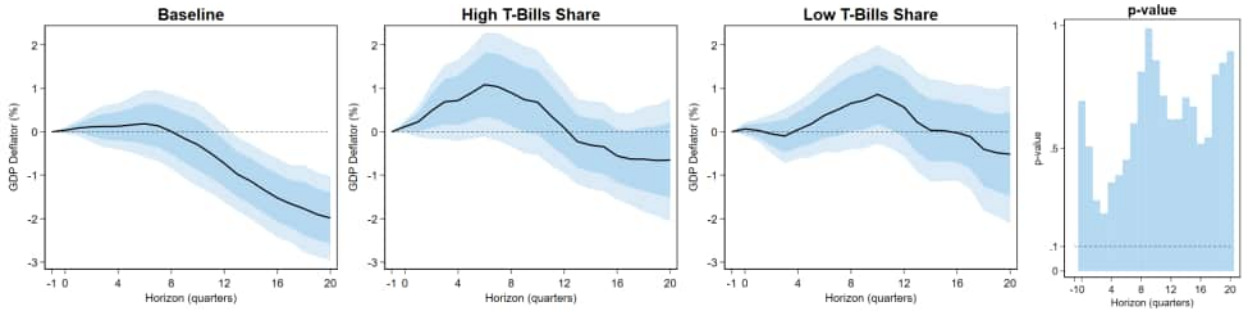
Panel A. Real GDP



Panel B. Real Private Fixed Investments



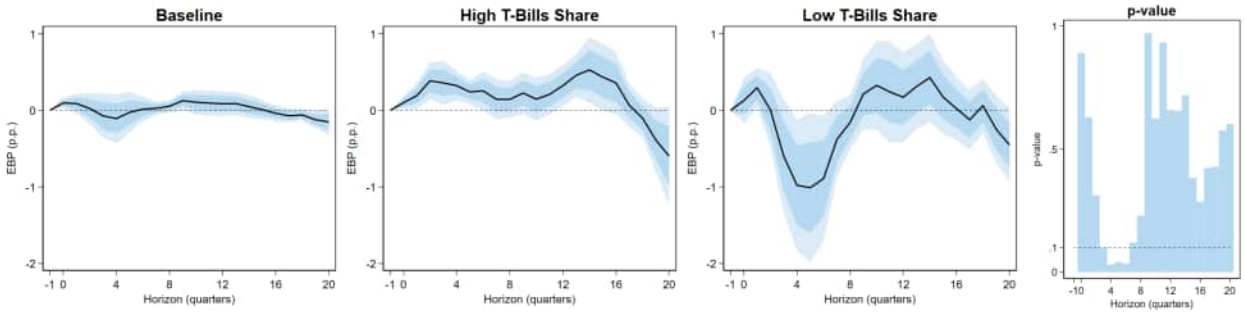
Panel C. GDP Deflator



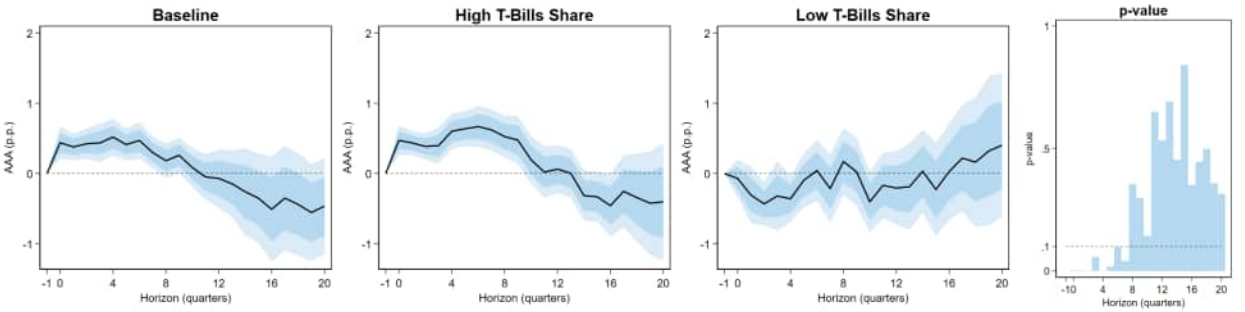
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t\big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \iota^{(h)}\big(F(r_{t-1}) - \overline{F(r_{t-1})}\big) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial level of debt held by the FED. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A116: ROBUSTNESS CHECK, CONTROLLING FOR QE (STF).
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

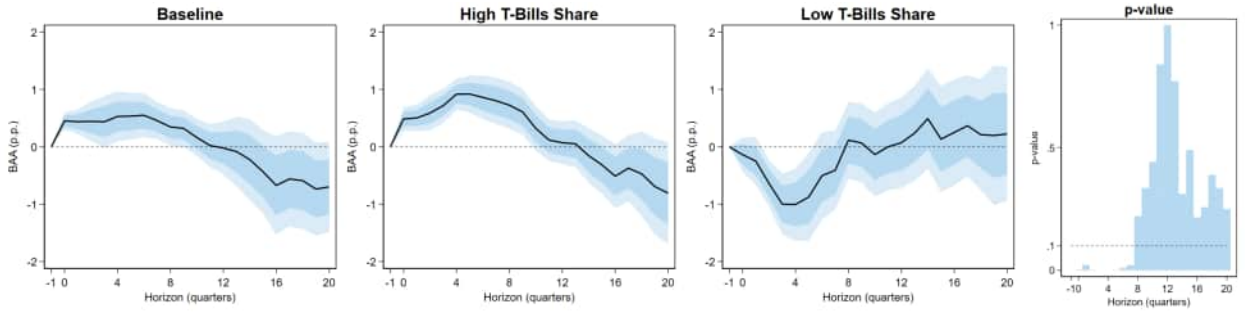
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



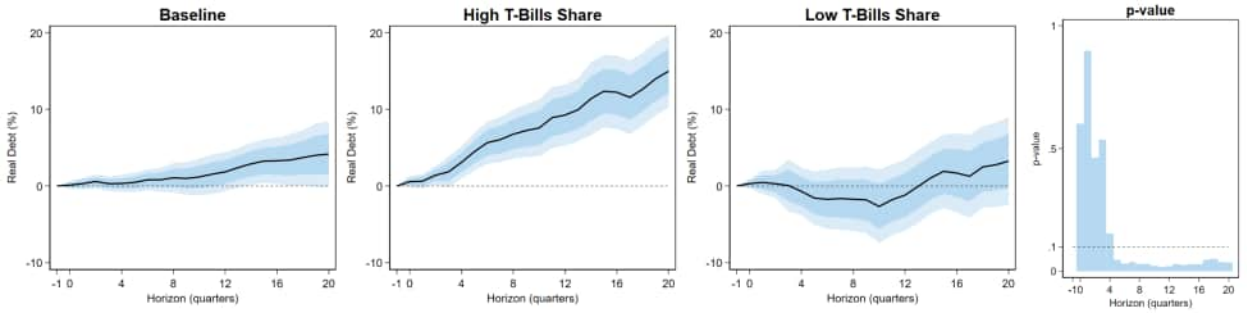
Panel C. Baa Corporate Bond Yield



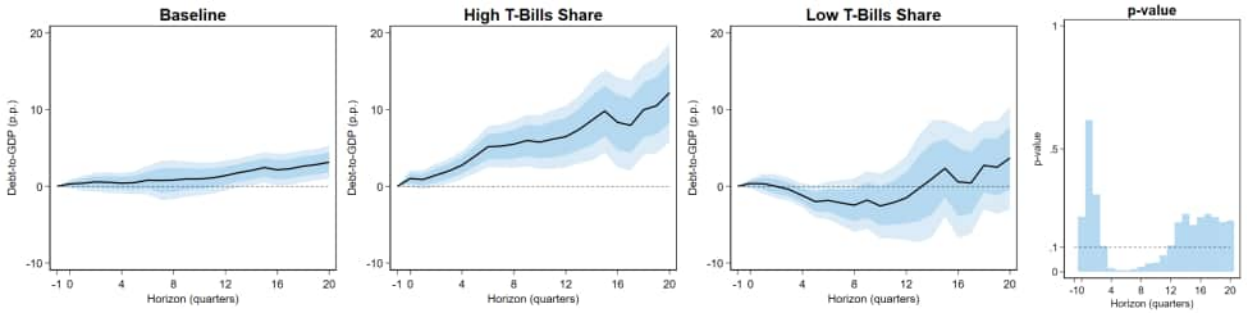
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t\left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)}\left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial level of debt held by the FED. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A117: ROBUSTNESS CHECK, CONTROLLING FOR QE (STF, HIGH-FREQ. SHOCK). THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



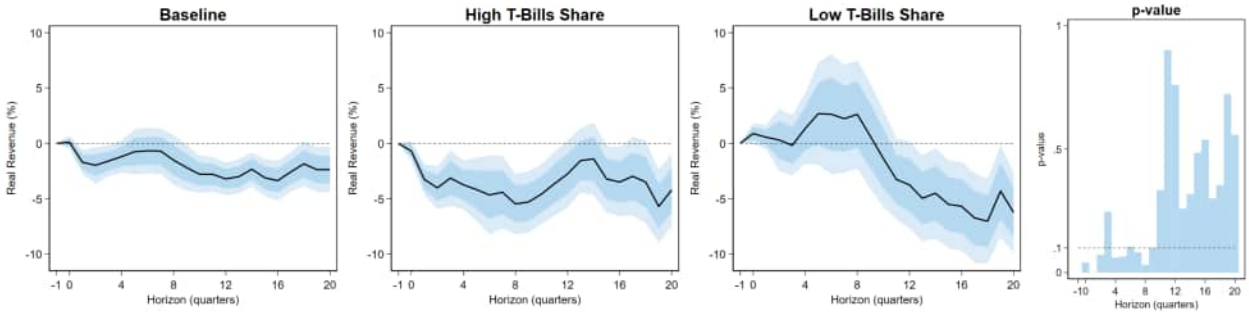
Panel B. Debt-to-GDP



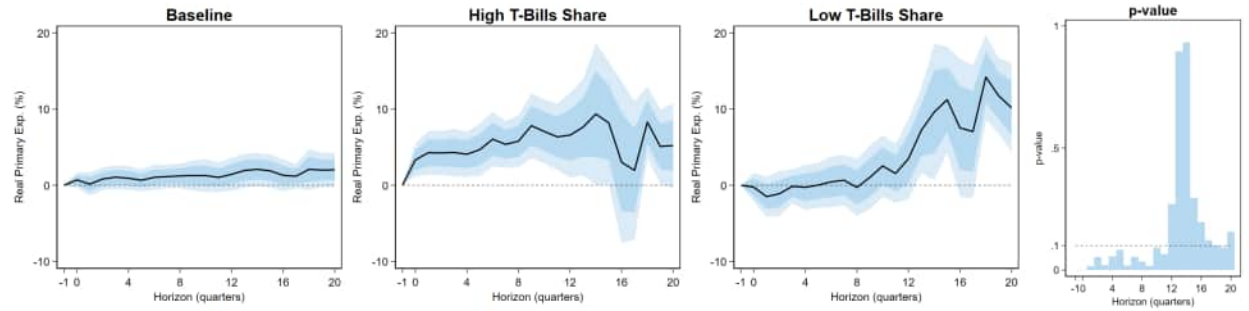
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1963q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial level of debt held by the FED. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A118: ROBUSTNESS CHECK, CONTROLLING FOR QE (STF, HIGH-FREQ. SHOCK). THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

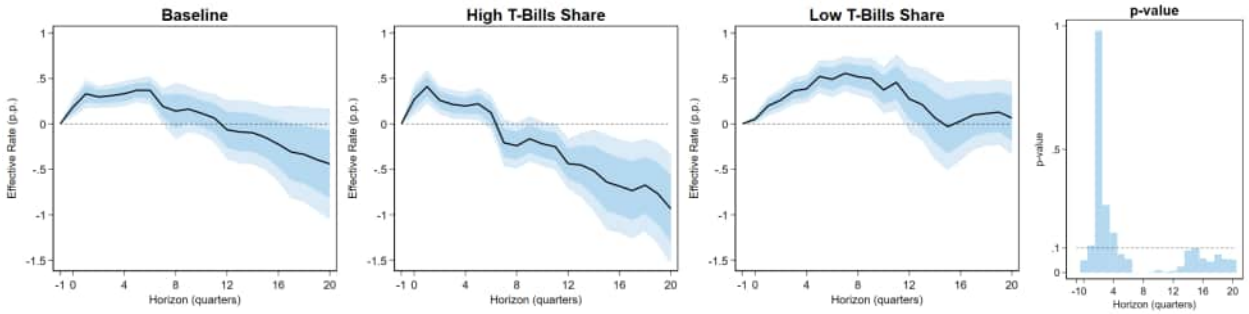
Panel A. Real Revenue



Panel B. Real Primary Expenditure



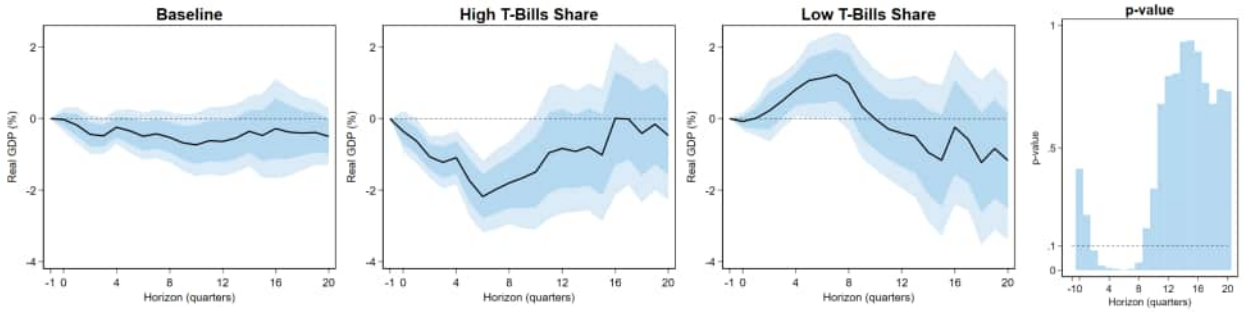
Panel C. Effective Interest Rate



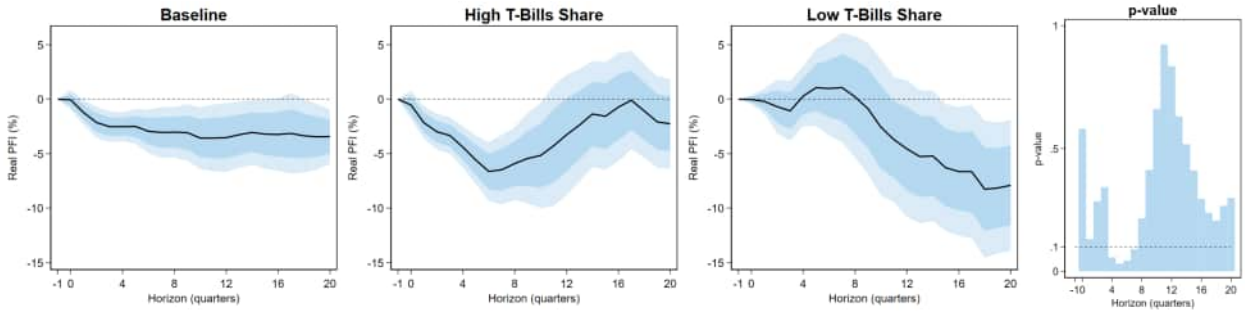
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1963q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)} \left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial level of debt held by the FED. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A119: ROBUSTNESS CHECK, CONTROLLING FOR QE (STF, HIGH-FREQ. SHOCK). THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

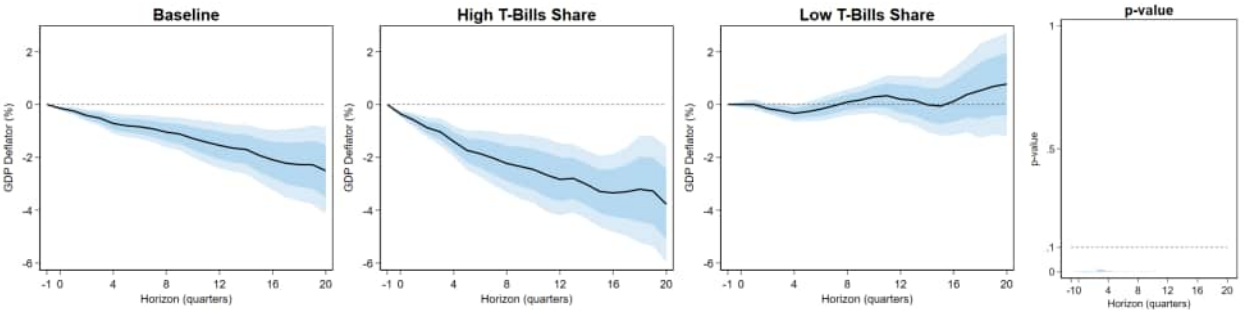
Panel A. Real GDP



Panel B. Real Private Fixed Investments



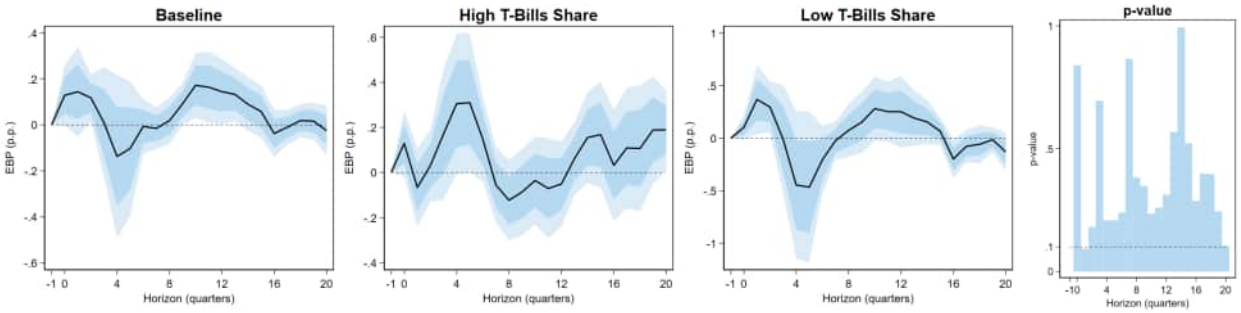
Panel C. GDP Deflator



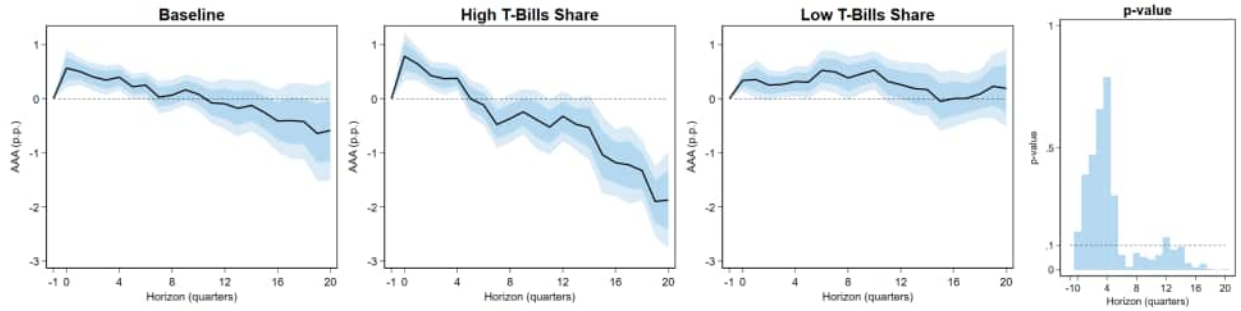
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1963q1-2025q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)} \left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial level of debt held by the FED. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A120: ROBUSTNESS CHECK, CONTROLLING FOR QE (STF, HIGH-FREQ. SHOCK). THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

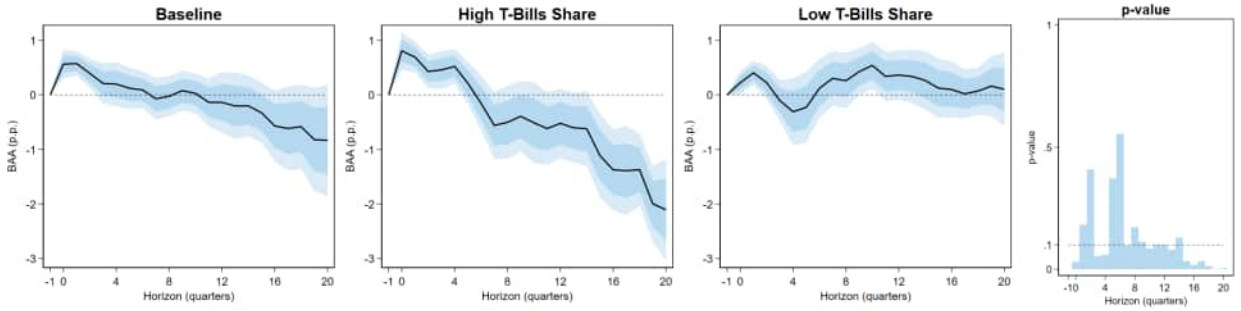
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



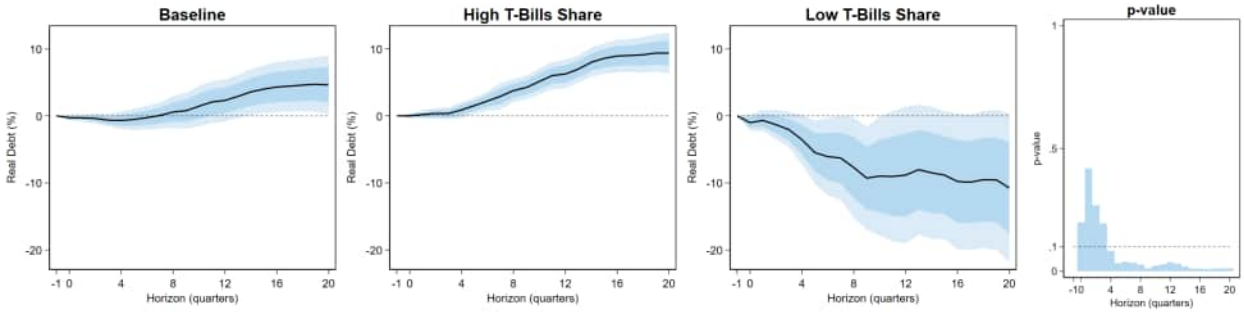
Panel C. Baa Corporate Bond Yield



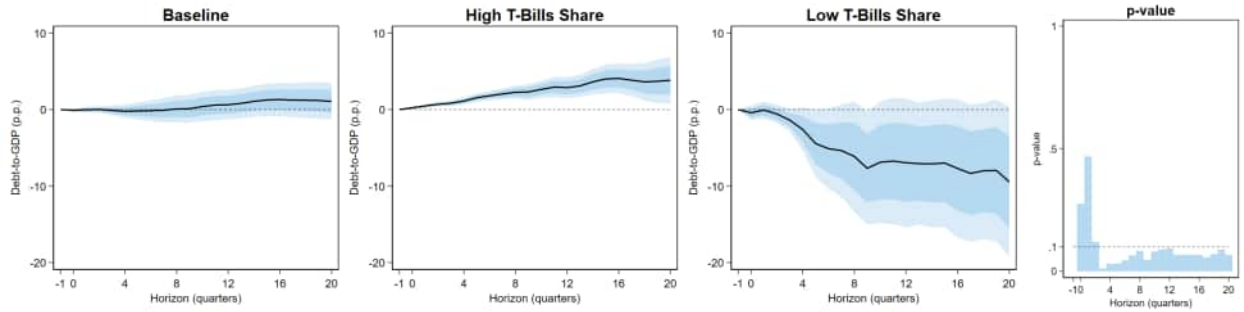
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1963q1-2025q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t\left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \iota^{(h)}\left(F(r_{t-1}) - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial level of debt held by the FED. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A121: ROBUSTNESS CHECK, CONTROLLING FOR QE (DUMMY).
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



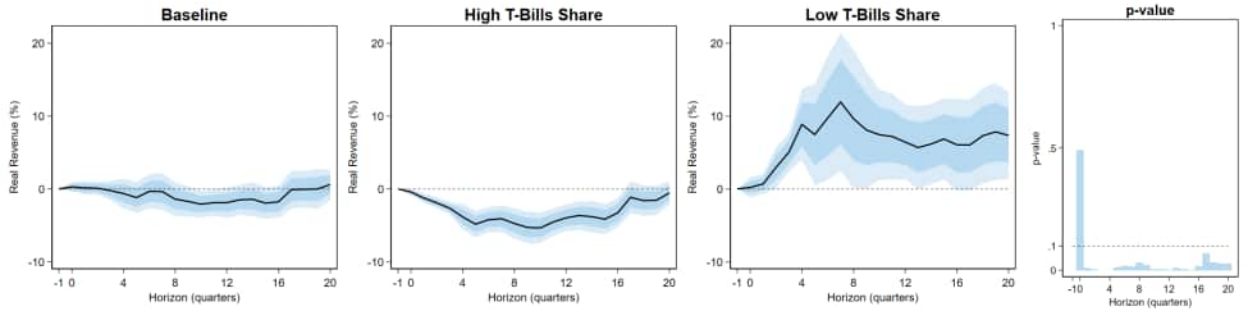
Panel B. Debt-to-GDP



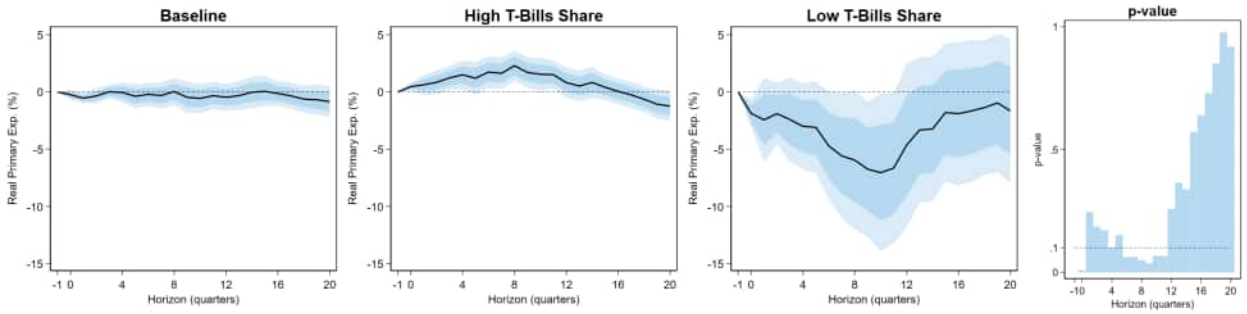
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \big(1 - F(z_{t-1})\big)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the Quantitative Easing dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A122: ROBUSTNESS CHECK, CONTROLLING FOR QE (DUMMY).
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

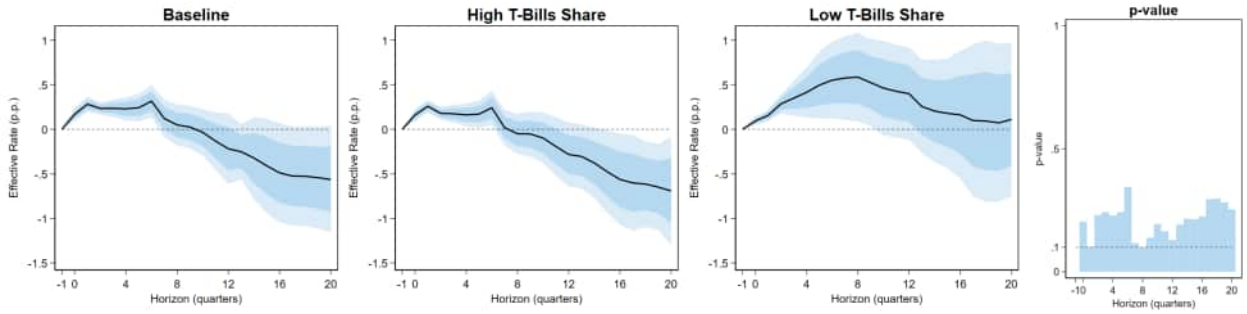
Panel A. Real Revenue



Panel B. Real Primary Expenditure



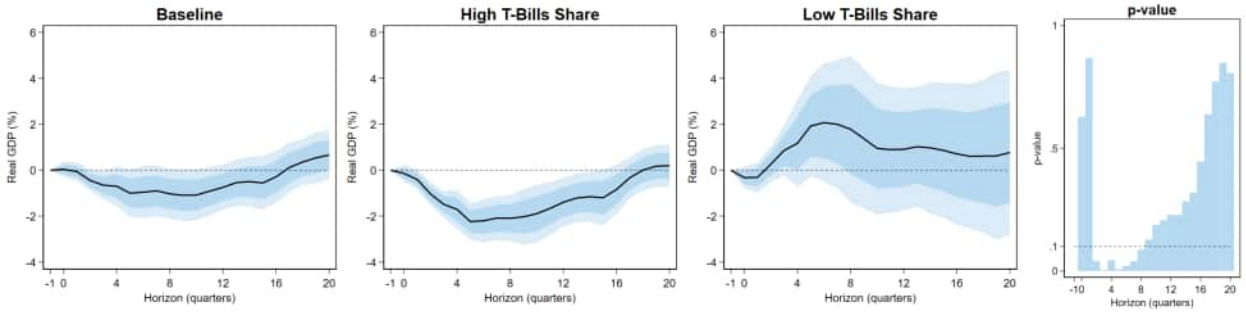
Panel C. Effective Interest Rate



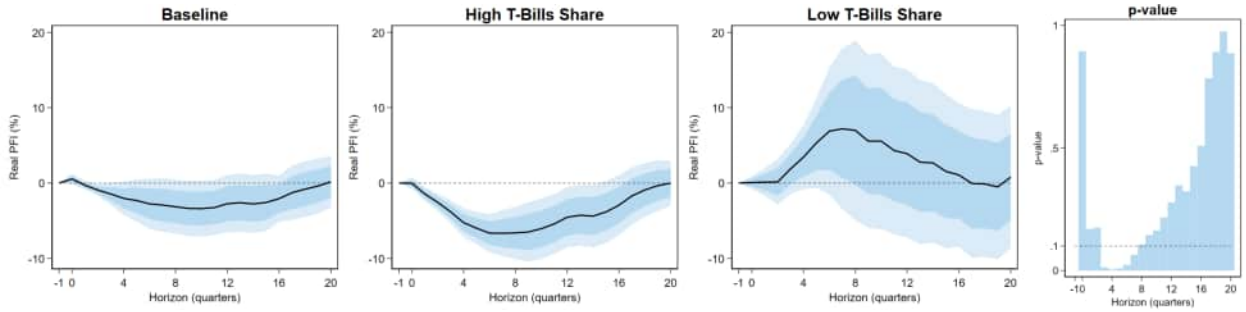
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the Quantitative Easing dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A123: ROBUSTNESS CHECK, CONTROLLING FOR QE (DUMMY).
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

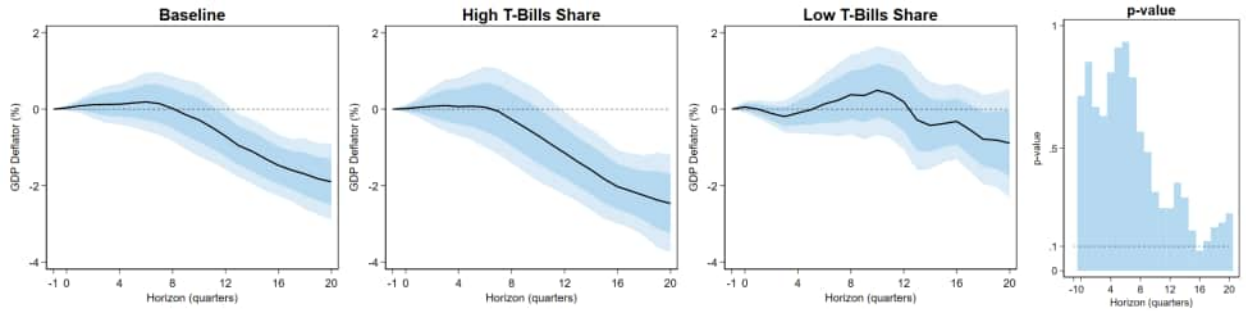
Panel A. Real GDP



Panel B. Real Private Fixed Investments



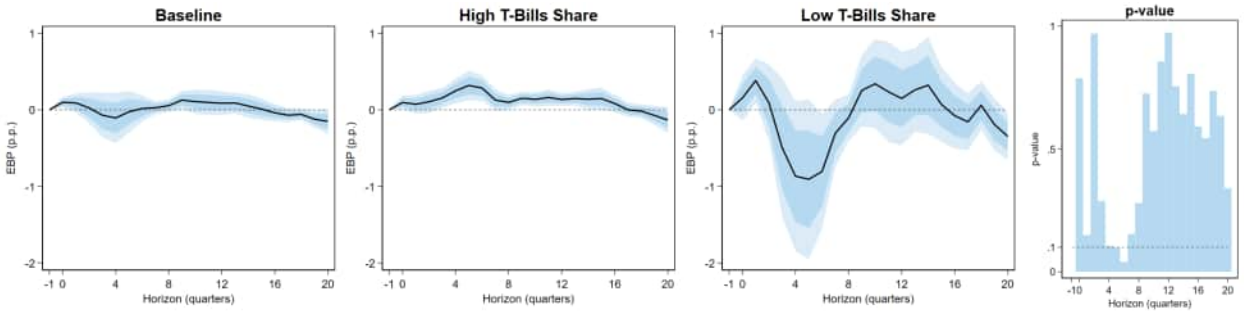
Panel C. GDP Deflator



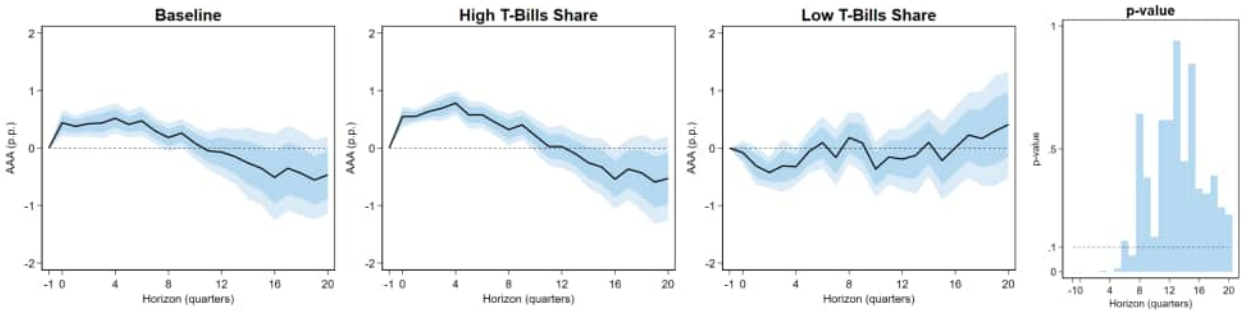
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the Quantitative Easing dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A124: ROBUSTNESS CHECK, CONTROLLING FOR QE (DUMMY).
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

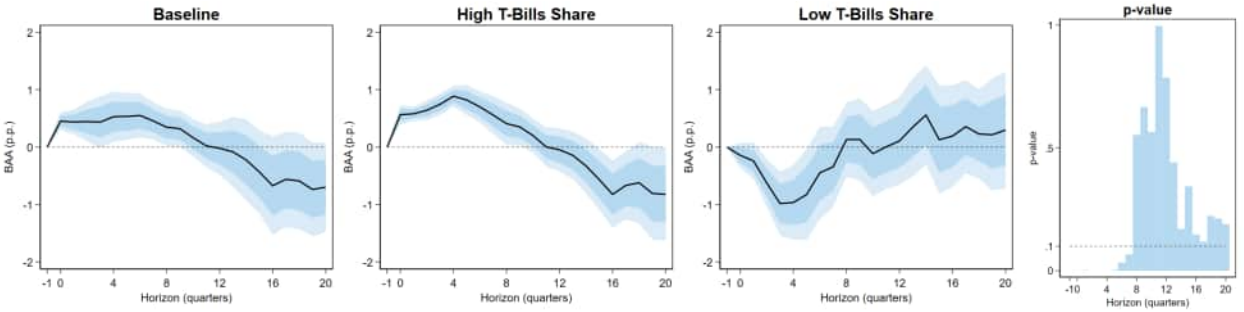
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



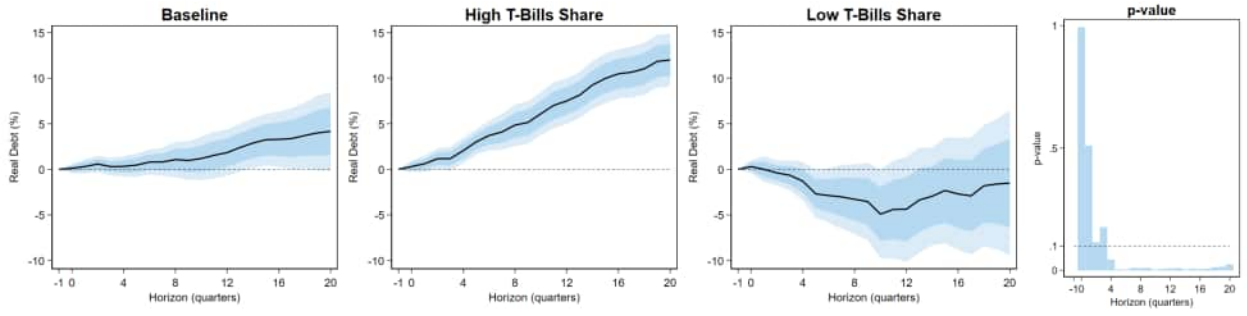
Panel C. Baa Corporate Bond Yield



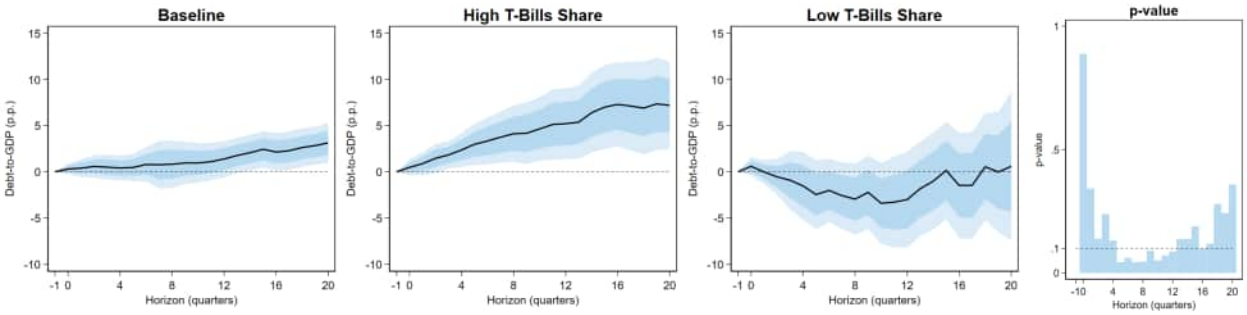
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the Quantitative Easing dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A125: ROBUSTNESS CHECK, CONTROLLING FOR QE (DUMMY, HIGH-FREQ. SHOCK). THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



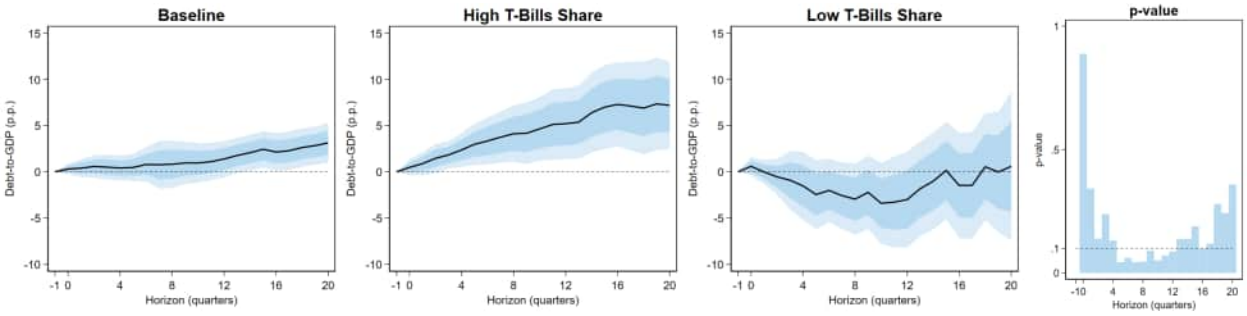
Panel B. Debt-to-GDP



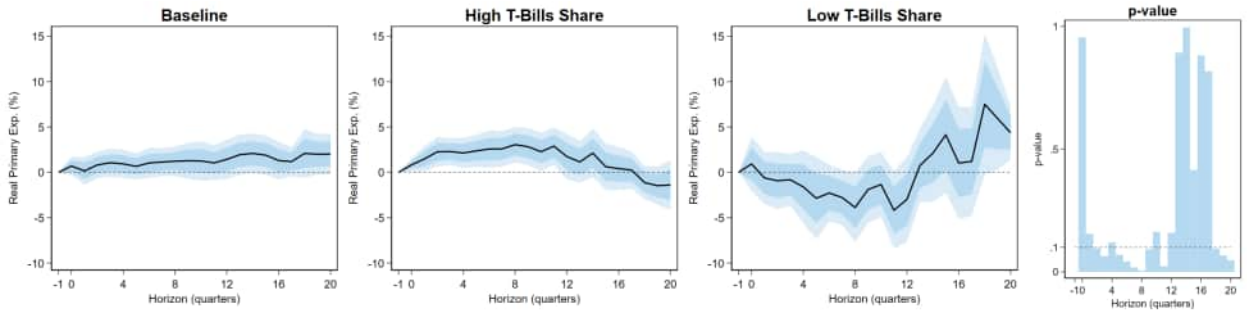
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1963q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the Quantitative Easing dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A126: ROBUSTNESS CHECK, CONTROLLING FOR QE (DUMMY, HIGH-FREQ. SHOCK). THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

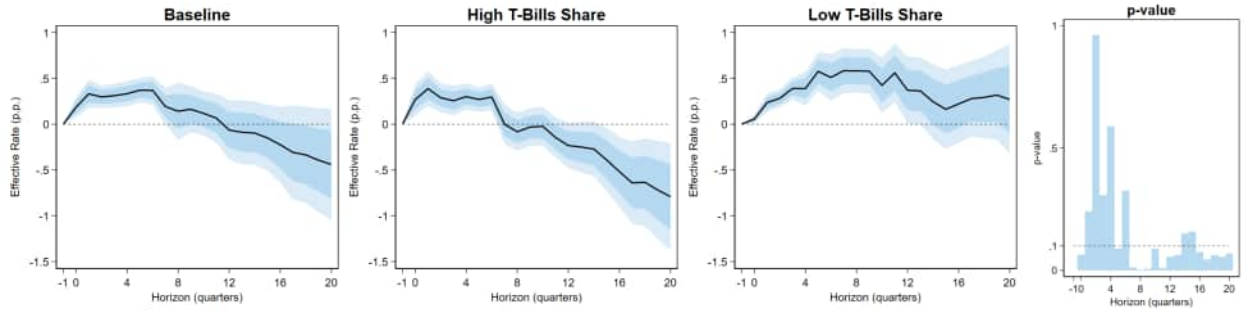
Panel A. Real Revenue



Panel B. Real Primary Expenditure



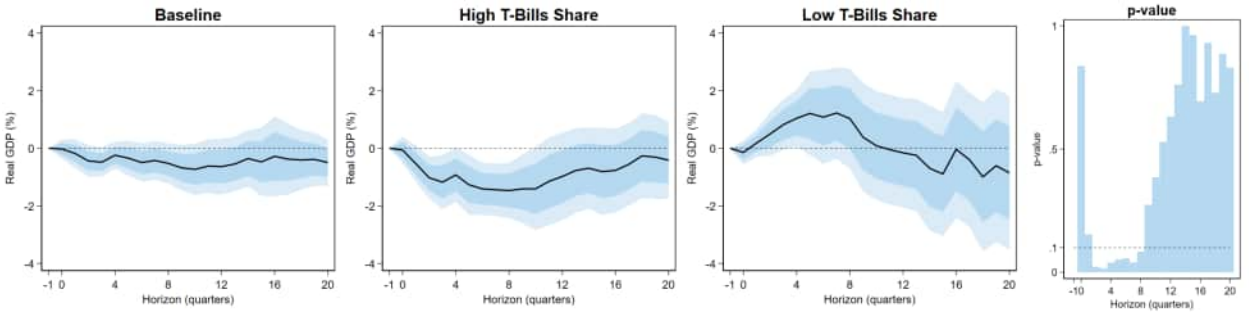
Panel C. Effective Interest Rate



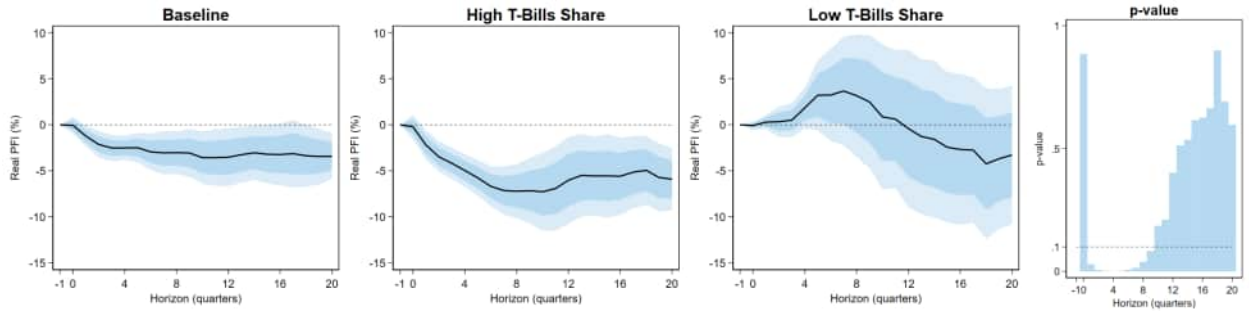
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1963q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the Quantitative Easing dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A127: ROBUSTNESS CHECK, CONTROLLING FOR QE (DUMMY, HIGH-FREQ. SHOCK). THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

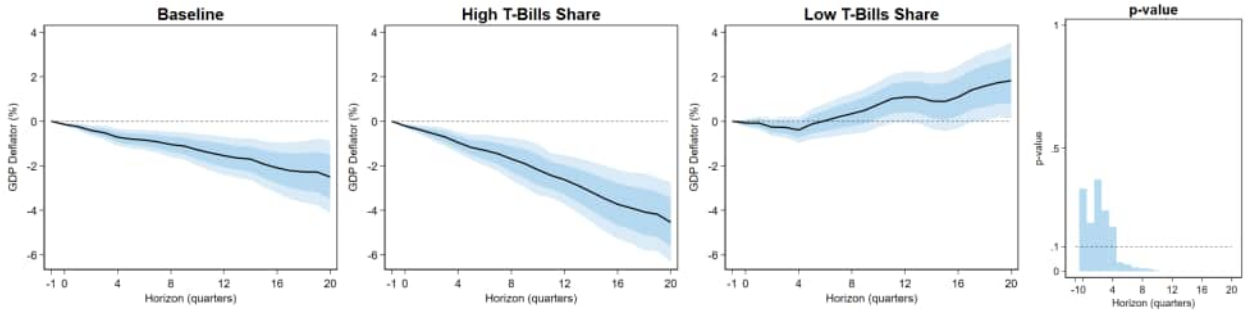
Panel A. Real GDP



Panel B. Real Private Fixed Investments



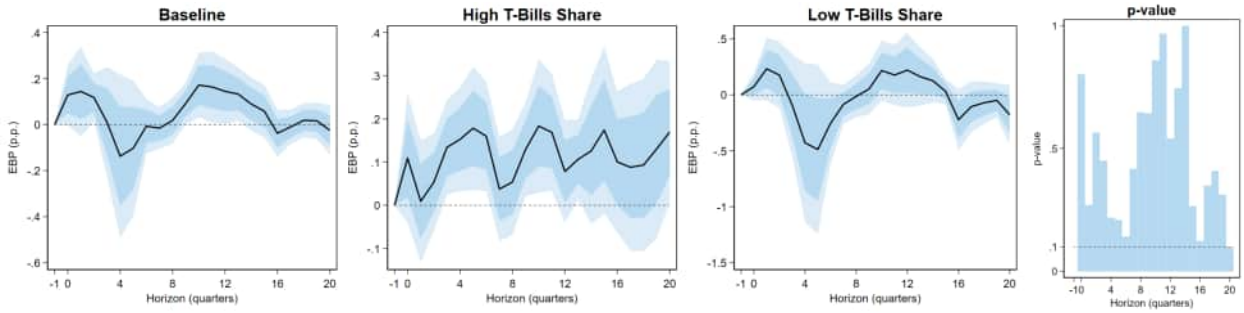
Panel C. GDP Deflator



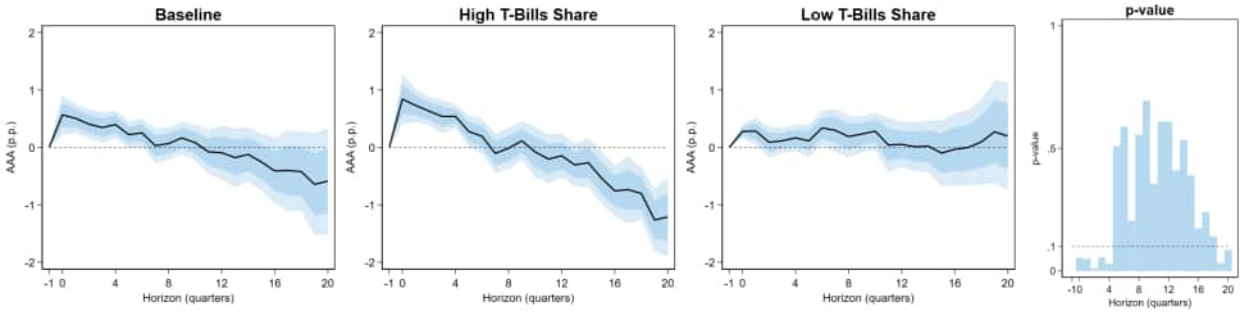
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the Quantitative Easing dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A128: ROBUSTNESS CHECK, CONTROLLING FOR QE (DUMMY, HIGH-FREQ. SHOCK). THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

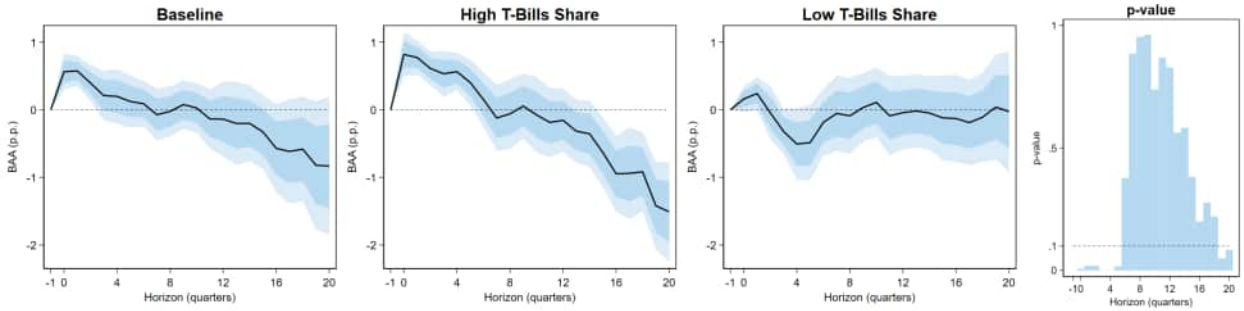
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



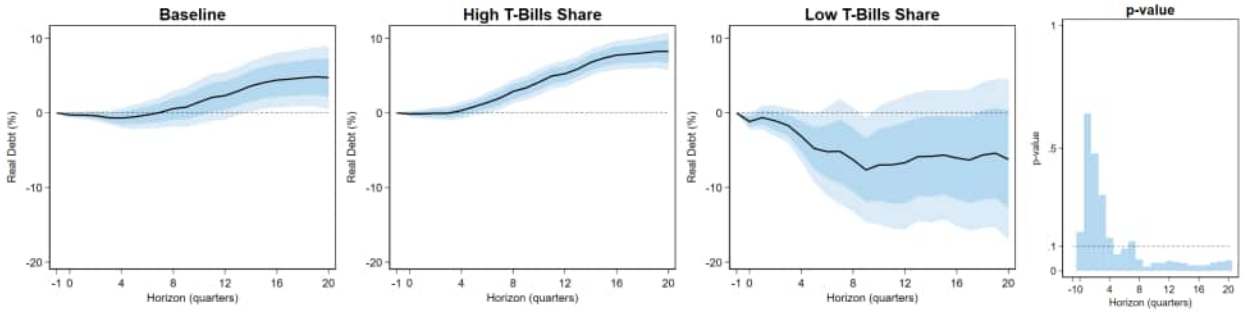
Panel C. Baa Corporate Bond Yield



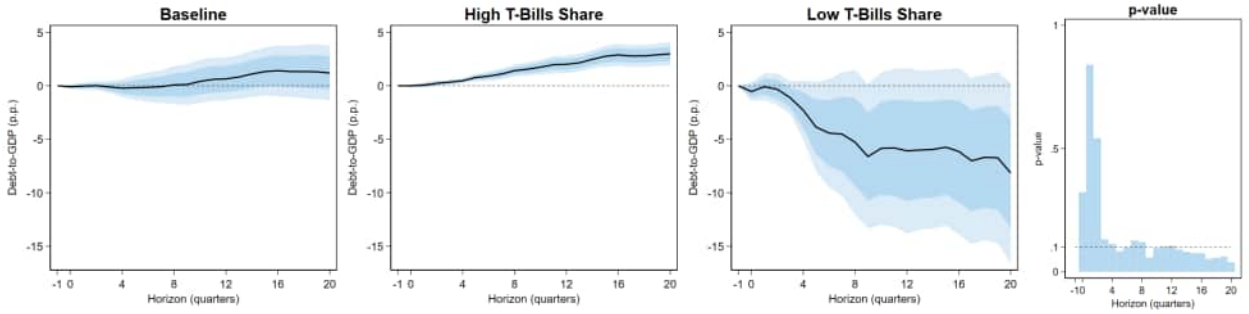
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1963q1-2025q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2025q3), using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t (D_{t-1} - \bar{D}) + \iota^{(h)} (D_{t-1} - \bar{D}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and D_{t-1} is the Quantitative Easing dummy. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A129: ROBUSTNESS CHECK, CONTROLLING FOR BUDGET BALANCE.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



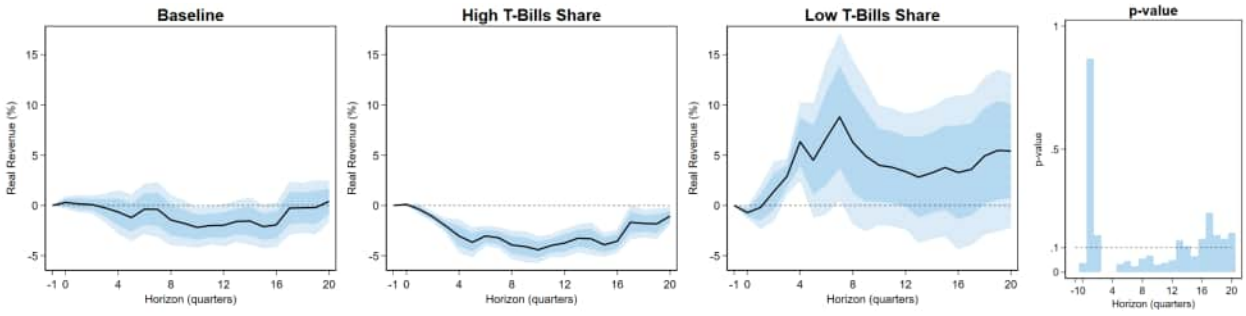
Panel B. Debt-to-GDP



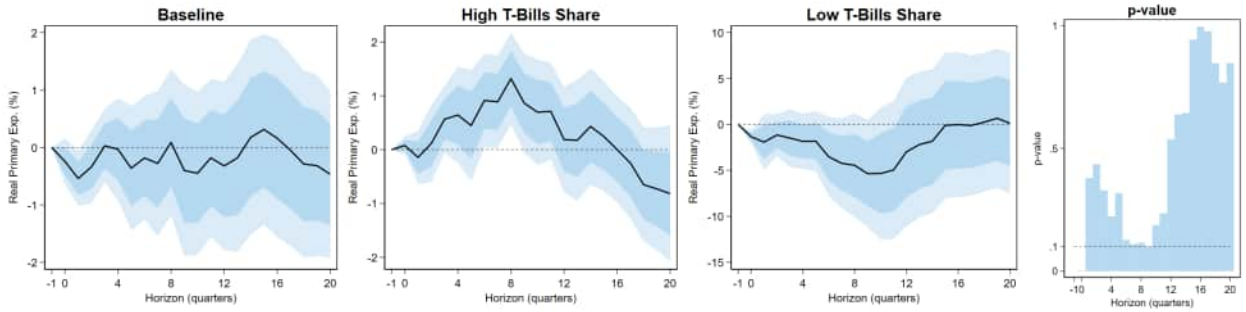
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial federal funds rate level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A130: ROBUSTNESS CHECK, CONTROLLING FOR BUDGET BALANCE.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

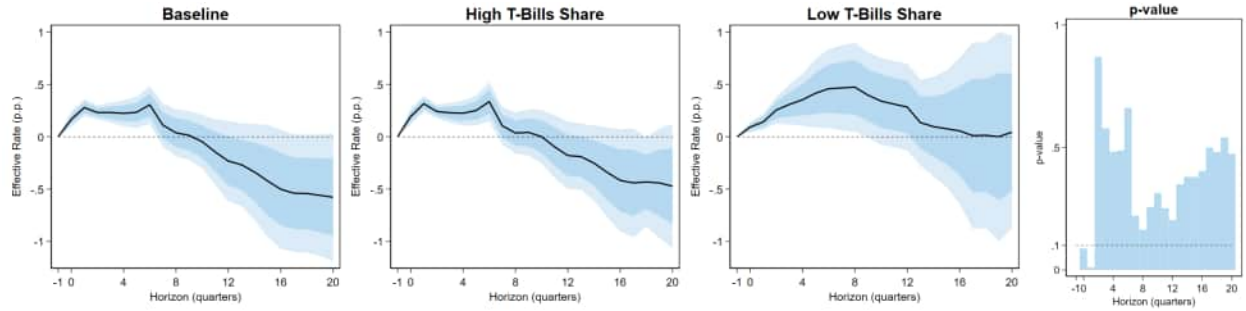
Panel A. Real Revenue



Panel B. Real Primary Expenditure



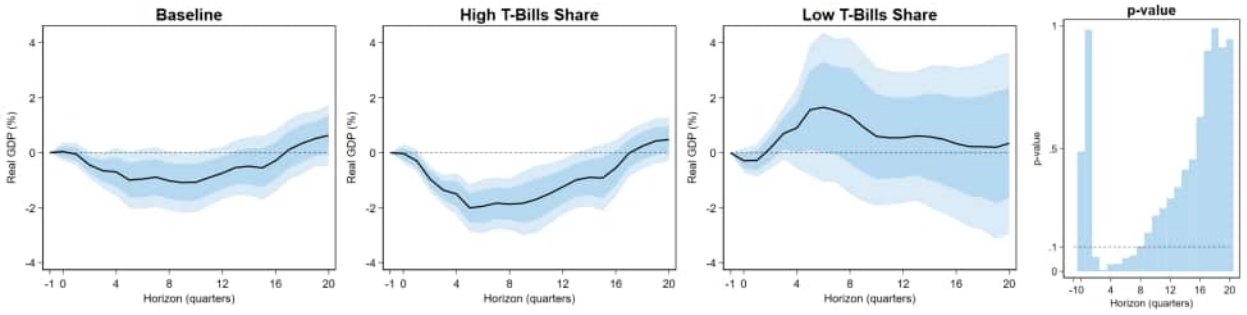
Panel C. Effective Interest Rate



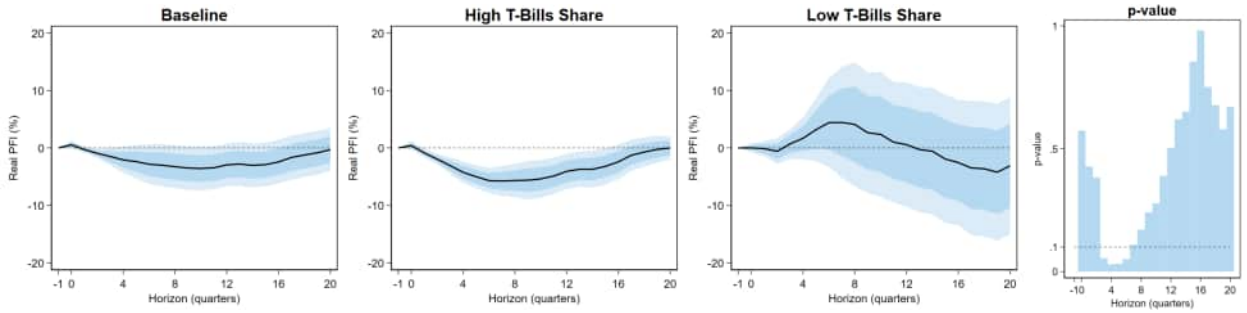
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial federal funds rate level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A131: ROBUSTNESS CHECK, CONTROLLING FOR BUDGET BALANCE.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

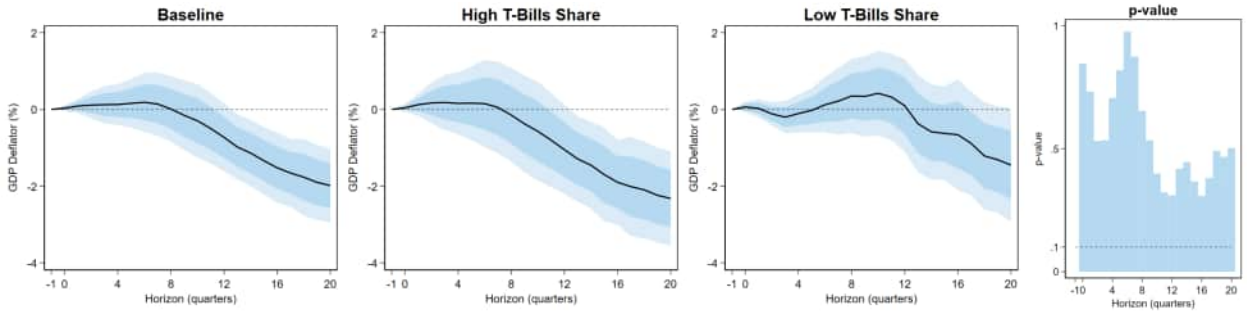
Panel A. Real GDP



Panel B. Real Private Fixed Investments



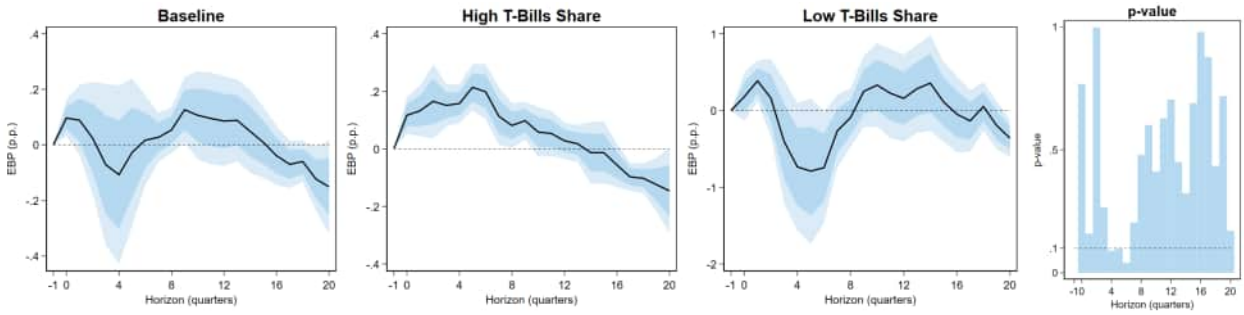
Panel C. GDP Deflator



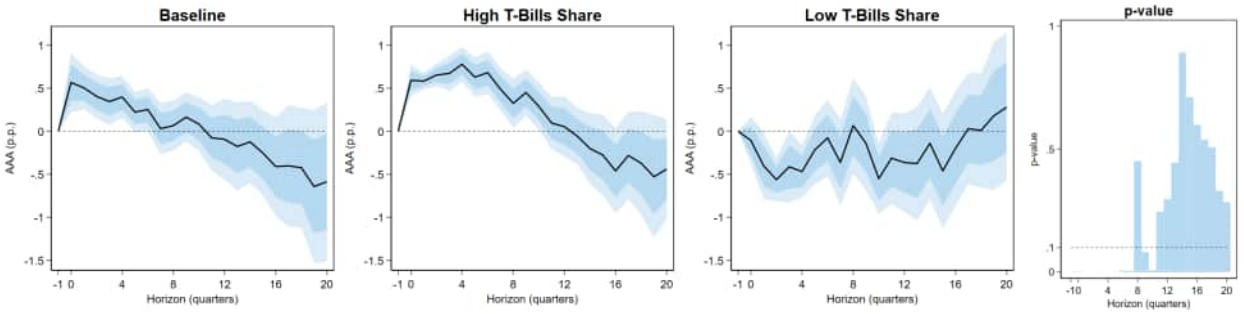
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t \big(F(r_{t-1} - \overline{F(r_{t-1})}\big) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial federal funds rate level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A132: ROBUSTNESS CHECK, CONTROLLING FOR BUDGET BALANCE.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

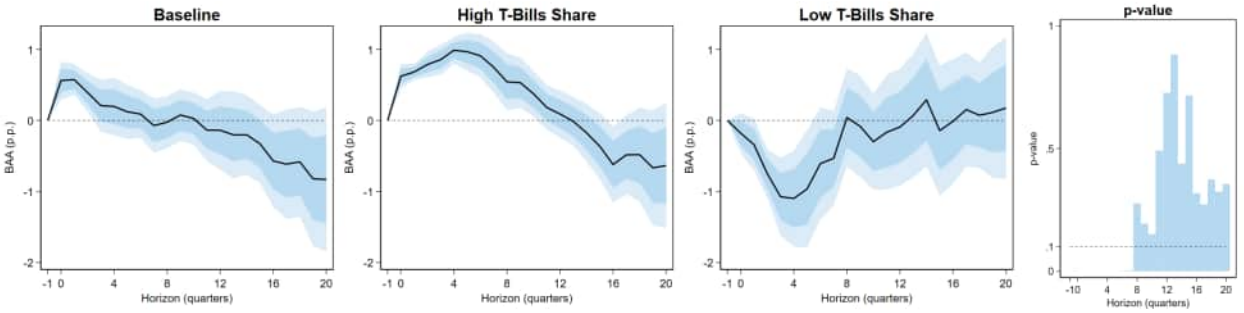
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



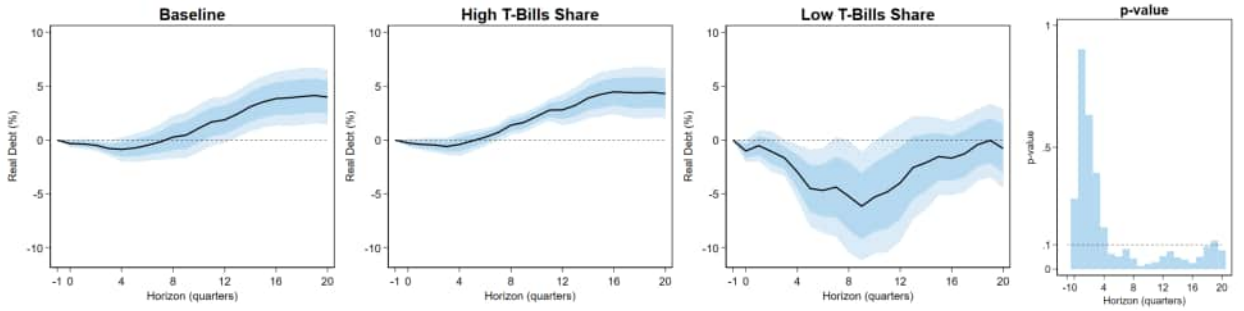
Panel C. Baa Corporate Bond Yield



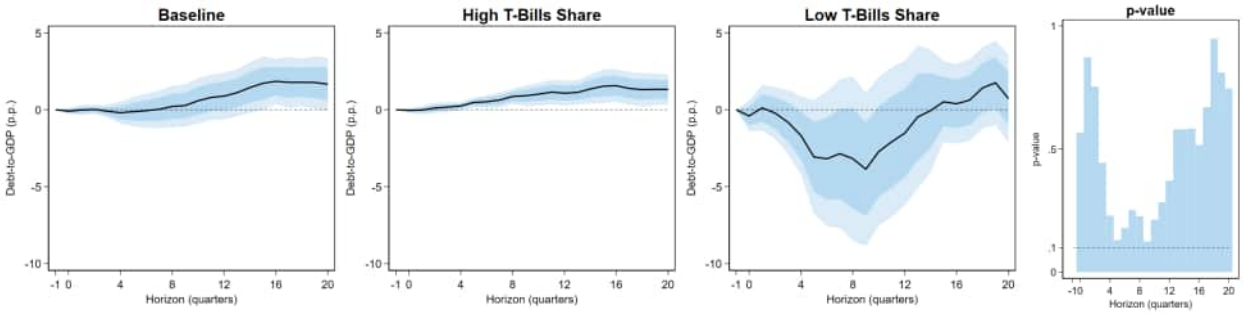
Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell}^{\prime(h)} W_{t-\ell} + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \psi^{(h)} MPS_t\left(F(r_{t-1} - \overline{F(r_{t-1})}\right) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime and $F(r_{t-1})$ is an indicator of the initial federal funds rate level. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A133: ROBUSTNESS CHECK, CONTROLLING FOR TIME TREND.
THE EFFECTS OF MONETARY POLICY SHOCKS ON DEBT DYNAMICS

Panel A. Real Debt



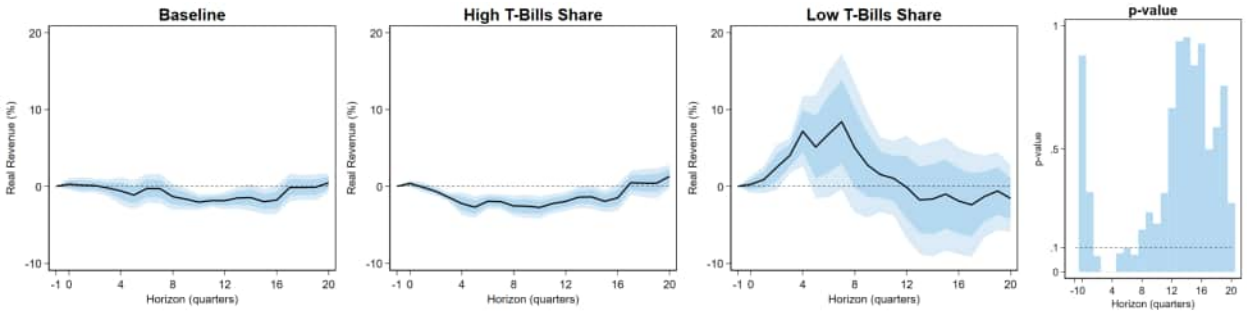
Panel B. Debt-to-GDP



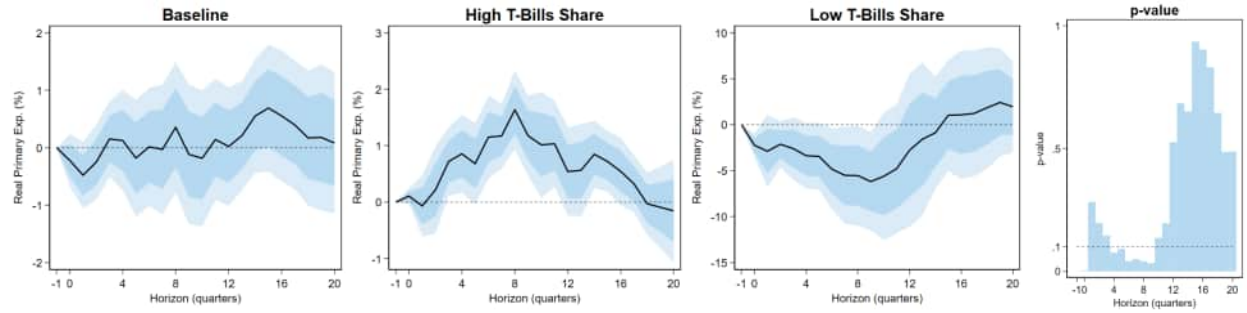
Note: The figure plots the impulse responses of Real Debt (Panel A) and Debt-to-GDP (Panel B) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \gamma^{(h)}(t-\bar{t})_t + \iota^{(h)}(t^2 - \bar{t^2})_t + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \gamma^{(h)}(t-\bar{t})_t + \iota^{(h)}(t^2 - \bar{t^2})_t + \gamma_{\text{high}}^{(h)}(t-\bar{t})_t F(z_{t-1}) + \iota_{\text{high}}^{(h)}(t^2 - \bar{t^2})_t F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A134: ROBUSTNESS CHECK, CONTROLLING FOR TIME TREND.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FISCAL DYNAMICS

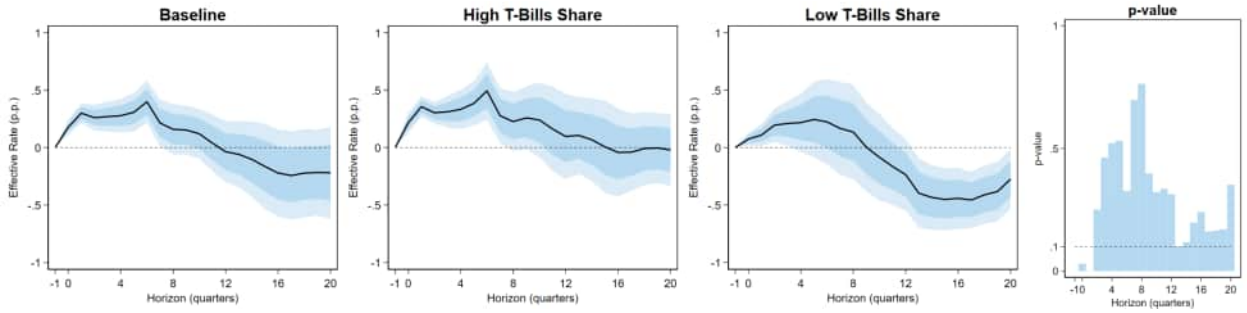
Panel A. Real Revenue



Panel B. Real Primary Expenditure



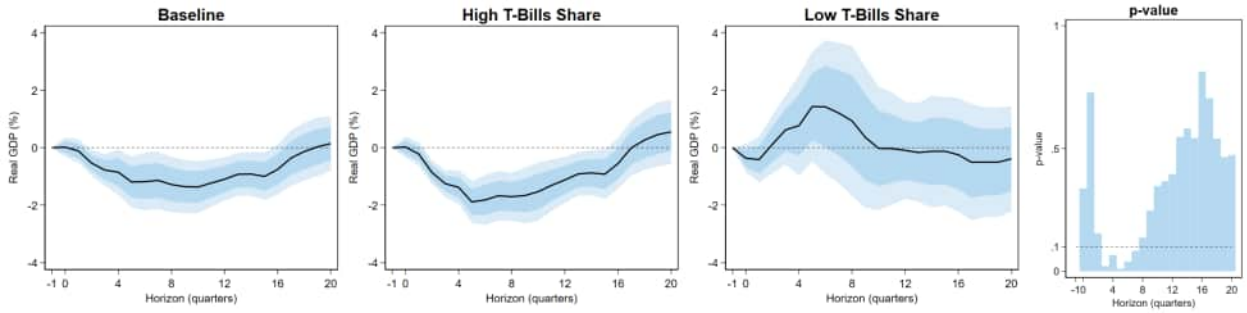
Panel C. Effective Interest Rate



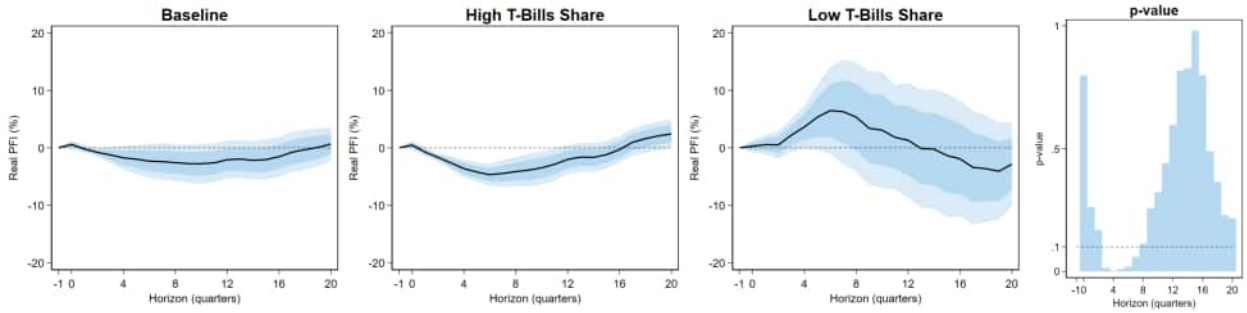
Note: The figure plots the impulse responses of Real Revenue (Panel A), Real Primary Expenditure (Panel B) and the effective interest rate (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \gamma^{(h)}(t-\bar{t})_t + \iota^{(h)}(t^2 - \bar{t^2})_t + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \big(1 - F(z_{t-1})\big)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \gamma^{(h)}(t-\bar{t})_t + \iota^{(h)}(t^2 - \bar{t^2})_t + \gamma_{\text{high}}^{(h)}(t-\bar{t})_t F(z_{t-1}) + \iota_{\text{high}}^{(h)}(t^2 - \bar{t^2})_t F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A135: ROBUSTNESS CHECK, CONTROLLING FOR TIME TREND.
THE EFFECTS OF MONETARY POLICY SHOCKS ON ECONOMIC ACTIVITY

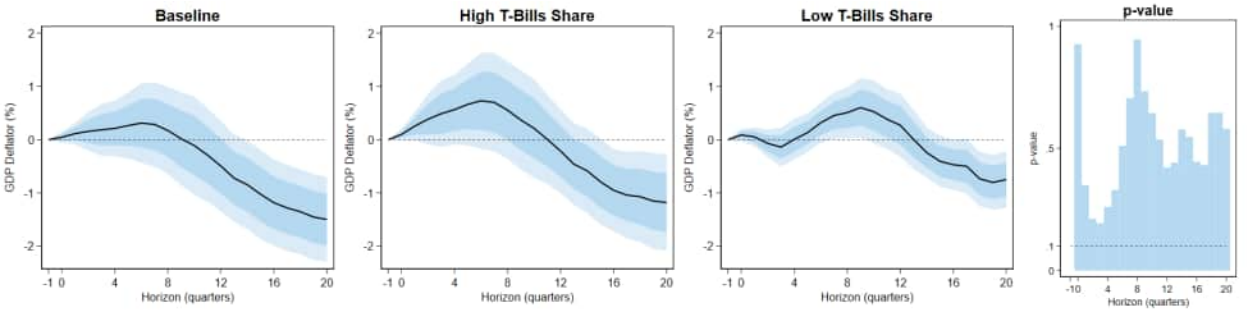
Panel A. Real GDP



Panel B. Real Private Fixed Investments



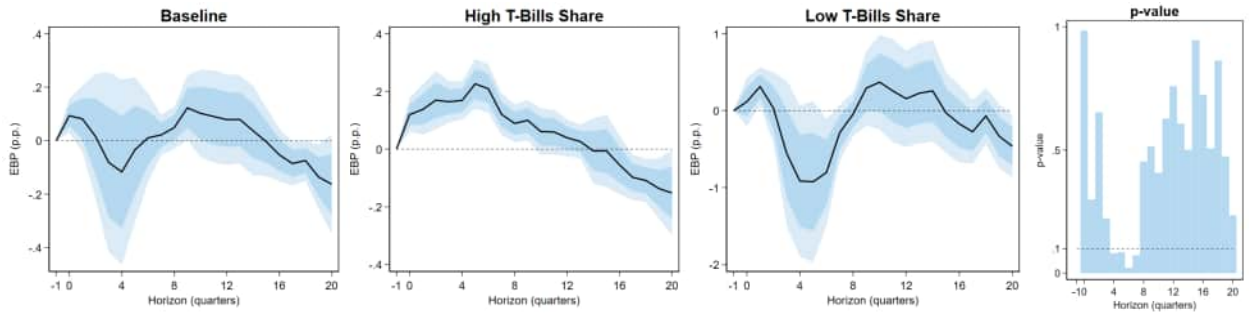
Panel C. GDP Deflator



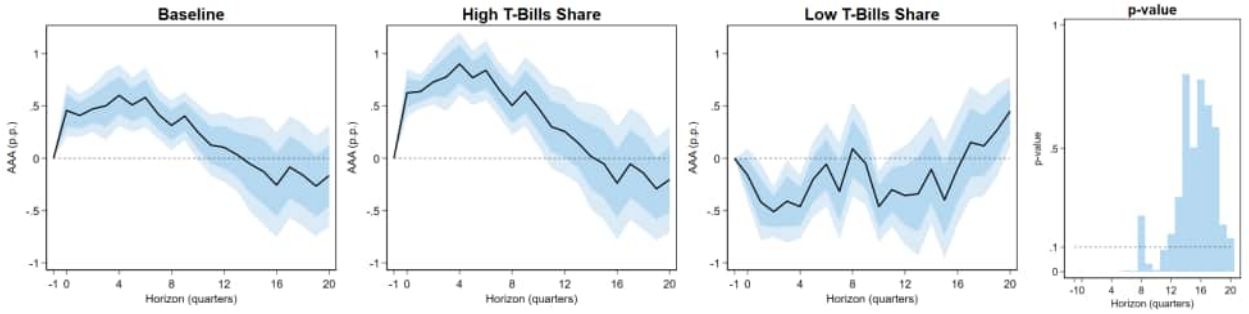
Note: The figure plots the impulse responses of Real GDP (Panel A), Real Private Investments (Panel B) and GDP Deflator (Panel C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \gamma^{(h)}(t-\bar{t})_t + \iota^{(h)}(t^2 - \bar{t^2})_t + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\Big(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\Big) + \big(1 - F(z_{t-1})\big)\Big(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\Big) + \delta^{(h)} F(z_{t-1}) + \gamma^{(h)}(t-\bar{t})_t + \iota^{(h)}(t^2 - \bar{t^2})_t + \gamma_{\text{high}}^{(h)}(t-\bar{t})_t F(z_{t-1}) + \iota_{\text{high}}^{(h)}(t^2 - \bar{t^2})_t F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

FIGURE A136: ROBUSTNESS CHECK, CONTROLLING FOR TIME TREND.
THE EFFECTS OF MONETARY POLICY SHOCKS ON FINANCIAL CONDITIONS

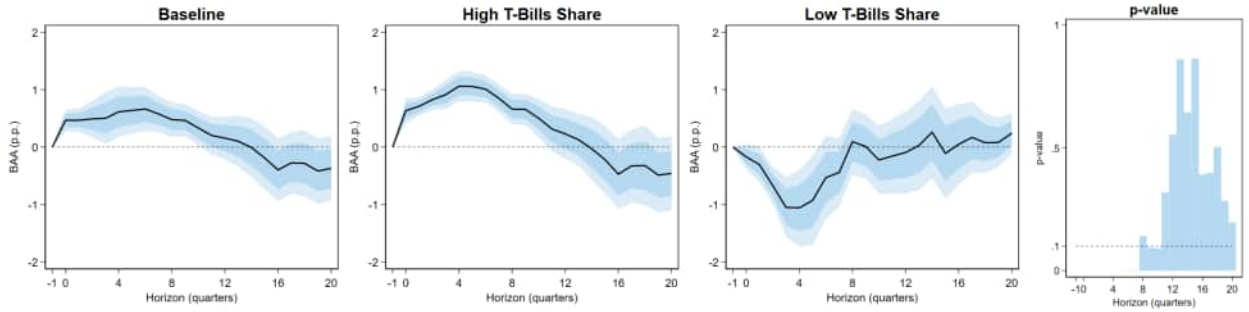
Panel A. Excess Bond Premium



Panel B. Aaa Corporate Bond Yield



Panel C. Baa Corporate Bond Yield



Note: The figure plots the impulse responses of EBP (Panel A), Aaa and Baa Corporate Bond Yields (Panel B and C) to a 100 bps monetary policy tightening shock. In Column 1, the solid black line shows the response in a state-independent model, while darker and lighter shaded areas are, respectively, 68% and 90% Newey-West HAC confidence intervals. Estimates are obtained via OLS on a sample of U.S. data (1969q1-2019q4), using local projections: $y_{t+h} = \alpha^{(h)} + \beta^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_\ell^{\prime(h)} W_{t-\ell} + \gamma^{(h)}(t-\bar{t})_t + \iota^{(h)}(t^2 - \bar{t^2})_t + \epsilon_{t+h}^{(h)}$, where y_{t+h} is the variable of interest h quarters after the shock, MPS_t is the monetary policy shock realized at time t, and $W_{t-\ell}$ is a vector of control variables with four lags ℓ of dependent variable and shock. Columns 2-3 show the impulse response in a state-dependent model, where the responses differ across high- and low-T-bill-share regimes. Estimates are obtained via OLS on the same sample, using local projections: $y_{t+h} = \alpha^{(h)} + F(z_{t-1})\left(\beta_{\text{high}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{high}}^{(h)} W_{t-\ell}\right) + \left(1 - F(z_{t-1})\right)\left(\beta_{\text{low}}^{(h)} MPS_t + \sum_{\ell=1}^{4} \theta_{\ell,\text{low}}^{(h)} W_{t-\ell}\right) + \delta^{(h)} F(z_{t-1}) + \gamma^{(h)}(t-\bar{t})_t + \iota^{(h)}(t^2 - \bar{t^2})_t + \gamma_{\text{high}}^{(h)}(t-\bar{t})_t F(z_{t-1}) + \iota_{\text{high}}^{(h)}(t^2 - \bar{t^2})_t F(z_{t-1}) + \varepsilon_{t+h}^{(h)}$, where $F(z_{t-1})$ is an indicator of the regime. The fourth column shows p-values testing the hypothesis $\beta_{\text{high}}^{(h)} = \beta_{\text{low}}^{(h)}$.

A Nonlinear equilibrium system

A.1 Households, wages, labor

$$\Lambda_t(1+\tau^c_t) = \tfrac{1}{C_t - bC_{t-1}} - \beta b\,\mathbb{E}_t\big[\tfrac{1}{C_{t+1}-bC_t}\big] \tag{A.1}$$

$$M_{t,t+1} = \beta\Lambda_{t+1}/\Lambda_t \tag{A.2}$$

$$1 = \mathbb{E}_t\big[M_{t,t+1}R_t/\Pi_{t+1}\big] \tag{A.3}$$

$$R^{LM}_{t+1} = (1+\kappa q^L_{t+1})/q^L_t \tag{A.4}$$

$$\zeta^L_t = \mathbb{E}_t\big[M_{t,t+1}R^{LM}_{t+1}/\Pi_{t+1}\big] \tag{A.5}$$

$$\zeta^L_t = \bar\zeta^L\big(\tfrac{q^L_t B^L_t/Y_t}{\bar q^L \bar B^L/\bar Y}\big)^{\eta_d} \tag{A.6}$$

$$\textstyle\sum_{k\ge0}(\beta\theta_w)^k\mathbb{E}_t\big\{H_{t+k|t}\big[\Lambda_{t+k}(1-\tau^\ell_{t+k})\frac{W^*_t}{P_{t+k}} - \frac{\epsilon_w}{\epsilon_w-1}\chi_H H^{1/\eta}_{t+k|t}\big]\big\} = 0, \quad H_{t+k|t} = \big(\tfrac{W^*_t}{W_{t+k}}\big)^{-\epsilon_w}H_{t+k} \tag{A.7}$$

$$W^{1-\epsilon_w}_t = (1-\theta_w)(W^*_t)^{1-\epsilon_w} + \theta_w W^{1-\epsilon_w}_{t-1} \tag{A.8}$$

$$\Pi^w_t = W_t/W_{t-1}, \qquad w_t = w_{t-1}\Pi^w_t/\Pi_t \tag{A.9}$$

A.2 Government bond and the carrying basis

$$R^L_t = \kappa + 1/q^L_t \tag{A.10}$$

$$A^{LB}_t = \tfrac{\kappa}{\Pi_t}A^{LB}_{t-1} + q^L_t\big(B^{LF}_t - \tfrac{\kappa}{\Pi_t}B^{LF}_{t-1}\big) \tag{A.11}$$

$$R^{LB}_t = \kappa + B^{LF}_t/A^{LB}_t \tag{A.12}$$

$$U_t = q^L_t B^{LF}_t - A^{LB}_t \tag{A.13}$$

$$i^{\text{eff}}_t = (1-\kappa)R^L_t + \kappa\, i^{\text{eff}}_{t-1} \tag{A.14}$$

Every unit of $\mathcal{B}_t$ is a claim to one currency unit next period followed by a geometrically decaying tail, so every unit is worth q^L_t whenever it was issued; the stock need not be revalued vintage by vintage. New purchases enter the book at the price paid and therefore carry no gain, so the entire market-to-book gap sits in the legacy. Combining (A.13) and (A.11), $U_t = (\kappa/\Pi_t)(q^L_t B^{LF}_{t-1} - A^{LB}_{t-1})$.

A.3 Financial intermediaries

$$\Omega_t = 1 - \sigma_F + \sigma_F\nu_t \tag{A.15}$$

$$\widetilde{M}_{t,t+1} = M_{t,t+1}\Omega_{t+1} \tag{A.16}$$

$$V_t = \nu_t N^F_t + \xi_t U_t \tag{A.17}$$

$$\mathbb{E}_t\big[\widetilde{M}_{t,t+1}(R^P_{t+1} - R_t)/\Pi_{t+1}\big] = \tfrac{\lambda_t}{1+\lambda_t}\theta \tag{A.18}$$

$$(R^L_t - R_t)\mathbb{E}_t\big[\widetilde{M}_{t,t+1}/\Pi_{t+1}\big] + \Xi_t = \tfrac{\lambda_t}{1+\lambda_t}\theta\Delta \tag{A.19}$$

$$\Xi_t = \tfrac{\sigma_F\kappa}{q^L_t}\mathbb{E}_t\big[M_{t,t+1}\xi_{t+1}(q^L_{t+1} - q^L_t)/\Pi_{t+1}\big] \tag{A.20}$$

$$\phi_t = \mathbb{E}_t\big[\widetilde{M}_{t,t+1}R_t/\Pi_{t+1}\big]\big/\big(\theta - \mathbb{E}_t\big[\widetilde{M}_{t,t+1}(R^P_{t+1} - R_t)/\Pi_{t+1}\big]\big) \tag{A.21}$$

$$\nu_t = \theta\phi_t \tag{A.22}$$

$$X^P_t + \Delta A^{LB}_t = \phi_t N^F_t + \big(\tfrac{\xi_t}{\theta} - \Delta\big)U_t \tag{A.23}$$

$$\xi_t = -(1+\lambda_t)\mathbb{E}_t\big[\widetilde{M}_{t,t+1}\tfrac{\kappa - R_t}{\Pi_{t+1}}\big] + (1+\lambda_t)\sigma_F\mathbb{E}_t\big[M_{t,t+1}\tfrac{\kappa}{\Pi_{t+1}}\xi_{t+1}\big] \tag{A.24}$$

$$N^F_t = \tfrac{\sigma_F}{\Pi_t}\big[R^P_t X^P_{t-1} + R^{LB}_{t-1}A^{LB}_{t-1} - R_{t-1}D_{t-1}\big] + X^F \tag{A.25}$$

$$D_t = X^P_t + A^{LB}_t - N^F_t \tag{A.26}$$

Adding ΔU_t to both sides of (A.23) gives $X_t^P + \Delta q_t^L B_t^{LF} = \phi_t N_t^F + (\xi_t/\theta)U_t$: divertible assets are valued at *market* on the left, while the continuation value on the right is built on *book* net worth. That asymmetry is the accounting channel.

A.4 Entrepreneurs

$$G(\bar\psi) = \textstyle\int_0^{\bar\psi} \psi\, dF(\psi), \quad \Gamma(\bar\psi) = \bar\psi[1 - F(\bar\psi)] + G(\bar\psi), \quad \Upsilon(\bar\psi) = \Gamma(\bar\psi) - \mu G(\bar\psi) \tag{A.27}$$

$$L_t = Q_t K_t - N_t \tag{A.28}$$

$$R_{t+1}^k = \big[r_{t+1}^k + (1-\delta)Q_{t+1}\big]/Q_t \tag{A.29}$$

$$R_{t+1}^K = R_{t+1}^k \Pi_{t+1} \tag{A.30}$$

$$\bar\psi_{t+1} R_{t+1}^K Q_t K_t = Z_t L_t \tag{A.31}$$

$$\mathbb{E}_t\big[\Upsilon(\bar\psi_{t+1}) R_{t+1}^K\big] Q_t K_t = \mathbb{E}_t\big[R_{t+1}^P\big] L_t \tag{A.32}$$

$$\mathbb{E}_t R_{t+1}^K / \mathbb{E}_t R_{t+1}^P = \mathcal{P}(Q_t K_t / N_t) \tag{A.33}$$

$$R_{t+1}^P = \Upsilon(\bar\psi_{t+1}) R_{t+1}^K Q_t K_t / L_t \tag{A.34}$$

$$N_t = \gamma\big[(1 - \mu G(\bar\psi_t)) R_t^k Q_{t-1} K_{t-1} - \tfrac{R_t^P}{\Pi_t} L_{t-1}\big] + W_t^e \tag{A.35}$$

$$C_t^E = (1-\gamma)\big[(1 - \mu G(\bar\psi_t)) R_t^k Q_{t-1} K_{t-1} - \tfrac{R_t^P}{\Pi_t} L_{t-1}\big] \tag{A.36}$$

$$\mathcal{M}_t = \mu G(\bar\psi_t) R_t^k Q_{t-1} K_{t-1} \tag{A.37}$$

$$W_t^e = (1-\alpha)(1-\omega) Y_t / X_t \tag{A.38}$$

A.5 Production, capital, price setting

$$Y_t = A_t \tilde H_t^{1-\alpha} K_{t-1}^{\alpha}, \quad \tilde H_t = H_t^{\omega} (H_t^e)^{1-\omega}, \qquad H_t^e = 1 \tag{A.39}$$

$$w_t = (1-\alpha)\omega\, Y_t/(X_t H_t), \qquad r_t^k = \alpha Y_t/(X_t K_{t-1}) \tag{A.40}$$

$$x_t = I_t / I_{t-1}, \qquad K_t = (1-\delta)K_{t-1} + [1 - S(x_t)] I_t, \quad S(x) = \tfrac{\phi_s}{2}(x-1)^2 \tag{A.41}$$

$$Q_t\big[1 - S(x_t) - x_t S'(x_t)\big] + \mathbb{E}_t\big[M_{t,t+1} Q_{t+1} S'(x_{t+1}) x_{t+1}^2\big] = 1 \tag{A.42}$$

$$V_t^n = \tfrac{Y_t}{X_t} + \beta\theta_p \mathbb{E}_t\Big[\tfrac{\Lambda_{t+1}}{\Lambda_t} \Pi_{t+1}^{\epsilon_p} V_{t+1}^n\Big] \tag{A.43}$$

$$V_t^d = Y_t + \beta\theta_p \mathbb{E}_t\Big[\tfrac{\Lambda_{t+1}}{\Lambda_t} \Pi_{t+1}^{\epsilon_p - 1} V_{t+1}^d\Big] \tag{A.44}$$

$$P_t^*/P_t = \tfrac{\epsilon_p}{\epsilon_p - 1} V_t^n / V_t^d, \quad 1 = (1-\theta_p)(P_t^*/P_t)^{1-\epsilon_p} + \theta_p \Pi_t^{\epsilon_p - 1} \tag{A.45}$$

A.6 Government, policy, market clearing

$$q_t^L B_t^L + T_t^\ell + T_t^c + T_t^{LS} = G_t + \tfrac{1+\kappa q_t^L}{\Pi_t} B_{t-1}^L \tag{A.46}$$

$$T_t^\ell = \tau_t^\ell w_t H_t, \qquad T_t^c = \tau_t^c C_t \tag{A.47}$$

$$G_t/\bar{G} = (Y_t/\bar{Y})^{-\alpha_G} e^{e_t^g} \tag{A.48}$$

$$\tfrac{\tau_t^\ell}{\bar{\tau}^\ell} = \big(\tfrac{q_{t-1}^L B_{t-1}^L}{\bar{q}^L \bar{B}^L}\big)^{\tau_b^\ell} \big(\tfrac{Y_{t-1}}{\bar{Y}}\big)^{\tau_y^\ell} \tag{A.49}$$

$$\tfrac{T_t^{LS}}{\bar{T}^{LS}} = \big(\tfrac{q_{t-1}^L B_{t-1}^L}{\bar{q}^L \bar{B}^L}\big)^{\tau_b} \big(\tfrac{Y_{t-1}}{\bar{Y}}\big)^{\tau_y} \tag{A.50}$$

$$\tfrac{R_t}{\bar{R}} = \big(\tfrac{R_{t-1}}{\bar{R}}\big)^{\rho_r} \big[\big(\tfrac{\Pi_t}{\bar{\Pi}}\big)^{\theta_\pi} \big(\bar{X}/X_t\big)^{\theta_y}\big]^{1-\rho_r} e^{\varepsilon_t^m} \tag{A.51}$$

$$B_t^L = B_t^{LF} + B_t^{LH}, \qquad X_t^P = L_t \tag{A.52}$$

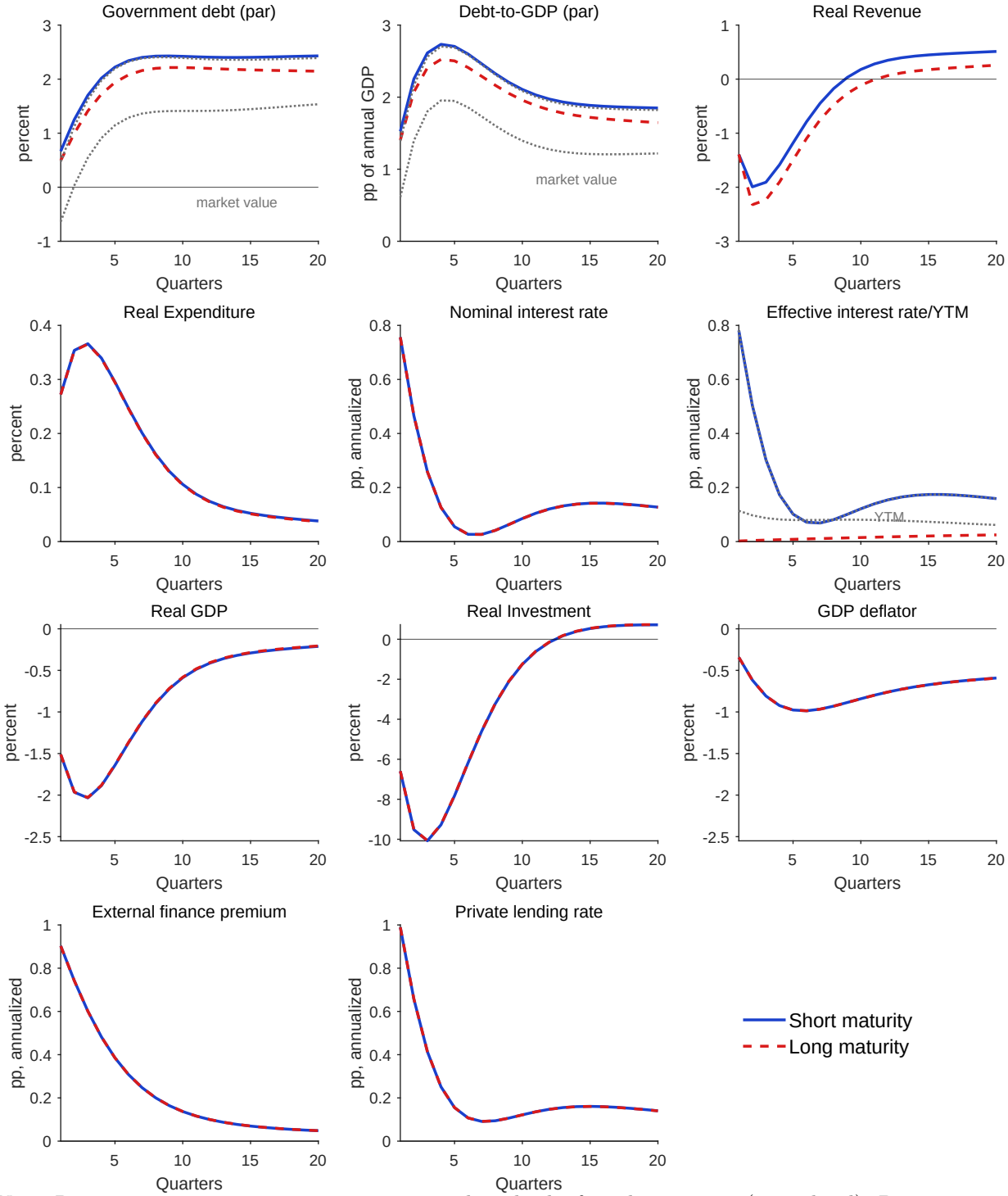
$$Y_t = C_t + C_t^E + I_t + G_t + \mathcal{M}_t \tag{A.53}$$

$$\ln A_t = (1-\rho_a)\ln \bar{A} + \rho_a \ln A_{t-1} + e_t^a \tag{A.54}$$

B Additional results from theoretical model

B.1 No Gertler and Karadi (2013)'s Friction

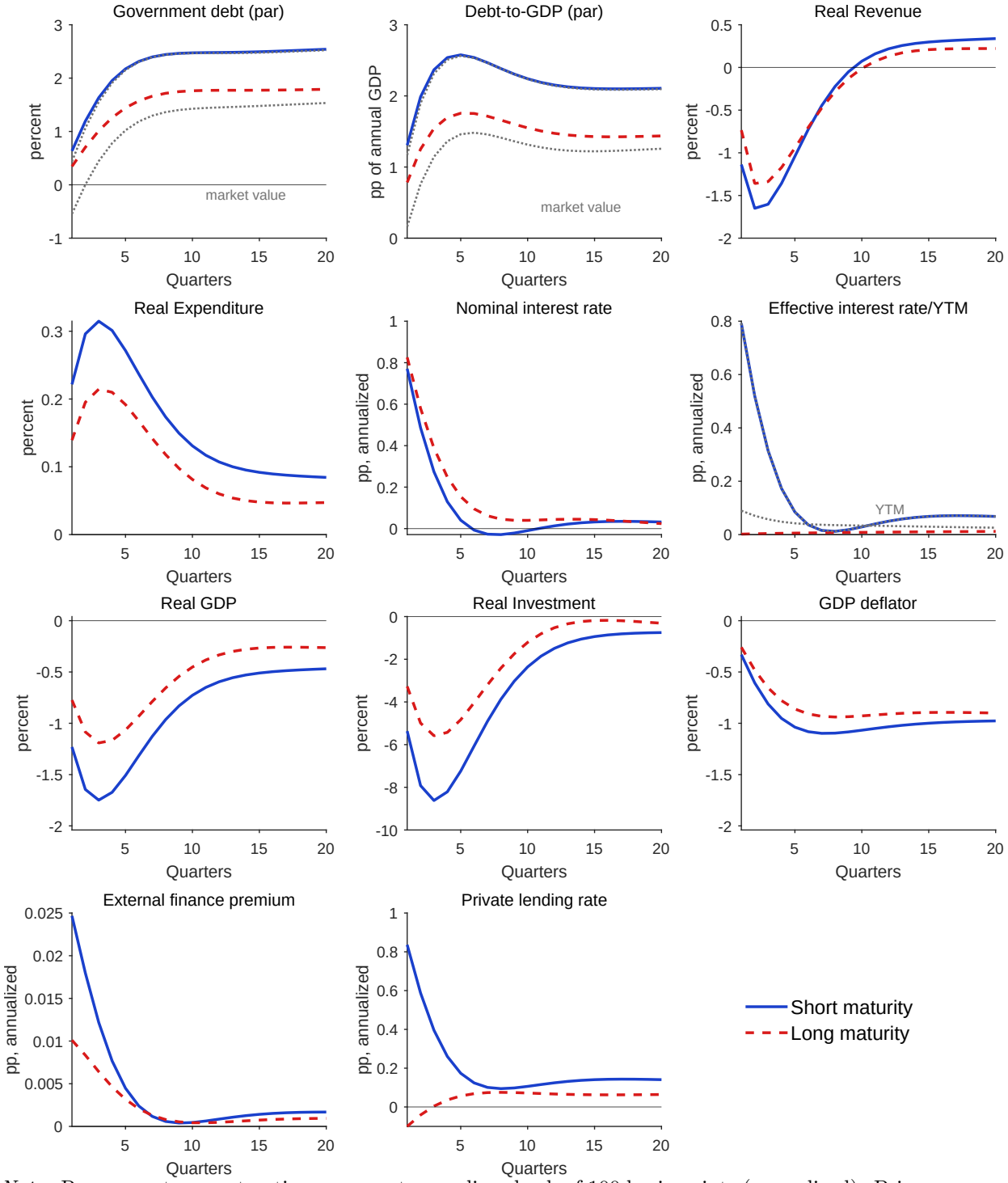
FIGURE A137: EFFECTS OF MONETARY POLICY SHOCKS: NO GK FRICTION



Note: Responses to a contractionary monetary policy shock of 100 basis points (annualized). Price response is the cumulated response of inflation. Debt is at par value; the dotted line is market value. YTM: Yield to maturity.

B.2 No Bernanke et al. (1999)'s Friction

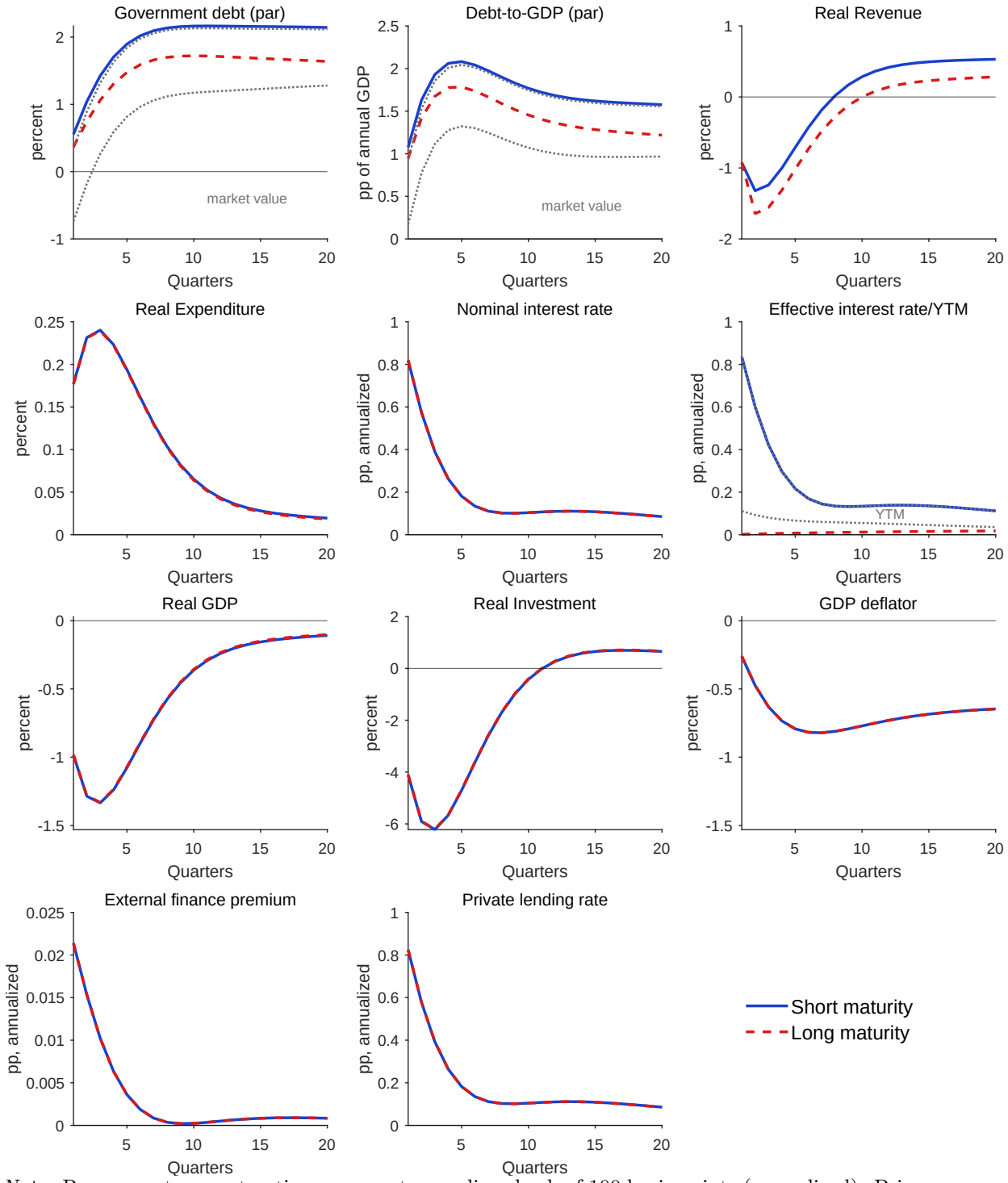
FIGURE A138: EFFECTS OF MONETARY POLICY SHOCKS: NO BGG FRICTION



Note: Responses to a contractionary monetary policy shock of 100 basis points (annualized). Price response is the cumulated response of inflation. Debt is at par value; the dotted line is market value. YTM: Yield to maturity.

B.3 No Frictions

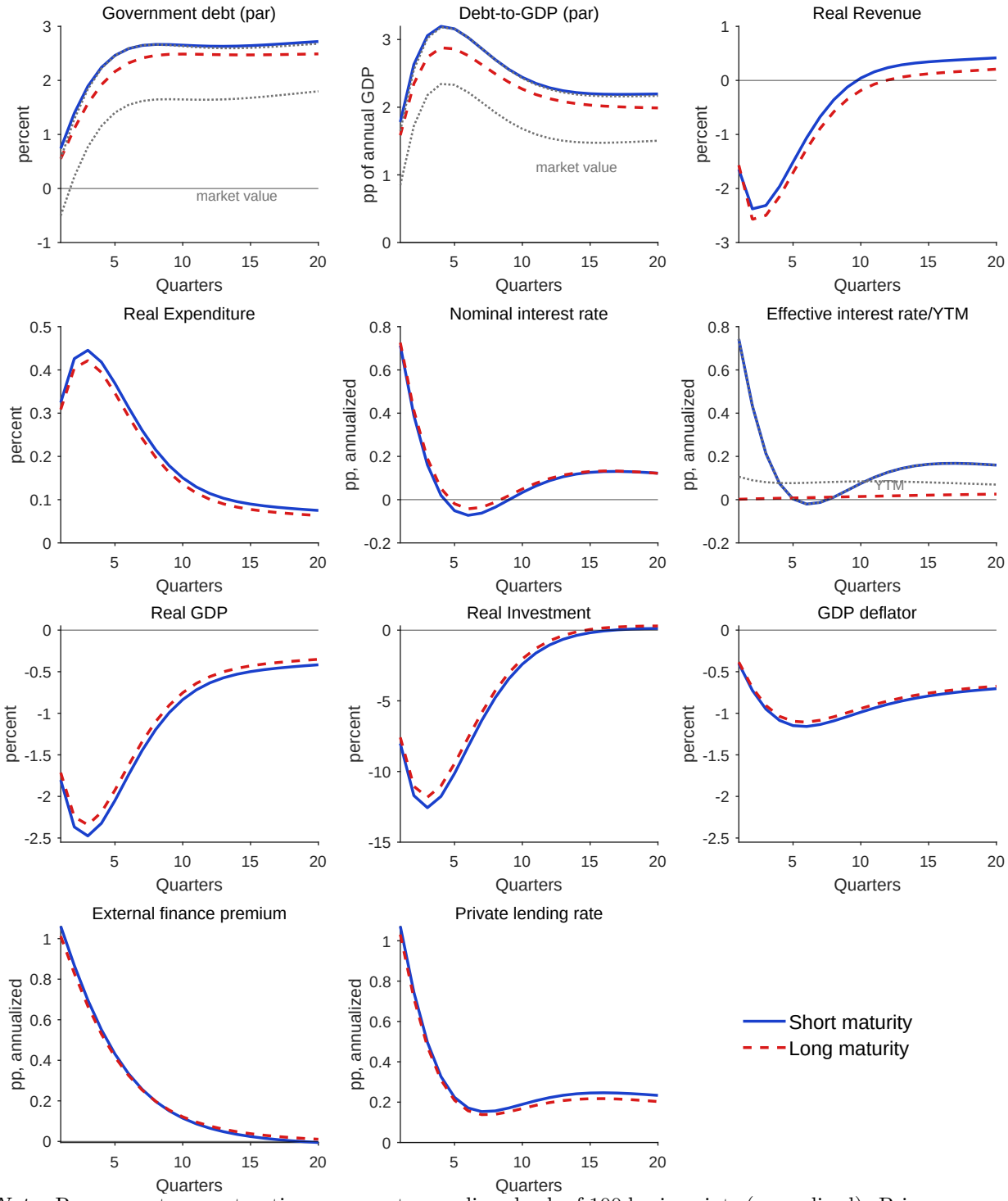
FIGURE A139: EFFECTS OF MONETARY POLICY SHOCKS: NO FRICTIONS



Note: Responses to a contractionary monetary policy shock of 100 basis points (annualized). Price response is the cumulated response of inflation. Debt is at par value; the dotted line is market value. YTM: Yield to maturity.

B.4 Mark-to-Market Convention

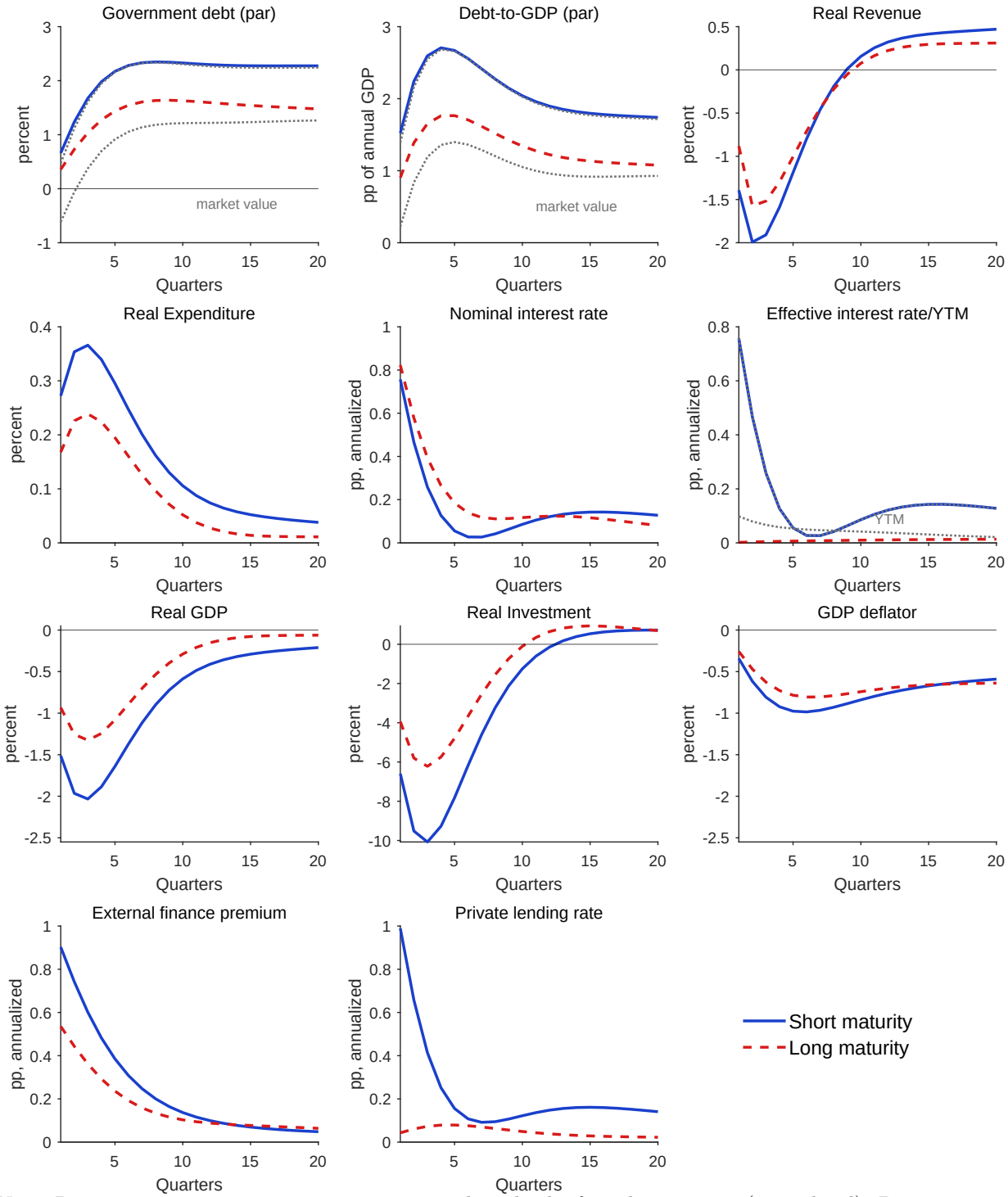
FIGURE A140: EFFECTS OF MONETARY POLICY SHOCKS: MARK-TO-MARKET



Note: Responses to a contractionary monetary policy shock of 100 basis points (annualized). Price response is the cumulated response of inflation. Debt is at par value; the dotted line is market value. YTM: Yield to maturity.

B.5 No Debt-Supply Wedge

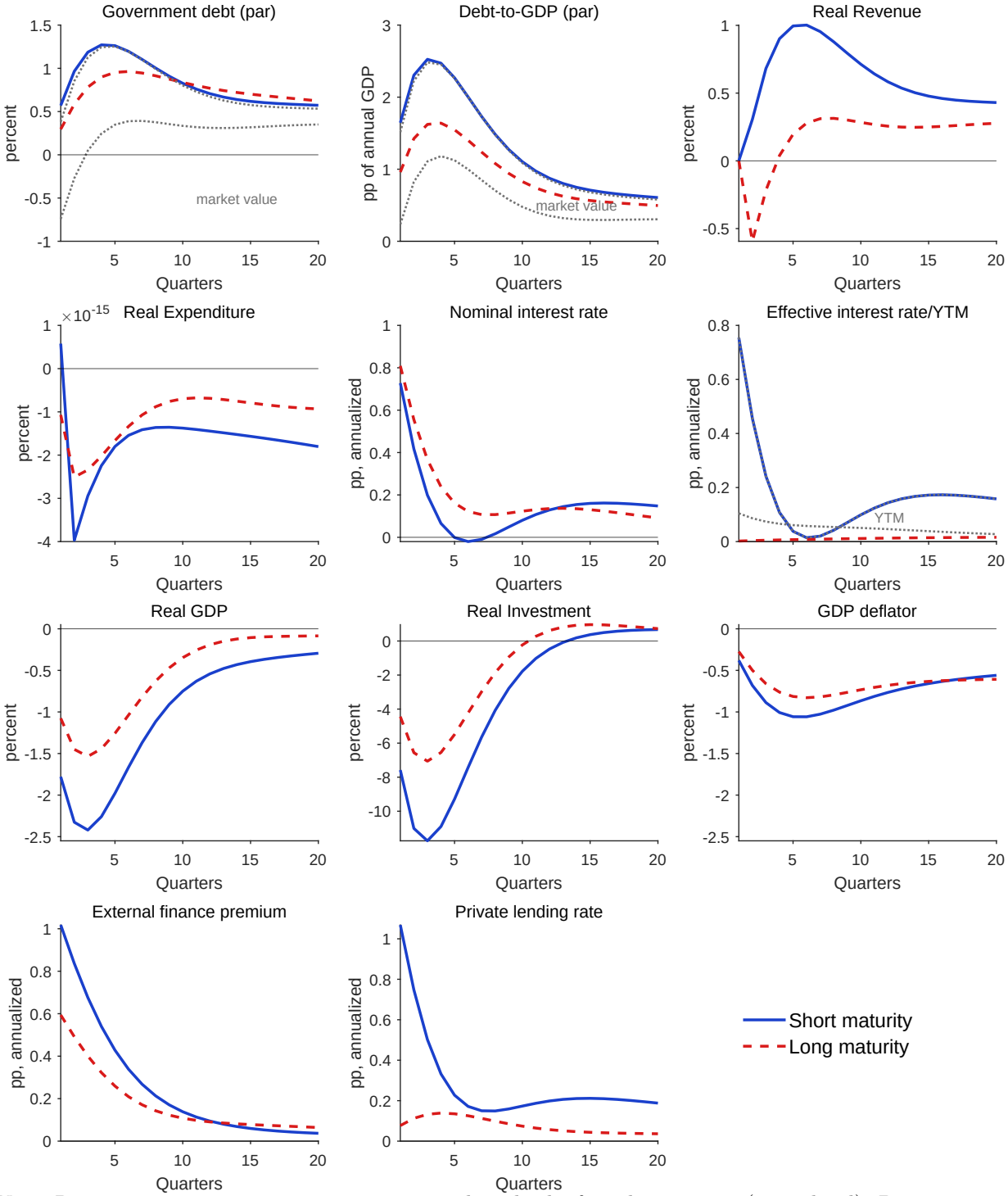
FIGURE A141: EFFECTS OF MONETARY POLICY SHOCKS: NO DEBT-SUPPLY WEDGE



Note: Responses to a contractionary monetary policy shock of 100 basis points (annualized). Price response is the cumulated response of inflation. Debt is at par value; the dotted line is market value. YTM: Yield to maturity.

B.6 No Fiscal Reactions: Tax/Spending

FIGURE A142: EFFECTS OF MONETARY POLICY SHOCKS: EXO. TAX/SPENDING



Note: Responses to a contractionary monetary policy shock of 100 basis points (annualized). Price response is the cumulated response of inflation. Debt is at par value; the dotted line is market value. YTM: Yield to maturity.



PUBLICATIONS

Fiscal-Monetary Interactions: The Role of Debt Maturity

Working Paper No. WP/2026/219